Into the Secret Garden

An Introduction to Schemes and Algebraic Geometry

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[...] *Oscar Zariski bewitched me. When he spoke the words "algebraic variety", there was a certain resonance in his voice that said distinctly that he was looking into a secret garden. I immediately wanted to be able to do this too. It led me to 25 years of struggling to make this world tangible and visible.*

David Mumford

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0 | Before we start

About this course

These notes supplement the 60-hour course (3 cycles for SISSA students, 48 compulsory hours for UNITS master's students) *Algebraic Geometry* (533SM).

The exam consists of an oral presentation by the student about a topic mutually agreed upon by the student and the instructor, plus a few questions regarding the material covered in the course.

For UNITS students: the exam can only be scheduled within the official exam session, see the academic calendar.

Prerequisites

Familiarity with basic theory of commutative rings and modules is of great help, but not necessary. The relevant notions will be recalled as we need them. We have, however, included Appendix B to cover the basic commutative algebra constructions we will be using (and much more), and Appendix A to cover the basics of category theory as well. Category theory is, up to a certain point, used mainly as a tool, or rather a language, and is "invoked" in many proofs that would otherwise be too verbose or, as a matter of fact, obscure. The reader is, however, encouraged to *not* look at category theory as a black box, but rather embrace it as part of the language that shapes modern Algebraic Geometry.

Conventions

We list here a series of conventions that will be used throughout this text.

- The axiom of choice (or Zorn's Lemma) is assumed; so, for instance, every nonzero ring has a maximal ideal, and a poset $(P, \leqslant)$ in which every chain has an upper bound admits a maximal element.
- Given two sets A and B, the phrase '$A \subset B$' means that A is contained in B, *possibly equal* to B.

- We write '$A \simeq B$' to indicate a bijection of sets. We reserve '$A \cong B$' for isomorphisms involving objects with some structure, and '$A = B$' for straight equality or — occasionally — a canonical isomorphism that should be clear from the context.
- A *ring* is a commutative, unitary ring. The zero ring (the one where $1 = 0$) is allowed (and in fact needed), but we always assume our rings are nonzero unless we explicitly mention it. Ring homomorphisms preserve the identity.
- Some results will be valid over an arbitrary field, some over an algebraically closed one. Therefore, we will make the following notational distinction:

$$\mathbb{F} = \text{an arbitrary field}, \qquad \mathbf{k} = \text{an algebraically closed field}.$$

- An open cover of a topological space U is the datum of a set I, and an open subset $U_i \subset U$ for every $i \in I$, such that $U = \bigcup_{i \in I} U_i$. We set $U_{ij} = U_i \cap U_j$. If $I = \emptyset$, then $U = \emptyset$.
- To say that x is an object of a category $\mathscr{C}$ we simply write '$x \in \mathscr{C}$' instead of $x \in \mathrm{Ob}(\mathscr{C})$.

Proofs

This text is based on the following principle. Newcomers to algebraic geometers need to (1) learn the basics, and (2) learn the main techniques. For this reason, the first chapters, building up the basic material from scratch, include proofs for essentially *all* the statements presented. In later chapters, some proofs are replaced by appropriate references. This happens especially when the main argument was already given in a similar form for an earlier result, or when it is not harmful for pedagogical purposes to just refer the reader to one of the excellent references that are available.

Exercises

This text contains *lots* of exercises, of varying difficulty. They are marked with the symbol



and, as the reader will notice, they are integrated within the main body of the text, and not confined to the end of every chapter. This choice may seem disadvantageous, but the reader should not see the exercises as something "interrupting the flow" of the exposition or, more realistically, breaking the author's delirious stream of consciousness made of definitions and theorems coming non-stop one after the other. The exercises are "put there" to suggest what students *can already solve* on their own at any given stage of the journey, having learnt only what comes earlier in the text.

Finally, and this might seem confusing at first, the author would like to remind all students that the key point of the exercises is not quite to *solve* them (although this would be spectacular), but mainly *work on* them — on as many as possible — possibly in front of a blackboard or with a piece of paper and *together with other human beings* (not machines, please!).

Warnings

This text also contains *lots* of warning signs. They are marked with the symbol



and they mean to indicate the subtleties of the theory. Such a sign means: stop for a second and check your intuition! The reader is encouraged to think about these subtleties and form their own examples and counterexamples to what might have seemed, at first sight, something completely innocent and intuitive, but turned out false for some reason.

References

In tennis, the *Open Era* is the period beginning in 1968, and continuing to the present day, in which both amateurs and professionals are allowed to compete in all tournaments. Somewhat counterintuitively, it was the professionals that were "invited" to the amateurs' land, not the other way around.

Also the subject of Algebraic Geometry is now living its own *Open Era* which, remarkably, also started in the late 60s! That is the time when Grothendieck and Dieudonné published the various *tomes* of foundations of Algebraic Geometry, namely the much celebrated *Éléments de Géométrie Algébrique* (EGA), written in the span of 1960 through 1967. This is and always will be *the Bible* of Algebraic Geometry.

However, the first book on the subject with a clear intent of spreading algebraic geometry to a wide community in a self-contained, general-enough-but-not-overwhelming way, is Hartshorne's *Algebraic geometry,* appeared in 1977 [4]. And that was just the beginning.

The present text would not exist without the many books on Algebraic Geometry that have been written since the start of Algebraic Geometry's Open Era. It would not exist without the things I learnt from my teachers and from my collaborators.

We list below just four bibliographical references that supplement these notes. Each has influenced, in some way more or less visible, the author's view of the subject. But there are many more available nowadays, that we cite here and there in this text: the newcomer is encouraged to read everything!

- Q. Liu, *Algebraic geometry and arithmetic curves* [5],
- R. Hartshorne, *Algebraic geometry* [4],
- R. Vakil, *The rising sea* [12],
- D. Eisenbud, *Commutative Algebra: With a View Toward Algebraic Geometry* [2],

A final remark: the reader will notice that many times references, especially in later chapters, point to *The stacks project* [10]. We believe it is a wonderful, constantly evolving reference for the foundational material that "everybody uses" in their work of professional algebraic geometers; also, it has the advantage over Grothendieck's EGA of being more easily navigable, thanks to the efforts of its Creators. We encourage every student to get familiar with *The stacks project*, because that is now becoming more than a tool: it is becoming *The Bible 2.0*.

Caveat

These notes are written by a human. As such, they are imperfect, at best. You are kindly encouraged to let the author know if you find mistakes, inaccuracies, typos.

1 | Sheaves and $\mathscr{O}$-modules

The notion of a sheaf originates in work of Leray, developed in part during his years as a prisoner in Austria during World War II.

Sheaves are a key notion present in the toolbox of every mathematician keen to understand the "nature" of a *geometric space*. They incarnate one of the basic principles that will be unraveled throughout this course, which can be stated as the slogan

geometric spaces are determined by functions on them.

In algebraic geometry, a space is studied together with its functions on every open subset: the resulting sheaf of functions is part of the structure of the space itself.

1.1 Key example: smooth functions

Before diving into precise definitions, we explore a key, well-known example of a sheaf.

Let X be a smooth manifold. For each open subset $U \subset X$, we have a ring (actually, an $\mathbb{R}$-algebra)

$$C^\infty(U,\mathbb{R}) = \{\text{ smooth functions } U \to \mathbb{R}\}.$$

Indeed, smooth functions with the same source can naturally be added and multiplied pointwise using the ring structure on $\mathbb{R}$. If $V \hookrightarrow U$ is an open subset, we have a restriction map

$$C^\infty(U,\mathbb{R}) \xrightarrow{\rho_{UV}} C^\infty(V,\mathbb{R}), \quad f \mapsto f|_V,$$

which is an $\mathbb{R}$-algebra homomorphism, and if $W \hookrightarrow V \hookrightarrow U$ is a chain of open subsets of X, we have a commutative diagram

$$\begin{array}{ccccc} C^\infty(U,\mathbb{R}) & \xrightarrow{\rho_{UV}} & C^\infty(V,\mathbb{R}) & \xrightarrow{\rho_{VW}} & C^\infty(W,\mathbb{R}). \\ & \searrow^{\rho_{UW}} & & \nearrow & \end{array}$$

So far, we have just observed that the assignment $U \mapsto C^\infty(U,\mathbb{R})$ is (contravariantly) *functorial* in U, i.e. it defines a functor from the opposite category of the category of open subsets of X to the category of $\mathbb{R}$-algebras. The two distinguished features of the assignment $U \mapsto C^\infty(U,\mathbb{R})$, which make it into a *sheaf* of $\mathbb{R}$-algebras on X, are the following:

(i) Fix an open subset $U \subset X$ and an open cover $U = \bigcup_{i\in I} U_i$. If $f, g \in C^\infty(U, \mathbb{R})$ are smooth functions such that $f|_{U_i} = g|_{U_i}$ for every $i \in I$, then $f = g$. In other words, a smooth function is determined by its restriction to the open subsets forming a covering. This is the *locality axiom*.

(ii) Fix an open subset $U \subset X$ and an open cover $U = \bigcup_{i\in I} U_i$. Given a smooth function $f_i \in C^\infty(U_i, \mathbb{R})$ on each U_i, such that $f_i|_{U_i\cap U_j} = f_j|_{U_i\cap U_j}$ for every $(i, j) \in I \times I$, there is a smooth function $f \in C^\infty(U, \mathbb{R})$ such that $f_i = f|_{U_i}$ for every $i \in I$. In other words, functions glue along an open cover. This is the *gluing axiom*.

A sheaf is an abstract notion formalising this "ability to glue local data". The formal definition will be given in Important Definition 1.2.0.5. Note that the result of the gluing in Condition (ii) is *unique* by Condition (i).

Let us continue with our example. Let $x \in X$ be a point. Consider the ring

$$C^\infty_{X,x} = \left\{ (U, f) \,\middle|\, x \in U,\, f \in C^\infty(U, \mathbb{R}) \right\} \Big/ \sim$$

where $(U, f) \sim (V, g)$ whenever there exists an open subset $W \subset U \cap V$, containing x, such that $f|_W = g|_W$. Note that $C^\infty_{X,x}$ is indeed a ring, with addition and multiplication

$$[U, f] + [U', f'] = [U \cap U', f + f']$$
$$[U, f] \cdot [U', f'] = [U \cap U', f f'].$$

This ring, which is in fact an $\mathbb{R}$-algebra via $c \mapsto [X, c]$, for all $c \in \mathbb{R}$, is called the *stalk* of the sheaf $C^\infty(-, \mathbb{R})$ at x (cf. Important Definition 1.4.0.1), and it receives a natural $\mathbb{R}$-algebra homomorphism from $C^\infty(U, \mathbb{R})$ for every open subset U of X such that $x \in U$, sending $f \mapsto [U, f]$. The image of f along this map is called the *germ of f at x*. The subset

$$\mathfrak{m}_x = \left\{ [U, f] \in C^\infty_{X,x} \,\middle|\, f(x) = 0 \right\} \subset C^\infty_{X,x}$$

forms an ideal, which is a *maximal* ideal, being the kernel of the (surjective) evaluation map

$$C^\infty_{X,x} \longrightarrow\!\!\!\!\!\rightarrow \mathbb{R}$$
$$[U, f] \longmapsto f(x).$$

In fact, $\mathfrak{m}_x$ is the *unique* maximal ideal of $C^\infty_{X,x}$. To see this, it is enough to check that every element of $C^\infty_{X,x} \setminus \mathfrak{m}_x$ is invertible. But this is true, since a smooth function that is nonzero in a neighbourhood of x is invertible there.

The upshot is, then, that the pair $(C^\infty_{X,x}, \mathfrak{m}_x)$ defines a *local ring* with residue field $\mathbb{R}$. The geometric spaces X one deals with in algebraic geometry, namely *schemes*, have precisely this property: they come with a sheaf of rings $\mathscr{O}_X$ such that each stalk $\mathscr{O}_{X,x}$ is a local ring. These spaces $(X, \mathscr{O}_X)$ actually form a larger category, that of *locally ringed spaces* (cf. Section 1.11). Schemes are particular instances of locally ringed spaces.

1.2 Presheaves, sheaves, morphisms

Let $\mathscr{C}$ be a concrete category (Definition A.3.0.10) with a final object $0 \in \mathscr{C}$. The concreteness assumption means that part of the structure is the datum of a faithful functor $F\colon \mathscr{C} \to \mathsf{Sets}$. A further assumption we make is that F sends the final object 0 to a singleton (i.e. a final object) in Sets. This just means that the final object has a one-element underlying set. To fix ideas, $\mathscr{C}$ should be thought of as any of the following familiar categories:

- $\mathscr{C} = \mathsf{Sets}$,
- $\mathscr{C} = \mathsf{Rings}$,
- $\mathscr{C} = \mathsf{Ab} = \mathsf{Mod}_{\mathbb{Z}}$,
- $\mathscr{C} = \mathsf{Mod}_R$, where R is a ring.

If X is a topological space, we denote by τ_X the category of open subsets of X. The set $\mathrm{Hom}_{\tau_X}(V,U)$ between two open sets $V, U \subset X$ is just the empty set if $V \not\subset U$, or the singleton $\{\, V \hookrightarrow U \,\}$ in case V is contained in U. Thus the opposite category τ_X^{op} satisfies

$$\mathrm{Hom}_{\tau_X^{\mathrm{op}}}(U,V) = \begin{cases} \{\, V \hookrightarrow U \,\} & \text{if } V \subset U \\ \emptyset & \text{if } V \not\subset U \end{cases}$$

and a functor $\mathcal{F}\colon \tau_X^{\mathrm{op}} \to \mathscr{C}$ (i.e. a contravariant functor $\tau_X \to \mathscr{C}$) determines a map

$$\mathrm{Hom}_{\tau_X^{\mathrm{op}}}(U,V) \longrightarrow \mathrm{Hom}_{\mathscr{C}}(\mathcal{F}(U), \mathcal{F}(V)),$$

which is nothing but a choice of an element $\rho_{UV} \in \mathrm{Hom}_{\mathscr{C}}(\mathcal{F}(U), \mathcal{F}(V))$ for any inclusion of open subsets $V \subset U$.

Definition 1.2.0.1 (Presheaf, take I)**.** A *presheaf* on a topological space X, with values in $\mathscr{C}$, is a contravariant functor $\mathcal{F}$ from τ_X to $\mathscr{C}$, i.e. an object of the functor category $\mathsf{Fun}(\tau_X^{\mathrm{op}}, \mathscr{C})$.

For those who do not like the categorical definition, here is an equivalent definition, which just unravels the definition of a functor (cf. Definition A.1.0.9).

Definition 1.2.0.2 (Presheaf, take II)**.** A *presheaf* on a topological space X, with values in $\mathscr{C}$, is the assignment $U \mapsto \mathcal{F}(U)$ of an object $\mathcal{F}(U) \in \mathscr{C}$ for each open subset $U \subset X$, and of a morphism $\rho_{UV}\colon \mathcal{F}(U) \to \mathcal{F}(V)$ in $\mathscr{C}$ for each inclusion $V \hookrightarrow U$, such that

(1) $\rho_{UU} = \mathrm{id}_{\mathcal{F}(U)}$ for every $U \in \tau_X$, and

(2) $\rho_{UW} = \rho_{VW} \circ \rho_{UV}$ for every chain of inclusions $W \hookrightarrow V \hookrightarrow U$.

Terminology 1.2.0.3. Elements of $\mathcal{F}(U)$ are often called 'sections of $\mathcal{F}$ over U', or (somewhat more vaguely) 'local sections' when $U \subsetneq X$. Elements of $\mathcal{F}(X)$ are called 'global sections', or just 'sections'. Possible alternative notations for $\mathcal{F}(U)$ are $\Gamma(U,\mathcal{F})$ and $\mathrm{H}^0(U,\mathcal{F})$. The maps ρ_{UV} are often called 'restriction maps' (from U to V, the larger set being U).

Notation 1.2.0.4. Motivated by Terminology 1.2.0.3, we shall often write $s|_V$ for the image of a section $s \in \mathcal{F}(U)$ along the restriction map ρ_{UV}.

Important Definition 1.2.0.5 (Sheaf, take I)**.** A *sheaf* on a topological space X, with values in $\mathscr{C}$, is a presheaf $\mathcal{F}$ such that the following two conditions hold:

(loc) Fix an open subset $U \subset X$, an open cover $U = \bigcup_{i\in I} U_i$, and two sections $s, t \in \mathcal{F}(U)$ satisfying $s|_{U_i} = t|_{U_i}$ for all $i \in I$. Then $s = t$.

(glue) Fix an open subset $U \subset X$, an open cover $U = \bigcup_{i\in I} U_i$ and a tuple $(s_i)_{i\in I}$ of sections $s_i \in \mathcal{F}(U_i)$ such that $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$ for all $(i,j) \in I\times I$. Then there exists a section $s \in \mathcal{F}(U)$ such that $s_i = s|_{U_i}$.

Conditions (loc) and (glue) generalise the conditions (i) and (ii), respectively, anticipated with the example $\mathcal{F} = C^\infty(-,\mathbb{R})$ in Section 1.1.

Terminology 1.2.0.6. A presheaf $\mathcal{F}$ is called *separated* if Condition (loc) holds. Sometimes this condition is called *locality axiom*. Condition (glue), on the other hand, is called the *gluing condition* (or *gluing axiom*).

Remark 1.2.0.7. Let $\mathcal{F}$ be a sheaf. Then, the section $s \in \mathcal{F}(U)$ in the gluing condition (glue) is necessarily unique because $\mathcal{F}$ is separated. In fact, the two sheaf conditions could be replaced by a single condition, identical to (glue), but imposing uniqueness of s.

Example 1.2.0.8 (Trivial sheaf)**.** Recall that we assumed $\mathscr{C}$ to admit a final object $0 \in \mathscr{C}$ and the faithful functor $F\colon \mathscr{C} \to \mathsf{Sets}$ to send 0 to a singleton. Under these assumptions, the presheaf defined by $U \mapsto 0$ for every U is a sheaf and is called the *trivial sheaf* (or sometimes the *zero sheaf*). It is simply denoted by '0'.

Example 1.2.0.9 (Restriction to an open)**.** Let $U \subset X$ be an open subset, $\mathcal{F}$ a presheaf on X. Then, setting $\mathcal{F}|_U(V) = \mathcal{F}(V)$ for V an open subset of U, defines a presheaf $\mathcal{F}|_U$ on U, which is a sheaf as soon as $\mathcal{F}$ is. It is called the *restriction of $\mathcal{F}$ to U*.

Definition 1.2.0.10 (Morphism of (pre)sheaves)**.** Let X be a topological space. A *morphism* between two presheaves $\mathcal{F},\mathcal{G}$ on X is a natural transformation $\eta\colon \mathcal{F} \Rightarrow \mathcal{G}$, i.e. a morphism in the functor category $\mathsf{Fun}(\tau_X^{\mathrm{op}}, \mathscr{C})$. A morphism of sheaves is just a morphism between the underlying presheaves.

Let us unravel the definition of natural transformation (cf. Definition A.3.0.1), to translate Definition 1.2.0.10 in naive terms.

To give a morphism of (pre)sheaves, one has to assign a homomorphism

$$\eta_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$$

in $\mathscr{C}$ to each $U \in \tau_X$, such that for every inclusion $V \hookrightarrow U$ of open subsets of X, the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\eta_U} & \mathcal{G}(U) \\ {\scriptstyle \rho^{\mathcal{F}}_{UV}}\downarrow & & \downarrow{\scriptstyle \rho^{\mathcal{G}}_{UV}} \\ \mathcal{F}(V) & \xrightarrow{\eta_V} & \mathcal{G}(V) \end{array} \tag{1.2.1}$$

commutes. For the sake of clarity, we have emphasised the relevant sheaf in the restriction maps notation, but we will not be doing that systematically.

Notation 1.2.0.11. It is clear that presheaves on X with values in $\mathscr{C}$ form a category $\mathsf{pSh}(X, \mathscr{C})$, tautologically defined as

$$\mathsf{pSh}(X, \mathscr{C}) \coloneqq \mathsf{Fun}(\tau_X^{\mathrm{op}}, \mathscr{C}).$$

By Definition 1.2.0.10, sheaves form a full subcategory, denoted $\mathsf{Sh}(X, \mathscr{C})$. We denote by

$$j_{X,\mathscr{C}} : \mathsf{Sh}(X, \mathscr{C}) \hookrightarrow \mathsf{pSh}(X, \mathscr{C}) \tag{1.2.2}$$

the (fully faithful) inclusion functor.

An isomorphism of (pre)sheaves is an isomorphism in $\mathsf{pSh}(X, \mathscr{C})$, i.e. a *natural isomorphism*, i.e. a natural transformation $\eta : \mathcal{F} \Rightarrow \mathcal{G}$ such that η_U is an isomorphism in $\mathscr{C}$ for every $U \in \tau_X$ (cf. Definition A.3.0.2).

Notation 1.2.0.12. Since (pre)sheaves form a genuine category, from now on we shall use the classical arrow notation '$\mathcal{F} \to \mathcal{G}$' (instead of $\mathcal{F} \Rightarrow \mathcal{G}$) to denote a morphism of (pre)sheaves.

The following definition makes sense, because $\mathscr{C}$ is assumed to be a concrete category.

Definition 1.2.0.13 (Injective map of presheaves)**.** A morphism of (pre)sheaves $\eta : \mathcal{F} \to \mathcal{G}$ is *injective* if η_U is injective for every U. We denote this by writing η as '$\mathcal{F} \hookrightarrow \mathcal{G}$'.

We close this section with a few examples and exercises.

Example 1.2.0.14 (Smooth functions)**.** Let X be a smooth manifold. Then, sending $U \subset X$ to the set $C^\infty(U, \mathbb{R})$ of smooth functions $U \to \mathbb{R}$, defines a sheaf $C^\infty(-, \mathbb{R})$ with values in the category of $\mathbb{R}$-algebras.

Example 1.2.0.15 (Holomorphic functions). Let X be a complex manifold. Then, sending $U \subset X$ to the set $\mathscr{O}_X^{\mathrm{h}}(U)$ of holomorphic functions on U, defines a sheaf $\mathscr{O}_X^{\mathrm{h}}$ with values in the category of $\mathbb{C}$-algebras. Sending U to the set $\mathscr{O}_X^{\mathrm{h},\times}(U)$ of nowhere-zero holomorphic functions defines a sheaf of abelian groups on X (the group structure being given by pointwise multiplication of functions).

Example 1.2.0.16 (Continuous functions are a sheaf). Let X, Y be topological spaces. For $U \subset X$ open, define

$$\mathcal{F}(U) = \{\text{ continuous functions } U \to Y\}.$$

Then $\mathcal{F}$ is a sheaf of sets on X.

Example 1.2.0.17 (Separated presheaf, not a sheaf, take I). Set $X = \mathbb{C}$. Then, sending $U \subset X$ to the subset

$$\mathcal{F}(U) = \left\{ f \in \mathscr{O}_X^{\mathrm{h}}(U) \,\middle|\, f = g^2 \text{ for some } g \in \mathscr{O}_X^{\mathrm{h}}(U) \right\}$$

defines a (separated) presheaf. However, $\mathcal{F}$ is not a sheaf: the function $f(z) = z$ on the annulus

$$U_\varepsilon = \{ z \in \mathbb{C} \mid 1 - \varepsilon < |z| < 1 + \varepsilon \} \subset \mathbb{C},$$

where $0 < \varepsilon < 1$, has a square root in any neighbourhood of any point $x \in U_\varepsilon$, but there is no global $\sqrt{z}$ defined on the whole of U_ε, so Condition (glue) fails.



Exercise 1.2.0.18 (Separated presheaf, not a sheaf, take II). Let $X = \mathbb{R}$, with the standard topology. Show that

$$U \longmapsto \mathrm{B}(U) = \{\text{ bounded continuous functions } U \to \mathbb{R}\}$$

is a separated presheaf on X, but not a sheaf (i.e. Condition (glue) fails).

Example 1.2.0.19 (Constant presheaf). Work with $\mathscr{C} = \mathsf{Ab} = \mathsf{Mod}_{\mathbb{Z}}$, the category of abelian groups, and fix $G \neq 0$ in this category. Fix a topological space X, and define

$$\underline{G}_X^{\mathrm{pre}}(U) = \begin{cases} G & \text{if } U \neq \emptyset, \\ 0 & \text{if } U = \emptyset. \end{cases}$$

As for the restriction maps, set $\rho_{UV} = \mathrm{id}_G$ if both U and V are nonempty. This is a presheaf, which happens to be a sheaf only in precise circumstances (cf. Exercise 1.2.0.21). For instance, suppose $X = U_1 \amalg U_2$ is a disjoint union of two nonempty open subsets. Then $\underline{G}_X^{\mathrm{pre}}(X) = G = \underline{G}_X^{\mathrm{pre}}(U_i)$ for $i = 1, 2$. Now, $X = U_1 \amalg U_2$ is an open cover. Pick two *distinct* sections $s_i \in G = \underline{G}_X^{\mathrm{pre}}(U_i)$ for $i = 1, 2$. Then, $s_1|_{U_1 \cap U_2} = s_1|_\emptyset = 0 = s_2|_\emptyset = s_2|_{U_1 \cap U_2}$, but there is no section $s \in \underline{G}_X^{\mathrm{pre}}(X) = G$ such that $s|_{U_i} = s_i$ since $\rho_{XU_i} = \mathrm{id}_G$ for $i = 1, 2$ and $s_1 \neq s_2$ by assumption. Hence Condition (glue), i.e. the gluing axiom, fails (whereas Condition (loc) is trivially satisfied). We will see in Example 1.5.0.3 that $\underline{G}_X^{\mathrm{pre}}$ can be "transformed" into a sheaf by a canonical procedure.



Exercise 1.2.0.20. Provide examples of presheaves which satisfy the gluing axiom but not the separation axiom. (**Hint**: Try a presheaf which has two global sections that become equal on every proper open subset).



Exercise 1.2.0.21. Show that the constant presheaf $\underline{G}_X^{\mathrm{pre}}$ of Example 1.2.0.19 is a sheaf if and only if every nonempty open subset $U \subset X$ is connected.



Exercise 1.2.0.22 (Presheaves kernel and cokernel). Let $\mathscr{C} = \mathsf{Mod}_A$, so that every morphism has a kernel and a cokernel. Let $\eta\colon \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves with values in $\mathscr{C}$. Consider the assignments

$$U \longmapsto (\ker_{\mathrm{pre}} \eta)(U) = \ker(\eta_U),$$

$$U \longmapsto (\mathrm{coker}_{\mathrm{pre}} \eta)(U) = \mathrm{coker}(\eta_U).$$

Prove the following assertions.

(i) Both $\ker_{\mathrm{pre}} \eta$ and $\mathrm{coker}_{\mathrm{pre}} \eta$ are presheaves.

(ii) There is a canonical morphism $\ker_{\mathrm{pre}} \eta \to \mathcal{F}$ (resp. $\mathcal{G} \to \mathrm{coker}_{\mathrm{pre}} \eta$), and it satisfies the universal property of the kernel (resp. the cokernel) in $\mathsf{pSh}(X, \mathscr{C})$.

(iii) The presheaf $\ker_{\mathrm{pre}} \eta$ is in fact a sheaf, denoted $\ker(\eta)$, as soon as η is a morphism of *sheaves*.

(iv) If η is a morphism of sheaves, then $\ker(\eta) \hookrightarrow \mathcal{F}$ satisfies the universal property of the kernel in $\mathsf{Sh}(X, \mathscr{C})$, and η is injective if and only if $\ker(\eta) = 0$.[1]

Example 1.2.0.23 ($\mathrm{coker}_{\mathrm{pre}} \eta$ may not be a sheaf). Let $X = \mathbb{C}$ and $\mathscr{C} = \mathsf{Ab}$. Consider the morphism of sheaves

$$\mathscr{O}_X^{\mathrm{h}} \xrightarrow{\exp} \mathscr{O}_X^{\mathrm{h},\times}, \quad f \mapsto \exp(f),$$

where $\mathscr{O}_X^{\mathrm{h},\times}$ is the sheaf of nowhere-zero holomorphic functions on X (cf. Example 1.2.0.15). We have that the open subset $U = X \setminus \{0\} \subset X$ is covered by the two open subsets

$$U_1 = X \setminus [0, +\infty) \subset X, \quad U_2 = X \setminus (-\infty, 0] \subset X.$$

The function $g(z) = z$ viewed in $\mathscr{O}_X^{\mathrm{h},\times}(U)$ is not of the form $\exp(f)$ for any $f \in \mathscr{O}_X^{\mathrm{h}}(U)$. Thus the image $\widetilde{g}$ of g along

$$\mathscr{O}_X^{\mathrm{h},\times}(U) \longrightarrow \mathrm{coker}_{\mathrm{pre}}(\exp)(U)$$

is nonzero. However, U_1 and U_2 are simply connected, thus every function $h_i \in \mathscr{O}_X^{\mathrm{h},\times}(U_i)$ is of the form $\exp(f_i)$ for some $f_i \in \mathscr{O}_X^{\mathrm{h}}(U_i)$. Thus $\mathrm{coker}_{\mathrm{pre}}(\exp)(U_i) = 0$ for $i = 1, 2$. In particular, the restrictions $g|_{U_i}$ have this property, namely they go (necessarily) to 0 in $\mathrm{coker}_{\mathrm{pre}}(\exp)(U_i)$. If $\mathrm{coker}_{\mathrm{pre}}(\exp)$ were a sheaf, the gluing axiom would force $\widetilde{g} = 0$, which is not true.

[1] See Exercise 1.6.0.6 for the corresponding property of cokernels.

1.3 The sheaf condition via equalisers

We now present an alternative way to define sheaves. We will repeatedly use this reinterpretation throughout these notes.

Let $\mathscr{C}$ be a category with limits (cf. Definition A.5.0.1). In particular, $\mathscr{C}$ has products, equalisers, and a final object (cf. Appendix A.5 for full details). The reader may imagine $\mathscr{C}$ to be, for instance, any of the following categories: sets, groups, rings, algebras over a fixed ring, modules over a fixed ring. Fix a presheaf $\mathcal{F}$ with values in $\mathscr{C}$ on a topological space X. Let $\{U_i\}_{i\in I}$ be a family of open subsets of X, and set $U = \bigcup_{i\in I} U_i$. Then, by our assumption on $\mathscr{C}$, one can consider the map

$$\mathcal{F}(U) \xrightarrow{\rho} \prod_{i\in I} \mathcal{F}(U_i), \quad s \mapsto (s|_{U_i})_{i\in I},$$

as well as the family of maps

$$\mu_{ij}\colon \prod_{i\in I} \mathcal{F}(U_i) \to \mathcal{F}(U_i) \to \mathcal{F}(U_i \cap U_j), \qquad (s_i)_{i\in I} \mapsto s_i|_{U_i\cap U_j}$$
$$\nu_{ij}\colon \prod_{i\in I} \mathcal{F}(U_i) \to \mathcal{F}(U_j) \to \mathcal{F}(U_i \cap U_j), \qquad (s_i)_{i\in I} \mapsto s_j|_{U_i\cap U_j}$$

which, taking products over $(i,j) \in I \times I$, can be assembled into two maps

$$\prod_{i\in I} \mathcal{F}(U_i) \underset{\nu}{\overset{\mu}{\rightrightarrows}} \prod_{(i,j)\in I\times I} \mathcal{F}(U_i \cap U_j).$$

Definition 1.3.0.1 (Sheaf, take II)**.** Let $\mathscr{C}$ be a category with limits, and let X be a topological space. A presheaf $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ is a *sheaf* if for every family of open subsets $\{U_i\}_{i\in I}$, with $U = \bigcup_{i\in I} U_i$, the diagram

$$\mathcal{F}(U) \xrightarrow{\rho} \prod_{i\in I} \mathcal{F}(U_i) \underset{\nu}{\overset{\mu}{\rightrightarrows}} \prod_{(i,j)\in I\times I} \mathcal{F}(U_i \cap U_j)$$

is an equaliser diagram in $\mathscr{C}$.

Informally, being an equaliser means that ρ is injective and its image agrees with the set of tuples $(s_i)_{i\in I}$ such that $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$ for all pairs (i,j).

Note that Definition 1.3.0.1 is *element-free*. However, let us check that it agrees with Important Definition 1.2.0.5 when $\mathscr{C}$ is concrete: in this case the injectivity of ρ, implied by the equaliser condition, coincides with separatedness; the fact that the set-theoretic image of ρ coincides with the collection of tuples of sections $(s_i)_{i\in I}$ such that $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$ is precisely the gluing condition.

Remark 1.3.0.2. Let $\mathcal{F}$ be a sheaf. Then, one has $\mathcal{F}(\emptyset) \cong 0$ canonically, where 0 is the final object in $\mathscr{C}$. Thus from now on we will simply write $\mathcal{F}(\emptyset) = 0$ for a sheaf $\mathcal{F}$. Note that this is sometimes listed as an axiom defining a (pre)sheaf, but it does in fact follow from our assumptions (cf. Exercise A.5.0.2).

Example 1.3.0.3. Let $\mathcal{F}$ be a sheaf on X. If $U = \coprod_{i\in I} U_i$ is a *disjoint* union of open subsets $U_i \subset U$, then ρ is an isomorphism, i.e. $s \mapsto (s|_{U_i})_{i\in I}$ defines an isomorphism

$$\rho : \mathcal{F}(U) \xrightarrow{\sim} \prod_{i\in I} \mathcal{F}(U_i).$$

Example 1.3.0.4. Let $\mathscr{C} = \mathsf{Mod}_R$.[2] Then a presheaf $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ is a sheaf if and only if for every family of open subsets $\{U_i\}_{i\in I}$, with $U = \bigcup_{i\in I} U_i$, the sequence

$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{\rho} \prod_{i\in I} \mathcal{F}(U_i) \xrightarrow{\mu-\nu} \prod_{(i,j)\in I\times I} \mathcal{F}(U_i \cap U_j)$$

is exact, where the map denoted $\mu - \nu$ sends $(s_i)_{i\in I} \mapsto (s_i|_{U_i\cap U_j} - s_j|_{U_i\cap U_j})_{i,j}$.

The following lemma applies, for instance, to categories of groups, rings, algebras over a ring, and modules over a ring. It allows one to check the sheaf condition of Definition 1.3.0.1 in the category of sets.

LEMMA 1.3.0.5 ([10, Tag 0073]). *Let $\mathscr{C}$ be a category, $F : \mathscr{C} \to \mathsf{Sets}$ a faithful functor such that $\mathscr{C}$ has limits and F commutes with them. Assume that F reflects isomorphisms. Then a presheaf $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ is a sheaf if and only if the underlying presheaf of sets $F \circ \mathcal{F} : \tau_X^{\mathrm{op}} \to \mathsf{Sets}$ is a sheaf.*

At the beginning of this chapter we have defined (pre)sheaves of objects in an arbitrary concrete category $\mathscr{C}$. We still have to define a few things, though, e.g. stalks and sheafification. In order for everything to be well-defined and work well (but still be compatible with all we have discussed so far, including Definition 1.3.0.1), we need to add a few initial data. This is provided by the following definition.

Definition 1.3.0.6 ([10, Tag 007L])**.** A *type of algebraic structure* is a pair $(\mathscr{C}, F)$, where $\mathscr{C}$ is a category, $F : \mathscr{C} \to \mathsf{Sets}$ is a faithful functor, such that

(1) $\mathscr{C}$ has limits and F commutes with them,

(2) $\mathscr{C}$ has filtered colimits and F commutes with them,

(3) F reflects isomorphisms (i.e. F is *conservative*).

A few remarks are in order, before we go on.

- Equipping a category $\mathscr{C}$ with a faithful functor $F : \mathscr{C} \to \mathsf{Sets}$ is like saying that $\mathscr{C}$ is a *concrete category*, which we had already assumed in Section 1.2. Our earlier assumption that F preserves the final object is now implied by Condition (1).

[2] The example remains valid for *complete* abelian categories, i.e. abelian categories closed under arbitrary products.

- If we have a type of algebraic structure $(\mathscr{C}, F)$, then we can verify whether a presheaf is a sheaf in the category of sets, by Lemma 1.3.0.5.
- The condition that F be conservative implies that a bijective morphism in $\mathscr{C}$ is an isomorphism. This, as already mentioned, excludes categories such as the category of topological spaces.
- For every type of algebraic structure $(\mathscr{C}, F)$, one has the following properties:
  (i) $\mathscr{C}$ has a final object 0, and $F(0)$ is a final object in Sets (i.e. a singleton).
  (ii) $\mathscr{C}$ has products, fibre products, and equalisers — this follows from the examples in Appendix A.5. Moreover, F commutes with all of them.
- Examples of categories $\mathscr{C}$ having the additional structure of Definition 1.3.0.6 are:
  – monoids,
  – groups,
  – abelian groups,
  – rings,
  – modules over a ring.

  In all these cases, we take as the functor F the obvious forgetful functor. The reader is encouraged to just think of $\mathscr{C}$ as one of these familiar categories, and not bother too much about Definition 1.3.0.6. As a counterexample, however, consider the category Top of topological spaces: the forgetful functor to Sets exists but does not reflect isomorphisms (a continuous bijection need not be a homeomorphism).

1.4 Stalks, and what they (don't) tell us

Fix a type of algebraic structure $(\mathscr{C}, F\colon \mathscr{C} \to \mathsf{Sets})$ as in Definition 1.3.0.6. Let X be a topological space, $x \in X$ a point. The collection of open subsets $U \subset X$ containing x forms a directed system (the partial order $\succeq$ being the inclusion relation, i.e. $V \succeq U$ if and only if $V \subset U$). Indeed, given two open neighbourhoods U and V of x, there is always a third open neighbourhood of x contained in both U and V, namely $U \cap V$ or any smaller open subset containing x. In fancier language, the subcategory

$$\mathsf{Ngb}_x = \{\, U \in \tau_X \mid x \in U \,\}^{\mathrm{op}} \xhookrightarrow{\iota_x} \tau_X^{\mathrm{op}}$$

is a filtered category (see Definition A.5.0.10).

Important Definition 1.4.0.1 (Stalks)**.** Let $x \in X$ be a point, $\mathcal{F}$ a presheaf. The *stalk of* $\mathcal{F}$ *at* x is the filtered colimit

$$\mathcal{F}_x = \varinjlim_{\mathsf{Ngb}_x} \mathcal{F} \circ \iota_x = \varinjlim_{U \ni x} \mathcal{F}(U) \in \mathscr{C}.$$

Because F commutes with filtered colimits, the underlying set $F(\mathcal{F}_x)$, still denoted $\mathcal{F}_x$, is

$$\mathcal{F}_x = \{(U,s) \mid x \in U,\, s \in \mathcal{F}(U)\} / \sim$$

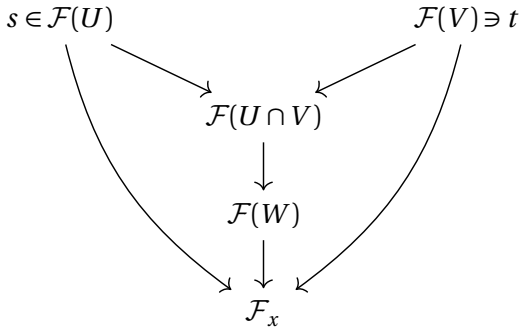
where $(U,s) \sim (V,t)$ whenever there is an open neighbourhood $W \subset U \cap V$ of x such that $s|_W = t|_W$. We denote by

$$s_x = [U,s] \in \mathcal{F}_x$$

the equivalence class of the pair (U,s). It is called the *germ of s at x*. By definition of direct limit, there are natural homomorphisms

$$\mathcal{F}(U) \longrightarrow \mathcal{F}_x, \quad s \mapsto s_x,$$

in $\mathscr{C}$, for every open neighbourhood U of x. The diagram



illustrates the fact that two sections $s \in \mathcal{F}(U)$ and $t \in \mathcal{F}(V)$ define the same element in the stalk $\mathcal{F}_x$ if and only if there is an intermediate open subset $W \subset U \cap V$ over which they agree.



Figure 1.1: A bunch of sheaves sitting in their natural habitat. The little tops of each leaf of corn are the stalks.

LEMMA 1.4.0.2. *If $\mathcal{F}$ is a separated presheaf of sets (e.g. a sheaf), then the natural map*

$$\sigma_U^{\mathcal{F}}\colon \mathcal{F}(U) \longrightarrow \prod_{x\in U} \mathcal{F}_x, \quad s \mapsto (s_x)_{x\in U} \tag{1.4.1}$$

is injective for every open subset U of X.

The lemma means, at an informal level, that sections are determined by their germs. In fact, one can prove that a presheaf $\mathcal{F}$ is separated if and only if the map $\sigma_U^{\mathcal{F}}$ is injective for all open subsets $U \subset X$.

Proof. If s and t are sections in $\mathcal{F}(U)$ such that $s_x = t_x$ in $\mathcal{F}_x$ for every $x \in U$, then for every $x \in U$ there is an open neighbourhood $U_x \subset U$ such that $s|_{U_x} = t|_{U_x}$. But this holds for every $x \in U$, and $U = \bigcup_{x\in U} U_x$ is an open covering, thus by the separation axiom we deduce $s = t$, i.e. $\sigma_U^{\mathcal{F}}$ is injective. □

Consider the following property of a tuple $(s_x)_{x\in U} \in \prod_{x\in U} \mathcal{F}_x$, for $U \subset X$ an open subset:

$$\begin{gathered} \text{for every } x \in U \text{ there exists a pair } (V_x, t^x), \\ \text{with } x \in V_x \subset U \text{ and } t^x \in \mathcal{F}(V_x), \\ \text{such that } t^x_y = s_y \text{ for all } y \in V_x. \end{gathered} \tag{1.4.2}$$

Definition 1.4.0.3 (Compatible germs)**.** Let $\mathcal{F}$ be a presheaf on X, and let $U \subset X$ be an open subset. We say that $(s_x)_{x\in U} \in \prod_{x\in U} \mathcal{F}_x$ is a *tuple of compatible germs* if Condition (1.4.2) is fulfilled.

We always have inclusions

$$\operatorname{im}(\sigma_U^{\mathcal{F}}) \subset \left\{ \text{tuples } (s_x)_{x\in U} \text{ of compatible germs} \right\} \subset \prod_{x\in U} \mathcal{F}_x \tag{1.4.3}$$

where the first inclusion is justified by taking $V_x = U$ and $t^x = s$ for every $x \in U$ as soon as $\sigma_U^{\mathcal{F}}(s) = (s_x)_{x\in U}$. If $\mathcal{F}$ is a sheaf, then tuples of compatible germs form precisely the image of the map (1.4.1), i.e. the first inclusion in (1.4.3) is an equality. Indeed, assume $(s_x)_{x\in U}$ consists of compatible germs. Let $\{(V_x, t^x) \mid x \in U\}$ be as in the displayed condition (1.4.2). By the compatibility condition, for every pair $(x, x') \in U \times U$ we have

$$t^x_y = t^{x'}_y, \quad y \in V_x \cap V_{x'}.$$

It follows from Lemma 1.4.0.2 that

$$t^x\big|_{V_x\cap V_{x'}} = t^{x'}\big|_{V_x\cap V_{x'}}. \tag{1.4.4}$$

Now, we have an open cover $U = \bigcup_{x\in U} V_x$, so by the gluing axiom, applicable by (1.4.4), the sections $t^x \in \mathcal{F}(V_x)$ glue to a (unique) section $t \in \mathcal{F}(U)$ such that $t|_{V_x} = t^x$. But $t^x_y = s_y$ for $y \in V_x$, and this holds for every $x \in U$, so $\sigma_U^{\mathcal{F}}(t) = (s_x)_{x\in U}$.

Summing up, when $\mathcal{F}$ is a sheaf, we have a *bijection*

$$\sigma_U^{\mathcal{F}}\colon \mathcal{F}(U) \xrightarrow{\sim} \{\text{tuples } (s_x)_{x\in U} \text{ of compatible germs}\}.$$

This also shows that *sections of a sheaf can always be identified with genuine functions!* Indeed, tuples $(s_x)_{x\in U}$ correspond to particular functions $U \to \coprod_{x\in U} \mathcal{F}_x$, sending $x \in U$ inside the corresponding stalk, and doing so in a compatible way.

LEMMA 1.4.0.4. *Let $s, t \in \mathcal{F}(X)$ be two global sections of a separated presheaf $\mathcal{F}$, such that $s_x = t_x \in \mathcal{F}_x$ for every $x \in X$. Then $s = t$.*

Proof. This is just a special case of Lemma 1.4.0.2. □



Exercise 1.4.0.5. Let $\mathcal{F}$ be a sheaf on X, and let $s, t \in \mathcal{F}(X)$ be two global sections. Show that

$$\{\, x \in X \mid s_x = t_x \,\} \subset X$$

is an open subset of X.

A morphism of presheaves $\eta\colon \mathcal{F} \to \mathcal{G}$ induces a morphism $\eta_x\colon \mathcal{F}_x \to \mathcal{G}_x$ at the level of stalks for every $x \in X$, defined by

$$s_x = [U, s] \longmapsto [U, \eta_U(s)] = (\eta_U(s))_x. \tag{1.4.5}$$



Exercise 1.4.0.6. Check that (1.4.5) is well-defined.

If $U \subset X$ is an open subset containing a point $x \in X$, then the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\eta_U} & \mathcal{G}(U) \\ \downarrow & & \downarrow \\ \mathcal{F}_x & \xrightarrow{\eta_x} & \mathcal{G}_x \end{array} \qquad \begin{array}{ccc} s & \xmapsto{\eta_U} & \eta_U(s) \\ \downarrow & & \downarrow \\ s_x & \xmapsto{\eta_x} & (\eta_U(s))_x \end{array}$$

commutes. What we have just said can be rephrased by saying that the association $\mathcal{F} \mapsto \mathcal{F}_x$ defines a functor

$$\mathsf{stalk}_x\colon \mathsf{pSh}(X, \mathscr{C}) \longrightarrow \mathscr{C}. \tag{1.4.6}$$

We will see that in reasonable circumstances the restriction of this functor to the category of sheaves is *exact* (cf. Proposition 1.6.0.9).

Definition 1.4.0.7. A morphism of (pre)sheaves $\eta\colon \mathcal{F} \to \mathcal{G}$ is *surjective* if η_x is surjective for every $x \in X$.



Warning 1.4.0.8. You may have noticed that surjectivity of a map of sheaves (cf. Definition 1.4.0.7) is defined differently than injectivity (cf. Definition 1.2.0.13)!

Let $\eta\colon \mathcal{F}\to\mathcal{G}$ be a morphism of sheaves. Then

$$\eta_x \text{ is surjective} \iff \begin{array}{c}\text{for every } t_x\in\mathcal{G}_x \text{ there exists an open neighbourhood}\\ U \text{ of } x \text{ and a section } s\in\mathcal{F}(U) \text{ such that } (\eta_U(s))_x = t_x.\end{array}$$

$$\eta \text{ is surjective} \iff \begin{array}{c}\text{for every open subset } U\subset X \text{ and for every}\\ t\in\mathcal{G}(U), \text{ there exists a covering } U=\bigcup_{i\in I}U_i\\ \text{such that } t|_{U_i} \text{ is in the image of } \eta_{U_i} \text{ for every } i.\end{array}$$

The second equivalence is obtained as follows.

Proof of '$\Rightarrow$'. Assume η is surjective, i.e. η_x is surjective for every $x\in X$. Fix $U\subset X$ open and a local section $t\in\mathcal{G}(U)$. For every $x\in U$, we have a commutative diagram

$$\begin{array}{ccc}\mathcal{F}(U) & \xrightarrow{\eta_U} & \mathcal{G}(U)\\ \downarrow & & \downarrow\\ \mathcal{F}_x & \xrightarrow{\eta_x} & \mathcal{G}_x\end{array}\qquad\qquad\begin{array}{c} t\\ \downarrow\\ t_x\end{array}$$

where $t_x\in\mathcal{G}_x$ can be lifted along η_x to an element $s_x\in\mathcal{F}_x$. Let (V_x,s) be a representative for s_x, so that in particular $s\in\mathcal{F}(V_x)$. The identity $\eta_x(s_x)=t_x$ implies that there is an open neighbourhood $x\in U_x\subset V_x\cap U$ such that the diagram

$$\begin{array}{ccccc}s\in\mathcal{F}(V_x) & \xrightarrow{\rho^{\mathcal{F}}_{V_xU_x}} & \mathcal{F}(U_x) & \xrightarrow{\eta_{U_x}} & \mathcal{G}(U_x)\\ & & \downarrow & & \downarrow\\ & & \mathcal{F}_x & \xrightarrow{\eta_x} & \mathcal{G}_x\end{array}\qquad\qquad\begin{array}{ccccc}s & \longmapsto & s|_{U_x} & \longmapsto & t|_{U_x}\\ & & \downarrow & & \downarrow\\ & & s_x & \longmapsto & t_x\end{array}$$

commutes, in particular $\eta_{U_x}(s|_{U_x}) = t|_{U_x}$. Now this holds for every $x\in U$, and the elements of $\{U_x \mid x\in U\}$ form a covering of U, thus we have proved the condition, using the cover $U=\bigcup_{x\in U}U_x$ along with the sections $s|_{U_x}\in\mathcal{F}(U_x)$.

Proof of '$\Leftarrow$'. Conversely, assuming the condition, let us prove surjectivity of η. Fix $x\in X$ along with a germ $t_x\in\mathcal{G}_x$. We need to prove that t_x has a preimage in $\mathcal{F}_x$. Let (U,t) be a representative of t_x, so that $t\in\mathcal{G}(U)$. By the condition we are assuming, there exists a covering $U=\bigcup_{i\in I}U_i$ such that $t|_{U_i}=\eta_{U_i}(s_i)$ for some $s_i\in\mathcal{F}(U_i)$, for every $i\in I$. If $x\in U_i$, we have a commutative diagram

$$\begin{array}{ccc}\mathcal{F}(U_i) & \xrightarrow{\eta_{U_i}} & \mathcal{G}(U_i)\\ \downarrow & & \downarrow\\ \mathcal{F}_x & \xrightarrow{\eta_x} & \mathcal{G}_x\end{array}\qquad\qquad\begin{array}{ccc}s_i & \longmapsto & t|_{U_i}\\ \downarrow & & \downarrow\\ \star & \longmapsto & t_x\end{array}$$

so the element $\star\in\mathcal{F}_x$ is a preimage of t_x. The equivalence is proved.

The next result incarnates the local nature of sheaves.

LEMMA 1.4.0.9. *Let $\eta\colon\mathcal{F}\to\mathcal{G}$ be a morphism of sheaves. The following are equivalent:*

(i) *η is an isomorphism,*

(ii) *η_x is an isomorphism for every $x \in X$,*

(iii) *η is injective and surjective.*

Proof. Recall that η is an isomorphism if and only if η_U is an isomorphism for every U.

Proof of (i) $\Rightarrow$ (ii). By functoriality of $\mathcal{F} \mapsto \mathcal{F}_x$, we have that if η is an isomorphism, then so is η_x for every $x \in X$.

Proof of (ii) $\Rightarrow$ (i). Suppose η_x is an isomorphism for every x. Let $U \subset X$ be an open subset: we need to show that η_U is an isomorphism. In doing so, we implicitly use Lemma 1.3.0.5 (indeed we just show η_U is bijective).

To see that η_U is injective, pick $s, t \in \mathcal{F}(U)$ such that $\eta_U(s) = \eta_U(t) \in \mathcal{G}(U)$. Then, for any $x \in U$, one has

$$\eta_x(s_x) = (\eta_U(s))_x = (\eta_U(t))_x = \eta_x(t_x),$$

which implies $s_x = t_x$ by injectivity of η_x. This holds for every $x \in U$ by assumption, thus $s = t$ by Lemma 1.4.0.4. Therefore, η_U is injective for every U (i.e. η is injective).

To see that η_U is surjective, pick $t \in \mathcal{G}(U)$. By surjectivity of η (which we have since η_x is surjective for every $x \in X$), we can find an open cover $U = \bigcup_{i \in I} U_i$ along with a collection of sections $s_i \in \mathcal{F}(U_i)$ such that $\eta_{U_i}(s_i) = t|_{U_i}$. But η is injective, so s_i and s_j agree on $U_i \cap U_j$. Therefore, since $\mathcal{F}$ is a sheaf, they glue to a section $s \in \mathcal{F}(U)$ such that $s_i = s|_{U_i}$. By construction, $\eta_U(s)|_{U_i} = \eta_{U_i}(s_i) = t|_{U_i}$, which implies $\eta_U(s) = t$ since $\mathcal{G}$ is a sheaf. Thus η_U is surjective.

Proof of (ii) $\Rightarrow$ (iii). The first paragraph of '(ii) $\Rightarrow$ (i)' already shows that if η_x is an isomorphism for every $x \in X$, then η_U is injective for all U, i.e. η is injective. Surjectivity follows from the definition.

Proof of (iii) $\Rightarrow$ (ii). We only need to show that if η_U is injective for every U, then η_x is injective for every $x \in X$. Consider $s_x = [U, s]$ and $s'_x = [U', s']$ two germs in $\mathcal{F}_x$ such that $\eta_x(s_x) = \eta_x(s'_x)$ in $\mathcal{G}_x$. Then there is an open subset $W \subset U \cap U'$ such that $\eta_U(s)|_W = \eta_{U'}(s')|_W$. But by compatibility of η_W with restrictions, this is equivalent to the identity $\eta_W(s|_W) = \eta_W(s'|_W)$, which by our assumption implies $s|_W = s'|_W$. But then $s_x = s'_x$. □



Warning 1.4.0.10. It is not true that two sheaves with isomorphic stalks are isomorphic: there may be no map between them! For instance, consider a topological space X consisting of two points x_0, x_1 where only x_0 is a closed point. Thus X and $U = X \setminus \{x_0\}$ are the only nonempty open subsets of X. Fix an abelian group $G \neq 0$ and define $\mathcal{F}(X) = G = \mathcal{F}(U)$. Then choose either $\rho_{XU} = \mathrm{id}_G$ or $\rho_{XU} = 0$ to define two distinct sheaves on X. They have the same stalks but they are not isomorphic.



Exercise 1.4.0.11. Show that Lemma 1.4.0.9 fails for presheaves. (**Hint**: Construct a nonzero presheaf all of whose stalks vanish).

Example 1.4.0.12 (Surjectivity is subtle). Let $\mathcal{F} = \mathscr{O}_X^{\mathrm{h}}$ be the sheaf of holomorphic functions on $X = \mathbb{C}\backslash\{0\}$, and let $\mathcal{G} = \mathcal{F}^\times$ be the sheaf of invertible holomorphic functions on X. The map $\exp\colon \mathcal{F} \to \mathcal{G}$ is surjective, but $\exp_X\colon \mathcal{F}(X) \to \mathcal{G}(X)$ is not surjective, e.g. the function $f(z) = z$ in $\mathcal{G}(X)$ is not the exponential of a holomorphic function (cf. Example 1.2.0.23).

Example 1.4.0.13 (Skyscraper sheaf). Let X be a topological space, G a nontrivial abelian group, $x \in X$ a point. The assignment

$$U \mapsto G_x(U) = \begin{cases} G & \text{if } x \in U \\ 0 & \text{if } x \notin U \end{cases}$$

defines a sheaf, choosing as restriction maps the identity of G or the zero map in the obvious way. This sheaf is called the *skyscraper sheaf* attached to (X, x, G). At the level of stalks, one has

$$(G_x)_y = \begin{cases} G & \text{if } y \in \overline{\{x\}} \\ 0 & \text{if } y \notin \overline{\{x\}}, \end{cases}$$

because if y is in the closure of x then every neighbourhood of y also contains x, whereas if y is *not* in the closure of x, then the neighbourhood $U = X \setminus \overline{\{x\}}$ of y does not contain x, and thus $(G_x)_y = 0$.

Observe that G_x has only one nonzero stalk (at x) if and only if x is a closed point: this is the case where the name 'skyscraper sheaf' for G_x fits best.



Exercise 1.4.0.14. Let $\mathcal{F}$ be a presheaf, $\mathcal{G}$ a sheaf, and let $\eta_1, \eta_2\colon \mathcal{F} \to \mathcal{G}$ be two morphisms of presheaves of sets such that $\eta_{1,x} = \eta_{2,x}$ for every $x \in X$. Show that $\eta_1 = \eta_2$. Show that it is in fact necessary to assume $\mathcal{G}$ to be a sheaf (**Hint**: Use a nonzero presheaf all of whose stalks vanish, as in Exercise 1.4.0.11). This exercise will be needed in Theorem 2.3.2.1.

1.5 Sheafification

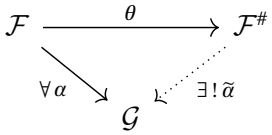
Fix a type of algebraic structure $(\mathscr{C}, F\colon \mathscr{C} \to \mathsf{Sets})$. Friendly translation: fix $\mathscr{C}$ to be either of the following categories:

$$\mathscr{C} \in \left\{ \mathsf{Sets}, \mathsf{Monoids}, \mathsf{Grp}, \mathsf{Ab}, \mathsf{Rings}, \mathsf{Mod}_A \right\}.$$

Let X be a topological space. Let $\mathcal{F}\colon \tau_X^{\mathrm{op}} \to \mathscr{C}$ be a presheaf. We next define a *sheaf* $\mathcal{F}^\#$, called the sheafification of $\mathcal{F}$, via an explicit universal property, and having precisely the same stalks as the initial presheaf $\mathcal{F}$.

Definition 1.5.0.1 (Sheafification of a presheaf)**.** Let $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ be a presheaf. A *sheafification* of $\mathcal{F}$ is a pair $(\mathcal{F}^\#, \theta)$, where $\mathcal{F}^\# \in \mathsf{Sh}(X, \mathscr{C})$ is a sheaf and $\theta\colon \mathcal{F} \to \mathcal{F}^\#$ is a

morphism of presheaves, such that for every other pair $(\mathcal{G}, \alpha)$ where $\mathcal{G}$ is a sheaf and $\alpha: \mathcal{F} \to \mathcal{G}$ is a morphism of presheaves, there exists a unique morphism of sheaves $\tilde{\alpha}: \mathcal{F}^{\#} \to \mathcal{G}$ such that $\alpha = \tilde{\alpha} \circ \theta$.



PROPOSITION 1.5.0.2. *Let $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ be a presheaf. Then a sheafification $(\mathcal{F}^{\#}, \theta)$ exists, and the map $\theta_x: \mathcal{F}_x \to \mathcal{F}^{\#}_x$ is an isomorphism for every $x \in X$.*

What follows immediately from Proposition 1.5.0.2 is that $\mathcal{F}^{\#}$ is unique up to a unique isomorphism, and moreover the canonical map $\theta: \mathcal{F} \to \mathcal{F}^{\#}$ is an isomorphism precisely when $\mathcal{F}$ is already a sheaf.

Proof. Let $U \subset X$ be an open subset. Define

$$\mathcal{F}^{\#}(U) = \left\{ \text{functions } U \xrightarrow{f} \coprod_{x \in U} \mathcal{F}_x \;\middle|\; \begin{array}{c} \text{for every } x \in U,\ f(x) \in \mathcal{F}_x \text{ and there exist an} \\ \text{open neighbourhood } V \subset U \text{ of } x \text{ and } s \in \mathcal{F}(V) \\ \text{such that } f(y) = s_y \text{ for every } y \in V \end{array} \right\}.$$

For a thorough proof that $\mathcal{F}^{\#}(U)$ defines a genuine object of $\mathscr{C}$, we refer to [10, Tag 0086].[3] Note that, since $\mathscr{C}$ has products, we can view a function f as above as a tuple

$$(f(x))_{x \in U} \in \prod_{x \in U} \mathcal{F}_x$$

and we can rephrase the definition of $\mathcal{F}^{\#}(U)$ by saying that

$$\mathcal{F}^{\#}(U) = \left\{ \text{tuples } (s_x)_{x \in U} \text{ of compatible germs} \right\}.$$

See Definition 1.4.0.3 for the definition of compatible germs. Functoriality of the assignment $U \mapsto \mathcal{F}^{\#}(U)$ is clear (functions restrict!), thus $\mathcal{F}^{\#}$ is a presheaf. The morphism $\theta_U: \mathcal{F}(U) \to \mathcal{F}^{\#}(U)$ defined by sending $s \in \mathcal{F}(U)$ to the function

$$U \xrightarrow{f_s} \coprod_{x \in U} \mathcal{F}_x, \qquad x \mapsto s_x = [U, s] \in \mathcal{F}_x$$

determines a morphism of presheaves, being compatible with restrictions. It is just the function $\sigma^{\mathcal{F}}_U$ introduced in (1.4.1)!

The presheaf $\mathcal{F}^{\#}$ is a sheaf: By Lemma 1.3.0.5, the sheaf condition can be verified in the category of sets. Fix an open cover $U = \bigcup_{i \in I} U_i$ of some open subset $U \subset X$ and a collection of sections $f_i \in \mathcal{F}^{\#}(U_i)$ such that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for every i and j. We need to find a unique $f \in \mathcal{F}^{\#}(U)$ such that $f|_{U_i} = f_i$. Define

$$f(x) = f_i(x) \in \mathcal{F}_x, \quad x \in U_i \subset U.$$

[3] The reader may easily verify this by hand for the familiar types of algebraic structure listed at the beginning of this section, which is all that really matters to us.

This is well-defined since, even though x can lie in more than one open U_i, by assumption we have $f_i(x) = f_j(x)$ as soon as $x \in U_i \cap U_j$. We need to check that f defines an element of $\mathcal{F}^\#(U)$, not just of the full product $\prod_{x\in U} \mathcal{F}_x$. But for every $i \in I$ we know the following: for every $x \in U_i$ there exist an open neighbourhood $x \in V_i \subset U_i$ and a section $s_i \in \mathcal{F}(V_i)$ such that $f(y) = f_i(y) = (s_i)_y$ for all $y \in V_i$. But V_i is also open in U, so the condition defining $\mathcal{F}^\#(U)$ also holds for f. Thus $f \in \mathcal{F}^\#(U)$ satisfies $f|_{U_i} = f_i$, and is clearly unique with this property.

The pair $(\mathcal{F}^\#, \theta)$ is the sheafification. Assume we have a sheaf $\mathcal{G}$ and a morphism of presheaves $\alpha\colon \mathcal{F} \to \mathcal{G}$. With a slight abuse of notation, we shall identify $\mathcal{G}$ and $\mathcal{G}^\#$ via the canonical isomorphism $\mathcal{G} \xrightarrow{\sim} \mathcal{G}^\#$. We need to define a morphism $\widetilde{\alpha}\colon \mathcal{F}^\# \to \mathcal{G}$ of presheaves such that $\alpha = \widetilde{\alpha} \circ \theta$. For every U open in X, we need to define a morphism $\widetilde{\alpha}_U : \mathcal{F}^\#(U) \to \mathcal{G}(U)$ in such a way that $\alpha_U = \widetilde{\alpha}_U \circ \theta_U$. Fix $s = (s_x)_{x\in U} \in \mathcal{F}^\#(U)$. The composition

$$\begin{array}{ccc} U & \xrightarrow{\ s\ } & \coprod_{x\in U} \mathcal{F}_x \\ & \searrow^{\widetilde{\alpha}_U(s)} & \downarrow{\scriptstyle \coprod_{x\in U} \alpha_x} \\ & & \coprod_{x\in U} \mathcal{G}_x \end{array}$$

defines a tuple of compatible germs for $\mathcal{G}$ over U, hence an element $\widetilde{\alpha}_U(s) \in \mathcal{G}^\#(U) = \mathcal{G}(U)$, using that $\mathcal{G}$ is a sheaf for this identity. This is the required morphism $\widetilde{\alpha}\colon \mathcal{F}^\# \to \mathcal{G}$.

The map θ is an isomorphism on stalks. The map θ, at the level of stalks, is defined by

$$\theta_x[U, s] = [U, f_s].$$

Injectivity: Suppose $\theta_x[U, s] = \theta_x[V, t]$ for two classes $[U, s], [V, t] \in \mathcal{F}_x$, i.e. assume $[U, f_s] = [V, f_t]$ in $\mathcal{F}^\#_x$. Then, by definition of germ, there exists an open neighbourhood $W \subset U \cap V$ of x such that $f_s|_W = f_t|_W$. But this means, by definition of f_s and f_t, that $s_y = t_y$ for all $y \in W$. Thus, in particular, $s_x = t_x$. But this is just the equality $[U, s] = [V, t]$ we were after.

Surjectivity: Pick a class $[U, f] \in \mathcal{F}^\#_x$ for some $f \in \mathcal{F}^\#(U)$ and open neighbourhood U of x. Then, for every $z \in U$, there exist an open neighbourhood $V \subset U$ of z and a section $s \in \mathcal{F}(V)$ such that $f(y) = s_y$ in $\mathcal{F}_y$ for every $y \in V$. Apply now this germ compatibility to the point $z = x$: we claim that $[U, f] = \theta_x(s_x)$, where $s_x = [V, s]$. Indeed, $\theta_x(s_x) \in \mathcal{F}^\#_x$ is the equivalence class of the map

$$V \xrightarrow{\ f_s\ } \coprod_{y\in V} \mathcal{F}_y, \quad y \mapsto s_y.$$

But this map agrees with the restriction of f to $V \subset U$ (by the condition $f(y) = s_y$ recalled above), i.e. $f_s = f|_V \in \mathcal{F}^\#(V)$. Since V is also an open neighbourhood of x, it follows that $(f|_V)_x = (f_s)_x = [V, f_s] = \theta_x(s_x) \in \mathcal{F}^\#_x$, but of course $(f|_V)_x = [U, f]$. Thus θ_x is surjective. □

Example 1.5.0.3 (Constant sheaf). Let G be a nontrivial abelian group. The *constant sheaf* on a topological space X, with values in G, is the sheafification $\underline{G}_X$ of the presheaf $\underline{G}_X^{\mathrm{pre}}$ defined in Example 1.2.0.19. This sheaf agrees with the sheaf whose sections over U are the locally constant functions $U \to G$. This, in turn, agrees with the following: endow G with the discrete topology and consider the assignment

$$U \longmapsto \{\text{continuous maps } U \to G\},$$

which we know is a sheaf by Example 1.2.0.16. If $U \subset X$ is a nonempty connected open subset, then $\underline{G}_X(U) = G$. By Proposition 1.5.0.2, at the level of stalks we have $\underline{G}_{X,x} = G$ for every $x \in X$, since the stalks of the constant presheaf are manifestly all equal to G.



Exercise 1.5.0.4. What is the sheafification of the presheaf of bounded continuous functions (cf. Exercise 1.2.0.18)?



Exercise 1.5.0.5. Let X be a topological space, x a point, G a nontrivial abelian group. Can you find a necessary and sufficient condition for the constant sheaf $\underline{G}_X$ being equal to the skyscraper sheaf G_x (cf. Example 1.4.0.13)?



Exercise 1.5.0.6. Show that $\mathcal{F} \mapsto \mathcal{F}^{\#}$ defines a functor $(-)^{\#}\colon \mathsf{pSh}(X,\mathscr{C}) \to \mathsf{Sh}(X,\mathscr{C})$, and that the forgetful functor $j_{X,\mathscr{C}}\colon \mathsf{Sh}(X,\mathscr{C}) \hookrightarrow \mathsf{pSh}(X,\mathscr{C})$ is a right adjoint. In other words (cf. Definition A.4.0.1), prove that there are bifunctorial bijections

$$\psi_{\mathcal{F},\mathcal{G}}\colon \mathrm{Hom}_{\mathsf{Sh}(X,\mathscr{C})}(\mathcal{F}^{\#},\mathcal{G}) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{pSh}(X,\mathscr{C})}(\mathcal{F},\mathcal{G}), \quad \widetilde{\alpha} \mapsto \widetilde{\alpha} \circ \theta$$

for any presheaf $\mathcal{F}$ and any sheaf $\mathcal{G}$. (**Hint**: the universal property of the sheafification!).

1.6 Subsheaves, Quotient sheaves, and exactness

We have essentially already proved the following general result.

PROPOSITION 1.6.0.1 ([10, Tag 007S]). *Let X be a topological space. Let $\mathcal{F},\mathcal{G} \in \mathsf{Sh}(X,\mathsf{Sets})$ be sheaves of sets, $\eta\colon \mathcal{F} \to \mathcal{G}$ a morphism. Then, the following are equivalent:*

(a) *η is a monomorphism,*

(b) *$\eta_x\colon \mathcal{F}_x \to \mathcal{G}_x$ is injective for all $x \in X$,*

(c) *$\eta_U\colon \mathcal{F}(U) \to \mathcal{G}(U)$ is injective for all open subsets $U \subset X$ (i.e. η is injective).*

Furthermore, the conditions

(i) *η is an epimorphism,*

(ii) *$\eta_x\colon \mathcal{F}_x \to \mathcal{G}_x$ is surjective for all $x \in X$ (i.e. η is surjective),*

are equivalent, and are implied (but not equivalent to, cf. Example 1.4.0.12!) by the condition

(iii) $\eta_U : \mathcal{F}(U) \to \mathcal{G}(U)$ *is surjective for all open subsets* $U \subset X$.

If $(\mathscr{C}, F)$ is a type of algebraic structure where $\mathscr{C}$ is abelian and has exact filtered colimits (e.g. Mod_A for a fixed ring A), then Proposition 1.6.0.1 holds replacing Sets with $\mathscr{C}$. In this setup, the condition that η_x be injective, resp, surjective, can safely be replaced by η_x being a monomorphism, resp. an epimorphism in $\mathscr{C}$.

Definition 1.6.0.2 (Subsheaf, quotient sheaf)**.** If there exists a morphism of sheaves $\eta : \mathcal{F} \to \mathcal{G}$ such that either of the equivalent conditions (a), (b) or (c) holds, we say that $\mathcal{F}$ is a *subsheaf* of $\mathcal{G}$ (and we may denote this by '$\mathcal{F} \subset \mathcal{G}$'). If either of the equivalent conditions (i) or (ii) holds, we say that $\mathcal{G}$ is a *quotient sheaf* of $\mathcal{F}$.

Example 1.6.0.3 (Quotient sheaf)**.** Let $(\mathscr{C}, F)$ be a type of algebraic structure such that $\mathscr{C}$ is an abelian category. If $\mathcal{F} \subset \mathcal{G}$ is a subsheaf (with values in $\mathscr{C}$), then sending

$$U \longmapsto \mathcal{G}(U)/\mathcal{F}(U) \tag{1.6.1}$$

is a presheaf on X, because the restriction maps respect the inclusions $\mathcal{F}(U) \hookrightarrow \mathcal{G}(U)$, and thus pass to the quotients. Its sheafification $\mathcal{G}/\mathcal{F}$ is called the *quotient sheaf of* $\mathcal{G}$ *by* $\mathcal{F}$. There is a natural morphism of sheaves $\mathcal{G} \to \mathcal{G}/\mathcal{F}$.

Definition 1.6.0.4 (Sheaf image, sheaf cokernel)**.** Let $(\mathscr{C}, F)$ be a type of algebraic structure with $\mathscr{C}$ abelian, $\eta : \mathcal{F} \to \mathcal{G}$ a morphism of sheaves (with values in $\mathscr{C}$), so that $\ker(\eta) \hookrightarrow \mathcal{F}$ is a subsheaf by Exercise 1.2.0.22. The sheafification $\mathrm{im}(\eta)$ of the presheaf

$$U \longmapsto \mathrm{im}_{\mathrm{pre}}(U) = \mathrm{im}(\eta_U) = \mathcal{F}(U)/\ker(\eta_U)$$

is called the *image of* η. It is a special case of Example 1.6.0.3 and defines a subsheaf

$$\mathrm{im}(\eta) = \mathcal{F}/\ker(\eta) \hookrightarrow \mathcal{G}.$$

The quotient sheaf

$$\mathrm{coker}(\eta) = \mathcal{G}/\mathrm{im}(\eta),$$

again a special case of Example 1.6.0.3, is called the *sheaf cokernel*.

Remark 1.6.0.5. Set $\mathscr{C} = \mathsf{Mod}_A$, or any Grothendieck abelian category supporting a type of algebraic structure. By definition, $\mathscr{C}$ has axiom (AB5), i.e. filtered colimits exist and are (also *left*) exact. Let $\mathcal{F} \subset \mathcal{G}$ be a subsheaf, $x \in X$ a point. Then

$$(\mathcal{G}/\mathcal{F})_x = \mathcal{G}_x/\mathcal{F}_x \tag{1.6.2}$$

in Mod_A. This follows from the fact that $(\mathcal{G}/\mathcal{F})_x$ agrees with the stalk of the *presheaf* (1.6.1), and from right exactness of filtered colimits. Moreover, if $\eta : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves and $x \in X$ is a point, then

$$\begin{aligned} \ker(\eta)_x &= \ker(\eta_x) \\ \mathrm{im}(\eta)_x &= \mathrm{im}(\eta_x) \\ \mathrm{coker}(\eta)_x &= \mathrm{coker}(\eta_x). \end{aligned} \tag{1.6.3}$$

The first identity in (1.6.3) follows from the fact that filtered colimits are *also left exact* in Mod_A, thus

$$\begin{aligned}\ker\left(\mathcal{F}_x \xrightarrow{\eta_x} \mathcal{G}_x\right) &= \ker\left(\varinjlim_{U\ni x} \mathcal{F}(U) \to \varinjlim_{U\ni x} \mathcal{G}(U)\right)\\ &= \varinjlim_{U\ni x} \ker(\mathcal{F}(U)\to\mathcal{G}(U))\\ &= \ker(\eta)_x.\end{aligned}$$

The last two identities in (1.6.3) are a special case of (1.6.2).

Exercise 1.6.0.6. Set $\mathscr{C} = \mathsf{Mod}_A$, or any Grothendieck abelian category supporting a type of algebraic structure. By definition, $\mathscr{C}$ has axiom (AB5), i.e. filtered colimits exist and are (also *left*) exact. Let $\eta\colon \mathcal{F}\to\mathcal{G}$ be a morphism of sheaves with values in $\mathscr{C}$. Show that the composition

$$\mathcal{G} \longrightarrow \operatorname{coker}_{\mathrm{pre}}\eta \longrightarrow \operatorname{coker}(\eta),$$

where the first morphism is given by the natural maps $\mathcal{G}(U)\twoheadrightarrow\mathcal{G}(U)/\operatorname{im}(\eta_U)$ and the last morphism is the sheafification, is a cokernel in the category $\mathsf{Sh}(X,\mathscr{C})$. Finally, show that η is an epimorphism if and only if $\operatorname{coker}(\eta)=0$, and if $\mathscr{C}=\mathsf{Mod}_A$ this is also equivalent to stalkwise surjectivity.

THEOREM 1.6.0.7 ([3, §10]). *If $\mathscr{C}$ is a Grothendieck abelian category, then $\mathsf{Sh}(X,\mathscr{C})$ is a Grothendieck abelian category.*

Definition 1.6.0.8. A *short exact sequence of sheaves* with values in a Grothendieck abelian category $\mathscr{C}$ is a short exact sequence

$$0 \longrightarrow \mathcal{F} \xrightarrow{\iota} \mathcal{G} \xrightarrow{\theta} \mathcal{H} \longrightarrow 0$$

of objects in the abelian category $\mathsf{Sh}(X,\mathscr{C})$. Explicitly, exactness means that ι is a monomorphism, θ is an epimorphism and $\operatorname{im}(\iota)=\ker(\theta)$.

PROPOSITION 1.6.0.9. *Let $\mathscr{C}$ be a Grothendieck abelian category. A sequence*

$$0 \longrightarrow \mathcal{F} \xrightarrow{\iota} \mathcal{G} \xrightarrow{\theta} \mathcal{H} \longrightarrow 0$$

of objects in $\mathsf{Sh}(X,\mathscr{C})$ is a short exact sequence if and only if

$$0 \longrightarrow \mathcal{F}_x \xrightarrow{\iota_x} \mathcal{G}_x \xrightarrow{\theta_x} \mathcal{H}_x \longrightarrow 0$$

is a short exact sequence in $\mathscr{C}$ for every $x\in X$.

Proof. Combine Remark 1.6.0.5 and Lemma 1.4.0.9 with one another. □



Exercise 1.6.0.10. Let $\eta\colon \mathcal{F}\to\mathcal{G}$ be a morphism of sheaves of A-modules, where A is a ring. Prove that there is an exact sequence of sheaves

$$0 \longrightarrow \ker(\eta) \longrightarrow \mathcal{F} \xrightarrow{\eta} \mathcal{G} \longrightarrow \operatorname{coker}(\eta) \longrightarrow 0.$$

In particular, if η is injective, this reduces to

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{G}/\mathcal{F} \longrightarrow 0.$$



Exercise 1.6.0.11. Let A be a ring. For a nonempty open subset U of a topological space X, consider the functor $\Gamma(U,-)\colon \mathsf{Sh}(X,\mathsf{Mod}_A)\to\mathsf{Mod}_A$ sending $\mathcal{F}\mapsto\mathcal{F}(U)$. Show that it is left exact. That is, it transforms an exact sequence of sheaves $0\to\mathcal{F}\to\mathcal{G}\to\mathcal{H}$ into an exact sequence of A-modules

$$0 \longrightarrow \mathcal{F}(U) \longrightarrow \mathcal{G}(U) \longrightarrow \mathcal{H}(U).$$

When $U=X$, this functor takes $\mathcal{F}\mapsto\mathcal{F}(X)$ and is thus called the *global section functor*. Another notation used for it in the literature is $\mathrm{H}^0(X,-)$, cf. Terminology 1.2.0.3.

1.7 Supports

Let A be a ring. Let $\mathcal{F}\in\mathsf{Sh}(X,\mathsf{Mod}_A)$ be a sheaf of A-modules on a topological space X. Let $U\subset X$ be an open subset, and fix a section $s\in\mathcal{F}(U)$. We have two notions of support: the support of $\mathcal{F}$, and the support of s, defined respectively as

$$\begin{aligned}\mathsf{Supp}(\mathcal{F}) &= \{\, x\in X \mid \mathcal{F}_x\neq 0\,\},\\ \mathsf{Supp}(s) &= \{\, x\in U \mid s_x\neq 0 \text{ in } \mathcal{F}_x\,\}.\end{aligned} \tag{1.7.1}$$

If $s_x=0$, then there is an open neighbourhood $x\in V\subset U$ such that $s|_V=0\in\mathcal{F}(V)$. Thus $V\subset U\setminus\mathsf{Supp}(s)$ and hence $\mathsf{Supp}(s)\subset U$ is closed. In fact, this follows from (or solves) Exercise 1.4.0.5. In general, however, $\mathsf{Supp}(\mathcal{F})\subset X$ is *not* closed, as the two next examples show.

Example 1.7.0.1 ($\mathsf{Supp}(\mathcal{F})$ need not be closed, take I)**.** Let X be an irreducible topological space. This means that any two nonempty open subsets of X intersect. Fix a nontrivial abelian group G, a point $x\in X$, and for $U\in\tau_X$ define

$$\mathcal{F}(U)=\begin{cases}0 & \text{if } U=\emptyset \text{ or } x\in U\\ G & \text{otherwise.}\end{cases}$$

Let $\rho_{UV}\in\{\mathrm{id}_G,0\}$ be chosen in the obvious way for all $U,V\in\tau_X$. Then $\mathcal{F}$ is a sheaf of abelian groups on X, with stalks

$$\mathcal{F}_y=\begin{cases}0 & \text{if } y\in\overline{\{x\}}\\ G & \text{otherwise.}\end{cases}$$

It follows that

$$\mathsf{Supp}(\mathcal{F}) = X \setminus \overline{\{x\}},$$

which is not closed in X as soon as $\overline{\{x\}} \hookrightarrow X$ is not open.

Example 1.7.0.2 ($\mathsf{Supp}(\mathcal{F})$ need not be closed, take II)**.** Let $j : U \hookrightarrow X$ be the inclusion of an open subset U of a topological space X. Let $\mathcal{F} \in \mathsf{Sh}(U, \mathsf{Ab})$ be a sheaf. Define $j_!\mathcal{F} \in \mathsf{Sh}(X, \mathsf{Ab})$ to be the sheafification of the presheaf $j_!^{\mathrm{pre}}\mathcal{F}$ defined by

$$j_!^{\mathrm{pre}}\mathcal{F}(V) = \begin{cases} \mathcal{F}(V) & \text{if } V \subset U \\ 0 & \text{otherwise} \end{cases}$$

on X. One has

$$(j_!\mathcal{F})_x = \begin{cases} \mathcal{F}_x & \text{if } x \in U \\ 0 & \text{otherwise,} \end{cases}$$

so that $\mathsf{Supp}(j_!\mathcal{F}) = \mathsf{Supp}(\mathcal{F})$. Fix $G \neq 0$ in Ab and consider the constant sheaf on U (cf. Example 1.5.0.3). We have $\mathsf{Supp}(j_!\underline{G}_U) = \mathsf{Supp}(\underline{G}_U) = U$. In particular, $\mathsf{Supp}(j_!\underline{G}_U) \subset X$ is not closed as soon as U is not closed in X.

If $\mathcal{F}$ is a sheaf of rings, the notions of support defined in (1.7.1) still make sense, and one has $\mathsf{Supp}(\mathcal{F}) = \mathsf{Supp}(1)$, where $1 \in \mathcal{F}(X)$ is the ring identity (recall that the '0 ring' is the one where $1 = 0$). Thus $\mathsf{Supp}(\mathcal{F})$ *is* in fact closed in this case.

1.8 Sheaves = sheaves on a base

Fix a type of algebraic structure $(\mathscr{C}, F : \mathscr{C} \to \mathsf{Sets})$.

Definition 1.8.0.1 ($\mathsf{Base\ of\ open\ sets}$)**.** Let X be a topological space. A *base of open sets* for X is a collection of open subsets $\mathcal{B} \subset \tau_X$ satisfying the following requirements:

(a) $\mathcal{B}$ is stable under finite intersections,

(b) every $U \in \tau_X$ can be written as a union of open sets belonging to $\mathcal{B}$.

The terminology just introduced is slightly nonstandard, in the sense that condition (a) is usually omitted. For convenience, we use the term *base* only for bases stable under finite intersections: this stronger convention will be particularly convenient for the basis of distinguished open subsets of an affine scheme.

Definition 1.8.0.2 ($\mathcal{B}$-sheaf)**.** A *$\mathcal{B}$-presheaf* (resp. *$\mathcal{B}$-sheaf*) is an assignment

$$U \xmapsto{\mathcal{F}_{\mathcal{B}}} \mathcal{F}_{\mathcal{B}}(U) \in \mathscr{C}, \text{ for each } U \in \mathcal{B},$$

such that the presheaf conditions (1)–(2) of Definition 1.2.0.2 (resp. those conditions and the sheaf conditions (loc)–(glue) of Important Definition 1.2.0.5) hold, considering only open sets belonging to $\mathcal{B}$.

Note that restriction maps

$$\rho_{UV}\colon \mathcal{F}_{\mathcal{B}}(U) \longrightarrow \mathcal{F}_{\mathcal{B}}(V)$$

are part of the data of a $\mathcal{B}$-(pre)sheaf whenever $V \subset U$ is an inclusion of open sets both belonging to $\mathcal{B}$. Note, also, that condition (a) in Definition 1.8.0.1 ensures that open subsets of the form $U \cap V$ still belong to $\mathcal{B}$ if $U, V \in \mathcal{B}$. In particular, as in Section 1.3, a $\mathcal{B}$-presheaf $\mathcal{F}_{\mathcal{B}}$ is a sheaf precisely when the following condition is fulfilled: for every open subset $U \in \mathcal{B}$ and for any open cover $U = \bigcup_{i\in I} U_i$ with all $U_i \in \mathcal{B}$, the diagram

$$\mathcal{F}_{\mathcal{B}}(U) \xrightarrow{\rho} \prod_{i\in I} \mathcal{F}_{\mathcal{B}}(U_i) \underset{\nu}{\overset{\mu}{\rightrightarrows}} \prod_{(i,j)\in I\times I} \mathcal{F}_{\mathcal{B}}(U_i \cap U_j)$$

is an equaliser diagram in $\mathscr{C}$.

Remark 1.8.0.3. Let $x \in X$ be a point. The collection of open neighbourhoods

$$\mathcal{B}_x = \{\, B \in \mathcal{B} \mid x \in B \,\}^{\mathrm{op}} \subset \tau_X^{\mathrm{op}}$$

is a fundamental system of open neighbourhoods of x, also called a local basis at x (i.e. for any $W \in \mathsf{Ngb}_x$ there exists $B \in \mathcal{B}_x$ such that $B \subset W$). In more technical terms, one may say that the filtered categories Ngb_x and $\mathcal{B}_x$ are *cofinal*, i.e. the inclusion $\mathcal{B}_x \hookrightarrow \mathsf{Ngb}_x$ is a cofinal functor. We will not use this terminology.

The stalk

$$\mathcal{F}_{\mathcal{B},x} = \varinjlim_{B\in\mathcal{B}_x} \mathcal{F}_{\mathcal{B}}(B) \in \mathscr{C} \tag{1.8.1}$$

of a $\mathcal{B}$-(pre)sheaf $\mathcal{F}_{\mathcal{B}}$ at a point $x \in X$ is well-defined as an object of $\mathscr{C}$. It receives, by definition of direct limit, canonical morphisms

$$\mathcal{F}_{\mathcal{B}}(B) \longrightarrow \mathcal{F}_{\mathcal{B},x}, \qquad B \in \mathcal{B}_x.$$

We denote by $s_x \in \mathcal{F}_{\mathcal{B},x}$, as ever, the image of $s \in \mathcal{F}_{\mathcal{B}}(B)$ under this morphism.

Moreover, if $B \in \mathcal{B}$ and $\mathcal{F}_{\mathcal{B}}$ is a $\mathcal{B}$-sheaf, the natural map

$$\sigma_B^{\mathcal{F}_{\mathcal{B}}}\colon \mathcal{F}_{\mathcal{B}}(B) \longrightarrow \prod_{x\in B} \mathcal{F}_{\mathcal{B},x}, \qquad s \mapsto (s_x)_{x\in B}, \tag{1.8.2}$$

is injective (as in Lemma 1.4.0.2), and its image agrees with the collections of compatible germs; to be more precise, we should now call them '$\mathcal{B}$-compatible', for they are, by definition, those tuples

$$(s_x)_{x\in B} \in \prod_{x\in B} \mathcal{F}_{\mathcal{B},x}$$

such that for every $x \in B$ there is a pair (V_x, t^x), where $V_x \in \mathcal{B}_x$ satisfies $V_x \subset B$ and $t^x \in \mathcal{F}_{\mathcal{B}}(V_x)$ satisfies $t^x_y = s_y$ for every $y \in V_x$. Note that, because $\mathcal{B}_x$ is a local basis at x, the notion of a tuple of $\mathcal{B}$-compatible germs makes sense for an arbitrary open subset $U \subset X$!

Definition 1.8.0.4 (Morphism of $\mathcal{B}$-sheaves)**.** A *morphism* of $\mathcal{B}$-(pre)sheaves

$$\eta_{\mathcal{B}}\colon \mathcal{F}_{\mathcal{B}} \longrightarrow \mathcal{G}_{\mathcal{B}} \tag{1.8.3}$$

is the datum of a collection of maps $\eta_B\colon \mathcal{F}_{\mathcal{B}}(B) \to \mathcal{G}_{\mathcal{B}}(B)$, one for each $B \in \mathcal{B}$, such that Diagram (1.2.1) commutes for all $B, C \in \mathcal{B}$ such that $C \subset B$.

With this definition, $\mathcal{B}$-sheaves form a category, denoted $\mathsf{Sh}_{\mathcal{B}}(X, \mathscr{C})$.[4]

Remark 1.8.0.5. Let X be a topological space, $\mathcal{B}$ a base of open subsets of X. A (pre)sheaf $\mathcal{F}$ on X is a $\mathcal{B}$-(pre)sheaf in a natural way. More precisely, there is (say, at the level of sheaves) a *restriction functor*

$$\mathsf{res}_{\mathcal{B}}\colon \mathsf{Sh}(X, \mathscr{C}) \longrightarrow \mathsf{Sh}_{\mathcal{B}}(X, \mathscr{C}), \tag{1.8.4}$$

defined on objects in the obvious way. Its functoriality is just a consequence of the definition of morphism of $\mathcal{B}$-sheaves, and is an easy routine check.

The next result says that the restriction functor (1.8.4) is essentially surjective.

LEMMA 1.8.0.6 (Extension from a base). *Let X be a topological space and let $\mathcal{B} \subset \tau_X$ be a base of open subsets. Let $\mathcal{F}_{\mathcal{B}} \in \mathsf{Sh}_{\mathcal{B}}(X, \mathscr{C})$ be a $\mathcal{B}$-sheaf. Then there exists a sheaf $\mathcal{F} \in \mathsf{Sh}(X, \mathscr{C})$ together with an isomorphism of $\mathcal{B}$-sheaves*

$$\alpha\colon \mathcal{F}_{\mathcal{B}} \xrightarrow{\sim} \mathsf{res}_{\mathcal{B}}(\mathcal{F}).$$

In particular, the restriction functor (1.8.4) *is essentially surjective.*

Proof. Recall from (1.8.1) the definition of the stalks $\mathcal{F}_{\mathcal{B},x}$ of a $\mathcal{B}$-sheaf. Let $U \in \tau_X$ be an arbitrary open set. Define

$$\mathcal{F}(U) = \Big\{ \text{tuples } (s_x)_{x\in U} \text{ of } \mathcal{B}\text{-compatible germs} \Big\} \subset \prod_{x\in U} \mathcal{F}_{\mathcal{B},x}.$$

As noted above, this makes sense for an arbitrary U, and confirms that $\mathcal{F}$ is a sheaf, for the same reason why the sheaf property is fulfilled by the sheafification of a presheaf (see the proof of Proposition 1.5.0.2). For every $B \in \mathcal{B}$, the natural map

$$\alpha_B\colon \mathcal{F}_{\mathcal{B}}(B) \xrightarrow{\sim} \mathcal{F}(B), \qquad s \mapsto (s_x)_{x\in B}$$

is an isomorphism (since the injective map $\sigma_B^{\mathcal{F}_{\mathcal{B}}}$ of (1.8.2) hits precisely the tuples of $\mathcal{B}$-compatible germs), and these maps are compatible with restriction. Hence they define an isomorphism of $\mathcal{B}$-sheaves $\alpha\colon \mathcal{F}_{\mathcal{B}} \xrightarrow{\sim} \mathsf{res}_{\mathcal{B}}(\mathcal{F})$ as required. □

We shall now prove that the restriction functor (1.8.4) is in fact an equivalence. A quasi-inverse is given precisely by Lemma 1.8.0.6 above for objects and by Theorem 1.8.0.8 below for morphisms.

[4] Also $\mathcal{B}$-presheaves form a category, but it is not as well-behaved as $\mathsf{Sh}_{\mathcal{B}}(X, \mathscr{C})$, and we do not need it, so we shall ignore it.

Remark 1.8.0.7. In the notation of Lemma 1.8.0.6, we have a canonical isomorphism $\mathcal{F}_{\mathcal{B},x} \xrightarrow{\sim} \mathcal{F}_x$ for all $x \in X$. This follows directly from Remark 1.8.0.3 (cofinal filtered categories have isomorphic colimits).

A (stronger) analogue of Lemma 1.8.0.6 for *morphisms* is the following.

THEOREM 1.8.0.8. *Let X be a topological space, $\mathcal{B} \subset \tau_X$ a base of open sets and let $\mathcal{F}$, $\mathcal{G} \in \mathsf{Sh}(X, \mathscr{C})$ be two sheaves on X. Suppose given a morphism*

$$\eta_{\mathcal{B}} \colon \mathsf{res}_{\mathcal{B}}(\mathcal{F}) \longrightarrow \mathsf{res}_{\mathcal{B}}(\mathcal{G})$$

between the underlying $\mathcal{B}$-sheaves. Then $\eta_{\mathcal{B}}$ extends uniquely to a sheaf homomorphism $\eta \colon \mathcal{F} \to \mathcal{G}$. Furthermore, if η_U is surjective (or injective, or an isomorphism) for every $U \in \mathcal{B}$, then so is η.

Proof. Before we start: By Lemma 1.3.0.5 we may perform the construction on the underlying sheaves of sets; locally the resulting maps agree with the morphisms $\eta_{\mathcal{B},B}$, hence they are morphisms in $\mathcal{C}$.

To ease notation, we set $\mathcal{F}_{\mathcal{B}} = \mathsf{res}_{\mathcal{B}}(\mathcal{F})$ and $\mathcal{G}_{\mathcal{B}} = \mathsf{res}_{\mathcal{B}}(\mathcal{G})$. Clearly we have $\mathcal{F}_{\mathcal{B}}(B) = \mathcal{F}(B)$ for $B \in \mathcal{B}$, and similarly for $\mathcal{G}$. Given $\eta_{\mathcal{B}}$, for every $B \in \mathcal{B}$ we already have a morphism

$$\eta_B \coloneqq \eta_{\mathcal{B},B} \colon \mathcal{F}_{\mathcal{B}}(B) \longrightarrow \mathcal{G}_{\mathcal{B}}(B),$$

compatible with restriction maps. We construct the extension $\eta \colon \mathcal{F} \to \mathcal{G}$ as follows. Let $U \subset X$ be an arbitrary open subset and let $s \in \mathcal{F}(U)$. Choose a covering

$$U = \bigcup_{i \in I} B_i, \qquad B_i \in \mathcal{B}.$$

For every i, consider the section

$$t_i = \eta_{B_i}(s|_{B_i}) \in \mathcal{G}(B_i).$$

We claim that the sections t_i agree on pairwise intersections. Indeed, since $\mathcal{B}$ is stable under finite intersections, $B_i \cap B_j \in \mathcal{B}$, and the compatibility of $\eta_{\mathcal{B}}$ with restriction maps gives

$$t_i|_{B_i \cap B_j} = \eta_{B_i \cap B_j}\big(s|_{B_i \cap B_j}\big) = t_j|_{B_i \cap B_j}.$$

Since $\mathcal{G}$ is a sheaf, there exists a unique section $t \in \mathcal{G}(U)$ such that $t|_{B_i} = t_i$ for every $i \in I$. We define

$$\eta_U(s) = t \in \mathcal{G}(U).$$

This definition is independent of the chosen covering. Indeed, if

$$U = \bigcup_{j \in J} B'_j$$

is another covering by elements of $\mathcal{B}$, then $B_i \cap B'_j \in \mathcal{B}$, and the two resulting sections of $\mathcal{G}(U)$ have the same restriction to every $B_i \cap B'_j$. Since these open subsets cover U, the locality axiom for $\mathcal{G}$ implies that the two sections coincide.

The maps η_U are compatible with restrictions. Indeed, let $V \subset U$ be an open subset and let $s \in \mathcal{F}(U)$. Cover V by elements $B_i \in \mathcal{B}$ satisfying $B_i \subset V$. Then

$$\eta_U(s)|_{B_i} = \eta_{B_i}(s|_{B_i}) = \eta_V(s|_V)|_{B_i} \in \mathcal{G}(B_i),$$

for every $i \in I$. Since the B_i cover V, the locality axiom for $\mathcal{G}$ gives

$$\eta_U(s)|_V = \eta_V(s|_V) \in \mathcal{G}(V).$$

Thus the collection $\{\eta_U\}_{U \in \tau_X}$ defines a morphism of sheaves

$$\eta \colon \mathcal{F} \longrightarrow \mathcal{G}.$$

By construction, $\eta_B = \eta_{\mathcal{B},B}$ as maps $\mathcal{F}(B) \to \mathcal{G}(B)$ for every $B \in \mathcal{B}$, so η extends $\eta_{\mathcal{B}}$.

Let us prove uniqueness. Suppose $\alpha \colon \mathcal{F} \to \mathcal{G}$ is another morphism extending $\eta_{\mathcal{B}}$. Fix an open $U \subset X$, a section $s \in \mathcal{F}(U)$, and a covering

$$U = \bigcup_{i \in I} B_i, \qquad B_i \in \mathcal{B}.$$

Then the section $\alpha_U(s) \in \mathcal{G}(U)$ satisfies

$$\alpha_U(s)|_{B_i} = \alpha_{B_i}(s|_{B_i}) = \eta_{B_i}(s|_{B_i}) = \eta_U(s)|_{B_i}$$

for every $i \in I$. Hence $\alpha_U(s) = \eta_U(s)$ by the locality axiom for $\mathcal{G}$. Thus $\alpha = \eta$, proving uniqueness.

We now prove the last assertions.

Suppose first that $\eta_B \colon \mathcal{F}(B) \to \mathcal{G}(B)$ is injective for every $B \in \mathcal{B}$. Let $U \subset X$ be open and let $s, t \in \mathcal{F}(U)$ satisfy

$$\eta_U(s) = \eta_U(t).$$

Choose a covering $U = \bigcup_i B_i$ by elements of $\mathcal{B}$. Then

$$\eta_{B_i}(s|_{B_i}) = \eta_U(s)|_{B_i} = \eta_U(t)|_{B_i} = \eta_{B_i}(t|_{B_i}).$$

Since η_{B_i} is injective, $s|_{B_i} = t|_{B_i}$ for every i, and therefore $s = t$. Thus η is injective.

Suppose now that every η_B is surjective. We claim that $\eta_x \colon \mathcal{F}_x \to \mathcal{G}_x$ is surjective for every $x \in X$. Let $t_x \in \mathcal{G}_x$ be a germ. Since the elements of $\mathcal{B}$ containing x form a local basis at x, we can represent t_x by a section $t \in \mathcal{G}(B)$ for some $B \in \mathcal{B}$ with $x \in B$. By surjectivity of η_B, there exists $s \in \mathcal{F}(B)$ such that $\eta_B(s) = t$. Passing to germs at x, we obtain

$$\eta_x(s_x) = t_x.$$

Hence η_x is surjective for every $x \in X$, so η is surjective.

Finally, if every η_B is an isomorphism, then η is both injective and surjective. Hence, by Lemma 1.4.0.9, we deduce that η is an isomorphism. □

COROLLARY 1.8.0.9. *The restriction functor of* (1.8.4),

$$\mathsf{res}_{\mathcal{B}}\colon \mathsf{Sh}(X,\mathscr{C}) \longrightarrow \mathsf{Sh}_{\mathcal{B}}(X,\mathscr{C}),$$

is an equivalence of categories.

Proof. Lemma 1.8.0.6 shows that $\mathsf{res}_{\mathcal{B}}$ is essentially surjective. On the other hand, Theorem 1.8.0.8 shows that, for every pair of sheaves $\mathcal{F},\mathcal{G}$, the map

$$\mathrm{Hom}_{\mathsf{Sh}(X,\mathscr{C})}(\mathcal{F},\mathcal{G}) \longrightarrow \mathrm{Hom}_{\mathsf{Sh}_{\mathcal{B}}(X,\mathscr{C})}\big(\mathsf{res}_{\mathcal{B}}(\mathcal{F}),\mathsf{res}_{\mathcal{B}}(\mathcal{G})\big)$$

is bijective. Thus $\mathsf{res}_{\mathcal{B}}$ is fully faithful and essentially surjective, hence an equivalence. □

1.9 Pushforward, inverse image

In this section we learn how to "move" sheaves from a topological space X to another topological space Y, in the presence of a continuous map between the two spaces.

$$\begin{array}{ccc} \mathcal{F} & \rightsquigarrow & f_*\mathcal{F} \\ \big| & & \big| \\ X & \xrightarrow{f} & Y \end{array} \qquad\qquad \begin{array}{ccc} f^{-1}\mathcal{G} & \leftsquigarrow & \mathcal{G} \\ \big| & & \big| \\ X & \xrightarrow{f} & Y \end{array}$$

1.9.1 Pushforward (or direct image)

Let $f\colon X \to Y$ be a continuous map of topological spaces, and let $\mathcal{F}$ be a presheaf on X. The assignment

$$V \longmapsto f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}V)$$

defines a presheaf $f_*\mathcal{F}$ on Y, called the *pushforward* (or *direct image*) of $\mathcal{F}$ by f. It is a sheaf as soon as $\mathcal{F}$ is, because if $V = \bigcup_{i\in I} V_i$ is an open covering of an open subset $V \subset Y$, then $f^{-1}V = \bigcup_{i\in I} f^{-1}(V_i)$ is an open covering of $f^{-1}V \subset X$.

Example 1.9.1.1. If X is arbitrary and $Y = \mathsf{pt}$, then $f_*\mathcal{F}(\mathsf{pt}) = \mathcal{F}(X)$, an object of $\mathscr{C}$. We will see in a minute that the direct image along any continuous map defines a functor. The direct image along the constant map $(X \to \mathsf{pt})_*\colon \mathsf{Sh}(X,\mathscr{C}) \to \mathscr{C}$ is also called the *global section functor*. If $\mathscr{C} = \mathsf{Mod}_A$, it is a left exact functor (you proved a more general statement in Exercise 1.6.0.11).

Example 1.9.1.2. If $f\colon X \hookrightarrow Y$ is the inclusion of a subspace, then $f_*\mathcal{F}$ is defined, for any open subset $V \subset Y$, by

$$f_*\mathcal{F}(V) = \mathcal{F}(V \cap X).$$

Example 1.9.1.3 (Skyscraper sheaf as a pushforward)**.** Let $x \in X$ be a point, G a nontrivial abelian group. Consider the constant sheaf $\underline{G}_{\{x\}}$ on $\{x\}$. Let $i_x\colon \{x\} \hookrightarrow X$ be

the inclusion. Then the skyscraper sheaf $G_x \in \mathsf{Sh}(X, \mathsf{Mod}_{\mathbb{Z}})$ defined in Example 1.4.0.13 can be described as

$$G_x = i_{x,*}\underline{G}_{\{x\}}.$$

Next, we observe that pushforward of sheaves is functorial, i.e. sending $\mathcal{F} \mapsto f_*\mathcal{F}$ defines functors

$$\begin{array}{ccc} \mathsf{Sh}(X, \mathscr{C}) & \xrightarrow{f_*} & \mathsf{Sh}(Y, \mathscr{C}) \\ \downarrow & & \downarrow \\ \mathsf{pSh}(X, \mathscr{C}) & \xrightarrow{f_*} & \mathsf{pSh}(Y, \mathscr{C}) \end{array}$$

where the vertical maps are the natural inclusions (1.2.2). Indeed, given a morphism of (pre)sheaves $\eta\colon \mathcal{F} \to \mathcal{G}$, we can construct a morphism of (pre)sheaves

$$f_*\eta\colon f_*\mathcal{F} \longrightarrow f_*\mathcal{G}$$

simply by setting

$$(f_*\eta)_V = \eta_{f^{-1}V}\colon \mathcal{F}(f^{-1}V) \longrightarrow \mathcal{G}(f^{-1}V)$$

for an open subset $V \subset Y$. The compatibility with restriction maps follows from those of η (and the obvious observation that if $V' \subset V$ then $f^{-1}V' \subset f^{-1}V$).

Moreover, $(-)_*$ is compatible with compositions of continuous maps, in the following sense: if $f\colon X \to Y$ and $g\colon Y \to Z$ are continuous maps of topological spaces, then, as functors, we have an equality $(g \circ f)_* = g_* \circ f_*$ on the nose (both for presheaves and for sheaves). In other words, the diagram

$$\begin{array}{ccc} \mathsf{Sh}(X, \mathscr{C}) & \xrightarrow{f_*} & \mathsf{Sh}(Y, \mathscr{C}) \\ & \searrow^{(g\circ f)_*} & \downarrow{\scriptstyle g_*} \\ & & \mathsf{Sh}(Z, \mathscr{C}) \end{array} \tag{1.9.1}$$

commutes. Indeed, if $\mathcal{F}$ is a (pre)sheaf on X, then for every open $W \subset Z$ one has

$$\begin{aligned} (g \circ f)_*\mathcal{F}(W) &= \mathcal{F}((g \circ f)^{-1}(W)) \\ &= \mathcal{F}(f^{-1}g^{-1}(W)) \\ &= f_*\mathcal{F}(g^{-1}(W)) \\ &= (g_*f_*\mathcal{F})(W) \\ &= (g_* \circ f_*)\mathcal{F}(W). \end{aligned}$$

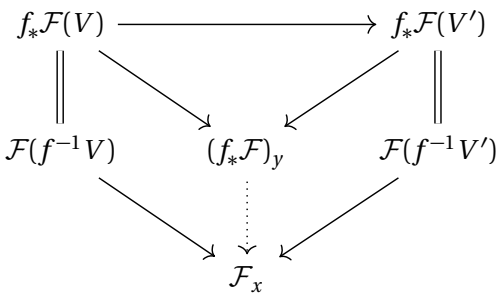
Note that no identifications are made here: all equalities are actual equalities!

LEMMA 1.9.1.4. *Let $f\colon X \to Y$ be a continuous map of topological spaces, and fix a sheaf $\mathcal{F} \in \mathsf{Sh}(X, \mathscr{C})$. Set $y = f(x)$. There is a canonical morphism*

$$(f_*\mathcal{F})_y \longrightarrow \mathcal{F}_x,$$

which is an isomorphism when f is the inclusion of a subspace $X \hookrightarrow Y$.

Proof. If $y \in V' \subset V \subset Y$, then $x \in f^{-1}V' \subset f^{-1}V \subset X$, and the commutative diagram



induces, via the universal property of the stalk

$$(f_*\mathcal{F})_y = \varinjlim_{V \ni y} \mathcal{F}(f^{-1}V),$$

a canonical morphism $(f_*\mathcal{F})_y \to \mathcal{F}_x$, as required.

Now, let us assume $f : X \hookrightarrow Y$ is the inclusion of a subspace, and let us take $y \in X$. Note that every neighbourhood $y \in U \subset X$ is of the form $U = V \cap X$ for some open neighbourhood $y \in V \subset Y$. Thus

$$(f_*\mathcal{F})_y = \varinjlim_{Y \supset V \ni y} \mathcal{F}(V \cap X) \xrightarrow{\sim} \varinjlim_{X \supset U \ni y} \mathcal{F}(U) = \mathcal{F}_y. \tag{1.9.2}$$

The proof is complete. □

Remark 1.9.1.5. We shall use Lemma 1.9.1.4 crucially with $\mathscr{C} = \mathrm{Rings}$, when defining morphisms of locally ringed spaces (cf. Definition 1.11.0.3 and the footnote there).



Caution 1.9.1.6. Even if $f : X \hookrightarrow Y$ is the inclusion of a subspace, it is not true that $(f_*\mathcal{F})_y = 0$ for all $y \in Y \setminus X$. This is nevertheless true when f is the inclusion of a *closed* subspace, cf. Remark 1.9.1.7.

Remark 1.9.1.7. If $f : X \hookrightarrow Y$ is the inclusion of a *closed* subspace, and $\mathcal{F}$ is a sheaf on X, then

$$(f_*\mathcal{F})_y = \begin{cases} \mathcal{F}_y & \text{if } y \in X \\ 0 & \text{if } y \notin X. \end{cases} \tag{1.9.3}$$

The case $y \in X$ is the computation (1.9.2). As for the case $y \notin X$, we use the definition

$$(f_*\mathcal{F})_y = \varinjlim_{Y \supset V \ni y} f_*\mathcal{F}(V) = \varinjlim_{Y \supset V \ni y} \mathcal{F}(V \cap X),$$

and the observation that, since X is closed in Y, there are arbitrarily small neighbourhoods V of y which are disjoint from X. For these, we have $\mathcal{F}(V \cap X) = \mathcal{F}(\emptyset) = 0$ since $\mathcal{F}$ is a sheaf (Remark 1.3.0.2). This causes the colimit to vanish.

Exactness of pushforward

We set $\mathscr{C} = \mathsf{Mod}_A$ in this subsection (for A a fixed ring), and we fix a continuous map $f: X \to Y$. We now say two (important) words on the exactness of the direct image functor

$$f_*\colon \mathsf{Sh}(X, \mathsf{Mod}_A) \longrightarrow \mathsf{Sh}(Y, \mathsf{Mod}_A).$$

It is important to remember that

f_* is always left exact, and it is exact if $f: X \hookrightarrow Y$ is a closed subspace.

Since f_* will turn out to be a right adjoint (Lemma 1.9.2.8), it is left exact by general category theory. However, we prove it directly here. Note that you have already proved the case $Y = \mathsf{pt}$ in Exercise 1.6.0.11. You will notice in the proof of the above slogan that this was essentially enough to handle the general case.

PROPOSITION 1.9.1.8. *Let A be a ring, $f: X \to Y$ a continuous map of topological spaces. The functor $f_*\colon \mathsf{Sh}(X, \mathsf{Mod}_A) \to \mathsf{Sh}(Y, \mathsf{Mod}_A)$ is left exact. If f is the inclusion of a closed subspace, then f_* is exact.*

Proof. Let us prove the first assertion. We need to show that an exact sequence

$$0 \longrightarrow \mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H}$$

in $\mathsf{Sh}(X, \mathsf{Mod}_A)$ induces an exact sequence

$$0 \longrightarrow f_*\mathcal{F} \xrightarrow{f_*\alpha} f_*\mathcal{G} \xrightarrow{f_*\beta} f_*\mathcal{H}$$

in $\mathsf{Sh}(Y, \mathsf{Mod}_A)$. We know by Exercise 1.6.0.11 that we have an exact sequence

$$0 \longrightarrow \mathcal{F}(f^{-1}V) \xrightarrow{\alpha_{f^{-1}V}} \mathcal{G}(f^{-1}V) \xrightarrow{\beta_{f^{-1}V}} \mathcal{H}(f^{-1}V) \tag{1.9.4}$$

for any open subset $V \subset Y$, by applying the functor $\Gamma(f^{-1}V, -)$ to the original sequence. In particular, $\alpha_{f^{-1}V} = (f_*\alpha)_V$ is injective for all V, which shows that $f_*\alpha$ is injective. There is an equality of presheaves

$$\mathrm{im}_{\mathrm{pre}}(f_*\alpha) = \ker(f_*\beta)$$

again thanks to exactness of (1.9.4) in the middle, ensuring precisely that $\mathrm{im}(\alpha_{f^{-1}V}) = \ker(\beta_{f^{-1}V})$. But $\ker(f_*\beta)$ is a sheaf, therefore we get exactness in the middle, i.e. $\mathrm{im}(f_*\alpha) = \ker(f_*\beta)$.

Let us show the second statement. Assume f is the inclusion of a closed subspace. By the first part of the proof, we only need to show that if $\eta: \mathcal{G} \twoheadrightarrow \mathcal{H}$ is surjective as a map of sheaves on X, then $f_*\mathcal{G} \twoheadrightarrow f_*\mathcal{H}$ is surjective as a map of sheaves on Y. If $y \in Y \setminus X$, then (using that X is closed, cf. Remark 1.9.1.7)

$$(f_*\mathcal{G})_y = 0 = (f_*\mathcal{H})_y. \tag{1.9.5}$$

Assume $y \in X$. Since $\mathcal{G}$ surjects onto $\mathcal{H}$, in the commutative diagram

$$\begin{array}{ccc} (f_*\mathcal{G})_y & \xrightarrow{(f_*\eta)_y} & (f_*\mathcal{H})_y \\ \| & & \| \\ \mathcal{G}_y & \xrightarrow{\eta_y} & \mathcal{H}_y \end{array}$$

the bottom map is surjective. The vertical equalities are given by Remark 1.9.1.7. Thus the top map is surjective as well. Hence $f_*\eta$ is surjective on all stalks, hence it is surjective. □

1.9.2 Inverse image

Let $f: X \to Y$ be a continuous map of topological spaces. Let $\mathcal{G}$ be a presheaf on Y. Given $U \subset X$, the collection of open subsets $V \subset Y$ containing $f(U)$ form a directed set via reverse inclusions. Sending

$$U \longmapsto (f_{\text{pre}}^{-1}\mathcal{G})(U) = \varinjlim_{V \supset f(U)} \mathcal{G}(V)$$

defines a presheaf on X. Indeed, assume $U' \subset U$ is an open subset. Then there is an inclusion $f(U') \subset f(U)$, inducing a map of directed systems

$$\{V \in \tau_Y \mid V \supset f(U)\} \hookrightarrow \{V \in \tau_Y \mid V \supset f(U')\},$$

which in turn induces a morphism

$$(f_{\text{pre}}^{-1}\mathcal{G})(U) = \varinjlim_{V \supset f(U)} \mathcal{G}(V) \longrightarrow \varinjlim_{V \supset f(U')} \mathcal{G}(V) = (f_{\text{pre}}^{-1}\mathcal{G})(U').$$

This is the restriction morphism for $f_{\text{pre}}^{-1}\mathcal{G}$.

Remark 1.9.2.1. If $f(U)$ is an open subset of Y, then $(f_{\text{pre}}^{-1}\mathcal{G})(U) = \mathcal{G}(f(U))$.

Now assume $\mathcal{G}$ is a sheaf. We define the *inverse image* of $\mathcal{G}$ by f to be the sheafification

$$f^{-1}\mathcal{G} = \left(f_{\text{pre}}^{-1}\mathcal{G}\right)^{\#}.$$

By Proposition 1.5.0.2, the canonical map $f_{\text{pre}}^{-1}\mathcal{G} \to f^{-1}\mathcal{G}$ of presheaves induces an isomorphism

$$(f_{\text{pre}}^{-1}\mathcal{G})_x \xrightarrow{\sim} (f^{-1}\mathcal{G})_x$$

on all the stalks.



Exercise 1.9.2.2. Both f_{pre}^{-1} and f^{-1} are functors.

Example 1.9.2.3. Let $\iota_y: \{y\} \hookrightarrow Y$ be the inclusion of a point $y \in Y$, and let $\mathcal{G}$ be a sheaf on Y. Then $\iota_y^{-1}\mathcal{G} = \mathcal{G}_y$, since $\iota_y^{-1}\mathcal{G}(\{y\}) = \varinjlim_{V \ni y} \mathcal{G}(V) = \mathcal{G}_y$. Thus ι_y^{-1} agrees with the stalk functor

$$\mathsf{stalk}_y: \mathsf{Sh}(Y, \mathscr{C}) \longrightarrow \mathscr{C}, \quad \mathcal{G} \mapsto \mathcal{G}_y.$$

Example 1.9.2.4. If $p: X \to \mathsf{pt}$ is the constant map, and $G \in \mathscr{C} \cong \mathsf{Sh}(\mathsf{pt}, \mathscr{C})$, then $p^{-1}G = \underline{G}_X$, the constant sheaf on X with values in the object G.

Example 1.9.2.5. Let $j: U \hookrightarrow Y$ be the inclusion of an open subset. Then $j_{\mathrm{pre}}^{-1}\mathcal{G} = \mathcal{G}|_U$ for any sheaf $\mathcal{G}$ on Y. The reason is that if U' is open in U, it is also open in Y, and thus

$$j_{\mathrm{pre}}^{-1}\mathcal{G}(U') = \varinjlim_{V \supset U'} \mathcal{G}(V) = \mathcal{G}(U') = \mathcal{G}|_U(U').$$

In particular, $j_{\mathrm{pre}}^{-1}\mathcal{G}$ is already a sheaf, and hence

$$\boxed{j^{-1}\mathcal{G} = \mathcal{G}|_U, \quad j: U \hookrightarrow Y \text{ open.}}$$

Remark 1.9.2.6. Despite Example 1.9.2.5, sheafifying f_{pre}^{-1} is in general necessary: consider a constant map $f: X = \{\star, \bullet\} \to \{*\} = Y$ from a discrete two point space, and fix a nontrivial abelian group G. The constant sheaf $\mathcal{G} = \underline{G}_Y$ has the property $f_{\mathrm{pre}}^{-1}\mathcal{G} = \underline{G}_X^{\mathrm{pre}}$, which is not a sheaf (cf. Example 1.2.0.19).

Functoriality (cf. Exercise 1.9.2.2) can be translated into a diagram of functors

$$\begin{array}{ccc} \mathsf{Sh}(Y, \mathscr{C}) & \xrightarrow{f^{-1}} & \mathsf{Sh}(X, \mathscr{C}) \\ {\scriptstyle j_{Y,\mathscr{C}}}\downarrow & & \downarrow\uparrow{\scriptstyle (-)^{\#}} \\ \mathsf{pSh}(Y, \mathscr{C}) & \xrightarrow{f_{\mathrm{pre}}^{-1}} & \mathsf{pSh}(X, \mathscr{C}) \end{array}$$

where $f^{-1} = (-)^{\#} \circ f_{\mathrm{pre}}^{-1} \circ j_{Y,\mathscr{C}}$.



Exercise 1.9.2.7. Let $f: X \to Y$ and $g: Y \to Z$ be continuous maps of topological spaces. Show that there is a canonical natural isomorphism[5]

$$f^{-1} \circ g^{-1} \xrightarrow{\sim} (g \circ f)^{-1}$$

of functors $\mathsf{Sh}(Z, \mathscr{C}) \to \mathsf{Sh}(X, \mathscr{C})$.

LEMMA 1.9.2.8 (Unit and counit maps). *For any pair of presheaves $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ and $\mathcal{G} \in \mathsf{pSh}(Y, \mathscr{C})$ there are canonical presheaf homomorphisms*

$$\mathcal{G} \xrightarrow{\mathsf{unit}^{\mathcal{G}}} f_* f_{\mathrm{pre}}^{-1}\mathcal{G}, \qquad f_{\mathrm{pre}}^{-1} f_* \mathcal{F} \xrightarrow{\mathsf{counit}^{\mathcal{F}}} \mathcal{F}.$$

Proof. We start with the map $\mathsf{unit}^{\mathcal{G}}$. The observation here is that there is, for any open subset $V \subset Y$, a canonical inclusion $f(f^{-1}V) \hookrightarrow V$. Thus $\mathcal{G}(V)$ appears in the colimit

$$\varinjlim_{W \supset f(f^{-1}V)} \mathcal{G}(W).$$

[5] Note: unlike the pushforward, this is not an equality on the nose, just a natural isomorphism.

This induces a canonical morphism

$$\mathsf{unit}_V^{\mathcal{G}}\colon \mathcal{G}(V) \longrightarrow \varinjlim_{W\supset f(f^{-1}V)} \mathcal{G}(W) = f_{\mathrm{pre}}^{-1}\mathcal{G}(f^{-1}V) = f_*f_{\mathrm{pre}}^{-1}\mathcal{G}(V)$$

which does define a natural transformation $\mathcal{G}\to f_*f_{\mathrm{pre}}^{-1}\mathcal{G}$ because if $V'\subset V$, then any open $W\subset Y$ containing $f(f^{-1}V)$ also contains $f(f^{-1}V')$, simply because $f(f^{-1}V')\subset f(f^{-1}V)$. Thus there is a natural morphism

$$\varinjlim_{W\supset f(f^{-1}V)} \mathcal{G}(W) \longrightarrow \varinjlim_{W\supset f(f^{-1}V')} \mathcal{G}(W)$$

and the induced diagram

$$\begin{array}{ccccc}
\mathcal{G}(V) & \xrightarrow{\mathsf{unit}_V^{\mathcal{G}}} & \varinjlim_{W\supset f(f^{-1}V)} \mathcal{G}(W) & = & f_*f_{\mathrm{pre}}^{-1}\mathcal{G}(V) \\
\downarrow & & \downarrow & & \\
\mathcal{G}(V') & \xrightarrow{\mathsf{unit}_{V'}^{\mathcal{G}}} & \varinjlim_{W\supset f(f^{-1}V')} \mathcal{G}(W) & = & f_*f_{\mathrm{pre}}^{-1}\mathcal{G}(V')
\end{array}$$

commutes. This defines the map $\mathsf{unit}^{\mathcal{G}}\colon \mathcal{G}\to f_*f_{\mathrm{pre}}^{-1}\mathcal{G}$ of presheaves.

To construct the map $\mathsf{counit}^{\mathcal{F}}$, one observes that for any open subset $U\subset X$ there is a canonical map

$$f_{\mathrm{pre}}^{-1}f_*\mathcal{F}(U) = \varinjlim_{V\supset f(U)} f_*\mathcal{F}(V) = \varinjlim_{V\supset f(U)} \mathcal{F}(f^{-1}V) \longrightarrow \mathcal{F}(U),$$

since if $V\supset f(U)$ inside Y, then $U\subset f^{-1}f(U)\subset f^{-1}V$ inside X. This map is also functorial in $U'\subset U$, thus the map $\mathsf{counit}^{\mathcal{F}}\colon f_{\mathrm{pre}}^{-1}f_*\mathcal{F}\to\mathcal{F}$ is defined. □

The usefulness of the homomorphisms $\mathsf{unit}^{\mathcal{G}}$ and $\mathsf{counit}^{\mathcal{F}}$ is that they make $(f_{\mathrm{pre}}^{-1}, f_*)$ into an adjoint pair of functors. More precisely, there are bijections

$$\varphi_{\mathcal{F},\mathcal{G}}\colon \mathrm{Hom}_{\mathsf{pSh}(Y,\mathscr{C})}(\mathcal{G}, f_*\mathcal{F}) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{pSh}(X,\mathscr{C})}(f_{\mathrm{pre}}^{-1}\mathcal{G}, \mathcal{F}),$$

functorial in both $\mathcal{F}$ and $\mathcal{G}$. Specifically, $\varphi_{\mathcal{F},\mathcal{G}}$ sends $\eta\colon\mathcal{G}\to f_*\mathcal{F}$ to

$$f_{\mathrm{pre}}^{-1}\mathcal{G} \xrightarrow{f_{\mathrm{pre}}^{-1}\eta} f_{\mathrm{pre}}^{-1}f_*\mathcal{F} \xrightarrow{\mathsf{counit}^{\mathcal{F}}} \mathcal{F}$$

with inverse sending $\zeta\colon f_{\mathrm{pre}}^{-1}\mathcal{G}\to\mathcal{F}$ to

$$\mathcal{G} \xrightarrow{\mathsf{unit}^{\mathcal{G}}} f_*f_{\mathrm{pre}}^{-1}\mathcal{G} \xrightarrow{f_*\zeta} f_*\mathcal{F}.$$

Using the adjunction

$$\mathsf{pSh}(Y,\mathscr{C}) \underset{f_*}{\overset{f_{\mathrm{pre}}^{-1}}{\rightleftarrows}} \mathsf{pSh}(X,\mathscr{C}) \tag{1.9.6}$$

it is immediate to show that also

$$\mathsf{Sh}(Y,\mathscr{C}) \underset{f_*}{\overset{f^{-1}}{\rightleftarrows}} \mathsf{Sh}(X,\mathscr{C}) \tag{1.9.7}$$

is an adjoint pair of functors. Indeed, for any pair of sheaves $\mathcal{F} \in \mathsf{Sh}(X,\mathscr{C})$ and $\mathcal{G} \in \mathsf{Sh}(Y,\mathscr{C})$, we have

$$\begin{aligned}
\mathrm{Hom}_{\mathsf{Sh}(Y,\mathscr{C})}(\mathcal{G}, f_*\mathcal{F}) &= \mathrm{Hom}_{\mathsf{pSh}(Y,\mathscr{C})}(\mathcal{G}, f_*\mathcal{F}) && j_{Y,\mathscr{C}} \text{ is fully faithful} \\
&\xrightarrow{\sim} \mathrm{Hom}_{\mathsf{pSh}(X,\mathscr{C})}(f_{\mathrm{pre}}^{-1}\mathcal{G}, \mathcal{F}) && \text{adjunction (1.9.6)} \\
&\xrightarrow{\sim} \mathrm{Hom}_{\mathsf{Sh}(X,\mathscr{C})}(f^{-1}\mathcal{G}, \mathcal{F}) && \text{Exercise 1.5.0.6.}
\end{aligned}$$

Remark 1.9.2.9. Let us revisit Exercise 1.9.2.7. The canonical isomorphism claimed there is in fact easily obtained as a consequence of uniqueness of adjoints, applied to the observation that both $f^{-1} \circ g^{-1}$ and $(g \circ f)^{-1}$ serve as left adjoints to $(g \circ f)_* = g_* \circ f_*$.

Remark 1.9.2.10. Fix $\mathcal{G} \in \mathsf{Sh}(Y,\mathscr{C})$. Once more, the adjunction (1.9.7) gives a canonical morphism $\mathcal{G} \to f_* f^{-1}\mathcal{G}$, corresponding to $\mathrm{id}_{f^{-1}\mathcal{G}}$ under

$$\mathrm{Hom}_{\mathsf{Sh}(Y,\mathscr{C})}(\mathcal{G}, f_* f^{-1}\mathcal{G}) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{Sh}(X,\mathscr{C})}(f^{-1}\mathcal{G}, f^{-1}\mathcal{G}).$$

Sending $\mathcal{G} \mapsto f_* f^{-1}\mathcal{G}$ is a functor $f_* f^{-1} \colon \mathsf{Sh}(Y,\mathscr{C}) \to \mathsf{Sh}(Y,\mathscr{C})$, and the naturality of this operation yields a natural transformation

$$\mathsf{unit} \colon \mathrm{Id}_{\mathsf{Sh}(Y,\mathscr{C})} \Longrightarrow f_* f^{-1}$$

of functors $\mathsf{Sh}(Y,\mathscr{C}) \to \mathsf{Sh}(Y,\mathscr{C})$, which is called the *unit* of the adjunction (1.9.7). Similarly, if $\mathcal{F} \in \mathsf{Sh}(X,\mathscr{C})$, there is a canonical morphism $f^{-1} f_* \mathcal{F} \to \mathcal{F}$ corresponding to $\mathrm{id}_{f_*\mathcal{F}}$ under

$$\mathrm{Hom}_{\mathsf{Sh}(Y,\mathscr{C})}(f_*\mathcal{F}, f_*\mathcal{F}) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{Sh}(X,\mathscr{C})}(f^{-1} f_*\mathcal{F}, \mathcal{F}).$$

Sending $\mathcal{F} \mapsto f^{-1} f_* \mathcal{F}$ is a functor $f^{-1} f_* \colon \mathsf{Sh}(X,\mathscr{C}) \to \mathsf{Sh}(X,\mathscr{C})$, and the naturality of this operation yields a natural transformation

$$\mathsf{counit} \colon f^{-1} f_* \Longrightarrow \mathrm{Id}_{\mathsf{Sh}(X,\mathscr{C})}$$

of functors $\mathsf{Sh}(X,\mathscr{C}) \to \mathsf{Sh}(X,\mathscr{C})$, which is called the *counit* of the adjunction (1.9.7).

The next lemma says that the stalk of the inverse image is somewhat easy to compute (unlike for the pushforward).

LEMMA 1.9.2.11 (Stalk of inverse image). *Let $f \colon X \to Y$ be a continuous map of topological spaces, $\mathcal{G}$ a sheaf on Y, and $x \in X$ a point. There is a canonical isomorphism*

$$(f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)}.$$

Proof. We have

$$\begin{aligned}(f^{-1}\mathcal{G})_x &= (\{x\} \hookrightarrow X)^{-1}(f^{-1}\mathcal{G}) && \text{by Example 1.9.2.3}\\ &\cong (\{x\} \hookrightarrow X \to Y)^{-1}\mathcal{G} && \text{by Exercise 1.9.2.7}\\ &= (\{f(x)\} \hookrightarrow Y)^{-1}\mathcal{G}\\ &= \mathcal{G}_{f(x)}\end{aligned}$$

where we have used Example 1.9.2.3 once more for the last identity. □

PROPOSITION 1.9.2.12. *Let $\mathscr{C} = \mathsf{Mod}_A$, for a ring A. Then the inverse image functor*

$$f^{-1}\colon \mathsf{Sh}(Y, \mathsf{Mod}_A) \longrightarrow \mathsf{Sh}(X, \mathsf{Mod}_A)$$

is exact.

Proof. Indeed, let

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow \mathcal{K} \longrightarrow 0$$

be an exact sequence in $\mathsf{Sh}(Y, \mathsf{Mod}_A)$. Then,

$$0 \longrightarrow \mathcal{G}_{f(x)} \longrightarrow \mathcal{H}_{f(x)} \longrightarrow \mathcal{K}_{f(x)} \longrightarrow 0$$

is exact in Mod_A by Proposition 1.6.0.9, for every $x \in X$. But this is, by Lemma 1.9.2.11, canonically isomorphic to the sequence

$$0 \longrightarrow (f^{-1}\mathcal{G})_x \longrightarrow (f^{-1}\mathcal{H})_x \longrightarrow (f^{-1}\mathcal{K})_x \longrightarrow 0.$$

Thus

$$0 \longrightarrow f^{-1}\mathcal{G} \longrightarrow f^{-1}\mathcal{H} \longrightarrow f^{-1}\mathcal{K} \longrightarrow 0$$

is exact, again by Proposition 1.6.0.9. □



Exercise 1.9.2.13. Show that if $f\colon X \hookrightarrow Y$ is the inclusion of a subspace, then the counit

$$f^{-1}f_*\mathcal{F} \xrightarrow{\sim} \mathcal{F}$$

is an isomorphism for every $\mathcal{F} \in \mathsf{Sh}(X, \mathscr{C})$. (**Hint**: check it on stalks).

PROPOSITION 1.9.2.14. *Let $f\colon X \hookrightarrow Y$ be the inclusion of a closed subspace. Let A be a ring.*

(1) *Let $\mathcal{G} \in \mathsf{Sh}(Y, \mathsf{Mod}_A)$ be a sheaf such that $\mathsf{Supp}(\mathcal{G}) \subset X$. Then the unit map*

$$\mathcal{G} \xrightarrow{\sim} f_*f^{-1}\mathcal{G}$$

is an isomorphism.

(2) *The functor f_* induces an equivalence of categories*

$$f_*\colon \mathsf{Sh}(X,\mathsf{Mod}_A) \xrightarrow{\sim} \mathsf{Sh}_X(Y,\mathsf{Mod}_A),$$

where $\mathsf{Sh}_X(Y,\mathsf{Mod}_A) \hookrightarrow \mathsf{Sh}(Y,\mathsf{Mod}_A)$ is the full subcategory of sheaves on Y with support contained in X.

Proof. To prove (1) it is enough to prove that the unit map is an isomorphism on all the stalks. If $y \in Y \setminus X$, we get $0 \xrightarrow{\sim} 0$, since $\mathcal{G}_y = 0$ by the assumption $\mathsf{Supp}(\mathcal{G}) \subset X$ and $(f_*f^{-1}\mathcal{G})_y = 0$ by (1.9.3). On the other hand, if $y \in X$, then $\mathcal{G}_y \to (f_*f^{-1}\mathcal{G})_y$ is nothing but the inverse of the isomorphism

$$(f_*f^{-1}\mathcal{G})_y \xrightarrow{\sim} (f^{-1}\mathcal{G})_y = \mathcal{G}_y$$

of Lemma 1.9.1.4.

To prove (2), observe first of all that f_* lands in the category $\mathsf{Sh}_X(Y,\mathsf{Mod}_A)$ by (1.9.3). Next, note that sending $\mathcal{G} \mapsto f^{-1}\mathcal{G}$ is an inverse to f_* by (1). In a little more detail, the equivalence (cf. Definition A.3.0.4) is set up by considering the pair of functors (f_*, f^{-1}) and exploiting the unit and counit *natural isomorphisms*

$$\mathsf{unit}\colon \mathrm{Id}_{\mathsf{Sh}_X(Y,\mathsf{Mod}_A)} \Longrightarrow f_*f^{-1} \qquad \mathsf{counit}\colon f^{-1}f_* \Longrightarrow \mathrm{Id}_{\mathsf{Sh}(X,\mathsf{Mod}_A)}$$

using (1) and Exercise 1.9.2.13. □



Exercise 1.9.2.15. Find examples of maps f and sheaves $\mathcal{G}$ such that $\mathcal{G} \to f_*f^{-1}\mathcal{G}$ is not an isomorphism. (**Hint**: Consider the inclusion of a suitable proper open subspace).

Remark 1.9.2.16. If $j\colon X \hookrightarrow Y$ is *open* and $\mathcal{G}$ is a sheaf on Y, then $j_*j^{-1}\mathcal{G}$ satisfies

$$j_*j^{-1}\mathcal{G}(V) = (j_*\mathcal{G}|_X)(V) = \mathcal{G}(V \cap X), \quad V \subset Y \text{ open.}$$

The natural map $\mathcal{G}(V) \to j_*j^{-1}\mathcal{G}(V)$ sends $s \mapsto s|_{V\cap X}$.

1.10 Gluing sheaves

The purpose of this section is to state the next theorem, which is of crucial importance.

THEOREM 1.10.0.1 ([10, Tag 00AK]). *Let X be a topological space, $X = \bigcup_{i\in I} U_i$ an open covering. Set $U_{ij} = U_i \cap U_j$ and similarly $U_{ijk} = U_{ij} \cap U_k$. Assume given a sheaf $\mathcal{F}_i$ on U_i for every $i \in I$, along with a collection of isomorphisms*

$$\varphi_{ij}\colon \mathcal{F}_i\big|_{U_{ij}} \xrightarrow{\sim} \mathcal{F}_j\big|_{U_{ij}}$$

such that $\varphi_{ii} = \mathrm{id}_{\mathcal{F}_i}$ for every i, and such that

$$\begin{array}{ccc} \mathcal{F}_i\big|_{U_{ijk}} & \xrightarrow{\varphi_{ij}|_{U_{ijk}}} & \mathcal{F}_j\big|_{U_{ijk}} \\ & \searrow^{\varphi_{ik}|_{U_{ijk}}} & \downarrow{\scriptstyle \varphi_{jk}|_{U_{ijk}}} \\ & & \mathcal{F}_k\big|_{U_{ijk}} \end{array}$$

commutes for every triple intersection. Then there exists a sheaf $\mathcal{F}$ on X, unique up to unique isomorphism compatible with the isomorphisms

$$\alpha_i : \mathcal{F}\big|_{U_i} \xrightarrow{\sim} \mathcal{F}_i$$

such that the diagrams

$$\begin{array}{ccc} \mathcal{F}\big|_{U_{ij}} & \xrightarrow{\alpha_j|_{U_{ij}}} & \mathcal{F}_j\big|_{U_{ij}} \\ & {\scriptstyle \alpha_i|_{U_{ij}}} \searrow \quad \nearrow {\scriptstyle \varphi_{ij}} & \\ & \mathcal{F}_i\big|_{U_{ij}} & \end{array}$$

commute for every $(i,j) \in I \times I$. The sheaf $\mathcal{F}$ is called the gluing of $(\mathcal{F}_i, \varphi_{ij})_{i,j}$ along the given covering.

1.11 Locally ringed spaces

Given our background on sheaves, we are ready for the definition of locally ringed space.

Definition 1.11.0.1 (Locally ringed space)**.** A *ringed space* is a pair $(X, \mathcal{O}_X)$ where X is a topological space and $\mathcal{O}_X$ is a sheaf of commutative rings on X. The sheaf $\mathcal{O}_X$ is called the *structure sheaf*. A *locally ringed space* is a ringed space such that the stalk $\mathcal{O}_{X,x}$ is a local ring for every $x \in X$.

Notation 1.11.0.2. Let $(X, \mathcal{O}_X)$ be a locally ringed space, $x \in X$ a point. We will write $\mathfrak{m}_x$ for the maximal ideal $\mathcal{O}_{X,x}$, and $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$ for the corresponding *residue field*.

Recall that, given two local rings $(B, \mathfrak{m}_B)$ and $(A, \mathfrak{m}_A)$, a local homomorphism between them is a ring homomorphism $h : B \to A$ such that $h^{-1}(\mathfrak{m}_A) = \mathfrak{m}_B$, or, equivalently, $h(\mathfrak{m}_B) \subset \mathfrak{m}_A$.

Definition 1.11.0.3 (Morphism of locally ringed spaces)**.** A morphism of ringed spaces, denoted

$$(X, \mathcal{O}_X) \xrightarrow{(f, f^\#)} (Y, \mathcal{O}_Y), \tag{1.11.1}$$

is a pair $(f, f^\#)$ where $f : X \to Y$ is a continuous map between the underlying topological spaces and $f^\# : \mathcal{O}_Y \to f_* \mathcal{O}_X$ is a homomorphism of sheaves of rings on Y. If $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ are locally ringed spaces, a morphism between them is a morphism of ringed spaces such that the canonical map

$$f_x^\# : \mathcal{O}_{Y, f(x)} \longrightarrow (f_* \mathcal{O}_X)_{f(x)} \longrightarrow \mathcal{O}_{X,x}$$

is a local homomorphism of local rings for every $x \in X$.[6]

[6] The second map in this composition is the one provided in Lemma 1.9.1.4.

Example 1.11.0.4. Assume $X = \mathsf{pt}$. Then a (locally) ringed space $(\mathsf{pt}, \mathscr{O}_{\mathsf{pt}})$ is just the choice of a (local) ring A. A morphism of (locally) ringed spaces $(\mathsf{pt}, A) \to (\mathsf{pt}, B)$ is the same as a (local) ring homomorphism $B \to A$.

Notation 1.11.0.5. In what follows, when there is no confusion possible, we shall omit the sheaf of rings from the notation, and simply write X to denote the (locally) ringed space $(X, \mathscr{O}_X)$, or $f : X \to Y$ to denote a morphism $(f, f^\#)$ of (locally) ringed spaces as in (1.11.1). When we want to emphasise the underlying topological space of $(X, \mathscr{O}_X)$, we write $|X|$.

Remark 1.11.0.6. Note that any morphism of locally ringed spaces $X \to Y$ induces, at every x, a field extension

$$\kappa(y) \hookrightarrow \kappa(x), \quad y = f(x).$$

Example 1.11.0.7. Let $(X, \mathscr{O}_X)$ be a (locally) ringed space, $U \subset X$ an open subset. Then $(U, \mathscr{O}_X|_U)$ is a (locally) ringed space. We shall *always* take $\mathscr{O}_X|_U$ as the structure sheaf of an open subset $U \subset X$ of a locally ringed space X. We denote it by $\mathscr{O}_U$.

The composition of two morphisms of locally ringed spaces is defined in a straightforward way (but you need to know that pushforward commutes with composition, see (1.9.1)). In a little more detail, consider two morphisms

$$(X, \mathscr{O}_X) \xrightarrow{(f, f^\#)} (Y, \mathscr{O}_Y), \quad (Y, \mathscr{O}_Y) \xrightarrow{(g, g^\#)} (Z, \mathscr{O}_Z)$$

and define their composition to be the morphism

$$(X, \mathscr{O}_X) \xrightarrow{(g \circ f, (g \circ f)^\#)} (Z, \mathscr{O}_Z)$$

where the map on sheaves $(g \circ f)^\#$ is the composition

$$\mathscr{O}_Z \xrightarrow{g^\#} g_* \mathscr{O}_Y \xrightarrow{g_* f^\#} g_* f_* \mathscr{O}_X = (g \circ f)_* \mathscr{O}_X.$$

Locally ringed spaces thus form a (large) category, denoted

$$\mathsf{LRS},$$

where isomorphisms are simply the invertible morphisms (those that compose to the identity both ways).

Remark 1.11.0.8. A morphism of locally ringed spaces $(f, f^\#)$ as in (1.11.1) is an isomorphism if and only if

- $f : X \to Y$ is a topological homeomorphism, and
- $f^\# : \mathscr{O}_Y \to f_* \mathscr{O}_X$ is an isomorphism of sheaves.

Definition 1.11.0.9 (Immersions)**.** Let $(f, f^{\#})\colon (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of locally ringed spaces. It is called an *open immersion* (resp. a *closed immersion*) if $f\colon X \to Y$ is a topological open immersion (resp. closed immersion) and $f^{\#}_x$ is an isomorphism (resp. surjective) for every $x \in X$. It is called an *immersion* (or a *locally closed immersion*) if it factors as a closed immersion followed by an open immersion.

Notation 1.11.0.10. We sometimes will denote an immersion by '$X \hookrightarrow Y$'.

It is clear that a morphism of locally ringed spaces $(f, f^{\#})\colon (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is an open immersion if and only if there exists an open subset $V \subset Y$ such that $(f, f^{\#})$ induces an isomorphism $(X, \mathscr{O}_X) \xrightarrow{\sim} (V, \mathscr{O}_Y|_V)$. It is also clear that the composition of two open (resp. closed) immersions is an open immersion (resp. a closed immersion).

1.12 Ideal sheaves = Closed immersions up to isomorphism

Unlike open immersions, closed immersions are quite subtle. They are characterised by *ideal sheaves*, in a way that we now describe.

Definition 1.12.0.1 (Ideal sheaf)**.** Fix a locally ringed space $(X, \mathscr{O}_X)$. An *ideal sheaf* (or a *sheaf of ideals*) is a subsheaf $\mathscr{I} \subset \mathscr{O}_X$ (as abelian groups) such that $\mathscr{I}(U) \subset \mathscr{O}_X(U)$ is an ideal for every open subset $U \subset X$.

Given an ideal sheaf $\mathscr{I}$, the subset

$$\mathrm{V}(\mathscr{I}) = \left\{ x \in X \;\middle|\; \mathscr{I}_x \neq \mathscr{O}_{X,x} \right\} \xhookrightarrow{j} X \tag{1.12.1}$$

is a closed subset. Indeed, for any $x \in X \setminus \mathrm{V}(\mathscr{I})$, i.e. for any x such that $\mathscr{I}_x = \mathscr{O}_{X,x}$, there is a neighbourhood U of x and a section $f \in \mathscr{I}(U)$ such that $f_x = 1 \in \mathscr{O}_{X,x}$. But this means that $f|_V = 1 \in \mathscr{O}_X(V)$ for some open subset $V \subset U$. Thus $V \subset X \setminus \mathrm{V}(\mathscr{I})$, and thus $X \setminus \mathrm{V}(\mathscr{I})$ is open.

The quotient sheaf $\mathscr{O}_X/\mathscr{I}$ is a sheaf of rings (because, by definition, it is the sheafification of a presheaf of rings), not just abelian groups. The pair

$$(\mathrm{V}(\mathscr{I}), j^{-1}(\mathscr{O}_X/\mathscr{I}))$$

defines a locally ringed space (indeed, for any $x \in \mathrm{V}(\mathscr{I})$, the stalk $(j^{-1}(\mathscr{O}_X/\mathscr{I}))_x = (\mathscr{O}_X/\mathscr{I})_{j(x)} = \mathscr{O}_{X,j(x)}/\mathscr{I}_{j(x)}$ is a local ring: we have used Lemma 1.9.2.11 and Remark 1.6.0.5), and the canonical surjection

$$j^{\#}\colon \mathscr{O}_X \twoheadrightarrow \mathscr{O}_X/\mathscr{I} = j_* j^{-1}(\mathscr{O}_X/\mathscr{I}) \tag{1.12.2}$$

turns $(j, j^{\#})$ into a closed immersion

$$(\mathrm{V}(\mathscr{I}), j^{-1}(\mathscr{O}_X/\mathscr{I})) \xhookrightarrow{(j, j^{\#})} (X, \mathscr{O}_X). \tag{1.12.3}$$

Note that we have used Proposition 1.9.2.14(1) for the identification $\mathscr{O}_X/\mathscr{I} = j_* j^{-1}(\mathscr{O}_X/\mathscr{I})$ in (1.12.2). So we have defined an assignment

$$\mathscr{O}_X \supset \mathscr{I} \qquad \longmapsto \qquad \mathrm{V}(\mathscr{I}) \hookrightarrow X.$$

Conversely, to a closed immersion $\iota\colon Y \hookrightarrow X$ one can associate an ideal sheaf, namely

$$\mathscr{I}_Y = \ker(\mathscr{O}_X \xrightarrow{\iota^\#} \!\!\!\rightarrow\ \iota_*\mathscr{O}_Y) \subset \mathscr{O}_X.$$

These two operations are inverse to each other "up to isomorphism", as the next proposition clarifies.

PROPOSITION 1.12.0.2 (Closed immersions and ideal sheaves). *Let*

$$(\iota, \iota^\#)\colon (Y, \mathscr{O}_Y) \hookrightarrow (X, \mathscr{O}_X)$$

be a closed immersion of locally ringed spaces. Set $\mathscr{I}_Y = \ker \iota^\# \subset \mathscr{O}_X$ and consider the associated closed immersion

$$(\mathrm{V}(\mathscr{I}_Y), j^{-1}(\mathscr{O}_X/\mathscr{I}_Y)) \xhookrightarrow{(j, j^\#)} (X, \mathscr{O}_X).$$

as in (1.12.3)*. There is a unique isomorphism of locally ringed spaces*

$$(g, g^\#)\colon (Y, \mathscr{O}_Y) \xrightarrow{\sim} (\mathrm{V}(\mathscr{I}_Y), j^{-1}(\mathscr{O}_X/\mathscr{I}_Y))$$

such that $(\iota, \iota^\#) = (j, j^\#) \circ (g, g^\#)$.

Moreover, an inclusion of ideal sheaves $\mathscr{I}_2 \hookrightarrow \mathscr{I}_1 \hookrightarrow \mathscr{O}_X$ determines, and is determined by (in the above sense) a chain of closed immersions $\mathrm{V}(\mathscr{I}_1) \hookrightarrow \mathrm{V}(\mathscr{I}_2) \hookrightarrow X$.

Proof. The last statement is immediate. We thus only prove the first.

Let us use the shorthand notation

$$(Z, \mathscr{O}_Z) = (\mathrm{V}(\mathscr{I}_Y), j^{-1}(\mathscr{O}_X/\mathscr{I}_Y)).$$

where $j\colon Z \hookrightarrow X$ is the topological closed embedding that first appeared above in (1.12.1). We need to find the isomorphism g as in the statement. By Remark 1.9.1.7, we have

$$(\iota_*\mathscr{O}_Y)_x = \begin{cases} \mathscr{O}_{Y,y} & \text{if } x = \iota(y) \\ 0 & \text{if } x \notin \iota(Y). \end{cases} \tag{1.12.4}$$

Combine the exact sequence

$$0 \longrightarrow \mathscr{I}_Y \longrightarrow \mathscr{O}_X \longrightarrow \iota_*\mathscr{O}_Y \longrightarrow 0$$

with Proposition 1.6.0.9 and Equation (1.12.4) to deduce that

$$\mathscr{I}_{Y,x} = \mathscr{O}_{X,x} \qquad \Longleftrightarrow \qquad x \notin \iota(Y).$$

This shows $Z = \mathrm{V}(\mathscr{I}_Y) = \iota(Y)$. Let $g\colon Y \xrightarrow{\sim} Z$ be the induced homeomorphism. We have a factorisation

$$\iota\colon Y \xrightarrow{\ g\ } Z \xhookrightarrow{\ j\ } X$$

as topological maps, and

$$j_*\mathscr{O}_Z = j_* j^{-1}(\mathscr{O}_X/\mathscr{I}_Y) \cong \mathscr{O}_X/\mathscr{I}_Y \cong \iota_*\mathscr{O}_Y = j_* g_* \mathscr{O}_Y$$

as sheaves of rings, the last identity being a consequence of Diagram (1.9.1). Therefore, we have

$$\mathscr{O}_Z \cong j^{-1} j_* \mathscr{O}_Z = j^{-1} j_* g_* \mathscr{O}_Y \cong g_* \mathscr{O}_Y.$$

This extends g to the desired isomorphism $(g, g^\#)$. It is straightforward to verify the identity

$$(\iota, \iota^\#) = (j, j^\#) \circ (g, g^\#)$$

as morphisms of locally ringed spaces.

It remains to prove uniqueness. The underlying topological map g is forced by the factorisation $\iota = j \circ g$, since j is an inclusion (cf. Exercise 1.9.2.13). Once g is fixed, the identity

$$\iota^\# = j_* g^\# \circ j^\#$$

determines $j_* g^\#$ uniquely, because $j^\#$ is surjective. Finally, j_* is fully faithful for the inclusion of a subspace, since $j^{-1} j_* \cong \mathrm{id}$. Hence $g^\#$ is uniquely determined. □

Proposition 1.12.0.2 will be used to characterise closed subschemes of a scheme X (cf. Definition 2.2.1.7) later in Proposition 2.4.1.4.

1.13 $\mathscr{O}_X$-modules and (quasi)coherent sheaves

Let $(X, \mathscr{O}_X)$ be a ringed space.

Definition 1.13.0.1 ($\mathscr{O}_X$-module)**.** An *$\mathscr{O}_X$-module* on the ringed space $(X, \mathscr{O}_X)$ is a sheaf of abelian groups $\mathcal{F}$ such that $\mathcal{F}(U)$ is an $\mathscr{O}_X(U)$-module for every open subset $U \subset X$, and such that for every inclusion $V \subset U$ of open subsets the diagram

$$\begin{array}{ccc} \mathscr{O}_X(U) \times \mathcal{F}(U) & \longrightarrow & \mathcal{F}(U) \\ \downarrow & & \downarrow \\ \mathscr{O}_X(V) \times \mathcal{F}(V) & \longrightarrow & \mathcal{F}(V) \end{array} \tag{1.13.1}$$

commutes. This means, explicitly, that $(a \cdot s)|_V = a|_V \cdot s|_V$ for every $a \in \mathscr{O}_X(U)$ and $s \in \mathcal{F}(U)$.

A morphism of $\mathscr{O}_X$-modules is a morphism of sheaves that preserves the linear structures. With this notion of morphisms, $\mathscr{O}_X$-modules form a category, denoted $\mathsf{Mod}(\mathscr{O}_X)$. We write $\mathrm{Hom}_{\mathscr{O}_X}(-,-)$ for the hom-sets in this category.

Example 1.13.0.2. Consider the following examples.

(1) An ideal sheaf $\mathscr{I} \subset \mathscr{O}_X$ is an $\mathscr{O}_X$-module, as well as the quotient $\mathscr{O}_X/\mathscr{I}$.

(2) The sheaf of 1-forms Ω^1_X on a smooth manifold $(X, \mathscr{C}^\infty_X)$ is a $\mathscr{C}^\infty_X$-module.

(3) The constant sheaf (cf. Example 1.5.0.3) $\mathcal{F} = \underline{G}_X$ is not an $\mathscr{O}_X$-module in general: for instance consider a smooth manifold $(X, \mathscr{C}^\infty_X)$ and set $G = \mathbb{Z}$. There is no natural action of $\mathscr{C}^\infty_X(U)$ on the set of continuous functions $U \to \mathbb{Z}$. The same issue occurs when $\mathcal{F} = \mathbb{Z}_x$ is a skyscraper sheaf (cf. Example 1.4.0.13) for a point $x \in X$. Again, say $x \in U \subset X$. How would one define an action of $\mathscr{C}^\infty_X(U)$ on $\mathbb{Z}$?

Example 1.13.0.3. Let $(X, \mathscr{O}_X)$ be a locally ringed space and $x \in X$. If $i_x \colon \{x\} \hookrightarrow X$ is the inclusion, then

$$i_{x*}\kappa(x)$$

is naturally an $\mathscr{O}_X$-module. Indeed, for every open U containing x, the action is induced by

$$\mathscr{O}_X(U) \longrightarrow \mathscr{O}_{X,x} \twoheadrightarrow \kappa(x).$$

More generally, every $\mathscr{O}_{X,x}$-module gives in the same way a skyscraper $\mathscr{O}_X$-module supported on $\overline{\{x\}}$.

Remark 1.13.0.4. Let $\mathcal{F}$ be an $\mathscr{O}_X$-module. There is a canonical identification

$$\mathcal{F}(X) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_X}(\mathscr{O}_X, \mathcal{F}).$$

It sends $s \in \mathcal{F}(X)$ to the $\mathscr{O}_X$-linear map $\mathscr{O}_X \to \mathcal{F}$ defined on an open $U \subset X$ as $f \mapsto f \cdot s|_U$. Conversely, an $\mathscr{O}_X$-linear map $\eta \colon \mathscr{O}_X \to \mathcal{F}$ determines the section $s = \eta_X(1) \in \mathcal{F}(X)$.

It is easy to check that the category $\mathsf{Mod}(\mathscr{O}_X)$ has arbitrary direct sums (categorical coproducts), kernels and cokernels. One can prove (cf. [10, Tag 01AG]) that $\mathsf{Mod}(\mathscr{O}_X)$ is in fact an abelian category, in which a complex

$$\mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H}$$

is exact in the middle if and only if

$$\mathcal{F}_x \longrightarrow \mathcal{G}_x \longrightarrow \mathcal{H}_x$$

is exact in the middle for every $x \in X$. In particular, a complex of $\mathscr{O}_X$-modules is exact if and only if it is exact as a complex of sheaves of abelian groups.

In $\mathsf{Mod}(\mathscr{O}_X)$, the *tensor product* $\mathcal{F} \otimes_{\mathscr{O}_X} \mathcal{G}$ of two objects $\mathcal{F}$ and $\mathcal{G}$ is a new $\mathscr{O}_X$-module, defined to be the sheafification of the presheaf $U \mapsto \mathcal{F}(U) \otimes_{\mathscr{O}_X(U)} \mathcal{G}(U)$. It satisfies

$$\begin{aligned}(\mathcal{F} \otimes_{\mathscr{O}_X} \mathcal{G})_x &\cong \varinjlim_{U \ni x} \mathcal{F}(U) \otimes_{\mathscr{O}_X(U)} \mathcal{G}(U) \\ &\cong \varinjlim_{U \ni x} \mathcal{F}(U) \otimes_{\varinjlim_{U \ni x} \mathscr{O}_X(U)} \varinjlim_{U \ni x} \mathcal{G}(U) \\ &= \mathcal{F}_x \otimes_{\mathscr{O}_{X,x}} \mathcal{G}_x\end{aligned} \tag{1.13.2}$$

for every $x \in X$. The first isomorphism is justified since stalks can be computed before sheafification, the second isomorphism follows since filtered colimits commute with tensor products (even when the base ring varies compatibly).

Moreover, the functor $-\otimes_{\mathcal{O}_X} \mathcal{G}$ is right exact for all $\mathcal{G} \in \mathsf{Mod}(\mathcal{O}_X)$.

The restriction $\mathcal{F}|_U$ of an $\mathcal{O}_X$-module $\mathcal{F}$ to an open subset $U \subset X$ is naturally an $\mathcal{O}_U$-module, where we set $\mathcal{O}_U = \mathcal{O}_X|_U$. Given $\mathcal{F}, \mathcal{G} \in \mathsf{Mod}(\mathcal{O}_X)$, one also has a new $\mathcal{O}_X$-module

$$\mathscr{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$$

called the *Hom sheaf*. Its sections over an open $U \subset X$ are given by

$$\mathscr{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})(U) = \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U).$$



Exercise 1.13.0.5. Show the following:

(i) $\mathscr{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is a sheaf of $\mathcal{O}_X$-modules.

(ii) As functors $\mathsf{Mod}(\mathcal{O}_X) \to \mathsf{Mod}(\mathcal{O}_X)$, we have that $\mathscr{H}om_{\mathcal{O}_X}(\mathcal{F}, -)$ is (covariant and) left exact, and $\mathscr{H}om_{\mathcal{O}_X}(-, \mathcal{G})$ is (contravariant and) left exact. (This is a shadow of a more general fact about Homs in abelian categories).

(iii) Prove the next lemma.

LEMMA 1.13.0.6 (Adjunction $(\otimes, \mathscr{H}om)$). *Given $\mathcal{F}, \mathcal{G}, \mathcal{H} \in \mathsf{Mod}(\mathcal{O}_X)$, there is an isomorphism of $\mathcal{O}_X$-modules*

$$\mathscr{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathscr{H}om_{\mathcal{O}_X}(\mathcal{G}, \mathcal{H})) \cong \mathscr{H}om_{\mathcal{O}_X}(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}, \mathcal{H}).$$



Exercise 1.13.0.7. Show that both Lemma 1.8.0.6 and Theorem 1.8.0.8 on recovering sheaves (or morphisms thereof) from sheaves on a base extend to the setup of $\mathcal{O}_X$-modules.

Definition 1.13.0.8 (Free, locally free, globally generated)**.** Let I be a set. An $\mathcal{O}_X$-module $\mathcal{F}$ isomorphic to $\mathcal{O}_X^{\oplus I}$ is called *free*. If I is finite and $|I| = r$, we say that $\mathcal{F}$ is *free of rank r*. We say that $\mathcal{F}$ is *locally free of rank r* if for every $x \in X$ there is an open neighbourhood U such that $\mathcal{F}|_U \cong \mathcal{O}_U^{\oplus r}$. A locally free $\mathcal{O}_X$-module of rank 1 is called an *invertible sheaf* (or a *line bundle*). Finally, we say that $\mathcal{F}$ is *globally generated* if it receives a surjection $\mathcal{O}_X^{\oplus I} \twoheadrightarrow \mathcal{F}$ from a free sheaf, for some set I.

Definition 1.13.0.9 (Quasicoherent sheaf)**.** An $\mathcal{O}_X$-module $\mathcal{F}$ is *quasicoherent* if it is locally the cokernel of a morphism of free sheaves, i.e. if every point $x \in X$ has an open neighbourhood $U \subset X$ such that there is an exact sequence

$$\mathcal{O}_X^{\oplus I}\big|_U \longrightarrow \mathcal{O}_X^{\oplus J}\big|_U \longrightarrow \mathcal{F}\big|_U \longrightarrow 0$$

for some sets I and J. We denote by $\mathsf{QCoh}(X) \subset \mathsf{Mod}(\mathcal{O}_X)$ the full subcategory of quasicoherent $\mathcal{O}_X$-modules on the ringed space $(X, \mathcal{O}_X)$.

Definition 1.13.0.10 (Finitely presented sheaf)**.** An $\mathcal{O}_X$-module $\mathcal{F}$ is *finitely presented* if it is locally the cokernel of a morphism of free sheaves of finite rank, in the sense of Definition 1.13.0.9.

Definition 1.13.0.11 (Finitely generated, coherent)**.** Let $(X, \mathcal{O}_X)$ be a ringed space. An $\mathcal{O}_X$-module $\mathcal{F}$ is said to be

- *finitely generated* (or *of finite type*) if every $x \in X$ has an open neighbourhood $U \subset X$ admitting a surjection $\mathcal{O}_X^{\oplus n}|_U \twoheadrightarrow \mathcal{F}|_U$ for some $n \geqslant 1$,
- *coherent* if it is finitely generated and for every morphism $\theta\colon \mathcal{O}_X^{\oplus n}|_U \to \mathcal{F}|_U$ on an open $U \subset X$ one has that $\ker\theta$ is also finitely generated.

We denote by $\mathsf{Coh}(X) \subset \mathsf{Mod}(\mathcal{O}_X)$ the full subcategory of coherent sheaves on $(X, \mathcal{O}_X)$.

Example 1.13.0.12. The structure sheaf $\mathcal{O}_X$ is quasicoherent. More generally, any finitely presented sheaf is quasicoherent.

Example 1.13.0.13. Consider the following examples.

(i) Take $(X, \mathcal{O}_X) = (\mathrm{pt}, A)$ as ringed space, where $A = \mathbb{F}[x_i \mid i \geqslant 1]/(x_i x_j \mid i, j \geqslant 1)$. Then $\mathcal{O}_X = A$ is not coherent, as the kernel of the homomorphism $A \to A$ sending $a \mapsto a x_1$, namely the ideal $\mathsf{Ann}_A(x_1) = (\overline{x}_i \mid i \geqslant 1)$, is not finitely generated. In particular, notice that locally free of finite rank does not imply coherent over an arbitrary ringed space.

(ii) Let $\alpha\colon \mathcal{O}_X^{\oplus m} \to \mathcal{O}_X^{\oplus n}$ be a morphism. Then $\mathcal{F} = \mathrm{coker}(\alpha)$ is finitely presented, hence quasicoherent. If $\mathcal{O}_X$ is coherent, then so is $\mathcal{F}$.

(iii) Take $(X, \mathcal{O}_X) = (\mathrm{pt}, \mathbb{Z})$ as ringed space, and consider the $\mathcal{O}_X$-module $\mathbb{Q}$. Then $\mathbb{Q}$ is quasicoherent, but not coherent since it is not finitely generated over $\mathbb{Z}$. In fact, on a one-point ringed space, every module is quasicoherent. Coherence reduces to the usual algebraic notion of a coherent module.

(iv) Take $(X, \mathcal{O}_X) = (\mathbb{R}, \underline{\mathbb{Z}}_{\mathbb{R}})$ as ringed space and set $\mathcal{F} = \underline{\mathbb{Z}/2\mathbb{Z}}_X$. Then $\mathcal{F}$ has a global presentation

$$\underline{\mathbb{Z}}_{\mathbb{R}} \xrightarrow{\cdot 2} \underline{\mathbb{Z}}_{\mathbb{R}} \longrightarrow \mathcal{F} \longrightarrow 0,$$

and indeed $\mathcal{F}$ is coherent: locally, the kernels appearing in the definition of coherence correspond to kernels of homomorphisms between finitely generated $\mathbb{Z}$-modules, and these are finitely generated.

(v) Let $\mathbb{F}$ be a field, and consider the ringed space $(X, \mathcal{O}_X) = (\mathbb{R}, \underline{\mathbb{F}}_{\mathbb{R}})$, where $\underline{\mathbb{F}}_{\mathbb{R}}$ is the constant sheaf associated with $\mathbb{F}$. Let $\mathcal{F} = \mathbb{F}_0$ be the skyscraper sheaf supported at $0 \in \mathbb{R}$. There is a natural surjection of $\mathcal{O}_X$-modules

$$\underline{\mathbb{F}}_{\mathbb{R}} \twoheadrightarrow \mathcal{F},$$

which sends a locally constant function to its value at 0 (on open sets containing 0), and to zero otherwise. Thus $\mathcal{F}$ is of finite type.

We claim that $\mathcal{F}$ is not quasicoherent. Indeed, suppose that it were. Then there would exist an open interval U containing 0 and an exact sequence

$$\mathscr{O}_U^{\oplus I} \longrightarrow \mathscr{O}_U^{\oplus J} \longrightarrow \mathcal{F}|_U \longrightarrow 0.$$

Since U is connected, every morphism between free $\underline{\mathbb{F}}_{\mathbb{R}}|_U$-modules is induced by a homomorphism of $\mathbb{F}$-vector spaces

$$\mathbb{F}^{\oplus I} \longrightarrow \mathbb{F}^{\oplus J}.$$

Consequently, its cokernel is a constant sheaf on U. In particular, all stalks of $\mathcal{F}|_U$ must be isomorphic as $\mathbb{F}$-vector spaces. But

$$\mathcal{F}_x = \begin{cases} \mathbb{F} & \text{if } x = 0, \\ 0 & \text{if } x \neq 0, \end{cases}$$

a contradiction. Therefore $\mathcal{F}$ is of finite type but not quasicoherent.

(vi) Let $\mathbb{F}$ be a field, and consider again the ringed space $(X, \mathscr{O}_X) = (\mathbb{R}, \underline{\mathbb{F}}_{\mathbb{R}})$. Define

$$\mathcal{F} = \bigoplus_{n \geqslant 1} \mathscr{O}_X.$$

Then $\mathcal{F}$ is quasicoherent, since it is a free $\mathscr{O}_X$-module (of infinite rank).

However, $\mathcal{F}$ is not coherent. Indeed, for every $x \in X$, taking stalks gives

$$\mathcal{F}_x \cong \bigoplus_{n \geqslant 1} \mathscr{O}_{X,x} \cong \bigoplus_{n \geqslant 1} \mathbb{F},$$

which is not finitely generated as a module over $\mathscr{O}_{X,x} \cong \mathbb{F}$.

Since a coherent sheaf is necessarily of finite type, and hence has finitely generated stalks, we conclude that $\mathcal{F}$ is not coherent.

Recall the notion of support, introduced in (1.7.1). We have the following fundamental lemma on its closedness.

LEMMA 1.13.0.14. *Let $(X, \mathscr{O}_X)$ be a ringed space and let $\mathcal{F}$ be an $\mathscr{O}_X$-module of finite type. Then $\mathsf{Supp}(\mathcal{F})$ is closed in X.*

Proof. Let $x \notin \mathsf{Supp}(\mathcal{F})$, so $\mathcal{F}_x = 0$. Shrinking around x, choose generators $s_1, \ldots, s_n$ of $\mathcal{F}|_U$. Since $(s_i)_x = 0$ for every i, after shrinking U further we may assume $s_i = 0$ for all i. Hence $\mathcal{F}|_U = 0$, so $U \cap \mathsf{Supp}(\mathcal{F}) = \emptyset$. Thus the complement of $\mathsf{Supp}(\mathcal{F})$ is open. □

We will revisit (quasi)coherent sheaves and the categories $\mathsf{Mod}(\mathscr{O}_X)$, $\mathsf{QCoh}(X)$ and $\mathsf{Coh}(X)$ in greater detail in the context of schemes in **??**.

1.14 Pushforward and pullback of $\mathcal{O}$-modules

Let $f: X \to Y$ be a morphism of ringed spaces, and consider the situation

$$\begin{array}{ccc} \mathcal{F} & & \mathcal{G} \\ | & & | \\ X & \xrightarrow{f} & Y \end{array}$$

where $\mathcal{F} \in \mathsf{Mod}(\mathcal{O}_X)$ and $\mathcal{G} \in \mathsf{Mod}(\mathcal{O}_Y)$. If these were sheaves with no additional $\mathcal{O}$-module structure, we would already know, in such a situation, how to go "back and forth" using f_* and f^{-1}. Going forward is, even for $\mathcal{O}$-modules, simpler than going back. To go back, we will first confirm that $f^{-1}(-)$ preserves $\mathcal{O}$-linearity along the morphism $f^{-1}\mathcal{O}_Y \to \mathcal{O}_X$ adjoint to $\mathcal{O}_Y \to f_*\mathcal{O}_X$, and then observe that there is a "better" operation than f^{-1}, called the *pullback*, that is denoted f^*. It should be seen as a global version of *extension of scalars*, whereas pushforward in the context of $\mathcal{O}$-modules should be seen as a *restriction of scalars*. Let us briefly review what this means in the local case.

Consider a ring homomorphism $\phi: A \to B$. Then a B-module N is naturally an A-module via $a \cdot n = \phi(a)n$. This A-module structure, called restriction of scalars along ϕ, is simply denoted by N again, or sometimes $\phi_* N$. On the other hand, if M is an A-module, then $M \otimes_A B$ is a B-module, via the rule

$$b \cdot z = \big(M \otimes_A B \xrightarrow{\mathrm{id}_M \otimes t_b} M \otimes_A B\big)(z), \quad z \in M \otimes_A B,$$

where $t_b: B \to B$ is multiplication by $b \in B$. This structure is called *extension of scalars*. Of course, one can make the same construction replacing B by an arbitrary B-module N. Then using $\mathrm{id}_M \otimes t_b$, with $t_b: N \to N$ being multiplication $b \in B$, yields a natural B-module structure on $M \otimes_A N$.



Exercise 1.14.0.1. Let $\phi: A \to B$ be a ring homomorphism. Show the following assertions.

(i) Both $M \mapsto M \otimes_A B$ and $N \mapsto \phi_* N$ are functors.

(ii) There is an adjunction

$$\mathsf{Mod}_A \underset{\phi_*}{\overset{-\otimes_A B}{\rightleftarrows}} \mathsf{Mod}_B .$$

That is, there are bifunctorial bijections

$$\mathrm{Hom}_B(M \otimes_A B, N) \cong \mathrm{Hom}_A(M, \phi_* N).$$

In the 'global' case, consider a morphism $f: X \to Y$ of ringed spaces, and let $f^{\#}: \mathcal{O}_Y \to f_*\mathcal{O}_X$ be the map on sheaves of rings on Y. Let $V \subset Y$ be an open set, so that

$$A = \mathcal{O}_Y(V) \xrightarrow{\phi} f_*\mathcal{O}_X(V) = \mathcal{O}_X(f^{-1}(V)) = B$$

is a ring homomorphism. For the pushforward: assume $\mathcal{F}$ is an $\mathcal{O}_X$-module, so that by definition $f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(V))$ is a module over $\mathcal{O}_X(f^{-1}(V)) = f_*\mathcal{O}_X(V)$. This says that $f_*\mathcal{F}$ is an $f_*\mathcal{O}_X$-module. But then, $f_*\mathcal{F}(V)$ is also an $\mathcal{O}_Y(V)$-module via restriction of scalars along the map $\phi = f_V^{\#}$, so

$$\boxed{f_*\mathcal{F} \text{ has a natural structure of } \mathcal{O}_Y\text{-module.}}$$

For the pullback: assume $\mathcal{G}$ is an $\mathcal{O}_Y$-module. At the presheaf level we have an action of

$$(f_{\mathrm{pre}}^{-1}\mathcal{O}_Y)(U) = \varinjlim_{V \supset f(U)} \mathcal{O}_Y(V)$$

on

$$(f_{\mathrm{pre}}^{-1}\mathcal{G})(U) = \varinjlim_{V \supset f(U)} \mathcal{G}(V)$$

defined as follows: if $a \in \mathcal{O}_Y(V)$, $s \in \mathcal{G}(W)$ and $U \subset X$ is an open such that $f(U) \subset V \cap W$, we define

$$[a] \cdot [s] = [a|_{V\cap W} \cdot s|_{V\cap W}].$$

This is well-defined and makes $f_{\mathrm{pre}}^{-1}\mathcal{G}$ into a module over $f_{\mathrm{pre}}^{-1}\mathcal{O}_Y$. After sheafification we obtain an $f^{-1}\mathcal{O}_Y$-module structure on $f^{-1}\mathcal{G}$. At the level of stalks, under the canonical identifications

$$(f^{-1}\mathcal{O}_Y)_x \cong \mathcal{O}_{Y,f(x)}, \qquad (f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)},$$

this is precisely the natural $\mathcal{O}_{Y,f(x)}$-module structure on $\mathcal{G}_{f(x)}$.

Now we need to extend scalars. This time we use the adjoint of $f^{\#}$, namely

$$f^{-1}\mathcal{O}_Y \longrightarrow \mathcal{O}_X,$$

allowing us to form the tensor product

$$\boxed{f^*\mathcal{G} = f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.}$$

This is, by definition, the sheaf associated to the presheaf

$$U \longmapsto (f^{-1}\mathcal{G})(U) \otimes_{(f^{-1}\mathcal{O}_Y)(U)} \mathcal{O}_X(U),$$

which in fact is canonically isomorphic to the sheafification of

$$U \longmapsto (f_{\mathrm{pre}}^{-1}\mathcal{G})(U) \otimes_{(f_{\mathrm{pre}}^{-1}\mathcal{O}_Y)(U)} \mathcal{O}_X(U).$$

Indeed, the canonical comparison is an isomorphism on every stalk: both sides have stalk

$$\mathcal{G}_{f(x)} \otimes_{\mathcal{O}_{Y,f(x)}} \mathcal{O}_{X,x}.$$

Summarising, we have used the two maps

$$\mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X, \qquad f^{-1}\mathcal{O}_Y \longrightarrow \mathcal{O}_X$$

for the following constructions:

- Since $f_*\mathcal{F}$ is an $f_*\mathcal{O}_X$-module, it is naturally an $\mathcal{O}_Y$-module. This yields a *pushforward functor*
$$f_*\colon \mathsf{Mod}(\mathcal{O}_X) \longrightarrow \mathsf{Mod}(\mathcal{O}_Y).$$
- Since $f^{-1}\mathcal{G}$ is an $f^{-1}\mathcal{O}_Y$-module, we can construct the $\mathcal{O}_X$-module
$$f^*\mathcal{G} = f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$
This yields a *pullback functor*
$$f^*\colon \mathsf{Mod}(\mathcal{O}_Y) \longrightarrow \mathsf{Mod}(\mathcal{O}_X).$$

Note that $f^*\mathcal{O}_Y \cong \mathcal{O}_X$ canonically, for every f. In fact, for every index set I one has, canonically, $f^*(\mathcal{O}_Y^{\oplus I}) \cong \mathcal{O}_X^{\oplus I}$ and the pullback of a locally free sheaf of rank r is again locally free of rank r.

Moreover, there is a canonical isomorphism
$$\boxed{(f^*\mathcal{G})_x \cong \mathcal{G}_{f(x)} \otimes_{\mathcal{O}_{Y,f(x)}} \mathcal{O}_{X,x}}$$
for every $x \in X$, combining (1.13.2) with Lemma 1.9.2.11.

Example 1.14.0.2. For an open immersion $j\colon U \hookrightarrow X$, one has
$$j^*\mathcal{F} = j^{-1}\mathcal{F} \otimes_{j^{-1}\mathcal{O}_X} \mathcal{O}_U \cong j^{-1}\mathcal{F} \otimes_{\mathcal{O}_U} \mathcal{O}_U \cong j^{-1}\mathcal{F} \cong \mathcal{F}|_U.$$
In other words, pullback generalises ordinary restriction.

Remark 1.14.0.3. By right exactness of tensor products, it is immediate to conclude that the pullback functor $f^*\colon \mathsf{Mod}(\mathcal{O}_Y) \to \mathsf{Mod}(\mathcal{O}_X)$ is right exact for every morphism $f\colon X \to Y$ of ringed spaces.

LEMMA 1.14.0.4 (Adjunction (f^*, f_*)). *Let $f\colon X \to Y$ be a morphism of ringed spaces. Then (f^*, f_*) is an adjoint pair of functors.*

Proof. Fix $\mathcal{G} \in \mathsf{Mod}(\mathcal{O}_Y)$ and $\mathcal{F} \in \mathsf{Mod}(\mathcal{O}_X)$. We have
$$\begin{aligned}\mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F}) &= \mathrm{Hom}_{\mathcal{O}_X}(f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X, \mathcal{F}) \\ &\cong \mathrm{Hom}_{f^{-1}\mathcal{O}_Y}(f^{-1}\mathcal{G}, \mathcal{F}) \\ &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F}).\end{aligned}$$
The first identity is just the definition; the first isomorphism is extension/restriction of scalars; the second is the adjunction (f^{-1}, f_*), restricted to linear maps. By naturality of these identifications, they are functorial in both $\mathcal{F}$ and $\mathcal{G}$, yielding the desired adjunction. □

2 | Schemes

The goal of this chapter is to introduce the category Aff of *affine schemes* and the larger category Sch of all *schemes*. They will arise as full subcategories

$$\mathsf{Aff} \subset \mathsf{Sch} \subset \mathsf{LRS}$$

of the larger category LRS of locally ringed spaces. We will also introduce the notions of affine and projective variety over a field.

2.1 Affine schemes

Let A be a commutative ring with identity. The set

$$\boxed{\operatorname{Spec} A = \{\, \mathfrak{p} \subset A \mid \mathfrak{p} \text{ is a prime ideal} \,\}}$$

is called the *prime spectrum* of A. If $A = 0$, there are no prime ideals, and hence $\operatorname{Spec} 0 = \emptyset$. For now, $\operatorname{Spec} A$ is just a set. We will endow it with a topology (cf. Corollary 2.1.1.3) and with a sheaf of rings $\mathscr{O}_{\operatorname{Spec} A}$ having local rings as stalks (cf. Theorem 2.1.4.3), to obtain a locally ringed space. Such a locally ringed space will be called an *affine scheme* (cf. Important Definition 2.1.5.2). General schemes are obtained by gluing affine schemes, just as a smooth manifold is obtained by gluing open subsets of $\mathbb{R}^m$.

Notation 2.1.0.1. We introduce the following notation that will be used throughout: given a ring B, the spectrum

$$\mathbb{A}^n_B = \operatorname{Spec} B[x_1, \ldots, x_n]$$

will be called *affine n-space* over B. If $n = 1$ (resp. $n = 2, 3$), we speak of *affine line* (resp. *affine plane, affine 3-space*) over B.

Before getting started, we recall a few basic tools from commutative algebra.

Radical of an ideal

Let A be a ring.

Definition 2.1.0.2 (Radical of an ideal)**.** The *radical* of an ideal $I \subset A$ is the subset

$$\sqrt{I} = \left\{ a \in A \,\middle|\, a^r \in I \text{ for some } r > 0 \right\} \subset A.$$

An ideal I is *radical* if $I = \sqrt{I}$.

The radical $\sqrt{I} \subset A$ is an ideal containing I, and satisfies

$$\sqrt{I} = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec} A \\ \mathfrak{p} \supset I}} \mathfrak{p}. \tag{2.1.1}$$

This implies that

a prime ideal contains I if and only if it contains $\sqrt{I}$.

Remark 2.1.0.3. A prime ideal $\mathfrak{p} \subset A$ is radical (reason: let $a \in A$ be an element such that $a^r \in \mathfrak{p}$, and choose $r \geqslant 1$ minimal; if $r = 1$ we are done, otherwise either $a \in \mathfrak{p}$ or $a^{r-1} \in \mathfrak{p}$ but the latter is excluded by minimality of r, thus $a \in \mathfrak{p}$), but the converse is false. For instance if $A = \mathbb{Z}$ and $m \neq 0$, we have $\sqrt{m\mathbb{Z}} = n\mathbb{Z}$, where $n = \prod_{p \mid m} p$. Thus if m is a product of distinct primes then $m\mathbb{Z}$ is radical but not prime.

Definition 2.1.0.4 (Nilradical of a ring)**.** The *nilradical* of a ring A is the ideal of nilpotent elements (the radical of the trivial ideal), namely

$$\operatorname{Nil}(A) = \sqrt{(0)} = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p} \subset A.$$

A ring A is *reduced* if $\operatorname{Nil}(A) = 0$, i.e. if it contains no nontrivial nilpotents.

Definition 2.1.0.5 (Minimal primes)**.** Let A be a ring, $\mathfrak{p} \subset A$ a prime ideal, $I \subsetneq A$ a proper ideal. We say that $\mathfrak{p}$ is a *minimal prime above* I if $I \subset \mathfrak{p}$ and for every other prime ideal $\mathfrak{q}$ such that $I \subset \mathfrak{q}$, one has $\mathfrak{q} \not\subset \mathfrak{p}$. A minimal prime above $I = (0)$ is simply called a *minimal prime*.

To say that $\mathfrak{p}$ is minimal above I is like saying that $\mathfrak{p}/I \subset A/I$ is a minimal prime.

LEMMA 2.1.0.6. *Let A be a reduced ring. The union of the minimal prime ideals in A coincides with the set of zero-divisors of A.*[1]

Proof. Let $a \in A$ be a zero-divisor. Let $b \in A \setminus 0$ be an element such that $ab = 0$. Assume, by contradiction, that $a \notin \mathfrak{p}$ for all minimal primes $\mathfrak{p}$. We have

$$ab = 0 \in (0) = \operatorname{Nil}(A) = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p} = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec} A \\ \text{minimal}}} \mathfrak{p},$$

so $ab \in \mathfrak{p}$ for every minimal prime $\mathfrak{p}$. But then $b \in \mathfrak{p}$ for every minimal prime $\mathfrak{p}$ by the assumption on a. Then $b = 0$, contradiction.

[1] See Exercise B.5.1.6 for a generalisation.

Conversely, assume $a \in \mathfrak{p}$ for some minimal prime $\mathfrak{p}$. Since A is reduced, so is the local ring $A_\mathfrak{p}$ by Lemma B.3.3.10. By minimality of $\mathfrak{p}$, there is a unique prime ideal in $A_\mathfrak{p}$, namely $\mathfrak{p}A_\mathfrak{p}$. Therefore

$$\mathfrak{p}A_\mathfrak{p} = \mathrm{Nil}(A_\mathfrak{p}) = (0).$$

But then, since $a \in \mathfrak{p}$, we have $a/1 = 0 \in \mathfrak{p}A_\mathfrak{p}$, i.e. there exists $s \in A \setminus \mathfrak{p}$ such that $sa = 0 \in A$. Since $s \notin \mathfrak{p}$, in particular $s \neq 0$. Thus a is a zero-divisor. □

Operations on ideals

Let $(I_\lambda)_{\lambda\in\Lambda}$ be an arbitrary family of ideals in a ring A. Then the intersection $\bigcap_{\lambda\in\Lambda} I_\lambda \subset A$ is again an ideal. Recall that the sum of ideals $\sum_{\lambda\in\Lambda} I_\lambda$ is, by definition, the ideal generated by (i.e. the smallest ideal containing) the union of the ideals in the family (which is not an ideal in general). It can be described as

$$\sum_{\lambda\in\Lambda} I_\lambda = \left\{ \sum_{\lambda\in F} a_\lambda i_\lambda \;\middle|\; a_\lambda \in A,\ i_\lambda \in I_\lambda,\ |F| < \infty \right\}. \tag{2.1.2}$$

If we have *finitely many* ideals $I_1, I_2, \ldots, I_m \subset A$, their product is the ideal generated by the products of the form $i_1 i_2 \cdots i_m$, where $i_k \in I_k$ for $k = 1, \ldots, m$. In symbols,

$$I_1 I_2 \cdots I_m = \left\{ \sum_{1\leqslant j\leqslant N} i_1^{(j)} i_2^{(j)} \cdots i_m^{(j)} \;\middle|\; i_k^{(j)} \in I_k,\ N < \infty \right\}.$$

In general, we have $I_1 I_2 \cdots I_m \subset I_1 \cap I_2 \cap \cdots \cap I_m$, with equality when $I_k + I_h = A$ for every pair (k,h) such that $k \neq h$ (if $I_k + I_h = A$ we say that I_k and I_h are *comaximal*).

2.1.1 The Zariski topology on $\mathrm{Spec}\, A$

For an arbitrary ideal $I \subset A$, set

$$\boxed{\mathrm{V}(I) = \{\, \mathfrak{p} \in \mathrm{Spec}\, A \mid \mathfrak{p} \supset I \,\} \subset \mathrm{Spec}\, A.}$$

In other words, if $I \neq A$, then $\mathrm{V}(I)$ consists of all the primes above I. If $I = A$, we have $\mathrm{V}(I) = \emptyset$. Note that there is a bijection

$$\mathrm{V}(I) \simeq \mathrm{Spec}\, A/I,$$

since (prime) ideals of A/I correspond precisely to (prime) ideals in A containing I.

If $I = (f) = fA \subset A$ (we will use both notations for principal ideals) for $f \in A$, we write $\mathrm{V}(f)$ instead of $\mathrm{V}(I)$, and define

$$\boxed{\mathrm{D}(f) = \mathrm{Spec}\, A \setminus \mathrm{V}(f) = \left\{ \mathfrak{p} \in \mathrm{Spec}\, A \;\middle|\; f \notin \mathfrak{p} \right\}.}$$

Example 2.1.1.1. Let $\mathbf{k}$ be an algebraically closed field. If $f \in \mathbf{k}[x]$ is nonzero, then $\mathrm{D}(f)$ consists of those prime ideals $\mathfrak{p} \subset \mathbf{k}[x]$ such that $f \notin \mathfrak{p}$. One such ideal is the trivial ideal (0), and the other ideals $\mathfrak{p}$ with this property are all the ideals of the form $\mathfrak{p} = (x - a)$, for $a \in \mathbf{k}$, such that $f(a) \neq 0 \in \mathbf{k}$.

Note that, for any ring A, one has

$$\operatorname{Spec} A = \mathrm{D}(1) = \mathrm{V}(0), \quad \emptyset = \mathrm{D}(0) = \mathrm{V}(1).$$

LEMMA 2.1.1.2 (V(—) gives closed sets axioms). *Let A be a ring.*

(1) *If $I, J \subset A$ are two ideals, then $\mathrm{V}(I) \cup \mathrm{V}(J) = \mathrm{V}(I \cap J)$.*

(2) *If $(I_\lambda)_{\lambda\in\Lambda}$ is an arbitrary family of ideals, then $\bigcap_{\lambda\in\Lambda} \mathrm{V}(I_\lambda) = \mathrm{V}\big(\sum_{\lambda\in\Lambda} I_\lambda\big)$.*

(3) *$\operatorname{Spec} A = \mathrm{V}(0)$ and $\emptyset = \mathrm{V}(1)$.*

Proof. We proceed step by step.

(1) If $\mathfrak{p} \subset A$ contains either I or J, then it contains the smaller ideal $I \cap J$, thus $\mathrm{V}(I) \cup \mathrm{V}(J) \subset \mathrm{V}(I \cap J)$. If $\mathfrak{p} \supset I \cap J$ but $\mathfrak{p} \not\supset I$, there is $i \in I$ such that $i \notin \mathfrak{p}$. If $j \in J$, then $ij \in I \cap J \subset \mathfrak{p}$, which implies $j \in \mathfrak{p}$ (because $\mathfrak{p}$ is prime), thus $J \subset \mathfrak{p}$. Therefore $\mathrm{V}(I \cap J) \subset \mathrm{V}(I) \cup \mathrm{V}(J)$.

(2) If $\mathfrak{p}$ contains the sum $\sum_\lambda I_\lambda$, then it contains each I_λ, therefore $\mathrm{V}\big(\sum_\lambda I_\lambda\big) \subset \bigcap_\lambda \mathrm{V}(I_\lambda)$. On the other hand, assume $\mathfrak{p} \supset I_\lambda$ for every index λ. Let $h = a_1 i_{\lambda_1} + \cdots + a_N i_{\lambda_N}$ as in Equation (2.1.2). Then $a_j i_{\lambda_j} \in \mathfrak{p}$ by assumption, thus $h \in \mathfrak{p}$ as well, i.e. $\mathfrak{p} \supset \sum_\lambda I_\lambda$.

(3) If $A = 0$, everything is empty and there is nothing to prove. If $A \neq 0$, every prime ideal $\mathfrak{p} \subset A$ contains $0 \in A$, and no prime ideal $\mathfrak{p} \subset A$ contains $1 \in A$. □

COROLLARY 2.1.1.3 (Zariski topology). *There exists a unique topology on $\operatorname{Spec} A$ whose closed sets are of the form $\mathrm{V}(I)$. Moreover, the sets $\mathrm{D}(f) \subset \operatorname{Spec} A$ form a base of open sets for this topology (according to Definition 1.8.0.1).*

Proof. The first statement is clear from Lemma 2.1.1.2 and the definition of a topology. The second one follows from these observations:

(i) $\mathrm{D}(f_1) \cap \mathrm{D}(f_2) = \mathrm{D}(f_1 f_2)$ for all $f_1, f_2 \in A$, and

(ii) an open subset $\operatorname{Spec} A \setminus \mathrm{V}(I)$ can be written as $\bigcup_{f\in I} \mathrm{D}(f)$. □

For instance,

$$\operatorname{Spec} A = \mathrm{D}(1) = \operatorname{Spec} A \setminus \mathrm{V}(1) = \bigcup_{f\in A} \mathrm{D}(f). \tag{2.1.3}$$

Important Definition 2.1.1.4 (Zariski topology)**.** The topology on $\operatorname{Spec} A$ given by Corollary 2.1.1.3 is called the *Zariski topology*.

Terminology 2.1.1.5. We call $\mathrm{D}(f)$ a *principal open set* in $\operatorname{Spec} A$, and $\mathrm{V}(f)$ a *principal closed set* in $\operatorname{Spec} A$.

Convention 2.1.1.6. When thinking of Spec A as a topological space, it will *always* be endowed with the Zariski topology.

Remark 2.1.1.7. Let $\mathbb{F}$ be a field. Consider the ideals $I_r = (x^r) \subset \mathbb{F}[x]$ for all $r > 0$. Then

$$\mathrm{V}(I_1) = \{\mathfrak{p} \subset \mathbb{F}[x] \mid \mathfrak{p} \supset (x)\} = \{(x)\} = \mathrm{V}(I_r)$$

for every r. Thus it may happen that

$$\mathrm{V}(I) = \mathrm{V}(J), \quad \text{with } I \neq J.$$

In general, by Equation (2.1.1), we have the set-theoretic identity

$$\mathrm{V}(I) = \mathrm{V}(\sqrt{I}) \subset \operatorname{Spec} A.$$

LEMMA 2.1.1.8. *Let $I, J \subset A$ be two ideals in a ring A. Then*

$$\mathrm{V}(I) \subset \mathrm{V}(J) \Longleftrightarrow J \subset \sqrt{I}.$$

Proof. This is a rephrasing of the identity $\sqrt{I} = \bigcap_{\mathfrak{p} \in \mathrm{V}(I)} \mathfrak{p}$, cf. Equation (2.1.1). □

Example 2.1.1.9. Let $f, g \in A$. We have

$$\begin{aligned} \mathrm{D}(g) \subset \mathrm{D}(f) &\Longleftrightarrow \mathrm{V}(f) \subset \mathrm{V}(g) \\ &\Longleftrightarrow g \in \sqrt{fA}. \end{aligned}$$

LEMMA 2.1.1.10 (Spec A is quasicompact). *Fix a subset $S = \{ f_k \mid k \in K \} \subset A$. Then, the following two conditions are equivalent.*

(i) $\operatorname{Spec} A = \bigcup_{k \in K} \mathrm{D}(f_k)$.

(ii) *There is a finite subset $F \subset K$ such that one can write $1 = \sum_{k \in F} a_k f_k$ for some elements $a_k \in A$.*

In particular, Spec A *equipped with the Zariski topology is quasicompact.*

Proof. Let $(-)^{\mathrm{c}}$ denote the complement of a subset of Spec A. The union

$$\bigcup_{k \in K} \mathrm{D}(f_k) = \bigcup_{k \in K} \operatorname{Spec} A \setminus \mathrm{V}(f_k) = \left(\bigcap_{k \in K} \mathrm{V}(f_k) \right)^{\mathrm{c}} = \mathrm{V}\left(\sum_{k \in K} f_k A \right)^{\mathrm{c}}$$

equals Spec A if and only if

$$\mathrm{V}\left(\sum_{k \in K} f_k A \right) = \emptyset = \mathrm{V}(1),$$

which by Lemma 2.1.1.8 happens if and only if $\sqrt{\sum_{k \in K} f_k A} = (1) = A$, which in turn means that $A = \sum_{k \in K} f_k A$. The first assertion then follows from the definition of sum of ideals. The last sentence in the statement follows from the first, setting $K = A$ and using (2.1.3). □

Remark 2.1.1.11 (Principal opens are quasicompact)**.** Every open subset of $\operatorname{Spec} A$ is a union of principal opens. Conversely, each principal open is quasicompact. Since principal opens form a basis, it is enough to consider a cover by principal opens. Suppose

$$\mathrm{D}(f) \subset \bigcup_{\lambda\in\Lambda} \mathrm{D}(g_\lambda).$$

Taking complements gives the inclusion $\mathrm{V}((g_\lambda \mid \lambda\in\Lambda)) \subset \mathrm{V}(f)$, and hence $f \in \sqrt{(g_\lambda \mid \lambda\in\Lambda)}$ by Lemma 2.1.1.8. Thus

$$f^N = \sum_{\lambda\in F} a_\lambda g_\lambda$$

for some $N>0$ and some finite subset $F\subset\Lambda$. It follows that $\mathrm{D}(f)\subset\bigcup_{\lambda\in F}\mathrm{D}(g_\lambda)$. Therefore an open subset $U\subset\operatorname{Spec} A$ is quasicompact if and only if it is a finite union of principal opens.



Warning 2.1.1.12. Not *every* open subset $U\subset\operatorname{Spec} A$ is quasicompact! For instance, consider $A=\mathbf{k}[x_i \mid i\in\mathbb{N}]$, and let $U\subset\operatorname{Spec} A$ be the complement of the origin (the point corresponding to the maximal ideal $(x_i \mid i\in\mathbb{N})\subset A$). Then the cover $U=\bigcup_{i\in\mathbb{N}}\mathrm{D}(x_i)$ has no finite subcover.

Remark 2.1.1.13 (Closed points = maximal ideals)**.** Let $\mathfrak{p}\in\operatorname{Spec} A$ be a closed point, i.e. a point such that $\{\mathfrak{p}\}\subset\operatorname{Spec} A$ is closed. Then $\{\mathfrak{p}\}=\mathrm{V}(I)=\{\mathfrak{q}\mid\mathfrak{q}\supset I\}$ for an ideal $I\subset A$. This says that $\mathfrak{p}$ is the only prime ideal above I. But every proper ideal is contained in a maximal ideal, and maximal ideals are prime. Thus $\mathfrak{p}$ is maximal. Conversely, if $\mathfrak{m}\subset A$ is maximal, then $\{\mathfrak{m}\}=\mathrm{V}(\mathfrak{m})$, in particular $\{\mathfrak{m}\}\subset\operatorname{Spec} A$ is closed, i.e. $\mathfrak{m}\in\operatorname{Spec} A$ is a closed point.

The previous remark can be generalised by the following lemma.

LEMMA 2.1.1.14. *Let $T\subset\operatorname{Spec} A$ be a subset, $\overline{T}\subset\operatorname{Spec} A$ its closure. Then*

$$\overline{T}=\mathrm{V}\left(\bigcap_{\mathfrak{p}\in T}\mathfrak{p}\right).$$

In particular, the closure of $\{\mathfrak{p}\}\subset\operatorname{Spec} A$ is precisely $\mathrm{V}(\mathfrak{p})$.

Proof. We have

$$\overline{T}=\bigcap_{\mathrm{V}(I)\supset T}\mathrm{V}(I)=\mathrm{V}\left(\sum_{\mathrm{V}(I)\supset T} I\right).$$

But by definition $\mathrm{V}(I)\supset T$ means that every $\mathfrak{p}\in T$ satisfies $\mathfrak{p}\supset I$, so the sum is over all $I\subset A$ such that $I\subset\bigcap_{\mathfrak{p}\in T}\mathfrak{p}$. The largest such ideal is precisely $\bigcap_{\mathfrak{p}\in T}\mathfrak{p}$, which concludes the proof. □

Remark 2.1.1.15 (Zariski topology has large opens)**.** Combining the previous topological observations, we find the slogan:

the Zariski topology on $\operatorname{Spec} A$ is almost never Hausdorff, or even T_1.

For instance, take an integral domain A that is not a field, so that $(0) \subset A$ is prime and the closure of the corresponding point $\xi \in \operatorname{Spec} A$ is equal to $V(0) = \operatorname{Spec} A$ by Lemma 2.1.1.14. Then any two nonempty open subsets intersect (thus $\operatorname{Spec} A$ is not Hausdorff), and in fact every open neighbourhood of a point $\mathfrak{p} \in \operatorname{Spec} A$ will also contain ξ (thus $\operatorname{Spec} A$ is not T_1).

2.1.2 Interlude: functions on $\operatorname{Spec} A$

The following slogan is important (and will be formalised in Theorem 2.1.4.3(b)):

$$\boxed{\text{elements of } A \text{ are functions on } \operatorname{Spec} A.}$$

The slogan is a bit premature, since by 'function on $\operatorname{Spec} A$' we actually mean 'regular function on *the scheme* $\operatorname{Spec} A$', and so far we only have a topological space (we haven't yet defined the sheaf of regular functions). However, it is worth explaining the slogan just to build some intuition.

Here is the explanation. To any $f \in A$, we can associate the *map*

$$\operatorname{Spec} A \xrightarrow{\theta_f} \coprod_{\mathfrak{p} \in \operatorname{Spec} A} A/\mathfrak{p}, \qquad \mathfrak{p} \longmapsto f \bmod \mathfrak{p}.$$

For instance, $f = 9 \in \mathbb{Z}$ takes the value $[1]$ in $\mathbb{Z}/2\mathbb{Z}$, and the value $[4]$ in $\mathbb{Z}/5\mathbb{Z}$. Its value in $\mathbb{Z}/0 = \mathbb{Z}$ is just... 9.[2] A possibly confusing thing here is that the ring where the function takes values depends on the point on which the function is evaluated! This confusion will be cleared out in Section 2.1.6.

Observe that we have a (tautological) notion of what it means for such a function to vanish at a point of $\operatorname{Spec} A$: we have

$$\boxed{\theta_f(\mathfrak{p}) = 0 \in A/\mathfrak{p} \text{ if and only if } f \in \mathfrak{p}.}$$

For instance, the function $f = 9 \in \mathbb{Z}$ vanishes on the point $(3) \in \operatorname{Spec} \mathbb{Z}$. Note, finally, that addition and multiplication of 'functions' works as one might expect, i.e. $\theta_{f+g}(\mathfrak{p}) = f + g \bmod \mathfrak{p} = \theta_f(\mathfrak{p}) + \theta_g(\mathfrak{p})$, and similarly $\theta_{fg}(\mathfrak{p}) = fg \bmod \mathfrak{p} = \theta_f(\mathfrak{p})\theta_g(\mathfrak{p})$. This is just a rephrasing of the fact that $A \to A/\mathfrak{p}$ is a ring homomorphism!

Example 2.1.2.1. Consider $\mathbb{C}[x]$, and the 'function' $f(x) = 2x^2 - x + 3 \in \mathbb{C}[x]$. The prime ideals of $\mathbb{C}[x]$ are $(0) \subset \mathbb{C}[x]$, and the maximal ideals $\mathfrak{m}_a = (x - a) \subset \mathbb{C}[x]$ for $a \in \mathbb{C}$. The value of f on the point $\mathfrak{m}_a \in \operatorname{Spec} \mathbb{C}[x]$ is just the evaluation of the polynomial $f(x)$ at $x = a$. Indeed,

$$\theta_f(\mathfrak{m}_a) = f \bmod \mathfrak{m}_a \in \mathbb{C}[x]/\mathfrak{m}_a,$$

corresponds to the element

$$f(a) = 2a^2 - a + 3 \in \mathbb{C} \cong \mathbb{C}[x]/\mathfrak{m}_a.$$

[2] Please come back here after reading about generic points in Section 2.2.3.

Spoiler 2.1.2.2 (The twist of nilpotents)**.** Consider $A = \mathbb{F}[t]/t^2$, and let $\overline{t} \in A$ be the image of $t \in \mathbb{F}[t]$ in A. If we see $\overline{t}$ as a function on $X = \operatorname{Spec}\mathbb{F}[t]/t^2$, we see that $\theta_{\overline{t}}$ evaluated on the point $(\overline{t}) \in X$ gives $0 \in A/\overline{t}$, i.e. the *nonzero* element $\overline{t} \in A$ determines a function that *vanishes at every point* of $\operatorname{Spec}\mathbb{F}[t]/t^2$. Here we encounter for the first time one of the magic aspects of scheme theory:

functions are not determined by their values on points!

This is due to the presence of nilpotents, which were not part of the game in the land of *classical* algebraic varieties. We will see that $\operatorname{Spec}\mathbb{F}[t]/t^2 \ncong \operatorname{Spec}\mathbb{F}$ *as affine schemes*, because their rings of functions are different: there is no ring isomorphism $\mathbb{F} \cong \mathbb{F}[t]/t^2$!

2.1.3 Zariski topology on $\operatorname{Spec} A$: first examples

In this subsection we analyse the Zariski topology on $\operatorname{Spec} A$ for a few interesting rings A.

Example 2.1.3.1 ($\operatorname{Spec}\mathbb{F}$, aka the point)**.** Let $\mathbb{F}$ be a field. The spectrum $\operatorname{Spec}\mathbb{F}$ consists of a single point corresponding to $(0) \subset \mathbb{F}$. Its 'functions' (in the sense of Section 2.1.2) are just the constants $\mathbb{F}$, as expected. For now, this (merely topological and hence dry) description is enough. However, when $\operatorname{Spec}\mathbb{F}$ is endowed with its scheme structure, things change: for instance, as we shall see, it is not true that the only morphism $\operatorname{Spec}\mathbb{F} \to \operatorname{Spec}\mathbb{F}$ is the identity! Such maps correspond (contravariantly) to field endomorphisms of $\mathbb{F}$, and fields such as $\mathbb{C}$ have many nontrivial endomorphisms, cf. Example 2.3.3.4. It is also not true that there exists a morphism $\operatorname{Spec}\mathbb{F} \to \operatorname{Spec}\mathbb{F}'$ for any pair of fields $\mathbb{F}$ and $\mathbb{F}'$, cf. Remark 2.3.3.5.

●

Figure 2.1: This is $\operatorname{Spec}\mathbb{F}$. Nothing more, nothing less.

Example 2.1.3.2 (The 'geometric' affine line $\mathbb{A}^1_{\mathbf{k}} = \operatorname{Spec}\mathbf{k}[x]$)**.** Let $\mathbf{k}$ be an algebraically closed field, such as $\mathbb{C}$. The ring $\mathbf{k}[x]$ is a principal ideal domain, whose prime ideals are (0) and $(x-a)$, one for each $a \in \mathbf{k}$. The spectrum

$$\mathbb{A}^1_{\mathbf{k}} = \operatorname{Spec}\mathbf{k}[x]$$

is called the *affine line* (over $\mathbf{k}$). Note that there is exactly one point, namely

$$\xi = (0) \in \mathbb{A}^1_{\mathbf{k}},$$

that is not closed. In fact, by Lemma 2.1.1.14, we have

$$\overline{\{\xi\}} = \mathrm{V}(0) = \mathbb{A}^1_{\mathbf{k}}.$$

This point was invisible in the land of *classical varieties*, where only closed points were allowed. It has a name: it is the *generic point* of the affine line. We will say a lot more about generic points later (cf. Section 2.2.3), but for now notice that the terminology is somewhat well chosen: if we think that $(x-a)$ corresponds to the 'classical' point $a \in \mathbb{C}$, then since $x - x = 0$ it is reasonable to think that the coordinate of this point has indeed stayed 'generic'. This is what an 'indeterminate' should be!

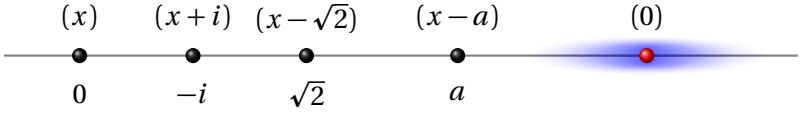


Figure 2.2: The topological space $\mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec}\mathbb{C}[x]$, with one closed point for every $a \in \mathbb{C}$. The generic point $\xi = (0)$ is 'dense', i.e. $\overline{\{\xi\}} = \mathbb{A}^1_{\mathbb{C}}$, since $0 \in \mathbb{C}[x]$ is in every prime ideal.

Note that, if $\mathbb{F}$ is an arbitrary field, not necessarily algebraically closed, $\mathbb{F}[x]$ is still a principal ideal domain, but now the closed points of $\mathbb{A}^1_{\mathbb{F}}$ correspond to those maximal ideals (f) generated by irreducible polynomials of degree possibly larger than 1. The composition

$$\mathbb{F} \hookrightarrow \mathbb{F}[x] \twoheadrightarrow \mathbb{F}[x]/(f)$$

is a finite extension of fields, of degree equal to $\deg f$ (cf. Example 2.1.3.4).

Example 2.1.3.3 (The 'arithmetic' affine line $\operatorname{Spec}\mathbb{Z}$). The spectrum $\operatorname{Spec}\mathbb{Z}$ is the arithmetic counterpart of $\operatorname{Spec}\mathbf{k}[x]$. It has one closed point for every nonzero prime ideal $(p) \subset \mathbb{Z}$. Additionally, there is precisely one non-closed point $\xi = (0) \in \operatorname{Spec}\mathbb{Z}$, which is called the generic point by virtue of the property $\overline{\{\xi\}} = \operatorname{Spec}\mathbb{Z}$.



Figure 2.3: The topological space $\operatorname{Spec}\mathbb{Z}$, with one closed point for every prime $p \in \mathbb{Z}$. The generic point $\xi = (0)$ is 'dense', i.e. $\overline{\{\xi\}} = \operatorname{Spec}\mathbb{Z}$, since $0 \in \mathbb{Z}$ is in every prime ideal.

Example 2.1.3.4 ($\mathbb{A}^1_{\mathbb{R}} = \operatorname{Spec}\mathbb{R}[x]$). The ring $\mathbb{R}[x]$ is a principal ideal domain. Its prime ideals are

$$(0), \quad (x-a), \quad (x^2+bx+c),$$

where $a \in \mathbb{R}$ and x^2+bx+c is irreducible and satisfies $b^2-4c<0$. The only prime ideal which is not maximal is, once more, $(0) \subset \mathbb{R}[x]$. However, we see here an important phenomenon arising when one considers fields that are not algebraically closed: we have, for the two types of *maximal* ideals,

$$\mathbb{R}[x]/(x-a) \cong \mathbb{R}, \quad \mathbb{R}[x]/(x^2+bx+c) \cong \mathbb{C}.$$

If f is an irreducible quadratic polynomial as above, the prime ideal $(f) \subset \mathbb{R}[x]$ defines one precise point in the spectrum, which, after base change to $\mathbb{C}$, splits into two conjugate points (depicted in green in Figure 2.4). For instance, the two points $(x \pm i)$ in $\mathbb{A}^1_{\mathbb{C}}$ give rise to the single point $(f) \in \mathbb{A}^1_{\mathbb{R}}$, where $f = x^2+1 = (x-i)(x+i)$. This is a simple shadow of the algebraic fact that in order to find a root for x^2+1 one need to leave $\mathbb{R}$ and go over $\mathbb{C}$. See Example 2.3.3.19 for more on this.

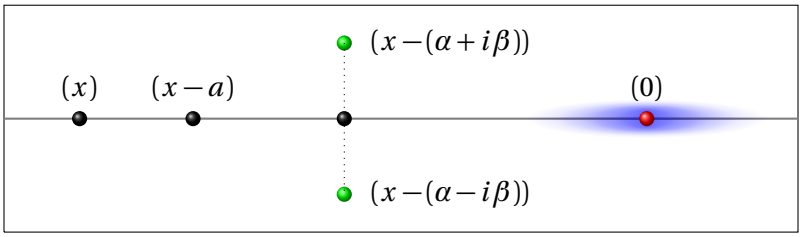


Figure 2.4: The real affine line $\mathbb{A}^1_{\mathbb{R}}$. Points $(f) \in \mathbb{A}^1_{\mathbb{R}}$ with $\mathbb{R}[x]/(f) \cong \mathbb{C}$ have "degree 2": they can be thought of as pairs of conjugate *complex* points in $\mathbb{A}^1_{\mathbb{C}}$ coming together.

Example 2.1.3.5 $(\mathbb{A}^2_{\mathbf{k}} = \operatorname{Spec} \mathbf{k}[x,y])$. Let $\mathbf{k}$ be an algebraically closed field. The spectrum $\mathbb{A}^2_{\mathbf{k}} = \operatorname{Spec} \mathbf{k}[x,y]$ is called the *affine plane* (over $\mathbf{k}$). The prime ideals in $\mathbf{k}[x,y]$ are

$$(0), \quad (x-a, y-b), \quad (f)$$

where $(a,b) \in \mathbf{k}^2$ and $f = f(x,y)$ is an irreducible polynomial. Maximal ideals are those of the form $(x-a, y-b)$, and correspond indeed to the 'classical' points (a,b) of $\mathbf{k}^2$. These are exactly the closed points of $\mathbb{A}^2_{\mathbf{k}}$ by Remark 2.1.1.13. Given an irreducible polynomial $f \in \mathbf{k}[x,y]$, we have

$$\mathrm{V}(f) = \left\{ \mathfrak{p} \in \mathbb{A}^2_{\mathbf{k}} \,\middle|\, f \in \mathfrak{p} \right\} = \left\{ (x-a, y-b) \,\middle|\, f(a,b) = 0 \right\} \cup \left\{ (f) \right\}.$$

The closure of $\{(f)\}$ is precisely $\mathrm{V}(f)$. The ideal (f) is the generic point of $\mathrm{V}(f)$, because it corresponds to the trivial ideal in the integral domain $\mathbf{k}[x,y]/(f)$, whereas the other points are closed points.

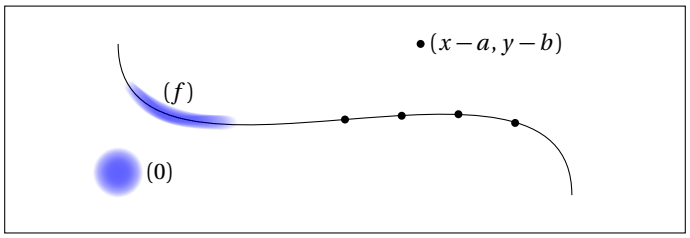


Figure 2.5: The affine plane $\mathbb{A}^2_{\mathbf{k}}$.

Definition 2.1.3.6. Let A be a commutative ring. We say that A is a *discrete valuation ring* (a DVR, for short) if the following equivalent conditions are satisfied.

(i) A is a principal ideal domain with exactly one nonzero maximal ideal $\mathfrak{m} \subset A$,

(ii) A is a local ring which is a principal ideal domain, and not a field,

(iii) A is a noetherian local ring whose unique maximal ideal is generated by a single non-nilpotent element.



Exercise 2.1.3.7. Prove that the conditions in Definition 2.1.3.6 are equivalent.

Example 2.1.3.8 (Spec of a DVR). Let A be a DVR. The maximal ideal $\mathfrak{m} \subset A$ is also prime, and there is precisely one other prime ideal, namely $\xi = (0) \subset A$. In other words,

$$\operatorname{Spec} A = \{ \xi, \mathfrak{m} \}$$

consists of two points, where $\mathfrak{m}$ is closed (cf. Remark 2.1.1.13) and hence ξ, corresponding to $(0) \subset A$, is open. Note that the Zariski topology is *not the discrete topology* on two points here, for the point $\mathfrak{m}$ is a *specialisation* of the point ξ, i.e. $\mathfrak{m}$ lies in the closure of ξ. We shall see later that, despite being a finite set, $\operatorname{Spec} A$ is a 1-dimensional scheme (or A is a 1-dimensional ring, cf. **??**), simply because $0 \in \mathfrak{m}$. An example of DVR is given by the ring of formal power series $\mathbb{F}[\![t]\!]$, where $\mathbb{F}$ is a field. In this case, the maximal ideal is just the ideal generated by t.

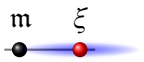


Figure 2.6: The black bullet represents the closed point $\mathfrak{m}$. The red point surrounded by the cloud, as usual, is the generic point.

Example 2.1.3.9 ($\operatorname{Spec} \mathbb{F}[t]/t^2$). First of all some terminology: the ring $A = \mathbb{F}[t]/t^2$ is called the *ring of dual numbers*[3] (over $\mathbb{F}$). Some people write $\mathbb{F}[\varepsilon]$ to denote this ring, being understood that $\varepsilon^2 = 0$. There is only one prime (and in fact maximal) ideal in A, namely

$$(\overline{t}) \subset A,$$

where $\overline{t}$ is the image of $t \in \mathbb{F}[t]$ under the projection $\mathbb{F}[t] \twoheadrightarrow A$. Thus, topologically, this space is the same as $\operatorname{Spec} \mathbb{F}$. However, it will be different (i.e. *not isomorphic* to $\operatorname{Spec} \mathbb{F}$) as an affine scheme. A first strong indication of this fact was already given, via our informal discussion on functions on the spectrum, in Spoiler 2.1.2.2.

[3] In case you care to know why they have this name, here is the answer directly from Wikipedia: Dual numbers were introduced in 1873 by William Clifford, and were used at the beginning of the twentieth century by the German mathematician Eduard Study, who used them to represent the dual angle which measures the relative position of two skew lines in space.

2.1.4 Sheaf associated to a module

Let A be a ring. Set $X = \operatorname{Spec} A$, equipped as always with the Zariski topology. For an A-module M, we now define an associated sheaf of abelian groups $\widetilde{M}$ on X. The case $M = A$ will yield, by definition, a sheaf of *rings* that we will denote by $\mathscr{O}_X$. For arbitrary M, the associated sheaf $\widetilde{M}$ will have a natural structure of $\mathscr{O}_X$-module, cf. Definition 1.13.0.1. The space X considered together with

$$\mathscr{O}_X \in \mathsf{Sh}(X, \mathsf{Rings}),$$

will be proved to define a locally ringed space, called an *affine scheme*. General schemes will be defined as those locally ringed spaces that are locally isomorphic to affine schemes.

Consider the base $\mathcal{B} = \{\mathrm{D}(f) \mid f \in A\}$ of principal opens. Define the assignment

$$\mathrm{D}(f) \longmapsto M_f = \left\{ \frac{m}{f^n} \;\middle|\; m \in M,\, n \geqslant 0 \right\}. \tag{2.1.4}$$

For an inclusion $\mathrm{D}(g) \subset \mathrm{D}(f)$ we now define a restriction homomorphism

$$M_f \xrightarrow{\rho_{f,g}} M_g.$$

We know (cf. Example 2.1.1.9) that

$$\mathrm{D}(g) \subset \mathrm{D}(f) \iff g \in \sqrt{fA} \iff \frac{f}{1} \in A_g^\times.$$

For the second equivalence, write $g^r = fb$ in A for some $b \in A$ and some $r \geqslant 1$. Thus

$$\frac{f}{1} \cdot \frac{b}{1} = \frac{fb}{1} = \frac{g^r}{1} \in A_g,$$

which is invertible in A_g. Therefore, by the universal property of localisation (cf. Appendix B.3.2), the canonical map $A \to A_g$ factors *uniquely* through A_f, i.e. we have a commutative diagram of rings

$$\begin{array}{ccc} A & \longrightarrow & A_f \\ \downarrow & \swarrow_{\rho_{f,g}} & \\ A_g & & \end{array} \tag{2.1.5}$$

and the uniqueness in the universal property shows that if $\mathrm{D}(f) = \mathrm{D}(g)$ then the maps $A_f \to A_g$ and $A_g \to A_f$ are inverse to each other (thus the assignment is well-defined), and moreover we have identities

$$\rho_{f,f} = \mathrm{id}_{M_f}, \qquad \rho_{g,h} \circ \rho_{f,g} = \rho_{f,h}$$

whenever $\mathrm{D}(h) \subset \mathrm{D}(g) \subset \mathrm{D}(f)$. Hence the assignment (2.1.4) defines a $\mathcal{B}$-presheaf of abelian groups for $M = A$. Applying $M \otimes_A -$ to Diagram (2.1.5), we obtain a canonical map

$$\rho^M_{f,g} \colon M_f \longrightarrow M_g$$

defined as $\rho^M_{f,g} = \mathrm{id}_M \otimes_A \rho_{f,g}$. This is our restriction map for the assignment $\mathrm{D}(f) \mapsto M_f$. The functoriality of this assignment is inherited from the one for $M = A$. We will use the notation $s|_{\mathrm{D}(g)} = \rho^M_{f,g}(s)$ for an element $s \in M_f$.

Remark 2.1.4.1. The map $\rho^M_{f,g}$ is explicit: say $g^r = f\,b$ as above, and fix an element $m/f^n \in M_f$ for some $m \in M$ and some $n \geqslant 0$. Then

$$\frac{m}{f^n} \xmapsto{\rho^M_{f,g}} \frac{m\,b^n}{g^{nr}}.$$

Remark 2.1.4.2. Note that the assignment (2.1.4) prescribes

$$\emptyset = \mathrm{D}(0) \longmapsto M_0 = 0.$$

THEOREM 2.1.4.3. *Let A be a ring. Set $X = \operatorname{Spec} A$, equipped with the Zariski topology.*

(a) *The rule (2.1.4) defines a sheaf of abelian groups on X. If $M = A$, this is a sheaf of rings that we will from now on denote by $\mathscr{O}_X$, and for general M the sheaf $\widetilde{M}$ has a natural structure of $\mathscr{O}_X$-module.*

(b) *We have $\widetilde{M}(X) = M$. In particular, $\mathscr{O}_X(X) = A$.*

(c) *The stalk of $\widetilde{M}$ at the point $\mathfrak{p} \in X$ is canonically isomorphic to the localisation $M_{\mathfrak{p}}$. In particular, the stalk $\mathscr{O}_{X,x}$ of $\mathscr{O}_X$ at the point $x \in X$ corresponding to $\mathfrak{p} \subset A$ is isomorphic to $A_{\mathfrak{p}}$ and is therefore a local ring.*

Proof. We proceed step by step.
Proof of (a). If $M = 0$ there is nothing to prove, so we assume $M \neq 0$. By Lemma 1.8.0.6 (cf. also Exercise 1.13.0.7), it is enough to prove that (2.1.4) defines a $\mathcal{B}$-sheaf: it will then extend uniquely to a sheaf, still denoted $\widetilde{M}$.

Consider an open covering $X = \bigcup_{i \in I} U_i$, where $U_i = \mathrm{D}(f_i)$ is principal. Set also $U_{ij} = U_i \cap U_j$ and recall that $U_{ij} = \mathrm{D}(f_i f_j)$ is also principal. We need to prove that

$$0 \longrightarrow M \longrightarrow \prod_{i \in I} M_{f_i} \xrightarrow{\delta} \prod_{(i,j) \in I \times I} M_{f_i f_j}$$

is an equaliser diagram, where δ is the map sending

$$(s_i)_{i \in I} \longmapsto \left(s_i|_{U_{ij}} - s_j|_{U_{ij}}\right)_{(i,j) \in I \times I}.$$

To be precise, we should also confirm the same conclusion with X replaced by an arbitrary principal open $U = \mathrm{D}(f)$. But this case is essentially the same as the $U = X$ case: replace A by A_f, M by M_f, and each f_i by its image in A_f.

By Lemma 2.1.1.10, the covering $X = \bigcup_{i \in I} \mathrm{D}(f_i)$ has a finite subcover indexed by a *finite* set $F \subset I$. Moreover, $1 \in (f_i \mid i \in F)$. In what follows, set $U_i = \mathrm{D}(f_i)$ and $U_{ij} = U_i \cap U_j = \mathrm{D}(f_i f_j)$.

Sheaf axiom (loc): Fix $s \in M_1 = M$ such that $s|_{U_i} = 0 \in M_{f_i}$ for all $i \in I$. This means

$$\frac{s}{1} = \frac{0}{1} \in M_{f_i},$$

i.e. there exists $k > 0$ such that $f_i^k s = 0 \in M$. Since F is finite, we can pick a uniform k which works for every f_i such that $i \in F$. Since

$$X = \bigcup_{i \in F} \mathrm{D}(f_i) = \bigcup_{i \in F} \mathrm{D}(f_i^k),$$

as before we have

$$1 \in (f_i^k \mid i \in F) = \sum_{i \in F} f_i^k A,$$

which implies

$$s = 1 \cdot s = \sum_{i \in F} a_i f_i^k s = 0,$$

as required.

Sheaf axiom (glue): By definition, $\widetilde{M}(U_{ij}) = M_{f_i f_j}$. Fix sections $s_i \in M_{f_i}$ such that $s_i|_{U_{ij}} = s_j|_{U_{ij}}$ for every i and j in I. That is, s_i and s_j have the same image along the maps

$$\begin{array}{ccc} M_{f_i} & & M_{f_j} \\ & \searrow \quad \swarrow & \\ & M_{f_i f_j} & \end{array}$$

Write (again for a uniform $k > 0$)

$$s_i = \frac{m_i}{f_i^k} \in M_{f_i}, \quad m_i \in M, \quad i \in F.$$

Now, $s_i|_{U_{ij}} = s_j|_{U_{ij}}$ means that there exists an integer $r > 0$ such that

$$(f_i f_j)^r (m_i f_j^k - m_j f_i^k) = 0 \in M \tag{2.1.6}$$

for all $i, j \in F$. As before,

$$1 \in (f_i^{k+r} \mid i \in F) = \sum_{i \in F} f_i^{k+r} A$$

yields

$$1 = \sum_{j \in F} a_j f_j^{k+r}, \quad a_j \in A. \tag{2.1.7}$$

Define a candidate gluing of the family $(s_i)_{i \in I}$ by setting

$$s = \sum_{j \in F} a_j f_j^r m_j \in M, \tag{2.1.8}$$

so that the chain of identities

$$\begin{aligned} f_i^{k+r} s &= \sum_{j\in F} a_j f_i^k (f_i f_j)^r m_j && \text{by (2.1.8)} \\ &= \sum_{j\in F} a_j f_j^k (f_i f_j)^r m_i && \text{by (2.1.6)} \\ &= \sum_{j\in F} a_j f_j^{k+r} f_i^r m_i && \text{rearranging} \\ &= f_i^r m_i && \text{by (2.1.7)} \end{aligned}$$

yields

$$f_i^r (m_i - f_i^k s) = 0 \in M.$$

But this in turn is equivalent to

$$\frac{s}{1} = \frac{m_i}{f_i^k} = s_i \in M_{f_i}, \qquad i \in F.$$

So we have proved $s|_{U_i} = s_i$ for every $i \in F$. It remains to compare s with the sections indexed by $I \setminus F$. Fix $\ell \in I$. The opens $U_\ell \cap U_i$, for $i \in F$, cover U_ℓ. On each of them,

$$(s|_{U_\ell})|_{U_\ell \cap U_i} = s|_{U_\ell \cap U_i} = (s|_{U_i})|_{U_\ell \cap U_i} = s_i|_{U_\ell \cap U_i} = s_\ell|_{U_\ell \cap U_i},$$

where the last identity is part of our initial assumption. The separation axiom applied to the cover $U_\ell = \bigcup_{i\in F} U_\ell \cap U_i$ therefore gives $s|_{U_\ell} = s_\ell$. Hence s glues the entire original family, as required.

We have proved that $\widetilde{M}$ is a sheaf of abelian groups. If $M = A$, we obtain a *sheaf of rings*, since A_f are rings for all $f \in A$, and the sheaf property can be verified at the level of abelian groups (or even sets).

Finally, to prove that $\widetilde{M}$ is an $\mathcal{O}_X$-module, first recall that $M_f \cong M \otimes_A A_f$ is naturally an A_f-module, where of course $A_f = \mathcal{O}_X(\mathrm{D}(f))$ by definition. The fact that the restriction map $M_f \to M_g$ is compatible with the restriction $A_f \to A_g$ (cf. Diagram (1.13.1)) follows at once from the canonical identifications $M_h = M \otimes_A A_h$.

Proof of (b). Obvious from the definition.

Proof of (c). Fix $\mathfrak{p} \in \operatorname{Spec} A$. For every $f \notin \mathfrak{p}$, there is (by the universal property of A_f) a canonical map $A_f \to A_\mathfrak{p} = \{\, a/h \mid h \notin \mathfrak{p} \,\}$ because f is invertible in $A_\mathfrak{p}$. This determines a map $\mu_f \colon M_f \to M_\mathfrak{p}$ by applying $M \otimes_A -$, so by the universal property of colimits, we get a canonical homomorphism of A-modules

$$\widetilde{M}_\mathfrak{p} = \varinjlim_{f\notin\mathfrak{p}} M_f \xrightarrow{\ \alpha\ } M_\mathfrak{p}.$$

An element of the form $m/h \in M_\mathfrak{p}$ for some $h \in A \setminus \mathfrak{p}$ lies in the image of $\alpha \circ \mu_h \colon M_h \to \widetilde{M}_\mathfrak{p} \to M_\mathfrak{p}$, therefore α is surjective. On the other hand, if $m/h^n \in M_h$ (for $h \in A \setminus \mathfrak{p}$ and

some $n > 0$[4]) maps to $0 = 0/1 \in M_{\mathfrak{p}}$, then by definition of localisation there exists $g \in A \setminus \mathfrak{p}$ such that $gm = 0 \in M$. Then the image of m/h^n in M_{gh} is

$$\frac{g^{n-1}gm}{(gh)^n} = 0 \in M_{gh},$$

so m/h^n goes to 0 in $\widetilde{M}_{\mathfrak{p}}$. We have confirmed that

$$\ker\left(M_h \xrightarrow{\mu_h} \widetilde{M}_{\mathfrak{p}} \xrightarrow{\alpha} M_{\mathfrak{p}}\right) = \ker\left(M_h \xrightarrow{\mu_h} \widetilde{M}_{\mathfrak{p}}\right) \tag{2.1.9}$$

for every $h \in A \setminus \mathfrak{p}$, which is enough to conclude that α is injective. Indeed, let $z \in \widetilde{M}_{\mathfrak{p}}$ be an element such that $\alpha(z) = 0 \in M_{\mathfrak{p}}$. There exist $m \in M$, $h \in A \setminus \mathfrak{p}$ and $n \geqslant 0$ such that $z = \mu_h(m/h^n)$. So $0 = \alpha(z) = \alpha(\mu_h(m/h^n))$ implies $m/h^n \in \ker(\alpha \circ \mu_h) = \ker \mu_h$, by the equality of kernels (2.1.9) just proved. It follows that $0 = \mu_h(m/h^n) = z$, as required. □

Example 2.1.4.4. Let $X = \operatorname{Spec} \mathbb{F}$, where $\mathbb{F}$ is a field. Then $\mathscr{O}_X = \underline{\mathbb{F}}_X$, the constant sheaf on X, cf. Example 1.5.0.3.

Remark 2.1.4.5. Constructing a sheaf via a sheaf on base leaves out the question: what exactly are the sections of the sheaf over an arbitrary open subset? One can answer abstractly via the equaliser sequence, but we now provide a clean answer in the case of the $\mathscr{O}_X$-module $\widetilde{M}$. Let $U \subset X = \operatorname{Spec} A$ be an open subset. Then, for an A-module M, one has the explicit identity

$$\widetilde{M}(U) = \left\{ U \xrightarrow{\phi} \coprod_{\mathfrak{p} \in U} M_{\mathfrak{p}} \;\middle|\; \begin{array}{c} \text{for every } \mathfrak{p} \in U,\ \phi(\mathfrak{p}) \in M_{\mathfrak{p}} \text{ and there exist} \\ \text{an open neighbourhood } V \subset U \text{ of } \mathfrak{p} \\ \text{and } m \in M, f \in A \text{ such that, for every } \mathfrak{q} \in V, \\ f \notin \mathfrak{q} \text{ and } \phi(\mathfrak{q}) = m/f \text{ in } M_{\mathfrak{q}} \end{array} \right\},$$

which of course recovers $\widetilde{M}(\mathrm{D}(f)) = M_f$ on principal opens. An independent proof that this assignment is a sheaf is verified in Exercise 2.1.4.6. Some details need to be filled in, but the fact that $U \mapsto \widetilde{M}(U)$ is a sheaf (and coincides with the sheaf $\widetilde{M}$ defined in Theorem 2.1.4.3(a)) is clear once one realises that the very definition of $\widetilde{M}(U)$ just rephrases the notion of compatible germs.



Exercise 2.1.4.6. Prove explicitly that the assignment $U \mapsto \widetilde{M}(U)$ is a sheaf of abelian groups on X.

We close this section with a few properties of the "tilde construction". The most important property to remember, at this stage, is that $\widetilde{M}$ is always a quasicoherent $\mathscr{O}_X$-module.

PROPOSITION 2.1.4.7. *Let A be a ring, and set $X = \operatorname{Spec} A$.*

(i) *If $(M_i)_i$ is a family of A-modules, then $(\bigoplus_i M_i)^{\sim} = \bigoplus_i \widetilde{M}_i$.*

[4]We may indeed assume $n > 0$, for we can always represent $m/1$ by mh/h in M_h.

(ii) *If $L \to M \to N$ is a sequence in Mod_A, then it is exact if and only if $\widetilde{L} \to \widetilde{M} \to \widetilde{N}$ is exact in $\mathsf{Mod}(\mathscr{O}_X)$.*

(iii) *$\widetilde{M}$ is quasicoherent, for any A-module M.*

(iv) *If M, N are A-modules, then $(M \otimes_A N)^{\sim} = \widetilde{M} \otimes_{\mathscr{O}_X} \widetilde{N}$.*

Proof. We proceed step by step.
Proof of (i). Fix $f \in A$. We have

$$\left(\bigoplus_i M_i\right)^{\sim}(\mathrm{D}(f)) = \left(\bigoplus_i M_i\right)_f = \bigoplus_i M_{i,f} = \bigoplus_i \widetilde{M}_i(\mathrm{D}(f)).$$

Proof of (ii). First assume $L \to M \to N$ is exact. Let $\mathfrak{p} \in \operatorname{Spec} A$. Then, since localisation is exact (cf. Proposition B.3.4.2) (see also **????**), the sequence of $A_{\mathfrak{p}}$-modules $L_{\mathfrak{p}} \to M_{\mathfrak{p}} \to N_{\mathfrak{p}}$ is exact. Then Proposition 1.6.0.9 implies that $\widetilde{L} \to \widetilde{M} \to \widetilde{N}$ is exact in $\mathsf{Mod}(\mathscr{O}_X)$.

Conversely, assume $\widetilde{L} \to \widetilde{M} \to \widetilde{N}$ is exact in $\mathsf{Mod}(\mathscr{O}_X)$. We have a commutative diagram

$$\begin{array}{ccccc} L & \xrightarrow{a} & M & \xrightarrow{b} & N \\ \downarrow & & \downarrow & & \downarrow \\ L_{\mathfrak{p}} & \longrightarrow & M_{\mathfrak{p}} & \longrightarrow & N_{\mathfrak{p}} \end{array}$$

where

- the upper row is obtained by taking global sections in $\widetilde{L} \to \widetilde{M} \to \widetilde{N}$,
- the vertical arrows are localisations,
- the lower row is exact, being the stalk of the exact sequence $\widetilde{L} \to \widetilde{M} \to \widetilde{N}$.

It follows from the last bullet that $(\ker b / \operatorname{im} a)_{\mathfrak{p}} = 0$ and since this holds for every $\mathfrak{p} \in \operatorname{Spec} A$ we have globally $\ker b = \operatorname{im} a$, proving that $L \to M \to N$ is exact.
Proof of (iii). Every $M \in \mathsf{Mod}_A$ admits an exact sequence

$$A^{\oplus J} \longrightarrow A^{\oplus I} \longrightarrow M \longrightarrow 0.$$

By parts (i) and (ii) combined, we have a corresponding exact sequence

$$\begin{array}{ccccccc} (A^{\oplus J})^{\sim} & \longrightarrow & (A^{\oplus I})^{\sim} & \longrightarrow & \widetilde{M} & \longrightarrow & 0 \\ \| & & \| & & & & \\ \mathscr{O}_X^{\oplus J} & & \mathscr{O}_X^{\oplus I} & & & & \end{array}$$

therefore $\widetilde{M}$ is quasicoherent.
Proof of (iv). For $f \in A$, there is a canonical A_f-bilinear map

$$M_f \times N_f \xrightarrow{\varphi_f} (M \otimes_A N)_f, \qquad \left(\frac{m}{f^r}, \frac{n}{f^s}\right) \longmapsto \frac{m \otimes n}{f^{r+s}}.$$

Such maps are compatible with restrictions to smaller principal opens $D(g) \subset D(f)$, and therefore they give rise to a global $\mathscr{O}_X$-linear map

$$\widetilde{M} \otimes_{\mathscr{O}_X} \widetilde{N} \xrightarrow{\varphi} (M \otimes_A N)^{\sim},$$

using the universal property of tensor product of sheaves. To check that φ is an isomorphism, it is enough to check it induces an isomorphism on all the stalks. If $\mathfrak{p} \in \operatorname{Spec} A$, then

$$\begin{aligned}
(\widetilde{M} \otimes_{\mathscr{O}_X} \widetilde{N})_{\mathfrak{p}} &\cong \widetilde{M}_{\mathfrak{p}} \otimes_{\mathscr{O}_{X,\mathfrak{p}}} \widetilde{N}_{\mathfrak{p}} && \text{by (1.13.2)} \\
&\cong M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}} && \text{by Theorem 2.1.4.3(c)} \\
&\cong (M \otimes_A N)_{\mathfrak{p}} && \text{by Proposition B.3.4.2} \\
&\cong ((M \otimes_A N)^{\sim})_{\mathfrak{p}} && \text{by Theorem 2.1.4.3(c)}
\end{aligned}$$

as desired. □

2.1.5 Definition of affine schemes and affine varieties

The following is now immediate from the definition of locally ringed space.

COROLLARY 2.1.5.1. *Let A be a ring, and set $X = \operatorname{Spec} A$. The pair $(X, \mathscr{O}_X)$ with $\mathscr{O}_X$ as in Theorem 2.1.4.3 defines a locally ringed space. For every $x \in X$, corresponding to a prime $\mathfrak{p} \subset A$, the local ring $(\mathscr{O}_{X,x}, \mathfrak{m}_x)$ is canonically isomorphic to the local ring $(A_{\mathfrak{p}}, \mathfrak{p}A_{\mathfrak{p}})$.*

The quotient $\kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is called the *residue field* at $\mathfrak{p}$.

Important Definition 2.1.5.2 (Affine scheme)**.** An *affine scheme* is a locally ringed space isomorphic (in the category of locally ringed spaces) to $(\operatorname{Spec} A, \mathscr{O}_{\operatorname{Spec} A})$ for some ring A.

As an important class of examples of affine schemes, we have the notion of affine algebraic variety. We will see a different notion, that of projective variety, in Important Definition 2.5.3.10.

Important Definition 2.1.5.3 (Affine variety)**.** An *affine variety* over a field $\mathbb{F}$ (also called an *affine $\mathbb{F}$-variety*) is an affine scheme of the form $\operatorname{Spec} A$, where A is a finitely generated $\mathbb{F}$-algebra (i.e. $A = \mathbb{F}[x_1, \ldots, x_n]/I$ for some n and some ideal I).



Caution 2.1.5.4. Some authors call what we call an affine variety an *affine algebraic scheme*, reserving the term "affine variety" for those affine algebraic schemes that further satisfy (either or both)

(i) A is reduced,

(ii) $\operatorname{Nil}(A) \subset A$ is prime.

We deliberately choose, in this text, to adopt the weaker notion as in Important Definition 2.1.5.3. In particular, a variety for us need not be reduced nor irreducible. See also Caution 2.4.1.2.



Exercise 2.1.5.5. Let A be a ring, $\mathfrak{p} \subset A$ a prime ideal. Set $\kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$. Show that there is a commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & A_{\mathfrak{p}} \\ \downarrow & & \downarrow \\ A/\mathfrak{p} & \hookrightarrow & \kappa(\mathfrak{p}) \end{array}$$

of rings, and that

$$\operatorname{Frac} A/\mathfrak{p} = \kappa(\mathfrak{p}).$$

In particular, if $\mathfrak{m} \subset A$ is maximal, then $\kappa(\mathfrak{m}) = A/\mathfrak{m}$.

LEMMA 2.1.5.6 (Density of closed points). *Let $A \neq 0$ be a finitely generated $\mathbb{F}$-algebra, for a field $\mathbb{F}$. Set $X = \operatorname{Spec} A$. Then the set*

$$X_{\mathsf{cl}} = \{\, \mathfrak{p} \in X \mid \mathfrak{p} \text{ is closed} \,\} \subset X$$

is dense.

Proof. Recall (cf. Remark 2.1.1.13) that the closed points of $X = \operatorname{Spec} A$ are precisely the maximal ideals of A. It is enough to prove that every $\emptyset \neq \mathrm{D}(f) \subset X$ contains a maximal ideal $\mathfrak{m} \in X$. Indeed, since the sets $\mathrm{D}(f)$ form a basis for the Zariski topology, every nonempty open subset will contain a closed point, and this implies that X_{cl} is dense in X.

By non-emptiness of $\mathrm{D}(f)$, the localisation A_f is nonzero. Since A_f is a nonzero ring, it has a maximal ideal $\mathfrak{n} \subset A_f$. Moreover, A_f is again a finitely generated $\mathbb{F}$-algebra, since $A_f \cong A[t]/(ft-1)$. Thus $A_f/\mathfrak{n}$ is a field which is finitely generated as an $\mathbb{F}$-algebra. By Zariski's lemma (see Lemma B.1.3.1), $A_f/\mathfrak{n}$ is a finite extension of $\mathbb{F}$.

Let $\ell : A \to A_f$ be the localisation map. Set

$$\mathfrak{m} = \ell^{-1}(\mathfrak{n}) \in \operatorname{Spec} A.$$

Then $A/\mathfrak{m}$ embeds into the field $A_f/\mathfrak{n}$. Hence $A/\mathfrak{m}$ is a finite-dimensional $\mathbb{F}$-algebra. It is also an integral domain, since $\mathfrak{m}$ is prime. A finite-dimensional domain over a field is itself a field (reason: if $0 \neq x \in A/\mathfrak{m}$, multiplication by x is an injective linear endomorphism of the finite-dimensional $\mathbb{F}$-vector space $A/\mathfrak{m}$, hence is surjective, so x has an inverse). Therefore $\mathfrak{m}$ is maximal. Furthermore, $f \notin \mathfrak{m}$, since f becomes a unit in A_f. Thus $\mathfrak{m} \in \mathrm{D}(f)$, which proves that every nonempty basic open subset of $\operatorname{Spec} A$ contains a closed point, as required. □

Example 2.1.5.7. Let $\mathbb{F}$ be a field, $0 \in X = \mathbb{A}^1_{\mathbb{F}} = \operatorname{Spec}\mathbb{F}[x]$ the point corresponding to $(x) \subset \mathbb{F}[x]$. (You are allowed, and in fact encouraged, to call this point 'the origin' of the affine line). The local ring of X at 0 is

$$\mathscr{O}_{X,0} = \mathbb{F}[x]_{(x)} = \left\{ \frac{f(x)}{g(x)} \,\middle|\, g(x) \notin (x) \right\} = \left\{ \frac{f(x)}{g(x)} \,\middle|\, g(0) \neq 0 \right\} \subset \operatorname{Frac}\mathbb{F}[x] = \mathbb{F}(x),$$

and the residue field is

$$\kappa(0) = \mathscr{O}_{X,0}/\mathfrak{m}_0 = \mathbb{F}[x]_{(x)}/(x)\mathbb{F}[x]_{(x)} \cong \mathbb{F}[x]/(x) \cong \mathbb{F}.$$

For the second-last isomorphism we used Exercise 2.1.5.5 (see also Proposition B.3.3.5). The same chain of isomorphisms holds replacing 0 with any other closed point of the form $(x-a)$, with $a \in \mathbb{F}$. If $\xi = (0)$, we have

$$\kappa(\xi) = \mathscr{O}_{X,\xi} = \mathbb{F}(x).$$

2.1.6 Functions on $\operatorname{Spec} A$ revisited

We already saw, but we need to emphasise, that

$$\boxed{\mathscr{O}_{\operatorname{Spec} A}(\operatorname{Spec} A) = A.}$$

That is, regular functions on $\operatorname{Spec} A$ are precisely the elements of A. In Section 2.1.2, when we only had a topology on $\operatorname{Spec} A$, we were left confused by the fact that a "function" on $\operatorname{Spec} A$ took values in different rings, *depending on the point* (!) on which it was evaluated. Now we can dissipate that confusion. Indeed, by Remark 2.1.4.5 and Exercise 2.1.4.6, we know we could have bypassed $\mathcal{B}$-sheaves and defined the sheaf of rings $\mathscr{O}_X$ on $X = \operatorname{Spec} A$ directly (as done in [4, Ch. 2]) by setting

$$\mathscr{O}_X(U) = \left\{ U \xrightarrow{s} \coprod_{\mathfrak{p} \in U} A_{\mathfrak{p}} \,\middle|\, \begin{array}{c} \text{for every } \mathfrak{p} \in U,\ s(\mathfrak{p}) \in A_{\mathfrak{p}} \text{ and there exist} \\ \text{an open neighbourhood } V \subset U \text{ of } \mathfrak{p} \\ \text{and } a, f \in A \text{ such that, for every } \mathfrak{q} \in V, \\ f \notin \mathfrak{q} \text{ and } s(\mathfrak{q}) = a/f \text{ in } A_{\mathfrak{q}} \end{array} \right\} \tag{2.1.10}$$

for every open subset $U \subset X$. Let us (partially) unravel the case $U = X = \operatorname{Spec} A$. Consider the map $\psi : A \to \mathscr{O}_X(X)$ sending $a \in A$ to the function

$$s_a : X \longrightarrow \coprod_{\mathfrak{p} \in X} A_{\mathfrak{p}}, \quad \mathfrak{p} \mapsto \text{image of } a \text{ along } A \to A_{\mathfrak{p}}.$$

This map ψ is bijective. To see it is injective, assume $s_a = s_b$ for $a, b \in A$. This means that for every $\mathfrak{p} \in X$, the elements a and b have the same image in $A_{\mathfrak{p}}$. Hence there is an element $r \in A \setminus \mathfrak{p}$ such that $r(a-b) = 0$ in A. Set $J = \operatorname{Ann}_A(a-b)$, so that $r \in J$. Thus $J \not\subset \mathfrak{p}$ for every $\mathfrak{p}$. But then, in particular, J is not contained in any maximal ideal, hence $J = A$. Thus $a - b = 1 \cdot (a-b) = 0$, i.e. ψ is injective.



Exercise 2.1.6.1. Show that the association $a \mapsto s_a$ is a ring isomorphism. Consult [4, Prop. 2.2.(b)] if in need of a hint.

If one wants a field-valued notion of regular function, one can instead do the following, starting from the definition in Equation (2.1.10). Let $a \in A = \mathscr{O}_X(X)$. Composing s_a with the quotient maps

$$A_{\mathfrak{p}} \longrightarrow\!\!\!\!\!\rightarrow \kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$$

one obtains the map

$$\widetilde{s}_a : X \longrightarrow \coprod_{\mathfrak{p}\in X} \kappa(\mathfrak{p}),$$

where the field $\kappa(\mathfrak{p})$ may (and will) vary from point to point (cf. Example 2.2.3.11). This allows one to make sense of the question:

What does it mean that a function s vanishes at $\mathfrak{p} \in X$?

The answer is: $\widetilde{s}_a(\mathfrak{p}) = 0 \in \kappa(\mathfrak{p})$, or equivalently, $s_a(\mathfrak{p}) \in \mathfrak{p}A_{\mathfrak{p}}$.

Remark 2.1.6.2. The closed set $\mathrm{V}(I) = \{\, \mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \supset I \,\} \subset \operatorname{Spec} A$ can be reinterpreted as

$$\mathrm{V}(I) = \{\, \mathfrak{p} \in \operatorname{Spec} A \mid \widetilde{s}_a(\mathfrak{p}) = 0 \in \kappa(\mathfrak{p}) \text{ for all } a \in I \,\}. \tag{2.1.11}$$

This explains once more the letter 'V', standing for 'vanishing'.[5]

2.2 Schemes: definition and first properties

2.2.1 The category of schemes

We are ready for the definition of schemes.

Important Definition 2.2.1.1 (Scheme)**.** A *scheme* is a locally ringed space $(X, \mathscr{O}_X)$ such that every point $x \in X$ has an open neighbourhood $x \in U \subset X$ such that $(U, \mathscr{O}_X|_U)$ is an affine scheme.

Terminology 2.2.1.2. Let $(X, \mathscr{O}_X)$ be a scheme. The structure sheaf $\mathscr{O}_X$ is referred to as the *sheaf of regular functions* of the scheme. The ring $\mathscr{O}_X(X)$ is called the ring of *regular functions* on X. We keep the notation $(\mathscr{O}_{X,x}, \mathfrak{m}_x, \kappa(x))$ for the local ring at a point $x \in X$.

Definition 2.2.1.3 (Morphism of schemes)**.** A *morphism of schemes* is a morphism in the category of locally ringed spaces. In particular, an *isomorphism of schemes* $(X, \mathscr{O}_X) \xrightarrow{\sim} (Y, \mathscr{O}_Y)$ is a morphism $(f, f^{\#})$ such that $f : X \xrightarrow{\sim} Y$ is a homeomorphism and $f^{\#} : \mathscr{O}_Y \xrightarrow{\sim} f_*\mathscr{O}_X$ is an isomorphism of sheaves of rings over Y.

[5] But if you encounter the letter 'Z' in the literature, know it stands for 'zero locus'!

Notation 2.2.1.4. Affine schemes (resp. schemes) form a category, denoted Aff (resp. Sch), where morphisms are just the morphisms in the larger category of locally ringed spaces.

We have thus a chain of full inclusions of categories

$$\mathsf{Aff} \hookrightarrow \mathsf{Sch} \hookrightarrow \mathsf{LRS}.$$

Most of the time, when no confusion is likely to arise, we shall denote a scheme $(X, \mathscr{O}_X)$ by X, and a morphism of schemes $(f, f^\#)\colon (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ by $f\colon X \to Y$.

Definition 2.2.1.5 (Immersions of schemes)**.** An open (resp. closed) immersion of schemes is an open (resp. closed) immersion in the category of locally ringed spaces (cf. Definition 1.11.0.9).

Definition 2.2.1.6 (Open subscheme)**.** An *open subscheme* of a scheme $(X, \mathscr{O}_X)$ is a scheme of the form $(U, \mathscr{O}_X|_U)$, where U is an open subset of X. We will often just write $\mathscr{O}_U$ instead of $\mathscr{O}_X|_U$.

Note that an open subscheme comes with an open immersion $U \hookrightarrow X$. We will see in Remark 2.3.1.5 that an open subset $U \subset X$ of a scheme X is naturally a scheme.

The definition of closed subscheme is more subtle.

Definition 2.2.1.7 (Closed subscheme)**.** Let X be a scheme. A *closed subscheme* of X is an equivalence class of closed immersions with target X, where two closed immersions $\iota\colon Z \hookrightarrow X$ and $\iota'\colon Z' \hookrightarrow X$ are isomorphic if there exists an isomorphism $\alpha\colon Z \xrightarrow{\sim} Z'$ such that $\iota' \circ \alpha = \iota$.

Recall that the definition of a closed immersion requires $\mathscr{O}_X \to \iota_* \mathscr{O}_Z$ to be surjective; we will see that its kernel is forced to be locally generated by sections. We spell out this condition precisely, and the resulting description of closed subschemes via ideal sheaves, in Proposition 2.4.1.4.

2.2.2 First topological properties of schemes

Recall the following definition from topology.

Definition 2.2.2.1 (Irreducible topological space)**.** A topological space X is *irreducible* if it cannot be written as a union $Z_1 \cup Z_2$ of two proper closed subsets $Z_i \subsetneq X$.

Equivalently, X is irreducible if and only if any two nonempty open subsets of X intersect. Moreover, any nonempty open subset U of an irreducible topological space is irreducible and dense, i.e. $\overline{U} = X$.

Definition 2.2.2.2 (Quasicompact, irreducible, connected)**.** A scheme $(X, \mathscr{O}_X)$ is said to be *quasicompact* (resp. *irreducible*, resp. *connected*) if the underlying topological space

X is. A morphism of schemes $f: X \to Y$ is called *quasicompact* if the preimage of every affine open subset of Y is quasicompact.

Translation: a scheme is quasicompact when it admits a finite open cover by affine schemes. For morphisms, we have the following.



Exercise 2.2.2.3. Show that a morphism of schemes $f: X \to Y$ is quasicompact if and only if Y has an affine open cover $Y = \bigcup_{i\in I} Y_i$ such that $f^{-1}(Y_i)$ is quasicompact for all i.

We already saw in Lemma 2.1.1.10 that an affine scheme is quasicompact. An irreducible scheme is in particular connected. There are, however, connected schemes which are reducible (i.e. not irreducible), see Example 2.2.3.2.

PROPOSITION 2.2.2.4 ([9, Prop. 4.1]). *If X is a nonempty quasicompact scheme, then X has a closed point.*



Exercise 2.2.2.5. Prove that a scheme X is the spectrum of a local ring if and only if it is quasicompact and has a unique closed point (**Hint**: an affine neighbourhood of the unique closed point must be all of X).



Exercise 2.2.2.6 (Connectedness and idempotents). Prove that a scheme X is connected if and only if $\mathcal{O}_X(X)$ has only the trivial idempotents $0, 1$.

2.2.3 Generic points

We start here our discussion around generic points. We cover, for now, only the case of irreducible schemes. You will notice that most arguments are entirely of topological nature.

The next result tells us when an affine scheme, or a closed subset thereof, is irreducible.

PROPOSITION 2.2.3.1. *Let A be a ring, and set $X = \operatorname{Spec} A$. Then a closed subset $\mathrm{V}(I) \subset X$ is irreducible if and only if $\sqrt{I} \subset A$ is prime. In particular,*

(a) *X is irreducible if and only if the nilradical $\sqrt{(0)} \subset A$ is prime.*

(b) *If A is an integral domain, then $\operatorname{Spec} A$ is irreducible.*

Proof. Assume $\mathrm{V}(I) = \{\, \mathfrak{p} \in X \mid \mathfrak{p} \supset I \,\} \subset X$ irreducible. To say that $\sqrt{I}$ is prime means that if $ab \in \sqrt{I}$ then either $a \in \sqrt{I}$ or $b \in \sqrt{I}$. Let us assume, by contradiction, that there are $a, b \in A \setminus \sqrt{I}$ such that $ab \in \sqrt{I}$. Set $X_a = \mathrm{V}(I) \cap \mathrm{V}(a)$ and $X_b = \mathrm{V}(I) \cap \mathrm{V}(b)$. Then $X_a \cup X_b \subset \mathrm{V}(I)$, and if $\mathfrak{p} \in \mathrm{V}(I) = \mathrm{V}(\sqrt{I})$ then $\mathfrak{p} \supset \sqrt{I} \ni ab$, so that either $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$, which proves $X_a \cup X_b = \mathrm{V}(I)$. Since $X_a \neq \mathrm{V}(I) \neq X_b$, we contradict irreducibility of $\mathrm{V}(I)$.

Conversely, assume $\sqrt{I} \subset A$ is prime.

Recall that $V(I)$ carries the subspace topology induced from $\operatorname{Spec} A$. Hence, if $Z \subset V(I)$ is closed, there exists an ideal $K \subset A$ such that $Z = V(I) \cap V(K) = V(I+K)$. Setting $J = I + K$, we obtain $J \supset I$ and $Z = V(J)$. Conversely, every ideal $J \supset I$ defines a closed subset $V(J) \subset V(I)$. Therefore closed subsets of $V(I)$ can be represented as $V(J)$ for $J \supset I$.

Now, if $V(I) = V(J_1) \cup V(J_2) = V(J_1 \cap J_2) = V(J_1 J_2)$ then $\sqrt{I} = \sqrt{J_1 J_2} \supset J_1 J_2$, thus either $J_1 \subset \sqrt{I}$ or $J_2 \subset \sqrt{I}$. But if, say, $J_1 \subset \sqrt{I}$, it follows that $V(I) = V(\sqrt{I}) \subset V(J_1)$, which yields $V(J_1) = V(I)$. Thus $V(I)$ is irreducible. □

Example 2.2.3.2. The closed subset

$$V(y - x^2) \subset \mathbb{A}^2_{\mathbf{k}}$$

is irreducible. The closed subset

$$V(xy) \subset \mathbb{A}^2_{\mathbf{k}}$$

is reducible, being equal to $V(x) \cup V(y)$.



Caution 2.2.3.3 (Irreducibility depends on the base field)**.** Consider the polynomial $f = x^2 + 1 \in \mathbb{R}[x] \subset \mathbb{C}[x]$. Then $V(f) \subset \mathbb{A}^1_{\mathbb{R}}$ is irreducible (a point), but $V(f) \subset \mathbb{A}^1_{\mathbb{C}}$ is reducible, being equal to $V(x - i) \cup V(x + i)$. The characteristic of the base field also influences irreducibility: for instance, in $\mathbb{A}^2_{\mathbf{k}}$, we have

$$V(y^2 - x^2) = \begin{cases} V(x-y) \cup V(x+y) & \text{if } \operatorname{char} \mathbf{k} \neq 2, \\ V(x+y) & \text{if } \operatorname{char} \mathbf{k} = 2. \end{cases}$$

The first case gives a connected reducible scheme, the second gives an irreducible scheme.

LEMMA 2.2.3.4. *Let X be an irreducible scheme. Then, there exists a unique point $\xi \in X$ such that $X = \overline{\{\xi\}}$.*

Proof. Let us show uniqueness first. Let ξ_1, ξ_2 be two points such that $\overline{\{\xi_1\}} = X = \overline{\{\xi_2\}}$. Then any nonempty open $U \subset X$ contains both ξ_1 and ξ_2. Pick a nonempty affine open subset $U = \operatorname{Spec} A \subset X$. Let $\mathfrak{p}_i \subset A$ be the prime ideal corresponding to ξ_i. Since X is irreducible, U is irreducible and dense, hence

$$U = \overline{\{\xi_i\}} = V(\mathfrak{p}_i), \quad i = 1, 2,$$

where the closure is taken in U. So when we write $\xi_2 \in U = V(\mathfrak{p}_1)$ we obtain $\mathfrak{p}_2 \supset \mathfrak{p}_1$, and when we write $\xi_1 \in U = V(\mathfrak{p}_2)$ we obtain $\mathfrak{p}_1 \supset \mathfrak{p}_2$. Thus $\mathfrak{p}_1 = \mathfrak{p}_2$, i.e. $\xi_1 = \xi_2$.[6]

Now for existence. If $U = \operatorname{Spec} A \subset X$ is a nonempty open affine subset, then U is irreducible, i.e. $\mathfrak{p} = \sqrt{(0)} \subset A$ is prime by Proposition 2.2.3.1(a), and hence $U = V(\sqrt{(0)}) = V(\mathfrak{p}) = \overline{\{\mathfrak{p}\}}$. But since U is dense in X, the closure of $\mathfrak{p}$ in X is X itself. □

[6] Alternatively, just note that $\mathfrak{p}_1 = \sqrt{\mathfrak{p}_1} = \sqrt{\mathfrak{p}_2} = \mathfrak{p}_2$.

Definition 2.2.3.5 (Generic point)**.** Let X be an irreducible scheme. The point $\xi \in X$ of Lemma 2.2.3.4 is called the *generic point* of X.

Remark 2.2.3.6. The proof of Lemma 2.2.3.4 actually works for an arbitrary irreducible closed subset X of an arbitrary scheme Y.

Remark 2.2.3.7. If A is a domain, then $(0) \subset A$ is the unique minimal prime, thus $\operatorname{Spec} A$ is irreducible by Proposition 2.2.3.1. For instance, if $\mathbb{F}$ is a field, the affine space $\mathbb{A}^n_{\mathbb{F}}$ is irreducible. However, the converse is false: for instance, $\operatorname{Spec} \mathbb{F}[t]/t^n$ is irreducible for every $n \geqslant 1$, but $\mathbb{F}[t]/t^n$ is not a domain as soon as $n > 1$.



Exercise 2.2.3.8. Decide whether the following affine schemes are irreducible (resp. connected):

(i) $\operatorname{Spec} \mathbb{C}[x,y]/(y^2 - x^2(x+1))$,

(ii) $\operatorname{Spec} \mathbb{C}[x,y]/(y^2 - x^3)$

(iii) $\operatorname{Spec} \mathbb{Z}[x]/(2x)$,

(iv) $\operatorname{Spec} \mathbb{C}[x,y]/(xy, y^2)$,

(v) $\operatorname{Spec} \mathbb{F}[x,y]/(x^2, xy, y^3)$, for a field $\mathbb{F}$,

(vi) $\operatorname{Spec}(A \times A')$, for two rings A and A',

(vii) $\operatorname{Spec} \mathbb{C}[x,y,z]/(xy - z^2)$,

(viii) $\operatorname{Spec} \mathbb{C}[x,y]/(x^2 + y^2 - 1)$.

It is clear from the last paragraph of the proof of Lemma 2.2.3.4 that if A is a domain then the generic point $\xi \in \operatorname{Spec} A$ is the point corresponding to $(0) \subset A$, which is manifestly the unique minimal prime. The following lemma clarifies the basic properties of the generic point in this special case.

LEMMA 2.2.3.9. *Let A be an integral domain with fraction field K. Let $\xi \in X = \operatorname{Spec} A$ be the point corresponding to $(0) \subset A$. Then, the following statements hold true.*

(i) *We have $\mathcal{O}_{X,\xi} = K$.*

(ii) *ξ belongs to every nonempty open subset $U \subset X$, and $\mathcal{O}_X(U) \to \mathcal{O}_{X,\xi}$ is injective.*

(iii) *For every nonempty open subset $V \hookrightarrow U$, the map $\mathcal{O}_X(U) \to \mathcal{O}_X(V)$ is injective.*

Proof. We proceed step by step.

(i) This follows from Theorem 2.1.4.3(c) and the observation that the localisation of an integral domain at the prime ideal (0) is precisely the fraction field of the domain, cf. Example B.3.3.7.

(ii) To say $\xi \in \operatorname{Spec} A \setminus \mathrm{V}(I) = \{\mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \not\supset I\}$ for every $(0) \not\subset I \subset A$ means precisely that $(0) \not\supset I$ for all I, a tautology. Write $U = \bigcup_{i \in I} \mathrm{D}(f_i)$ as a union of nonempty principal open subsets, and assume $s \in \mathscr{O}_X(U)$ goes to 0 in $K = \mathscr{O}_{X,\xi}$. Well, s goes to $s|_{\mathrm{D}(f_i)} \in \mathscr{O}_X(\mathrm{D}(f_i)) = A_{f_i}$ first, and $A_{f_i} \hookrightarrow K$ is injective. Thus $s|_{\mathrm{D}(f_i)} = 0$ for every $i \in I$, so $s = 0$ by the separation axiom for the sheaf $\mathscr{O}_X$.

(iii) Follows immediately from (ii) and the factorisation $\mathscr{O}_X(U) \to \mathscr{O}_X(V) \to \mathscr{O}_{X,\xi}$. □

Example 2.2.3.10. Let $A = \mathbb{F}[x_1, \ldots, x_n]$, and consider the generic point $\xi \in \mathbb{A}^n_{\mathbb{F}}$, i.e. the point corresponding to the ideal $(0) \subset A$. Then

$$\kappa(\xi) = \mathbb{F}(x_1, \ldots, x_n).$$

If $f \in A \setminus 0$ is an irreducible polynomial, then the generic point $\xi_f \in \operatorname{Spec} A/(f)$ satisfies

$$\kappa(\xi_f) = \operatorname{Frac} A/(f).$$

Example 2.2.3.11. Let $A = \mathbb{Z}$. Every open subset $U \subset \operatorname{Spec} \mathbb{Z}$ is principal, i.e. of the form $U = \mathrm{D}(f)$ for some $f \in \mathbb{Z}$. We have $\mathscr{O}_{\operatorname{Spec}\mathbb{Z}}(\mathrm{D}(f)) = \mathbb{Z}_f \subset \operatorname{Frac}(\mathbb{Z}) = \mathbb{Q}$, and a rational number $a/b \in \mathbb{Q}$ (with a, b coprime) belongs to $\mathscr{O}_{\operatorname{Spec}\mathbb{Z}}(\mathrm{D}(f))$ if and only if every prime p dividing b also divides f. As for the generic point $\xi = (0)$, we have $\kappa(\xi) = \mathscr{O}_{\operatorname{Spec}\mathbb{Z},\xi} = \mathbb{Q}$. If $x \in \operatorname{Spec} \mathbb{Z}$ corresponds to the maximal ideal $(p) \subset \mathbb{Z}$, then

$$\kappa(x) = \mathbb{Z}_{(p)}/(p) \cdot \mathbb{Z}_{(p)} \cong \mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$$

by Exercise 2.1.5.5. Therefore the residue fields at different points of a scheme can have different characteristic!

2.3 Main theorem on affine schemes

Let $\phi : A \to B$ be a ring homomorphism. Then we have a set-theoretic map

$$\begin{array}{ccc} \operatorname{Spec} B & \xrightarrow{f_\phi} & \operatorname{Spec} A \\ \mathfrak{q} & \longmapsto & \phi^{-1}(\mathfrak{q}). \end{array}$$

We now analyse some crucial properties of this map: our goal is to promote it to a morphism of affine schemes (cf. Proposition 2.3.1.6). This will turn Spec into a functorial operation; the main theorem of this section is then that Spec defines, in fact, an equivalence of categories

$$\operatorname{Spec}\colon \mathsf{Rings}^{\mathrm{op}} \xrightarrow{\sim} \mathsf{Aff}.$$

2.3.1 Morphisms of affine schemes

LEMMA 2.3.1.1. *Let $\phi : A \to B$ be a ring homomorphism. Then, the following statements hold true.*

(a) *f_ϕ is continuous.*

(b) *If ϕ is surjective, then f_ϕ induces a homeomorphism from* $\operatorname{Spec} B$ *onto the closed subset* $\mathrm{V}(\ker\phi) \subset \operatorname{Spec} A$.

(c) *If ϕ is a localisation $A \to S^{-1}A$, then f_ϕ induces a homeomorphism from* $\operatorname{Spec} S^{-1}A$ *onto the subset $Y_S = \{\mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \cap S = \emptyset\} \subset \operatorname{Spec} A$, endowed with the subspace topology.*

Proof. We proceed step by step.

(a) We prove that the preimage of a closed subset $\mathrm{V}(I) \subset \operatorname{Spec} A$ is closed. We have

$$\begin{aligned} f_\phi^{-1}(\mathrm{V}(I)) &= \{\mathfrak{q} \in \operatorname{Spec} B \mid \phi^{-1}(\mathfrak{q}) \in \mathrm{V}(I)\} \\ &= \{\mathfrak{q} \in \operatorname{Spec} B \mid I \subset \phi^{-1}(\mathfrak{q})\} \\ &= \{\mathfrak{q} \in \operatorname{Spec} B \mid \phi(I) \subset \mathfrak{q}\} \\ &= \{\mathfrak{q} \in \operatorname{Spec} B \mid IB \subset \mathfrak{q}\} \\ &= \mathrm{V}(IB). \end{aligned}$$

(b) We have $B = A/\ker\phi$ by assumption, and we know that the prime ideals of B are in bijection with the prime ideals of A containing $\ker\phi$. By (a), and by definition of $\mathrm{V}(-)$, we then know that f_ϕ factors through a continuous bijection $\operatorname{Spec} B \to \mathrm{V}(\ker\phi)$, still denoted f_ϕ. To conclude it is a homeomorphism, it is enough to check the map is closed. Let then $J \subset B$ be an ideal. Then

$$\begin{aligned} f_\phi(\mathrm{V}(J)) &= \{\mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} = \phi^{-1}(\mathfrak{q}) \text{ for some } \mathfrak{q} \supset J\} \\ &= \{\mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \supset \phi^{-1}(J)\} \\ &= \mathrm{V}(\phi^{-1}(J)). \end{aligned}$$

(c) The existence of a continuous bijection $\operatorname{Spec} S^{-1}A \to Y_S \subset \operatorname{Spec} A$, sending a principal open $\mathrm{D}(a/1)$ to $Y_S \cap \mathrm{D}(a)$, is obtained combining (a) with Lemma B.3.3.1. As before, to see that the map is closed, fix an ideal $J \subset S^{-1}A$. Then

$$\begin{aligned} f_\phi(\mathrm{V}(J)) &= \{\mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \cap S = \emptyset \text{ and } \mathfrak{p} = \phi^{-1}(\mathfrak{q}) \text{ for some } \mathfrak{q} \supset J\} \\ &= Y_S \cap \mathrm{V}(\phi^{-1}(J)), \end{aligned}$$

which is closed in Y_S, as required. □



Exercise 2.3.1.2. Let $\mathbb{F}$ be a field. Let $\phi : A \to B$ be a homomorphism of finitely generated $\mathbb{F}$-algebras. Show that $f_\phi : \operatorname{Spec} B \to \operatorname{Spec} A$ maps closed points to closed points.

Remark 2.3.1.3. Note that if $S = \{ g^m \mid m \geqslant 0 \}$ for some $g \in A$, then

$$Y_S = \{ \mathfrak{p} \in \operatorname{Spec} A \mid g \notin \mathfrak{p} \} = \mathrm{D}(g) \subset \operatorname{Spec} A.$$

In particular, in this case the map $f_\ell \colon \operatorname{Spec} A_g \to \operatorname{Spec} A$ corresponding to the localisation $\ell \colon A \to A_g$ is a topological open embedding.

PROPOSITION 2.3.1.4. *Let $X = \operatorname{Spec} A$, and fix $g \in A$. The localisation $\ell \colon A \to A_g$ induces an isomorphism of locally ringed spaces*

$$(f_\ell, f_\ell^\#) \colon (\operatorname{Spec} A_g, \mathscr{O}_{\operatorname{Spec} A_g}) \xrightarrow{\sim} (\mathrm{D}(g), \mathscr{O}_X|_{\mathrm{D}(g)}).$$

In particular, $(\mathrm{D}(g), \mathscr{O}_X|_{\mathrm{D}(g)})$ is an affine scheme.

Proof. A topological open embedding $f_\ell \colon \operatorname{Spec} A_g \to \operatorname{Spec} A$ with image $\mathrm{D}(g)$ is provided by Lemma 2.3.1.1(c), applied to the localisation $\ell \colon A \to A_g$. Let us denote by f the homeomorphism $\operatorname{Spec} A_g \to \mathrm{D}(g)$. We need to extend it to a morphism of locally ringed spaces and show the resulting map is an isomorphism.

To define a morphism of sheaves of rings

$$f^\# \colon \mathscr{O}_X|_{\mathrm{D}(g)} \longrightarrow f_* \mathscr{O}_{\operatorname{Spec} A_g}$$

it is enough to define it on a base of open subsets by Theorem 1.8.0.8. Let $\mathrm{D}(h) \subset \mathrm{D}(g)$ be a principal open, for $h \in A$. Let $\overline{h} = h/1 \in A_g$ be the image of h in A_g. Then, canonically,

$$\begin{aligned} \mathscr{O}_X|_{\mathrm{D}(g)}(\mathrm{D}(h)) = \mathscr{O}_X(\mathrm{D}(h)) &= A_h \\ &\xrightarrow{\sim} (A_g)_{\overline{h}} \\ &= \mathscr{O}_{\operatorname{Spec} A_g}(\mathrm{D}(\overline{h})) \\ &= \mathscr{O}_{\operatorname{Spec} A_g}(f^{-1}\mathrm{D}(h)) \\ &= f_* \mathscr{O}_{\operatorname{Spec} A_g}(\mathrm{D}(h)). \end{aligned}$$

The isomorphism $A_h \xrightarrow{\sim} (A_g)_{\overline{h}}$ follows by directly checking that $(A_g)_{\overline{h}}$ satisfies the universal property of A_h, which in turn follows from g becoming invertible in A_h. The above isomorphisms $\mathscr{O}_X|_{\mathrm{D}(g)}(\mathrm{D}(h)) \xrightarrow{\sim} f_* \mathscr{O}_{\operatorname{Spec} A_g}(\mathrm{D}(h))$ are canonical, hence compatible with restrictions to smaller principal opens, thus they determine an isomorphism of $\mathcal{B}$-sheaves, which in turn uniquely determines an isomorphism of sheaves by Theorem 1.8.0.8. □

Remark 2.3.1.5. Let $(X, \mathscr{O}_X)$ be a scheme, $U \subset X$ an open subset. We know that $(U, \mathscr{O}_U)$ is a locally ringed space. (Recall that we set $\mathscr{O}_U = \mathscr{O}_X|_U$ in such a situation). We claim that it is in fact a scheme. That is, every open subset of a scheme has a natural scheme structure. To see this, cover X with open affine subsets $U_i = \operatorname{Spec} A_i \subset X$, so that $U = \bigcup_i U \cap U_i$. But $U \cap U_i \subset U_i$ is open in an affine scheme, therefore it is a union of principal open subsets $\mathrm{D}(f_{ij}) \subset U_i$, for $f_{ij} \in A_i$. Thus U admits a covering $U = \bigcup_{i,j} \mathrm{D}(f_{ij})$, and each $\mathrm{D}(f_{ij})$ is an affine scheme by Proposition 2.3.1.4.

PROPOSITION 2.3.1.6. *Let $\phi: A \to B$ be a ring homomorphism. Then the continuous map $f_\phi: \operatorname{Spec} B \to \operatorname{Spec} A$ of Lemma 2.3.1.1 extends to a morphism of affine schemes $(f_\phi, f_\phi^\#)$ such that $f_\phi^\#(\operatorname{Spec} A) = \phi$.*

Proof. Set $f = f_\phi$. First of all, we have to construct the sheaf homomorphism

$$\mathcal{O}_{\operatorname{Spec} A} \xrightarrow{f^\#} f_* \mathcal{O}_{\operatorname{Spec} B}.$$

For $g \in A$, the preimage of the principal open $\mathrm{D}(g) \subset \operatorname{Spec} A$ is

$$\begin{aligned} f^{-1}(\mathrm{D}(g)) &= \{\mathfrak{q} \in \operatorname{Spec} B \mid \phi^{-1}(\mathfrak{q}) \in \mathrm{D}(g)\} \\ &= \{\mathfrak{q} \in \operatorname{Spec} B \mid g \notin \phi^{-1}(\mathfrak{q})\} \\ &= \{\mathfrak{q} \in \operatorname{Spec} B \mid \phi(g) \notin \mathfrak{q}\} \\ &= \mathrm{D}(\phi(g)). \end{aligned}$$

There is an induced commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow & & \downarrow \\ A_g & \xrightarrow{\phi_g} & B_{\phi(g)} \\ \| & & \| \\ \mathcal{O}_{\operatorname{Spec} A}(\mathrm{D}(g)) & & \mathcal{O}_{\operatorname{Spec} B}(\mathrm{D}(\phi(g))) = f_* \mathcal{O}_{\operatorname{Spec} B}(\mathrm{D}(g)) \end{array} \qquad \frac{a}{g^r} \mapsto \frac{\phi(a)}{\phi(g)^r}$$

allowing us to set $f^\#(\mathrm{D}(g)) = \phi_g$. These morphisms are canonical, hence compatible with restrictions to smaller principal opens, thus they give rise to a morphism of $\mathcal{B}$-sheaves on $\operatorname{Spec} A$, which in turn uniquely determines a morphism of sheaves $f^\#$ by Theorem 1.8.0.8. If we take global sections of $f^\#$ (i.e. we evaluate it on $\mathrm{D}(1) = \operatorname{Spec} A$), we get back our original map ϕ, by construction.

We are left with checking that $(f, f^\#)$ induces local homomorphisms of local rings at the level of stalks. Assume $y = f(x)$, where $x \in \operatorname{Spec} B$ corresponds to a prime ideal $\mathfrak{q} \subset B$ and $y \in \operatorname{Spec} A$ corresponds to $\mathfrak{p} = \phi^{-1}(\mathfrak{q}) \subset A$. Then the canonical map

$$A_\mathfrak{p} \longrightarrow B_\mathfrak{q}, \quad \frac{a}{s} \mapsto \frac{\phi(a)}{\phi(s)}$$

is a local homomorphism of local rings, which coincides with

$$f_x^\#: \mathcal{O}_{\operatorname{Spec} A, y} \longrightarrow (f_* \mathcal{O}_{\operatorname{Spec} B})_y \longrightarrow \mathcal{O}_{\operatorname{Spec} B, x},$$

as required. □

It is very easy to check that sending $A \mapsto \operatorname{Spec} A$ is a contravariant *functor* (cf. Definition A.1.0.9) from rings to affine schemes. Proposition 2.3.1.6 says 'what to do' on morphisms. The axioms defining a functor (identity goes to identity, and compositions are preserved) are elementary, and therefore left to the reader.

2.3.2 Proof that Spec is an equivalence

We can finally prove the main result of this chapter.

THEOREM 2.3.2.1. *The functor* Spec*, from rings to affine schemes, induces an equivalence*

$$\mathrm{Spec}\colon \mathsf{Rings}^{\mathrm{op}} \xrightarrow{\sim} \mathsf{Aff}, \quad A \mapsto \mathrm{Spec}\, A,$$

with quasi-inverse functor given by $X \mapsto \mathscr{O}_X(X)$*. In particular,* $\mathrm{Spec}\,\mathbb{Z}$ *is a final object in* Aff*.*

Proof. The Spec functor is essentially surjective, by definition of affine scheme. We need to show it is fully faithful (cf. Exercise A.3.0.9). Set $X = \mathrm{Spec}\, B$ and $Y = \mathrm{Spec}\, A$. We claim that the inverse of the natural map

$$\mathrm{Hom}_{\mathsf{Rings}^{\mathrm{op}}}(B, A) \longrightarrow \mathrm{Hom}_{\mathsf{Aff}}(X, Y), \quad \phi \mapsto f_\phi.$$

is the map

$$\rho_{X,Y}\colon \mathrm{Hom}_{\mathsf{Aff}}(X, Y) \longrightarrow \mathrm{Hom}_{\mathsf{Rings}}(A, B) \tag{2.3.1}$$

sending $f\colon X \to Y$ to $f^\#(Y)\colon A = \mathscr{O}_Y(Y) \to f_*\mathscr{O}_X(Y) = \mathscr{O}_X(X) = B$. We must show that (2.3.1) is bijective for any pair of affine schemes $X = \mathrm{Spec}\, B$ and $Y = \mathrm{Spec}\, A$. Note that we already know that $\rho_{X,Y}$ is surjective thanks to Proposition 2.3.1.6.

Fix $f \in \mathrm{Hom}_{\mathsf{Aff}}(X, Y)$. Set $\phi = \rho_{X,Y}(f) = f^\#(Y)$. We know by Proposition 2.3.1.6 that ϕ gives rise to a morphism of affine schemes $f_\phi\colon X \to Y$ such that $\rho_{X,Y}(f_\phi) = \phi = \rho_{X,Y}(f)$. It is thus enough to show that $f = f_\phi$.

We need to show that f and f_ϕ are the same map set-theoretically, and, once we know this, that $f_x^\# = (f_\phi)_x^\#$ are the same as local homomorphisms of local rings, for every $x \in X$. This will imply that $f^\# = f_\phi^\#$ by Exercise 1.4.0.14. Let's start. Let $\mathfrak{q} \subset B$ be the prime ideal corresponding to $x \in X = \mathrm{Spec}\, B$, and let $\mathfrak{p} \subset A$ be the prime ideal corresponding to $f(x) \in Y = \mathrm{Spec}\, A$. We have two commutative diagrams

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ {\scriptstyle \mathrm{loc}_{\mathfrak{p}}}\downarrow & & \downarrow{\scriptstyle \mathrm{loc}_{\mathfrak{q}}} \\ A_{\mathfrak{p}} & \xrightarrow{f_x^\#} & B_{\mathfrak{q}} \end{array} \qquad \begin{array}{ccc} A & \xrightarrow{\phi} & B \\ {\scriptstyle \mathrm{loc}_{\phi^{-1}(\mathfrak{q})}}\downarrow & & \downarrow{\scriptstyle \mathrm{loc}_{\mathfrak{q}}} \\ A_{\phi^{-1}(\mathfrak{q})} & \xrightarrow{(f_\phi)_x^\#} & B_{\mathfrak{q}} \end{array}$$

that we need to show are the same. The very existence of $f_x^\#$ (fitting in the left diagram) implies that whenever $s \notin \mathfrak{p}$ one must have $\phi(s) \notin \mathfrak{q}$, i.e.

$$\phi(A \setminus \mathfrak{p}) \subset B \setminus \mathfrak{q},$$

which implies $\phi^{-1}(\mathfrak{q}) \subset \mathfrak{p}$. But the local condition $(f_x^\#)^{-1}(\mathfrak{q}B_{\mathfrak{q}}) = \mathfrak{p}A_{\mathfrak{p}}$ (combined with the classical correspondence between prime ideals in a ring and in a localisation of it, cf. Lemma B.3.3.1) implies $\phi^{-1}(\mathfrak{q}) = \phi^{-1}\mathrm{loc}_{\mathfrak{q}}^{-1}(\mathfrak{q}B_{\mathfrak{q}}) = \mathrm{loc}_{\mathfrak{p}}^{-1}(\mathfrak{p}A_{\mathfrak{p}}) = \mathfrak{p}$.

Thus $f = f_\phi$ set-theoretically. However, there is only one possible commutative diagram as above: the one where the bottom map sends $a/s \mapsto \phi(a)/\phi(s)$. Thus $f_x^\# = (f_\phi)_x^\#$ as wanted. This concludes the proof that $\rho_{X,Y}$ is bijective.

The final statement now follows, since a ring A is a $\mathbb{Z}$-algebra $\mathbb{Z} \to A$ in a unique way (or, equivalently, $\mathbb{Z}$ is an initial object in Rings). □

Remark 2.3.2.2. The map (2.3.1) is functorial in the following sense: for any morphism of affine schemes $g\colon Z = \operatorname{Spec} C \to \operatorname{Spec} B = X$ the diagram

$$\begin{array}{ccc} \operatorname{Hom}_{\mathsf{Aff}}(X,Y) & \xrightarrow{\rho_{X,Y}} & \operatorname{Hom}_{\mathsf{Rings}}(A,B) \\ \downarrow{\scriptstyle f\mapsto f\circ g} & & \downarrow{\scriptstyle \phi\mapsto g^\#(X)\circ\phi} \\ \operatorname{Hom}_{\mathsf{Aff}}(Z,Y) & \xrightarrow{\rho_{Z,Y}} & \operatorname{Hom}_{\mathsf{Rings}}(A,C) \end{array}$$

commutes. This is just a rephrasing of the fact that morphisms of locally ringed spaces can be composed! In a little more detail, fix $f \in \operatorname{Hom}_{\mathsf{Aff}}(X,Y)$. The upper journey takes f to the map $g^\#(X) \circ f^\#(Y) \in \operatorname{Hom}_{\mathsf{Rings}}(A,C)$, whereas the lower journey takes f to $(f\circ g)^\#(Y)$. These maps are clearly the same, since $(f\circ g)^\#\colon \mathscr{O}_Y \to f_*g_*\mathscr{O}_Z$ is nothing but the composition $f_*g^\# \circ f^\#\colon \mathscr{O}_Y \to f_*\mathscr{O}_X \to f_*g_*\mathscr{O}_Z$.

2.3.3 Examples of affine schemes and their morphisms

In this section we collect some examples of affine schemes (and morphisms between them), besides those already considered in Section 2.1.3 at a purely topological level.

Recall that open (resp. closed) immersion of schemes are just open (resp. closed) immersions in the category of locally ringed spaces (cf. Definition 1.11.0.9). The next two examples are very important.

Key Example 2.3.3.1 (Principal open immersions)**.** Let A be a ring, $f \in A$. It follows from Proposition 2.3.1.4 and Definition 1.11.0.9 that the canonical morphism

$$\operatorname{Spec} A_f \longrightarrow \operatorname{Spec} A$$

is an open immersion of affine schemes.

Key Example 2.3.3.2 (Closed immersions)**.** Let A be a ring, $I \subset A$ an ideal. Set $B = A/I$. The canonical surjection $\phi\colon A \twoheadrightarrow B$ determines, and is determined by, a morphism of affine schemes

$$i\colon \operatorname{Spec} B \longrightarrow \operatorname{Spec} A$$

This morphism is a *closed immersion* according to Definition 1.11.0.9. Indeed, it is a homeomorphism onto $\mathrm{V}(I) \subset \operatorname{Spec} A$, and induces a surjective map of sheaves, because $i^\#(\mathrm{D}(g))$ is surjective for every $g \in A$ (it agrees with the canonical surjection $A_g \twoheadrightarrow B_{\phi(g)}$), and by Theorem 1.8.0.8 it is enough to check surjectivity on a base of open sets.

In what follows, we grant the following proposition, saying that *all* closed immersions into an affine scheme are as in Key Example 2.3.3.2.

PROPOSITION 2.3.3.3 ([5, Ch. 2, Prop. 3.20]). *Let $Y = \operatorname{Spec} A$ be an affine scheme, and let $\iota\colon Z \hookrightarrow Y$ be a closed immersion. Then Z is affine, and there is a unique ideal $I \subset A$ such that ι induces an isomorphism of schemes $Z \xrightarrow{\sim} \operatorname{Spec} A/I$.*

We use this proposition without proof; its ideal-sheaf interpretation is given in Proposition 2.4.1.4.

Example 2.3.3.4 (Many maps to the point!). One is used to think that there is only one map $\bullet \to \bullet$ from a one-point space to itself. However, this is not necessarily true in algebraic geometry: think of the identity $\mathbb{C} \to \mathbb{C}$, which is different from complex conjugation $\mathbb{C} \to \mathbb{C}$. By Theorem 2.3.2.1, they give rise to different maps $\operatorname{Spec}\mathbb{C} \to \operatorname{Spec}\mathbb{C}$. In fact, there are *infinitely many* morphisms $\operatorname{Spec}\mathbb{C} \to \operatorname{Spec}\mathbb{C}$. Indeed, the set $\operatorname{Hom}_{\mathsf{Rings}}(\mathbb{C},\mathbb{C})$ contains the Galois group of $\mathbb{C} \supset \mathbb{Q}$![7] Another example, in characteristic $p > 0$, is the *Frobenius morphism*, namely the map $\Phi_{\mathbb{F}}\colon \operatorname{Spec}\mathbb{F} \to \operatorname{Spec}\mathbb{F}$ induced by the field homomorphism $\phi_{\mathbb{F}}\colon \mathbb{F} \to \mathbb{F}$ sending $x \mapsto x^p$. A field $\mathbb{F}$ is *perfect* if and only if either $\mathbb{F}$ has characteristic 0 or $\phi_{\mathbb{F}}$ is surjective. In particular, the Frobenius morphism $\Phi_{\mathbb{F}}$ is an isomorphism if and only if $\mathbb{F}$ is perfect. For instance, all finite fields are perfect, but $\mathbb{F}_p(t)$ is not perfect, since t is (for degree reasons) not of the form $f(t)^p/g(t)^p$ for any two polynomials $f(t)$ and $g(t)$. For $\mathbb{F} = \mathbb{F}_p$ the Frobenius is the identity; for $\mathbb{F} = \mathbb{F}_{p^r}$ with $r > 1$ it is a nontrivial automorphism of order r. For a general field of characteristic $p > 0$ it need not be surjective.

Remark 2.3.3.5. Note that there is, on the other hand, *only one* morphism of schemes $\operatorname{Spec}\mathbb{R} \to \operatorname{Spec}\mathbb{R}$. This is because there is only one ring endomorphism $\mathbb{R} \to \mathbb{R}$, namely the identity. Can you prove it? (**Hint**: start by showing that a ring endomorphism $\mathbb{R} \to \mathbb{R}$ is order-preserving). If $\mathbb{F}$ has no nontrivial field endomorphisms (for example $\mathbb{Q}$ or $\mathbb{F}_p$), then $\operatorname{Hom}_{\mathsf{Sch}}(\operatorname{Spec}\mathbb{F}, \operatorname{Spec}\mathbb{F})$ is reduced to the identity. Also, note that there is *no morphism* $\operatorname{Spec}\mathbb{R} \to \operatorname{Spec}\mathbb{C}$, since there is no ring homomorphism $\mathbb{C} \to \mathbb{R}$.

Example 2.3.3.6 (Dual numbers). Let $A \neq 0$ be a reduced ring. Thanks to Theorem 2.3.2.1, we finally have made rigorous our claim (formulated in Example 2.1.3.9 with $A = \mathbb{F}$) that

$$\operatorname{Spec} A \ncong \operatorname{Spec} A[t]/t^2.$$

See Example 2.3.3.7 and Example 2.3.3.8 for generalisations.

Example 2.3.3.7 (Curvilinear schemes). Let $n > 0$ be an integer, and set $A_n = \mathbf{k}[t]/t^n$.

[7] The cardinality of the automorphism group of $\mathbb{C}$ is 2^c, where $c = 2^{\aleph_0}$. More generally, the cardinality of $\operatorname{Aut}(\mathbf{k})$ for $\mathbf{k}$ an *algebraically closed* field, is $2^{\operatorname{card}\mathbf{k}}$, see [1].

There is a closed immersion

$$\operatorname{Spec} A_n \hookrightarrow \mathbb{A}^1_{\mathbf{k}}.$$

Note that $\dim_{\mathbf{k}} A_n = n$. The affine scheme $\operatorname{Spec} A_n$ is called a *curvilinear scheme of length* n. All these schemes admit a (bijective) closed immersion $\operatorname{Spec}\mathbf{k} \hookrightarrow \operatorname{Spec} A_n$, which is never an isomorphism (unless $n = 1$). Thus $\operatorname{Spec} A_n$ can be seen as a 'thickening' of the origin in $\mathbb{A}^1_{\mathbf{k}}$.

Example 2.3.3.8 (Fat points). Let $\mathbf{k}$ be an algebraically closed field, $(A, \mathfrak{m})$ a local artinian $\mathbf{k}$-algebra with residue field $A/\mathfrak{m} = \mathbf{k}$. Then $\operatorname{Spec} A$ is topologically just one (closed) point, corresponding to the maximal ideal $\mathfrak{m} \subset A$. These affine schemes are called *fat points*. For instance, consider $A = \mathbf{k}[x, y]/(x^2, xy, y^2)$. Each fat point has a *length*, namely the number $\dim_{\mathbf{k}} A = \dim_{\mathbf{k}} \mathcal{O}_X(X)$. For instance, $\operatorname{Spec}\mathbf{k}[x, y]/(x^2, xy, y^2)$ has length 3, and is *not curvilinear*. The closed immersion $\operatorname{Spec} A/\mathfrak{m} \hookrightarrow \operatorname{Spec} A$ is a bijection (both schemes have precisely one point), but $\operatorname{Spec} A/\mathfrak{m} = \operatorname{Spec}\mathbf{k} \ncong \operatorname{Spec} A$ as schemes. Fat points encode nontrivial information in their structure sheaves!

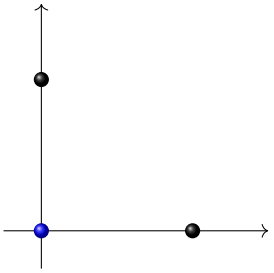


Figure 2.7: The length 3 fat point $\operatorname{Spec}\mathbf{k}[x, y]/(x^2, xy, y^2)$ arises as the 'collision' of two points (black bullets) running towards the origin. This has to be interpreted as a mere coefficientwise limit: it can be seen as a degeneration of the product ideal $(x - a, y) \cdot (x, y - b)$ for $a, b \to 0$. See also **??**.



Exercise 2.3.3.9. Show that the only fat point of length 3 which is not curvilinear is, up to isomorphism, precisely $\operatorname{Spec}\mathbf{k}[x, y]/(x^2, xy, y^2)$.

Example 2.3.3.10 (A non-affine scheme, take I). Let $X = \operatorname{Spec}\mathbb{Z}[x]$, and $z \in X$ the closed point corresponding to (p, x), where $p \in \mathbb{Z}$ is a prime number. Then $U = X \setminus \{z\} = \mathrm{D}(p) \cup \mathrm{D}(x)$ is not affine. Indeed, by Lemma 2.2.3.9(iii), we have $\mathcal{O}_X(U) \subset \mathcal{O}_X(\mathrm{D}(p)) \cap \mathcal{O}_X(\mathrm{D}(x)) = \mathbb{Z}[x, 1/p] \cap \mathbb{Z}[x, 1/x] = \mathbb{Z}[x] = \mathcal{O}_X(X)$, which readily implies $\mathcal{O}_X(U) = \mathcal{O}_X(X)$. If U were affine, the open immersion $U \hookrightarrow X$ would correspond to the identity on A, contradicting $U \neq X$. Note that this example also shows that the union of two affine schemes need not be affine.

Example 2.3.3.11 (A non-affine scheme, take II). Let $n > 1$ be an integer, $\mathbf{k}$ a field, $A = \mathbf{k}[x_1, \ldots, x_n]$. Consider the origin of $X = \mathbb{A}^n_{\mathbf{k}} = \operatorname{Spec} A$, namely the point $0 \in X$ corresponding to the maximal ideal $(x_1, \ldots, x_n) \subset A$. Form the open complement $U = X \setminus \{0\} \hookrightarrow X$. We now prove that the restriction map $\mathbf{k}[x_1, \ldots, x_n] = \mathscr{O}_X(X)\mathscr{O}_X(U)$ is the identity and then conclude as in Example 2.3.3.10 using that U is not isomorphic to $\mathbb{A}^n_{\mathbf{k}}$. We have $U = \bigcup_{1 \leqslant i \leqslant n} \mathrm{D}(x_i)$, so by Lemma 2.2.3.9(iii), we have

$$\mathscr{O}_X(U) \subset \bigcap_{1 \leqslant i \leqslant n} \mathscr{O}_X(\mathrm{D}(x_i)) = A_{x_1} \cap \cdots \cap A_{x_n}.$$

This can be proven directly to be equal to A. However, it also follows from the algebraic version of Hartog's Lemma below, combined with the fact that height 1 primes (see **??** for the definition of height of an ideal) in the (normal) domain A correspond to irreducible polynomials. The $n = 2$ case of this example can be seen as the geometric analogue of Example 2.3.3.10.

LEMMA 2.3.3.12 ([5, Ch. 4, Lemma 1.13]). *Let A be a normal noetherian domain of dimension at least 1. Then*

$$A = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec} A \\ \operatorname{ht}(\mathfrak{p}) = 1}} A_{\mathfrak{p}},$$

the intersection being taken inside $\operatorname{Frac} A$.

Example 2.3.3.13 (Affine line minus one point). If we take $n = 1$ in Example 2.3.3.11, we do in fact get an affine scheme $U = \mathbb{A}^1_{\mathbf{k}} \setminus \{0\} \hookrightarrow \mathbb{A}^1_{\mathbf{k}}$. Indeed, $U = \mathrm{D}(x) = \operatorname{Spec} \mathbf{k}[x]_x$. In fact, the ring $\mathbf{k}[x]_x = \{f(x)/x^r \mid r \geqslant 0\}$ is isomorphic to the $\mathbf{k}$-algebra $\mathbf{k}[x, x^{-1}] = \mathbf{k}[x, y]/(xy - 1)$, which yields a closed immersion $U \hookrightarrow \mathbb{A}^2_{\mathbf{k}}$.

Example 2.3.3.14 (Affine hypersurfaces). Let $\mathbb{F}$ be a field, $f \in \mathbb{F}[x_1, \ldots, x_n]$ a nonzero polynomial. Then

$$Y_f = \operatorname{Spec} \mathbb{F}[x_1, \ldots, x_n]/(f)$$

is called an *affine hypersurface* in $\mathbb{A}^n_{\mathbb{F}}$. The surjection $\mathbb{F}[x_1, \ldots, x_n] \twoheadrightarrow \mathbb{F}[x_1, \ldots, x_n]/(f)$ canonically determines a closed immersion

$$Y_f \hookrightarrow \mathbb{A}^n_{\mathbf{k}} = \operatorname{Spec} \mathbb{F}[x_1, \ldots, x_n].$$

Suppose (f) is a prime ideal in $\mathbb{F}[x_1, \ldots, x_n]$, so that $\mathbb{F}[x_1, \ldots, x_n]/(f)$ is an integral domain and Y_f is irreducible. Then (f) corresponds to the trivial (prime) ideal $(0) \subset \mathbb{F}[x_1, \ldots, x_n]/(f)$. This is the generic point of Y_f.

Example 2.3.3.15 (Affine quadric cone). As a special case of Example 2.3.3.14, consider $f = xy - z^2 \in \mathbb{C}[x, y, z]$. Its vanishing scheme

$$Y_f = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy - z^2) \hookrightarrow \mathbb{A}^3_{\mathbb{C}}$$

is called the *affine quadric cone*.

Example 2.3.3.16. Let R be a DVR with fraction field K, and set $X = \operatorname{Spec} R = \{x_0, \xi\}$ where x_0 is the closed point. Then $K = \mathscr{O}_{X,\xi}$. The open immersion $\{\xi\} = X \setminus \{x_0\} \hookrightarrow X$ corresponds to the canonical inclusion $R \hookrightarrow K$.

Example 2.3.3.17. Let A be a ring, and set $\mu_n = \operatorname{Spec} A[x]/(x^n - 1)$ for some $n > 1$. This is the scheme-theoretic version of the group of n-th roots of unity. One can prove that it is a group object in the category of A-schemes. Such objects are called *group schemes* (over $\operatorname{Spec} A$). As for μ_n, it comes with a natural closed immersion inside the affine line $\mathbb{A}^1_A = \operatorname{Spec} A[x]$.

Spoiler 2.3.3.18 (An n-fold ramified cover)**.** Consider the morphism $f : \mathbb{A}^1_{\mathbf{k}} \to \mathbb{A}^1_{\mathbf{k}}$ defined by the ring homomorphism $\mathbf{k}[t] \to \mathbf{k}[t]$ sending $t \mapsto t^n$, where $n \in \mathbb{Z}_{>1}$ is not a multiple of the characteristic of $\mathbf{k}$. This is the typical example of what we will call a *ramified* morphism. The intuition is the following: every closed point $x \in \mathbb{A}^1_{\mathbf{k}} \setminus 0$ is the image of precisely n closed points in the source (because $\mathbf{k}$ is algebraically closed), but there is only one preimage over the origin $0 \in \mathbb{A}^1_{\mathbf{k}}$. Over this point, the morphism is 'fully ramified'. We shall see that the *scheme-theoretic fibre* of f over 0 is the curvilinear scheme $\operatorname{Spec} \mathbf{k}[t]/t^n$. If we restrict f to $\mathbb{A}^1_{\mathbf{k}} \setminus 0 \to \mathbb{A}^1_{\mathbf{k}} \setminus 0$, it becomes *unramified*, and in fact *étale*. These notions are extremely important and will be treated in later chapters.

Example 2.3.3.19. The inclusion $\mathbb{R}[x] \hookrightarrow \mathbb{C}[x]$ induces a morphism of affine schemes

$$\mathbb{A}^1_{\mathbb{C}} \longrightarrow \mathbb{A}^1_{\mathbb{R}},$$

sending the generic point $(0) \subset \mathbb{C}[x]$ to the generic point $(0) \subset \mathbb{R}[x]$. For any $c \in \mathbb{R} \subset \mathbb{C}$, the maximal ideal $(x - c) \subset \mathbb{R}[x]$ is the preimage of the maximal ideal $(x - c) \subset \mathbb{C}[x]$, so $\mathbb{A}^1_{\mathbb{C}} \to \mathbb{A}^1_{\mathbb{R}}$ sends the closed point $(x - c) \in \mathbb{A}^1_{\mathbb{C}}$ to the closed point $(x - c) \in \mathbb{A}^1_{\mathbb{R}}$. On the other hand, if $c = \alpha + i\beta \in \mathbb{C} \setminus \mathbb{R}$, then both ideals

$$\mathfrak{p}_1 = (x - c),\ \mathfrak{p}_2 = (x - \overline{c}) \subset \mathbb{C}[x]$$

viewed as closed points of $\mathbb{A}^1_{\mathbb{C}}$, map to the closed point

$$\mathfrak{q} = (f) \in \mathbb{A}^1_{\mathbb{R}}, \quad f = (x - c)(x - \overline{c}).$$

However, note that this closed point has 'degree 2', for

$$\kappa(\mathfrak{q}) = \frac{\mathbb{R}[x]_{(f)}}{(f)\mathbb{R}[x]_{(f)}} \cong \frac{\mathbb{R}[x]}{(f)} \cong \mathbb{C},$$

since $\deg f = 2$. This does not happen for the other points $(x - c) \in \mathbb{A}^1_{\mathbb{R}}$, in the sense that

$$\kappa(x - c) = \frac{\mathbb{R}[x]_{(x-c)}}{(x - c)\mathbb{R}[x]_{(x-c)}} \cong \frac{\mathbb{R}[x]}{(x - c)} \cong \mathbb{R}.$$

As for $\xi = (0) \in \operatorname{Spec} \mathbb{R}[x]$, we have

$$\kappa(\xi) = \frac{\mathbb{R}[x]_{(0)}}{(0)} = \mathbb{R}[x]_{(0)} = \operatorname{Frac} \mathbb{R}[x] = \mathbb{R}(x).$$

We have used Exercise 2.1.5.5 in the last three displayed equations. The elements of $\kappa(\xi)$ are 'rational functions' g/h, not defined everywhere but *almost everywhere*, away from the (finitely many) zeros of $h \in \mathbb{R}[x]$.

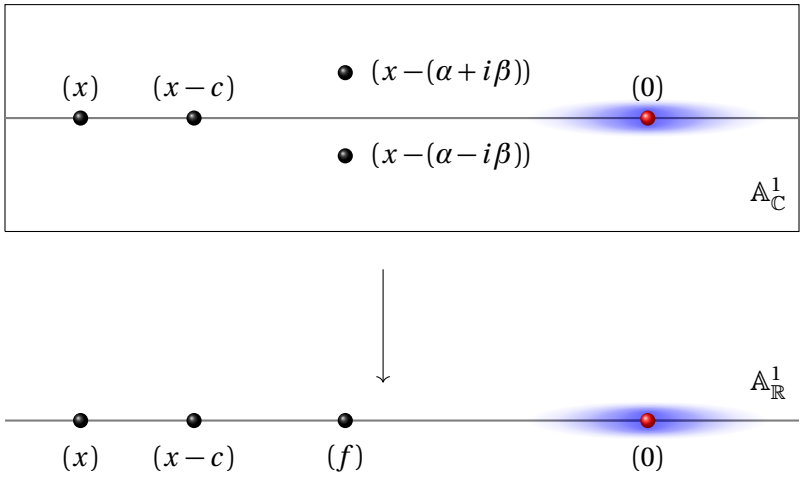


Figure 2.8: The morphism $\mathbb{A}^1_{\mathbb{C}} \to \mathbb{A}^1_{\mathbb{R}}$ induced by $\mathbb{R}[x] \hookrightarrow \mathbb{C}[x]$.

Example 2.3.3.20. This example is the arithmetic analogue of Example 2.3.3.19. Consider the inclusion of rings

$$\phi : \mathbb{Z} \lhook\joinrel\longrightarrow \mathbb{Z}[i] = \mathbb{Z}[x]/(x^2+1), \quad i^2 = -1.$$

Here $\mathbb{Z}[i]$ is the ring of *Gaussian integers*, which is an euclidean domain, in particular a principal ideal domain. We will recall some basic number theory in this example, in order to study the induced morphism

$$f : \operatorname{Spec}\mathbb{Z}[i] \longrightarrow \operatorname{Spec}\mathbb{Z}.$$

The algebraic question is: what happens to a prime number $p \in \mathbb{Z}$ when one adds in the imaginary unit? More precisely, is the principal ideal

$$p\mathbb{Z}[i] \subset \mathbb{Z}[i]$$

still a prime ideal? If this happens we say that p is *inert*, otherwise there are two possibilities: either p *splits* (i.e. $p\mathbb{Z}[i]$ is a product of two distinct prime ideals) or it *ramifies* (i.e. $p\mathbb{Z}[i]$ is the square of a prime ideal). By Fermat's theorem on sums of two squares, one has that $p > 2$ is a sum of squares if and only if $p \equiv 1 \bmod 4$. In this case, one can write $p = a^2 + b^2 = (a+ib)(a-ib)$ for some integers $a, b \in \mathbb{Z}$. Such primes are then those that split. On the other hand, if $p \equiv 3 \bmod 4$, then $p\mathbb{Z}[i]$ is inert, i.e. p 'stays prime' in $\mathbb{Z}[i]$. For $p = 2$, one has that $2 = (1+i)(1-i)$, but $(1+i) = (1-i)$ as ideals in $\mathbb{Z}[i]$, since $i(1-i) = i+1$, thus $p = 2$ *ramifies*, in that $2\mathbb{Z}[i] = (1+i)^2$, and no other prime ramifies. The first prime that splits is $5 = (2+i)(2-i)$, followed by $13 = (3+2i)(3-2i)$ (since 3, 7

and 11 are inert). Primes (larger than 2) that split correspond, geometrically, to those points $(p) \in \operatorname{Spec}\mathbb{Z}$ having more than one preimage along f. The prime $p=2$ should be thought of as having one, but slightly thick, preimage.

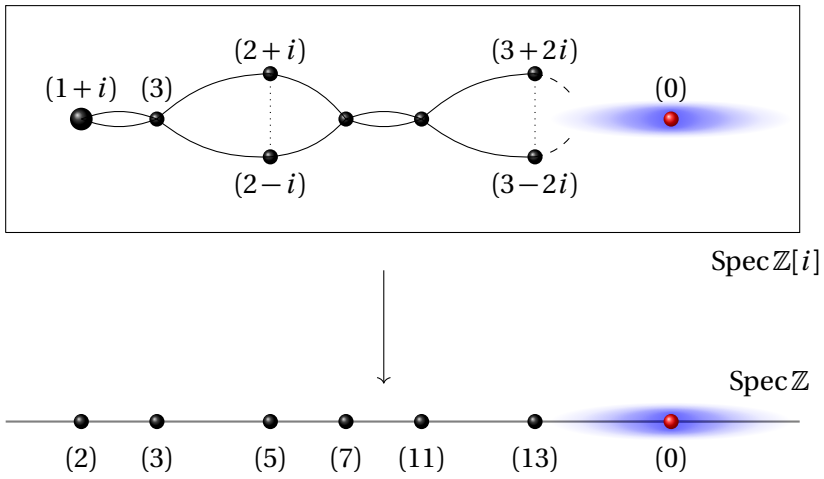


Figure 2.9: The morphism $f\colon \operatorname{Spec}\mathbb{Z}[i] \to \operatorname{Spec}\mathbb{Z}$ induced by the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]$. The point corresponding to $(1+i)$ is drawn bigger since $p=2$ ramifies. The points aligned horizontally correspond to the inert primes downstairs, the others, aligned vertically, correspond to split primes.

Example 2.3.3.21 (The 'arithmetic surface'). Here is another arithmetic example. Consider the inclusion of rings $\mathbb{Z} \hookrightarrow \mathbb{Z}[x]$. We want to study the induced morphism

$$\operatorname{Spec}\mathbb{Z}[x] \longrightarrow \operatorname{Spec}\mathbb{Z}.$$

A pictorial description of $\operatorname{Spec}\mathbb{Z}[x]$ was given by Mumford [7], see Figure 2.10.

First let us list all prime ideals in $\mathbb{Z}[x]$.

- $(0) \subset \mathbb{Z}[x]$ is a prime ideal, since $\mathbb{Z}[x]$ is an integral domain. It corresponds to the generic point of $\operatorname{Spec}\mathbb{Z}[x]$. The residue field is $\kappa(0) = \operatorname{Frac}\mathbb{Z}[x] = \mathbb{Q}(x)$.
- $(p) \subset \mathbb{Z}[x]$ is a prime ideal, for any prime number $p \in \mathbb{Z}$, since the quotient

  $$\mathbb{Z}[x]/(p) \cong \mathbb{F}_p[x]$$

  is an integral domain. These points are *not closed*. Note that each point (p) is precisely the generic point of the affine line

  $$\mathbb{A}^1_{\mathbb{F}_p} = \operatorname{Spec}\mathbb{F}_p[x] = \operatorname{Spec}\mathbb{Z}[x]/(p) \hookrightarrow \operatorname{Spec}\mathbb{Z}[x].$$

  These lines are drawn as vertical lines in Figure 2.10. The residue field of $\operatorname{Spec}\mathbb{Z}[x]$ at these points is

  $$\kappa(p) = \mathbb{Z}[x]_{(p)}/(p)\mathbb{Z}[x]_{(p)} = \operatorname{Frac}\mathbb{Z}[x]/(p) = \operatorname{Frac}\mathbb{F}_p[x] = \mathbb{F}_p(x).$$

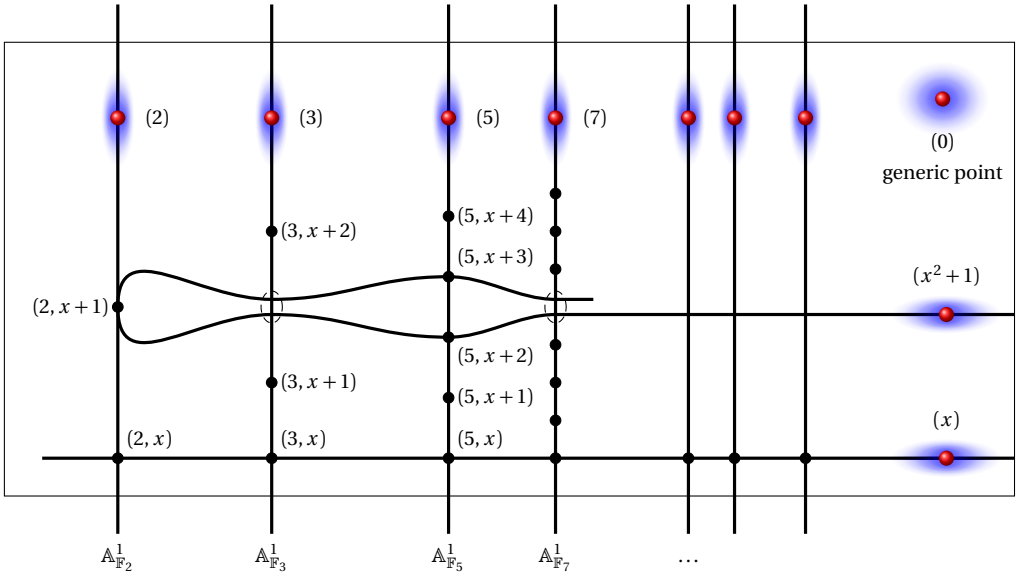


Figure 2.10: The code for this figure is adapted from code on Pieter Belmans's website. This picture was originally drawn by David Mumford in [7], where he called $\operatorname{Spec}\mathbb{Z}[x]$ an *arithmetic surface*.

- $(f)\subset\mathbb{Z}[x]$, where $f\in\mathbb{Z}[x]$ is a nonconstant irreducible polynomial.[8] Each such polynomial draws an "arithmetic curve"

$$\operatorname{Spec}\mathbb{Z}[x]/(f)\hookrightarrow\operatorname{Spec}\mathbb{Z}[x],$$

  depicted as a horizontal curve for $f=x$ and $f=x^2+1$ in Figure 2.10. Because f is irreducible, the point (f) is exactly the generic point of such arithmetic curve.

- $(p,f)\subset\mathbb{Z}[x]$, where p is a prime number and the image $\overline{f}$ of $f\in\mathbb{Z}[x]$ in $\mathbb{F}_p[x]$ is an irreducible polynomial. These are all the closed points of $\operatorname{Spec}\mathbb{Z}[x]$. The residue fields of these points are finite extensions of $\mathbb{F}_p$.

Explicitly, one has set-theoretic identities

$$\left|\mathbb{A}^1_{\mathbb{F}_p}\right|=\left\{(p,g)\,\middle|\,\overline{g}\in\mathbb{F}_p[x]\text{ is monic and irreducible}\right\}\cup\left\{(p)\right\},$$
$$\left|\operatorname{Spec}\mathbb{Z}[x]/(f)\right|=\left\{(p,g)\,\middle|\,\overline{g}\in\mathbb{F}_p[x]\text{ is monic and irreducible, }\overline{g}\mid\overline{f}\right\}\cup\left\{(f)\right\},$$

and the schematic intersection between a horizontal curve and a vertical line is

$$\operatorname{Spec}\mathbb{F}_p[x]/(\overline{f}),\qquad\overline{f}=\text{ image of }f\text{ in }\mathbb{F}_p[x]. \tag{2.3.2}$$

Its underlying set of points is

$$\left|\operatorname{Spec}\mathbb{F}_p[x]/(\overline{f})\right|=\left\{(p,g)\,\middle|\,\overline{g}\text{ is a monic irreducible divisor of }\overline{f}\right\}.$$

[8] By Gauss' lemma, a nonconstant polynomial $f\in\mathbb{Z}[x]$ is irreducible if and only if it is primitive (the gcd of its coefficients is 1) and irreducible over $\mathbb{Q}[x]$.

Repeated factors of $\overline{f}$ do not create additional points: their multiplicities are retained by the affine scheme (2.3.2). Let us see this via some examples.

One such (horizontal) arithmetic curve is for instance the one "cut out by $x = 0$", consisting of a copy of $\operatorname{Spec}\mathbb{Z}$ itself, for

$$\operatorname{Spec}\mathbb{Z}[x]/(x) = \operatorname{Spec}\mathbb{Z} \hookrightarrow \operatorname{Spec}\mathbb{Z}[x].$$

Another curve is

$$\operatorname{Spec}\mathbb{Z}[x]/(x^2+1) = \operatorname{Spec}\mathbb{Z}[i] \hookrightarrow \operatorname{Spec}\mathbb{Z}[x].$$

Let us now analyse Figure 2.10 carefully.

- $\mathbb{A}^1_{\mathbb{F}_2}$. Two "classical points" of this vertical line are the closed points $(2, x)$ and $(2, x+1)$, corresponding to the points with coordinates 0 and 1, respectively, in the affine line $\mathbb{A}^1_{\mathbb{F}_2} \subset \operatorname{Spec}\mathbb{Z}[x]$. These two points are drawn as black bullets.

  But this affine line also intersects the arithmetic curve $\operatorname{Spec}\mathbb{Z}[x]/(x^2+1)$, since

  $$\mathrm{V}(x^2+1) \cap \mathrm{V}(2) = \{(2, x+1)\}, \quad \text{as } x^2+1 = (x+1)^2.$$

  Scheme-theoretically, the intersection within $\operatorname{Spec}\mathbb{Z}[x]$ is

  $$\operatorname{Spec}\mathbb{F}_2[x]/(x^2+1) = \operatorname{Spec}\mathbb{F}_2[x]/(x+1)^2 \cong \operatorname{Spec}\mathbb{F}_2[t]/(t^2),$$

  which is a double point: the geometric incarnation of the algebraic splitting $x^2+1 = (x+1)^2$. Pictorially, this explains why the curve $\operatorname{Spec}\mathbb{Z}[x]/(x^2+1)$ is depicted *tangent* to the affine line $\mathbb{A}^1_{\mathbb{F}_2}$.

  Of course there are many other curves $\operatorname{Spec}\mathbb{Z}[x]/(f)$ meeting $\mathbb{A}^1_{\mathbb{F}_2}$. In other words, $\mathbb{A}^1_{\mathbb{F}_2}$ has many other points of the form $(2, g)$, with $\overline{g}$ dividing $\overline{f}$ in $\mathbb{F}_2[x]$. The closed points of $\mathbb{A}^1_{\mathbb{F}_2}$ are indexed by monic irreducible polynomials over $\mathbb{F}_2$. For instance,

  $$f = x^2 + x + 1$$

  has no roots over $\mathbb{F}_2$, and if we denote by α a root of f we have a splitting

  $$x^2 + x + 1 = (x+\alpha)(x+\alpha+1)$$

  over the larger field $\mathbb{F}_2[\alpha] = \{0, 1, \alpha, \alpha+1\} \supset \mathbb{F}_2$. We thus have two *different* residue fields

  $$\begin{aligned} \kappa(2, x+1) &= \mathbb{F}_2 \\ \kappa(2, x^2+x+1) &= \mathbb{F}_2[x]/(x^2+x+1) = \mathbb{F}_2[\alpha] \end{aligned}$$

  for these two different types of points of $\mathbb{A}^1_{\mathbb{F}_2} \subset \operatorname{Spec}\mathbb{Z}[x]$.

- $\mathbb{A}^1_{\mathbb{F}_3}$. The polynomial x^2+1 is irreducible over $\mathbb{F}_3$ (having no roots), so the point $(3, x^2+1)$ is not a 'classical' point of $\mathbb{A}^1_{\mathbb{F}_3}$. Let α be a root of x^2+1. Then

$$x^2+1=(x-\alpha)(x-2\alpha)$$

over $\mathbb{F}_3[\alpha]\supset\mathbb{F}_3$. In this larger field, the two points $(3, x-\alpha)$ and $(3, x-2\alpha)$ would be 'separated' and would be depicted as two classical points.

The point $(3, x^2+1)$ representing the physical intersection $V(3)\cap V(x^2+1)$ is depicted as a small dotted circle. The curve $V(x^2+1)$ passes through this circle, but in the picture the two branches of the curve remain separated: this reflects the fact that the 'separation' of the roots happens over the larger field $\mathbb{F}_3[\alpha]\supset\mathbb{F}_3$. The residue field of $(3, x^2+1)$ is

$$\kappa(3, x^2+1)=\mathbb{F}_3[x]/(x^2+1)=\mathbb{F}_3[\alpha],$$

a degree 2 extension of $\mathbb{F}_3$. Scheme-theoretically, the intersection is given as

$$\operatorname{Spec}\mathbb{F}_3[x]/(x^2+1)=\operatorname{Spec}\mathbb{F}_3[\alpha].$$

- $\mathbb{A}^1_{\mathbb{F}_5}$. The polynomial x^2+1 factors as $(x+2)(x+3)$ over $\mathbb{F}_5$, so we have two classical points $(5, x+2)$ and $(5, x+3)$, both with residue field equal to $\mathbb{F}_5$.

- $\mathbb{A}^1_{\mathbb{F}_7}$. The situation here is similar to that of $\mathbb{A}^1_{\mathbb{F}_3}$.



Exercise 2.3.3.22 ($\mathbb{A}^1_{\mathbb{F}}$ is always infinite). We have, in the previous example, unconsciously confirmed that $\mathbb{A}^1_{\mathbb{F}_p}=\operatorname{Spec}\mathbb{F}_p[x]$ has far more than the 'traditional' p points $(x),(x-1),\ldots,(x-(p-1))$ corresponding to the coordinates $0,1,\ldots,p-1\in\mathbb{F}_p$. Show that this is always the case, by proving that $\mathbb{A}^1_{\mathbb{F}}=\operatorname{Spec}\mathbb{F}[x]$ is infinite for *any field* $\mathbb{F}$. (**Hint**: Euclid's proof of the infinitude of prime numbers!)

Example 2.3.3.23 (Nodal cubic). Let $C=\operatorname{Spec}\mathbb{C}[x,y]/(y^2-x^2(x+1))\hookrightarrow\mathbb{A}^2_{\mathbb{C}}$. Then the morphism

$$f_\phi:\mathbb{A}^1_{\mathbb{C}}\longrightarrow C$$

induced by the ring homomorphism $\phi:\mathbb{C}[x,y]/(y^2-x^2(x+1))\to\mathbb{C}[t]$ defined by $\phi(x)=t^2-1$ and $\phi(y)=t(t^2-1)$ is not an isomorphism. Indeed, the function $t=y/x$ is not regular at $(0,0)$, and as such it does not lie in the image of ϕ. Note that f_ϕ is not even bijective on closed points: the origin $(0,0)\in C$ has two preimages, corresponding to $t=\pm1$. One can check that, in fact, there is no ring isomorphism $\mathbb{C}[t]\cong\mathbb{C}[x,y]/(y^2-x^2(x+1))$ at all (this is for instance a consequence of the normalisation computation in **??**).

Example 2.3.3.24 (Cuspidal cubic). Let $C=\operatorname{Spec}\mathbb{C}[x,y]/(y^2-x^3)\hookrightarrow\mathbb{A}^2_{\mathbb{C}}$. Then the morphism

$$f_\phi:\mathbb{A}^1_{\mathbb{C}}\longrightarrow C$$

induced by the ring homomorphism $\phi : \mathbb{C}[x,y]/(y^2-x^3) \to \mathbb{C}[t]$ defined by $\phi(x) = t^2$ and $\phi(y) = t^3$ is a bijective morphism, but not an isomorphism. It sends the closed point $(t-a) \in \mathbb{A}^1_{\mathbb{C}}$ to the point of $\mathbb{A}^2_{\mathbb{C}}$ with coordinates (a^2, a^3). The morphism f_ϕ is called a *rational parametrisation* of the plane curve $C \hookrightarrow \mathbb{A}^2_{\mathbb{C}}$.

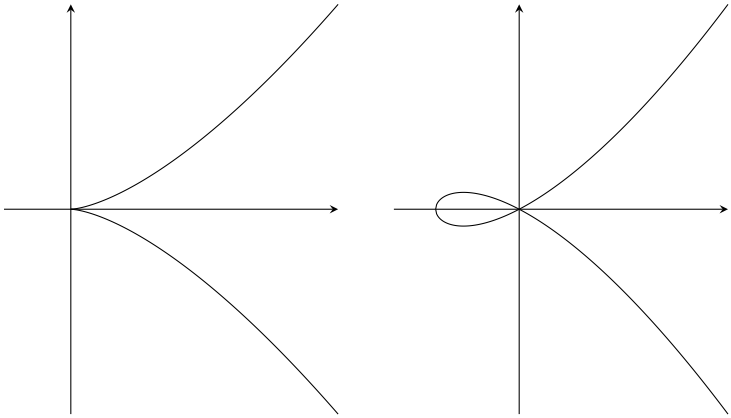


Figure 2.11: The (real points of the) cuspidal cubic curve $y^2 = x^3$ and the nodal cubic curve $y^2 = x^2(x+1)$.

We have learnt from several examples (including the Frobenius of $\operatorname{Spec}\mathbb{F}_p(t)$ in Example 2.3.3.4, the dual numbers in Example 2.3.3.6 and the cuspidal parametrisation in Example 2.3.3.24) that

a bijective morphism of schemes need not be an isomorphism.

Example 2.3.3.25. Consider the ring homomorphism

$$\phi : \mathbf{k}[x,y] \longrightarrow \mathbf{k}[x,y,z]/(xz-y)$$

sending $x \mapsto x$ and $y \mapsto y$. The corresponding morphism of affine schemes

$$f_\phi : \operatorname{Spec}\mathbf{k}[x,y,z]/(xz-y) \longrightarrow \mathbb{A}^2_{\mathbf{k}}$$

sends $(a, ab, b) \mapsto (a, ab)$, and the generic point maps to the generic point. The image of f_ϕ is $\mathrm{V}(x,y) \cup \mathrm{D}(x)$, which is neither open nor closed in $\mathbb{A}^2_{\mathbf{k}}$.

The image of the morphism f_ϕ in the previous example may look topologically weird. It is not that bad, though. Recall that a subset $T \subset X$ of a topological space X is *constructible* if it is a finite disjoint union of locally closed subsets. In general, a morphism of *algebraic varieties* (defined later in Important Definition 2.4.1.1 as those $\mathbf{k}$-schemes $X \to \operatorname{Spec}\mathbf{k}$ admitting an open cover by finitely many affine varieties) preserves constructible subsets.

THEOREM 2.3.3.26 (Chevalley). *Let $f : X \to Y$ be a morphism of algebraic varieties over $\mathbf{k}$. If $T \subset X$ is constructible, then $f(T) \subset Y$ is constructible. In particular, $f(X) \subset Y$ is constructible.*

2.4 Schemes over a scheme

In Important Definition 2.2.1.1 we defined a *scheme* as a locally ringed space $(X, \mathscr{O}_X)$ such that every point $x \in X$ has an open neighbourhood $x \in U \subset X$ such that $(U, \mathscr{O}_X|_U)$ is an affine scheme. Morphisms of schemes are defined as morphisms in the category of locally ringed spaces. As we saw (cf. Notation 2.2.1.4), schemes form a category, denoted Sch. It is by definition a full subcategory of LRS.

2.4.1 The category of S-schemes and algebraic varieties

For any scheme S, we can form the *category* Sch_S *of* S*-schemes* (also called *schemes over* S), whose objects are pairs (X, f), where X is a scheme and $f : X \to S$ is a morphism of schemes.[9] Morphisms

$$(X_1, f_1) \longrightarrow (X_2, f_2)$$

in Sch_S are commutative diagrams

$$\begin{array}{ccc} X_1 & \xrightarrow{\quad g \quad} & X_2 \\ & {\scriptstyle f_1}\searrow \quad \swarrow{\scriptstyle f_2} & \\ & S & \end{array}$$

i.e. morphisms of schemes $g : X_1 \to X_2$ such that $f_2 \circ g = f_1$. We call them *morphisms over* S or S*-morphisms* (or A-morphisms if $S = \operatorname{Spec} A$).

We use the notation

$$\operatorname{Hom}_S(X_1, X_2)$$

for the set of S-morphisms between S-schemes X_1 and X_2. If $S = \operatorname{Spec} A$ is affine, we write Sch_A, resp. Hom_A, instead of $\mathsf{Sch}_{\operatorname{Spec} A}$, resp. $\operatorname{Hom}_{\operatorname{Spec} A}$. For instance, when B is an A-algebra via a ring homomorphism $A \to B$, one says that $\operatorname{Spec} B \to \operatorname{Spec} A$ is an A-scheme via the canonical scheme morphism attached to $A \to B$.

Important Definition 2.4.1.1 (Algebraic variety)**.** Let $\mathbb{F}$ be a field, $X \to \operatorname{Spec} \mathbb{F}$ an $\mathbb{F}$-scheme. We say that $X \to \operatorname{Spec} \mathbb{F}$ is

(i) *locally of finite type* if every $x \in X$ has affine open neighbourhood $U \subset X$ such that $\mathscr{O}_X(U)$ is a finitely generated $\mathbb{F}$-algebra,[10]

(ii) *of finite type* if it is locally of finite type and X is quasicompact.

An $\mathbb{F}$-scheme $X \to \operatorname{Spec} \mathbb{F}$ of finite type is called an $\mathbb{F}$-*variety*, or an *algebraic variety over* $\mathbb{F}$. In other words, an $\mathbb{F}$-*variety* is an $\mathbb{F}$-scheme which admits an open cover by finitely many affine varieties over $\mathbb{F}$ (cf. Important Definition 2.1.5.3).

[9] This is an instance of a standard categorical construction, that of the *comma category* associated to an object Ω of a category $\mathscr{C}$.

[10] This notion will be generalised in **??**.



Caution 2.4.1.2. Different authors give different definitions of algebraic variety. Some authors call an *algebraic* $\mathbb{F}$*-scheme* what we call an $\mathbb{F}$-variety. Other (stronger) variants of Important Definition 2.4.1.1 include:

(1) reduced scheme of finite type over a field,

(2) reduced separated[11] scheme of finite type over a field,

(3) integral scheme of finite type over a field.

Probably the most common one is the strongest:

(4) integral separated scheme of finite type over a field.

Note that, with our definitions, fat points are considered to be (affine) algebraic varieties, even though they are not reduced. The nodal cubic (cf. Example 2.3.3.23) and the cuspidal cubic (cf. Example 2.3.3.24) are integral algebraic varieties; being affine, hence separated, they are also algebraic varieties in the sense of (4).



Caution 2.4.1.3. It is *not* true that if X is an algebraic variety, its ring of regular function $\mathscr{O}_X(X)$ is finitely generated! See [11].

Recall that *closed subschemes* of a scheme X were defined as equivalence classes of closed immersions into X, see Definition 2.2.1.7. We can now characterise closed immersions completely.

PROPOSITION 2.4.1.4 (Closed subschemes and ideal sheaves). *Let X be a scheme. There is an inclusion-reversing bijection*

$$\{\text{closed subschemes of } X\} \xrightarrow{\sim} \{\text{quasicoherent ideal sheaves } \mathscr{I} \subset \mathscr{O}_X\}.$$

On an affine open $U = \operatorname{Spec} A$ on which $\mathscr{I}|_U = \widetilde{J}$, the corresponding closed subscheme is

$$Z \cap U = \operatorname{Spec}(A/J) \hookrightarrow \operatorname{Spec} A.$$

In particular, for every ring A there is a bijection

$$\{\text{closed subschemes of } \operatorname{Spec} A\} \xrightarrow{\sim} \{\text{ideals } J \subset A\}.$$

Moreover, $Z \subset Z'$ as closed subschemes of X if and only if $\mathscr{I}_{Z'} \subset \mathscr{I}_Z$.

Proof. Let $\iota : Z \hookrightarrow X$ be a closed immersion. The induced ideal sheaf is $\mathscr{I}_Z = \ker(\mathscr{O}_X \to \iota_* \mathscr{O}_Z)$. By Proposition 1.12.0.2, the only thing left to prove to confirm the make sense of the correspondence is the claim that a closed immersion of schemes yields a quasicoherent ideal sheaf. But an ideal sheaf $\mathscr{J}$ on X is quasicoherent if and only if for every

[11] This notion will be introduced in **??**.

$x \in X$ there is an affine open neighbourhood $U \subset X$ along with sections $f_k \in \mathscr{J}(U)$, for k ranging in a (possibly infinite) index set K, such that $\mathscr{J}|_U = (f_k \mid k \in K)^\sim$.

If $U = \operatorname{Spec} A \subset X$ is affine, Proposition 2.3.3.3 gives an ideal $J \subset A$ and an isomorphism

$$i^{-1}(U) \cong \operatorname{Spec} A/J$$

over U. On every principal open $\mathrm{D}(f) \subset U$, the relevant map on sections is $A_f \twoheadrightarrow (A/J)_f$, whose kernel is J_f. Hence $\mathscr{I}_Z|_U = \widetilde{J}$, so $\mathscr{I}_Z$ is quasicoherent. Conversely, let $\mathscr{I} \subset \mathscr{O}_X$ be quasicoherent. Consider the closed subset $\mathrm{V}(\mathscr{I})$ already introduced (for locally ringed spaces) in Proposition 1.12.0.2. Locally this ringed space is $\operatorname{Spec} A/J$, so it is a scheme and the canonical morphism $\mathrm{V}(\mathscr{I}) \hookrightarrow X$ is a closed immersion. The two constructions are inverse to each other, again by Proposition 1.12.0.2.

As for the final claim, it is enough to observe that on affine opens one has that, for any two ideals I and J in a ring A, an inclusion $I \subset J$ corresponds to a closed immersion $\operatorname{Spec} A/J \hookrightarrow \operatorname{Spec} A/I$. □



Exercise 2.4.1.5. Ideal sheaves that are *not* quasicoherent do exist on schemes. Can you think of an example?

Spoiler 2.4.1.6. We will see in **??** that for every scheme X there is a 'nicest' closed subscheme $X_{\mathsf{red}} \hookrightarrow X$, called the *reduction* of X, which is topologically the same as X and is the smallest closed subscheme with this property.

place elsewhere better

2.4.2 What is a point?

Let $(X, \mathscr{O}_X)$ be a scheme. There are the following notions of a *point* of X.

Definition 2.4.2.1 (Points)**.** Let $(X, \mathscr{O}_X)$ be a scheme.

(1) A *point* of $(X, \mathscr{O}_X)$ is a point x of the underlying topological space X.

(2) A *closed point* of $(X, \mathscr{O}_X)$ is a point x such that $\{x\} \subset X$ is closed. Their set is denoted $X_{\mathsf{cl}} \subset X$.

(3) Let T be a scheme. A *T-valued point* of X is a morphism $T \to X$. The set of T-valued points of a scheme X is denoted $\mathsf{h}_X(T)$.

(4) Let $\mathbb{F}$ be a field, $X \to \operatorname{Spec}\mathbb{F}$ a scheme over $\mathbb{F}$. An *$\mathbb{F}$-rational point* (or simply a *rational point*) is an $\mathbb{F}$-morphism $\operatorname{Spec}\mathbb{F} \to X$.

(5) A morphism $\overline{x}\colon \operatorname{Spec}\Omega \to X$ from an algebraically closed field Ω is called a *geometric point* of X. We say that $\overline{x}$ *lies over* x if $x \in X$ is the image of $\overline{x}$.

Given a scheme X, its *functor of points* is the functor $\mathsf{h}_X\colon \mathsf{Sch}^{\mathrm{op}} \to \mathsf{Sets}$ sending

$$T \longmapsto \mathsf{h}_X(T) = \operatorname{Hom}_{\mathsf{Sch}}(T, X).$$

By Yoneda's Lemma, X is uniquely reconstructed from its functor of points. In other words, to "understand" X is the same as to understand all the sets $\mathsf{h}_X(T) = \mathrm{Hom}_{\mathsf{Sch}}(T, X)$ along with the maps $\mathrm{Hom}_{\mathsf{Sch}}(T, X) \to \mathrm{Hom}_{\mathsf{Sch}}(U, X)$ for every $U \to T$.

Definition 2.4.2.2. Let $\pi\colon X \to S$ be an S-scheme. View S as an S-scheme via $\mathrm{id}_S\colon S \to S$. Set

$$X(S) = \mathrm{Hom}_{\mathsf{Sch}_S}(S, X) = \left\{ S \xrightarrow{\sigma} X \;\middle|\; \pi \circ \sigma = \mathrm{id}_S \right\} \subset \mathsf{h}_X(S).$$

Elements in the set $X(S)$ are called *sections* of π. If $S = \mathrm{Spec}\, A$, we write $X(A)$ instead of $X(\mathrm{Spec}\, A)$.

If $S = \mathrm{Spec}\, \mathbb{F}$, then as in Definition 2.4.2.1(4), an element of

$$X(\mathbb{F}) = \mathrm{Hom}_{\mathbb{F}}(\mathrm{Spec}\, \mathbb{F}, X) \subset \mathsf{h}_X(\mathbb{F})$$

is called an $\mathbb{F}$-*rational point* (or simply a *rational point*) of X.

The next lemma indicates a link between valued points and rational points, for schemes defined over a field.

LEMMA 2.4.2.3. *Let $\mathbb{F}$ be a field, $X \to \mathrm{Spec}\, \mathbb{F}$ a scheme over $\mathbb{F}$. Then, there is a bijection*

$$X(\mathbb{F}) \simeq \{\, x \in X \mid \mathbb{F} \to \kappa(x) \text{ is an isomorphism} \,\}.$$

Proof. Take $\sigma \in X(\mathbb{F})$, i.e.

$$\begin{array}{ccc} \mathrm{Spec}\, \mathbb{F} & \xrightarrow{\sigma} & X \\ & \searrow & \downarrow \\ & & \mathrm{Spec}\, \mathbb{F} \end{array}$$

commutes. Let $x \in X$ be the image of σ. The homomorphism

$$\sigma^{\#}\colon \mathscr{O}_{X,x} \longrightarrow \mathbb{F}$$

induces the field homomorphism

$$\kappa(x) \longrightarrow \mathbb{F},$$

The structural map $\mathbb{F} \to \kappa(x)$ followed by $\kappa(x) \to \mathbb{F}$ is the identity. The latter map is a homomorphism of fields, hence injective, and its image contains $\mathbb{F}$. Its image is also contained in $\mathbb{F}$, so it is all of $\mathbb{F}$ and $\mathbb{F} \to \kappa(x)$ is an isomorphism.

Conversely, let $x \in X$ be a point such that $\mathbb{F} \xrightarrow{\sim} \kappa(x)$. We have a canonical surjection

$$\mathscr{O}_{X,x} \twoheadrightarrow \kappa(x) \xrightarrow{\sim} \mathbb{F}.$$

Recall that we have a canonical map $\mathrm{Spec}\, \mathscr{O}_{X,x} \to X$ (reminder: take any affine open subset $U \subset X$ containing x, so that we have a ring homomorphism $\mathscr{O}_X(U) \to \mathscr{O}_{X,x}$, inducing $\mathrm{Spec}\, \mathscr{O}_{X,x} \to \mathrm{Spec}\, \mathscr{O}_X(U) = U \hookrightarrow X$). The composition

$$\sigma\colon \mathrm{Spec}\, \mathbb{F} \xrightarrow{\sim} \mathrm{Spec}\, \kappa(x) \hookrightarrow \mathrm{Spec}\, \mathscr{O}_{X,x} \longrightarrow X$$

sends the unique point of $\operatorname{Spec}\mathbb{F}$ to x by construction: it is the required section of the structure morphism $X \to \operatorname{Spec}\mathbb{F}$. □

PROPOSITION 2.4.2.4 (Closed and rational points over a field). *Let $\mathbb{F}$ be a field and let $X \to \operatorname{Spec}\mathbb{F}$ be locally of finite type. A point $x \in X$ is closed if and only if $\kappa(x)$ is a finite extension of $\mathbb{F}$.*

Consequently, if $\mathbb{F} = \mathbf{k}$ is algebraically closed, then

$$X_{\mathsf{cl}} = X(\mathbf{k}).$$

Proof. Suppose first that x is closed. Choose an affine open neighbourhood $U = \operatorname{Spec} A$ of x, with A a finitely generated $\mathbb{F}$-algebra. The point x corresponds to a maximal ideal $\mathfrak{m} \subset A$, and Zariski's lemma (see Lemma B.1.3.1) shows that $\kappa(x) = A/\mathfrak{m}$ is finite over $\mathbb{F}$.

Conversely, suppose that $\kappa(x)/\mathbb{F}$ is finite. Let $y \in \overline{\{x\}}$ and choose an affine open neighbourhood $V = \operatorname{Spec} B$ of y. Since y is a specialisation of x, every open neighbourhood of y contains x, hence $x \in V$. Let $\mathfrak{p} \subset \mathfrak{q}$ be the primes of B corresponding to x and y. The domain $B/\mathfrak{p}$ embeds into $\kappa(x)$, so it is a finite-dimensional $\mathbb{F}$-vector space. Therefore $B/\mathfrak{p}$ is a field, and $\mathfrak{p}$ is maximal. It follows that $\mathfrak{p} = \mathfrak{q}$, so $y = x$. Thus x is closed.

If $\mathbb{F} = \mathbf{k}$ is algebraically closed, its only finite field extension is $\mathbf{k}$ itself. The last assertion now follows from Lemma 2.4.2.3. □

Example 2.4.2.5. To see that Proposition 2.4.2.4 fails for schemes that are not locally of finite type over a field, consider

$$X = \operatorname{Spec}\mathbb{C}(x) \xrightarrow{\pi} \operatorname{Spec}\mathbb{C}.$$

Then X has a unique closed point, but this point has residue field $\mathbb{C}(x)$, which is not finite over $\mathbb{C}$.



Caution 2.4.2.6. Mind the differences between the various notions of points. For instance, $X = \operatorname{Spec}\mathbb{C}$ admits infinitely many $\mathbb{C}$-valued points (there are (uncountably) many ring homomorphisms $\mathbb{C} \to \mathbb{C}$, cf. Example 2.3.3.4) and a fortiori infinitely many geometric points, but of course when viewed *as a $\mathbb{C}$-scheme* it has a unique $\mathbb{C}$-rational point (as there is *only one* $\mathbb{C}$-algebra homomorphism $\mathbb{C} \to \mathbb{C}$)!

2.4.3 Schemes over an affine scheme

A complete characterisation of morphisms *of affine schemes* was given, somewhat implicitly, in Theorem 2.3.2.1. Now we let X be an arbitrary scheme. Our goal is to characterise morphisms

$$X \longrightarrow Y = \operatorname{Spec} A.$$

We will show that the natural map

$$\rho_{X,Y} \colon \operatorname{Hom}_{\mathsf{Sch}}(X, Y) \longrightarrow \operatorname{Hom}_{\mathsf{Rings}}(A, \mathscr{O}_X(X)), \tag{2.4.1}$$

already introduced in (2.3.1) in the affine case, is a bijection. The map works just as in the affine case: a morphism $f: X \to Y$ is sent to $f^{\#}(Y)\colon A = \mathscr{O}_Y(Y) \to f_*\mathscr{O}_X(Y) = \mathscr{O}_X(X)$. Functoriality also holds, i.e. the diagram

$$\begin{array}{ccc} \mathrm{Hom}_{\mathsf{Sch}}(X,Y) & \xrightarrow{\rho_{X,Y}} & \mathrm{Hom}_{\mathsf{Rings}}(A, \mathscr{O}_X(X)) \\ \downarrow{\scriptstyle f \mapsto f\circ g} & & \downarrow{\scriptstyle \phi \mapsto g^{\#}(X)\circ\phi} \\ \mathrm{Hom}_{\mathsf{Sch}}(Z,Y) & \xrightarrow{\rho_{Z,Y}} & \mathrm{Hom}_{\mathsf{Rings}}(A, \mathscr{O}_Z(Z)) \end{array}$$

commutes for any morphism $g: Z \to X$ (for the same reason as in Remark 2.3.2.2).

We need the following preliminary result.

LEMMA 2.4.3.1. *Let X and Y be schemes. Then sending*

$$U \longmapsto \mathrm{Hom}_{\mathsf{Sch}}(U, Y) \in \mathsf{Sets}$$

for each open subset $U \subset X$ defines a sheaf of sets on X.

Proof. By Remark 1.2.0.7, we need to verify that given an open subset $U \subset X$, an open cover $U = \bigcup_{i\in I} U_i$ and a collection of morphisms $f_i: U_i \to Y$ such that $f_i|_{U_{ij}} = f_j|_{U_{ij}}$ (as morphisms of schemes!) for every $(i,j) \in I \times I$, there exists a unique $f: U \to Y$ such that $f|_{U_i} = f_i$. (We have used the usual notation $U_{ij} = U_i \cap U_j$). At the level of topological spaces, it is clear that there is a unique continuous map $f: U \to Y$ with the required property. We need to extend it (uniquely) to a morphism of *schemes*. So we need a well defined sheaf homomorphism $f^{\#}\colon \mathscr{O}_Y \to f_*\mathscr{O}_U$. Let $V \subset Y$ be an open subset. We define $f^{\#}(V)\colon \mathscr{O}_Y(V) \to \mathscr{O}_U(f^{-1}V)$ as follows.

First of all, each $f_i: U_i \to Y$ induces a map $\mathscr{O}_Y(V) \to \mathscr{O}_{U_i}(f_i^{-1}V) = \mathscr{O}_U(f_i^{-1}V)$. Moreover, $f^{-1}V = \bigcup_{i\in I} f_i^{-1}V$ is an open covering, and since $\mathscr{O}_U$ is a sheaf we have a diagram

$$\begin{array}{ccccc} & & \mathscr{O}_Y(V) & & \\ & \swarrow & \downarrow{\scriptstyle \tau} & & \\ \mathscr{O}_U(f^{-1}V) & \longrightarrow & \prod_{i\in I} \mathscr{O}_U(f_i^{-1}V) & \underset{\nu}{\overset{\mu}{\rightrightarrows}} & \prod_{(i,j)\in I\times I} \mathscr{O}_U(f_i^{-1}V \cap f_j^{-1}V) \end{array}$$

where the bottom row is an equaliser sequence. Saying that $f_i|_{U_{ij}} = f_j|_{U_{ij}}$ is like saying that $\mu\circ\tau = \nu\circ\tau$, thus by the universal property of equalisers there is precisely one way to fill in the dotted arrow to $\mathscr{O}_U(f^{-1}V)$. This is the definition of $f^{\#}(V)$.

Uniqueness in the gluing axiom shows that these maps are compatible with restriction when $W \subset V$. The same uniqueness, applied to sums and products, shows that every $f^{\#}(V)$ is a ring homomorphism. Thus $f^{\#}$ is a morphism of sheaves of rings. Near a point $x \in U_i$ it agrees with $f_i^{\#}$, so the induced map on the stalk at x is local. Consequently $(f, f^{\#})$ is a morphism of locally ringed spaces, and hence a morphism of schemes. The same local argument proves uniqueness. □

Remark 2.4.3.2. The statement of Lemma 2.4.3.1 remains true for locally ringed spaces or, more generally, ringed spaces.

THEOREM 2.4.3.3. *Let X be a scheme, $Y = \operatorname{Spec} A$ an affine scheme. Then the canonical map* (2.4.1) *is bijective.*

Proof. Fix a covering $X = \bigcup_{i\in I} U_i$, where $\iota_i : U_i = \operatorname{Spec} B_i \hookrightarrow X$ is an affine open subset. Since $U \mapsto \operatorname{Hom}_{\mathsf{Sch}}(U, Y)$ is a sheaf on X (cf. Lemma 2.4.3.1), the natural map

$$\operatorname{Hom}_{\mathsf{Sch}}(X, Y) \xrightarrow{\alpha} \prod_{i\in I} \operatorname{Hom}_{\mathsf{Sch}}(U_i, Y)$$

is injective. We have a new diagram

$$\begin{array}{ccc} \operatorname{Hom}_{\mathsf{Sch}}(X, Y) & \xrightarrow{\rho_{X,Y}} & \operatorname{Hom}_{\mathsf{Rings}}(A, \mathscr{O}_X(X)) \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \phi \mapsto (\iota_i^{\#}(X) \circ \phi)_{i\in I}} \\ \prod_{i\in I} \operatorname{Hom}_{\mathsf{Sch}}(U_i, Y) & \xrightarrow{\beta} & \prod_{i\in I} \operatorname{Hom}_{\mathsf{Rings}}(A, B_i) \end{array}$$

where β is a bijection as confirmed during the proof of Theorem 2.3.2.1. It follows that $\rho_{X,Y}$ is injective. We are left to prove its surjectivity. Fix $\phi \in \operatorname{Hom}_{\mathsf{Rings}}(A, \mathscr{O}_X(X))$, and consider its image $(\phi_i)_{i\in I} \in \prod_{i\in I} \operatorname{Hom}_{\mathsf{Rings}}(A, B_i)$. This corresponds to a unique tuple of morphisms $(f_i : U_i \to Y)_{i\in I}$. These have the property that $f_i|_V = f_j|_V$ for every affine open subset $V \subset U_i \cap U_j$. To see this, notice that for any $i \in I$ we have a commutative diagram

$$\begin{array}{ccc} \operatorname{Hom}_{\mathsf{Sch}}(U_i, Y) & \xrightarrow{\sim} & \operatorname{Hom}_{\mathsf{Rings}}(A, B_i) \\ \downarrow & & \downarrow{\scriptstyle \psi \mapsto j_i^{\#}(U_i) \circ \psi} \\ \operatorname{Hom}_{\mathsf{Sch}}(V, Y) & \xrightarrow{\sim} & \operatorname{Hom}_{\mathsf{Rings}}(A, \mathscr{O}_X(V)) \end{array}$$

where $j_i : V \hookrightarrow U_i$ is the open immersion. It is clear that the image of f_i in the set $\operatorname{Hom}_{\mathsf{Rings}}(A, \mathscr{O}_X(V))$ does not depend on i, being equal to the image of ϕ, namely its post-composition with $\mathscr{O}_X(X) \to \mathscr{O}_X(V)$. Therefore all the f_i map to the same element of $\operatorname{Hom}_{\mathsf{Sch}}(V, Y)$, which is what we wanted to confirm. Affine open subsets form a basis of $U_i \cap U_j$, so the equalities $f_i|_V = f_j|_V$ for every affine open subset $V \subset U_i \cap U_j$ imply $f_i|_{U_i\cap U_j} = f_j|_{U_i\cap U_j}$.

It now follows from Lemma 2.4.3.1 that $(f_i : U_i \to Y)_{i\in I}$ glue to a (unique) morphism $f : X \to Y$, which by construction maps to ϕ via $\rho_{X,Y}$. Thus $\rho_{X,Y}$ is surjective. □

COROLLARY 2.4.3.4. *Let B be a ring. To give a B-scheme $X \to \operatorname{Spec} B$ is the same as to give a scheme X along with a B-algebra structure on $\mathscr{O}_X(X)$.*

COROLLARY 2.4.3.5. *Let X be a scheme. There is a canonical morphism*

$$X \longrightarrow \operatorname{Spec} \mathscr{O}_X(X).$$

The affine scheme $\operatorname{Spec} \mathscr{O}_X(X)$ *is sometimes called the affinisation of X.*

Proof. Take $Y = \operatorname{Spec} \mathscr{O}_X(X)$ and consider the morphism corresponding to the identity $\mathrm{id} \in \mathrm{Hom}_{\mathsf{Rings}}(\mathscr{O}_X(X), \mathscr{O}_X(X))$ under $\rho_{X,Y}$. □

Remark 2.4.3.6. A possible translation of Theorem 2.4.3.3 is the following: if $\Gamma(-)$ denotes the functor taking a scheme X to the ring of its regular functions $\mathscr{O}_X(X)$, then the pair of functors $(\Gamma(-), \operatorname{Spec})$ is an adjoint pair

$$\mathsf{Sch} \underset{\operatorname{Spec}}{\overset{\Gamma(-)}{\rightleftarrows}} \mathsf{Rings}^{\mathrm{op}}$$

where of course Spec is now viewed as the composition $\mathsf{Rings}^{\mathrm{op}} \xrightarrow{\sim} \mathsf{Aff} \hookrightarrow \mathsf{Sch}$.



Exercise 2.4.3.7. Confirm that $\operatorname{Spec}\mathbb{Z}$ is a final object in the category of schemes, so that (in the notation of Section 2.4.1) in particular $\mathsf{Sch} = \mathsf{Sch}_{\mathbb{Z}}$.

2.4.4 Gluing schemes

You may have encountered interesting spaces such as *projective spaces* or *Grassmannians* before. For example, projective n-space over a field $\mathbb{F}$ can be defined as follows: consider the scaling action of $\mathbb{F}^\times$ on $\mathbb{F}^{n+1} \setminus 0$, sending $v \mapsto \lambda v$ for $\lambda \in \mathbb{F}^\times$, and set

$$\mathbb{P}^n(\mathbb{F}) = (\mathbb{F}^{n+1} \setminus 0)/\mathbb{F}^\times.$$

This is all good in the topological (or smooth) category, however we cannot make such a definition in algebraic geometry. Quotients exist (sometimes) and their theory has now become classical, but they are delicate to deal with.

We shall see *two ways* to define projective space in algebraic geometry. The first one is by gluing schemes (along open immersions). We now describe this procedure in full generality.

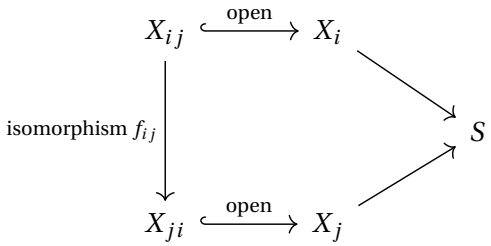
The input data are as follows:

(1) a scheme S,

(2) a family of S-schemes $\{X_i \to S \mid i \in I\}$,

(3) open subschemes $X_{ij} \subset X_i$ for every $(i, j) \in I \times I$,

(4) isomorphisms $f_{ij} \colon X_{ij} \xrightarrow{\sim} X_{ji}$ over S for every $(i, j) \in I \times I$.



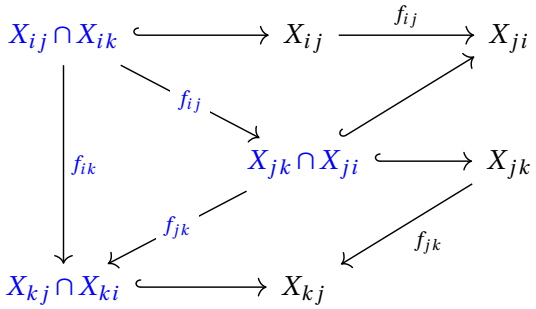
The assumptions on the input data are the following:

(i) $X_{ii} = X_i$ and $f_{ii} = \mathrm{id}_{X_i}$ for every $i \in I$,

(ii) $f_{ij}(X_{ij} \cap X_{ik}) = X_{jk} \cap X_{ji}$, for every $(i, j, k) \in I \times I \times I$

(iii) the *cocycle condition* holds: $f_{jk} \circ f_{ij} = f_{ik}$ on $X_{ij} \cap X_{ik}$.

The cocycle condition is the following compatibility:



THEOREM 2.4.4.1 (Gluing schemes). *Given the data* (1)–(4) *satisfying conditions* (i)–(iii)*, there exists an S-scheme X (unique up to isomorphism), along with open immersions $\theta_i : X_i \hookrightarrow X$ over S such that $\theta_j|_{X_{ji}} \circ f_{ij} = \theta_i|_{X_{ij}}$ and $X = \bigcup_{i \in I} \theta_i(X_i)$. Moreover, $\theta_i(X_i) \cap \theta_j(X_j) = \theta_i(X_{ij})$.*

Proof. See [5, Ch. 2, Lemma 3.33]. □

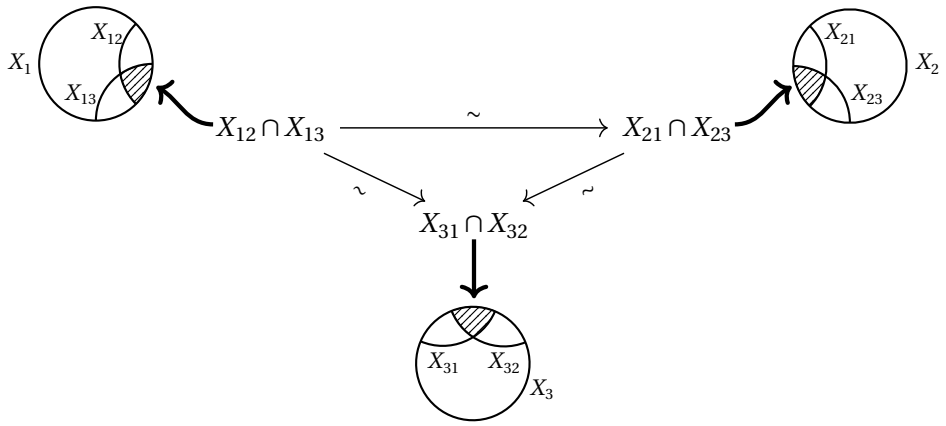


Figure 2.12: The gluing construction starting with 3 open subsets X_1, X_2, $X_3 \subset X$.

Definition 2.4.4.2. The *disjoint union* of a family of S-schemes $\{X_i \to S\}_{i \in I}$ is the gluing of the family along $X_{ij} = \emptyset$ (and empty maps f_{ij}) for all $i \neq j$. It is denoted $\coprod_{i \in I} X_i \to S$.

Next, we describe the construction of

$$\boxed{\mathbb{P}^n_A = \text{projective } n\text{-space over } A.}$$

Let A be a ring, $S = \operatorname{Spec} A$ the corresponding affine scheme, $n \geqslant 0$ an integer and $I = \{0, 1, \ldots, n\}$. Fix a variable x_i for every $i \in I$, and form the ring $R = A[x_0^{\pm}, x_1^{\pm}, \ldots, x_n^{\pm}]$. Consider the A-subalgebras

$$A_i = A\left[x_k x_i^{-1} \;\middle|\; 0 \leqslant k \leqslant n\right] \subset R, \quad i \in I.$$

Note that A_i is the homogeneous localisation of $A[x_0, x_1, \ldots, x_n]$ at the degree 1 element x_i, a polynomial ring in n variables (cf. Example 2.5.2.3). Each yields an A-scheme

$$X_i = \operatorname{Spec} A_i \longrightarrow \operatorname{Spec} A, \quad i \in I.$$

Now, for each $j \neq i$, the scheme X_i contains the (principal) affine open subscheme

$$X_{ij} = \mathrm{D}(x_j x_i^{-1}) = \operatorname{Spec} (A_i)_{x_j x_i^{-1}} \subset X_i.$$

But

$$(A_i)_{x_j x_i^{-1}} = (A_j)_{x_i x_j^{-1}}$$

are *equal* as subrings of R, and thus we have canonical isomorphisms $f_{ij} \colon X_{ij} \xrightarrow{\sim} X_{ji}$. Explicitly, after the identifications

$$(A_i)_{x_j x_i^{-1}} \cong A_i[t]/(t \cdot x_j x_i^{-1} - 1)$$
$$(A_j)_{x_i x_j^{-1}} \cong A_j[u]/(u \cdot x_i x_j^{-1} - 1),$$

we see that an isomorphism between the quotient rings on the right hand sides is given by sending

$$x_j x_i^{-1} \mapsto u, \quad t \mapsto x_i x_j^{-1}, \quad x_k x_i^{-1} \mapsto u x_k x_j^{-1} \text{ for } k \neq i, j.$$

The hypotheses of Theorem 2.4.4.1 are satisfied by our gluing data. The resulting A-scheme is called *projective n-space over A*, and is denoted $\mathbb{P}_A^n$. It has an open cover by $n+1$ affine open subsets isomorphic to affine spaces over A, namely $X_i = \operatorname{Spec} A_i \cong \mathbb{A}_A^n$. Indeed, the variables $\{ x_k x_i^{-1} \mid 0 \leqslant k \leqslant n,\ k \neq i \}$ are algebraically independent.

Remark 2.4.4.3. This construction shows that $\mathbb{P}_A^n$ is a quasicompact scheme. We shall see that it is not affine (unless $n = 0$).

Example 2.4.4.4 (Projective line)**.** The most explicit instance of the above construction of $\mathbb{P}_A^n$ arises for $n = 1$. In this case, our input data are two schemes $X_1 = \operatorname{Spec} A[t]$ and $X_2 = \operatorname{Spec} A[u]$, and the isomorphism $X_{12} = \mathrm{D}(t) \xrightarrow{\sim} \mathrm{D}(u) = X_{21}$ induced by the A-algebra isomorphism $A[u, u^{-1}] \xrightarrow{\sim} A[t, t^{-1}]$ sending $u \mapsto t^{-1}$. The gluing gives, by definition, the *projective line* $\mathbb{P}_A^1$. Similarly, Figure 2.12 can be seen as a pictorial construction of $\mathbb{P}_A^2$.

Example 2.4.4.5. Keep the notation of Example 2.4.4.4, but assume $A = \mathbb{F}$ is a field, for simplicity. Then, had we chosen the isomorphism $\mathbb{F}[u, u^{-1}] \xrightarrow{\sim} \mathbb{F}[t, t^{-1}]$ sending $u \mapsto t$, we would have of course identified the complements of the origin in the two

affine lines, but we also would have 'kept the origin twice'. The result of the gluing is called an *affine line with double origin*. We shall come back to this scheme, for it is the prototypical example of a non-separated scheme. Separatedness is, as we shall see, the scheme-theoretic analogue of the Hausdorff property, which we have already given up on (cf. Remark 2.1.1.15). Since affine schemes are separated (cf. **??**), this also gives another example (besides Example 2.3.3.10 and Example 2.3.3.11) of a non-affine scheme.

Figure 2.13: The affine line with two origins.

Since the schemes X_i form an open covering of the glued up scheme X, by Example 1.3.0.4 we have an exact sequence of abelian groups

$$0 \longrightarrow \mathscr{O}_X(X) \longrightarrow \prod_{i\in I} \mathscr{O}_X(X_i) \longrightarrow \prod_{(i,j)\in I\times I} \mathscr{O}_X(X_{ij}) \tag{2.4.2}$$

where the first map is restriction and the second map sends $(f_i)_i \mapsto (f_i|_{X_{ij}} - f_j|_{X_{ij}})_{i,j}$.



Exercise 2.4.4.6. Use the sequence (2.4.2) to show that $\mathscr{O}_{\mathbb{P}^n_A}(\mathbb{P}^n_A) = A$. Observe, then, that $\mathbb{P}^n_A$ is not affine (unless $n = 0$)!

2.5 Projective schemes

In this section we define an important class of schemes, including *projective schemes*. These are the closed subschemes of $\mathbb{P}^n_A$ for some given ring A and some $n \geqslant 0$. They are the natural upgrade of classical *projective varieties* over a field. The general construction is somewhat analogous (though exhibiting many differences as well, see e.g. Remark 2.5.1.10, Caution 2.5.3.1 and Caution 2.5.3.13) to the construction of Spec A starting from a ring A. The main difference is that now we have to work with *graded rings*. The ubiquity of gradings and homogeneous ideals can be 'explained' at an informal level as follows: points $p = (a_0 : a_1 : \cdots : a_n)$ of classical projective space $\mathbb{P}^n(\mathbb{C})$ have *homogeneous coordinates*, meaning e.g. that $(1 : 2)$ is the same point as $(-3 : -6)$ in $\mathbb{P}^1(\mathbb{C})$. As such, the evaluation of a polynomial $f \in \mathbb{C}[x_0, x_1, \ldots, x_n]$ at p is not well-defined. What *is* well-defined though, is the *vanishing* of f at p, so long as f is homogeneous, for

$$f(\lambda x_0, \lambda x_1, \ldots, \lambda x_n) = \lambda^d f(x_0, x_1, \ldots, x_n), \quad d = \deg f, \quad \lambda \in \mathbb{C}^\times.$$

We will see that the equation $f = 0$ defines a closed subscheme of $\mathbb{P}^n_{\mathbb{C}}$. This will be called a *hypersurface in* $\mathbb{P}^n_{\mathbb{C}}$.

2.5.1 Zariski topology on Proj B

Let A be a ring. A *graded A-algebra* is an A-algebra $A \to B$ equipped with a decomposition

$$B = \bigoplus_{d \geqslant 0} B_d,$$

where $B_d \subset B$ are subgroups satisfying $B_d B_e \subset B_{d+e}$ for each $d, e \geqslant 0$, and such that the image of $A \to B$ is contained in B_0. In this situation, we have that

(1) $B_0 \subset B$ is a subring, so that B is naturally a B_0-algebra, and

(2) each graded piece B_d is a B_0-module.

Elements of B_d are called *homogeneous of degree d* (and $0 \in B$ is considered homogeneous of any degree). Every $f \in B$ has a unique decomposition $f = \sum_{0 \leqslant i \leqslant e} f_i$ into homogeneous elements f_i. An A-algebra homomorphism $\phi : B \to C$ is called a *graded homomorphism* if there exists an integer $e > 0$ such that $\phi(B_d) \subset C_{ed}$ for every $d \geqslant 0$. Graded A-algebras thus form a category. If B is a graded A-algebra, the ideal

$$B_+ = \bigoplus_{d > 0} B_d \subset B$$

is called the *irrelevant ideal*, for reasons that will become clear soon (cf. Remark 2.5.1.8).

Example 2.5.1.1. Let $B = A[x_0, x_1, \ldots, x_n]$ be the polynomial ring with A-coefficients and with x_i in degree 1 for all i. Elements of B_d are the homogeneous polynomials of degree d in the classical sense. The irrelevant ideal is $B_+ = (x_0, x_1, \ldots, x_n) \subset B$.

Let $I \subset B$ be an ideal. Then, the following are equivalent:

- $I \subset B$ is a graded submodule,
- I can be generated by homogeneous elements,
- $I = \bigoplus_{d \geqslant 0} (I \cap B_d)$,
- If $f \in I$ has homogeneous decomposition $f = f_0 + f_1 + \cdots + f_k$, then $f_e \in I$ for all e.

If any of these equivalent conditions is fulfilled, we say that I is *homogeneous*.

Remark 2.5.1.2. The class of homogeneous ideals in B is closed under sum, product, intersection, and radical.

Remark 2.5.1.3. Let $I \subset B$ be a homogeneous ideal. The quotient B/I is naturally a graded A-algebra via $(B/I)_d = B_d/(I \cap B_d)$.

Let $A \to B$ be a graded A-algebra. The localisation of B at a multiplicative subset $S \subset B$ inherits a grading as soon as S consists of homogeneous elements. If $\mathfrak{p} \subset B$ is a homogeneous prime ideal, we may localise B at

$$S(\mathfrak{p}) = \left\{ b \in B \setminus \mathfrak{p} \;\middle|\; b \text{ is homogeneous} \right\}.$$

This localisation, denoted $B_{\mathfrak{p}}$ with a slight abuse of notation (see also Warning B.3.3.4), contains as a subring its degree 0 piece, denoted $B_{(\mathfrak{p})}$. It is a local ring with maximal ideal

$$\mathfrak{m}_{(\mathfrak{p})} = \left\{ \frac{a}{h} \;\middle|\; a \in \mathfrak{p},\, h \in S(\mathfrak{p}),\, \deg a = \deg h \right\}.$$

The local ring $(B_{(\mathfrak{p})}, \mathfrak{m}_{(\mathfrak{p})})$ may be called the homogeneous localisation of B at $\mathfrak{p}$. Another key example of homogeneous localisation is the *homogeneous principal localisation*.

Construction 2.5.1.4 (Homogeneous principal localisation)**.** Let $A \to B$ be a graded A-algebra. If $f \in B$ is homogeneous of degree e, then $B_f = \bigoplus_{d\in\mathbb{Z}} (B_f)_d$, where

$$(B_f)_d = \left\{ \frac{a}{f^n} \;\middle|\; a \in B_{d+ne} \right\} \subset B_f.$$

Such graded rings are the only ones (that we consider) with negative graded pieces. We set

$$B_{(f)} = (B_f)_0 = \left\{ \frac{a}{f^n} \;\middle|\; a \in B_{ne} \right\}.$$

It is called the *homogeneous principal localisation* (or simply *homogeneous localisation*) of B at f. This is a ring by the condition (1), and in fact, it is an A-subalgebra of B_f, by our key assumption that $A \to B$ lands in B_0. This can be seen via the diagram

$$\begin{array}{ccccc} A & \longrightarrow & B_0 & \xrightarrow{\text{subring}} & B \\ & & \vdots & & \downarrow \ell \\ & & B_{(f)} & \hookrightarrow & B_f \end{array} \tag{2.5.1}$$

and the fact that ℓ preserves the grading. We endow $B_{(f)}$ with the trivial grading. The inclusion $B_{(f)} \hookrightarrow B_f$ turns B_f into a graded $B_{(f)}$-algebra, and this gives a natural morphism of affine schemes

$$\operatorname{Spec} B_f \longrightarrow \operatorname{Spec} B_{(f)}.$$

To an arbitrary ideal $I \subset B$ we may associate a homogeneous ideal $I^{\mathrm{h}} = \bigoplus_{d\geqslant 0} (I \cap B_d)$. Note that $I^{\mathrm{h}} \subset I$, with equality if and only if I is homogeneous. Indeed, I^{h} is the largest homogeneous ideal contained in I.

LEMMA 2.5.1.5. *Let $I \subset B$ be a homogeneous ideal. Then I is prime if and only if whenever $ab \in I$ for homogeneous elements $a, b \in B$, one has that either $a \in I$ or $b \in I$.*

Proof. Let $a = \sum_{1\leqslant i\leqslant n} a_i$ and $b = \sum_{1\leqslant j\leqslant m} b_j$ be the homogeneous decompositions of two elements $a, b \in B$ such that $ab \in I$. Since I is homogeneous, it must contain all the homogeneous components of ab. Assume, by contradiction, that $a \notin I$ and $b \notin I$. Then, there is a largest d such that $a_d \notin I$ and a largest e such that $b_e \notin I$. We have $(ab)_{d+e} = \sum_{i+j=d+e} a_i b_j$, but every pair $(i,j) \neq (d,e)$ appearing in the sum satisfies either $i > d$ or $j > e$. Thus $a_i b_j \in I$ for every such pair. But since $(ab)_{d+e} \in I$ as well, we must have $a_d b_e \in I$. Thus, by our assumption, either $a_d \in I$ or $b_e \in I$. Contradiction. □

LEMMA 2.5.1.6. *If $I \subset B$ is a prime ideal, then the homogeneous ideal $I^{\mathrm{h}} \subset B$ is prime.*

Proof. We exploit Lemma 2.5.1.5. Let $a, b \in B$ be two homogeneous elements, say $a \in B_d$ and $b \in B_e$, such that $ab \in I^{\mathrm{h}}$. In fact, $ab \in I^{\mathrm{h}}_{d+e} = I \cap B_{d+e} \subset I$. Then, since I is prime, we have either $a \in I$ or $b \in I$. Thus either $a \in I \cap B_d \subset I^{\mathrm{h}}$, or $b \in I \cap B_e \subset I^{\mathrm{h}}$. □

Let $A \to B$ be a graded A-algebra. Define the *projective spectrum* of B to be the set

$$\boxed{\operatorname{Proj} B = \left\{ \mathfrak{p} \subset B \;\middle|\; \begin{array}{c} \mathfrak{p} \text{ is a homogeneous prime} \\ \text{ideal such that } \mathfrak{p} \not\supset B_+ \end{array} \right\}.}$$

Our goal is to put a structure of A-scheme on $\operatorname{Proj} B$. By Corollary 2.4.3.4, this amounts to construct a scheme $(\operatorname{Proj} B, \mathscr{O}_{\operatorname{Proj} B})$ along with an A-algebra structure on $\mathscr{O}_{\operatorname{Proj} B}$.

As in the affine case, we start from the topology on $\operatorname{Proj} B$. For a homogeneous ideal $I \subset B$, we define

$$\mathrm{V}_+(I) = \left\{ \mathfrak{p} \in \operatorname{Proj} B \;\middle|\; \mathfrak{p} \supset I \right\} \subset \operatorname{Proj} B.$$

These sets satisfy the axioms of closed subsets for a topology on $\operatorname{Proj} B$. The properties

(1) $\mathrm{V}_+(I) \cup \mathrm{V}_+(J) = \mathrm{V}_+(I \cap J) = \mathrm{V}_+(IJ)$

(2) $\bigcap_{\lambda \in \Lambda} \mathrm{V}_+(I_\lambda) = \mathrm{V}_+\left(\sum_{\lambda \in \Lambda} I_\lambda\right)$

(3) $\mathrm{V}_+(B) = \emptyset$ and $\mathrm{V}_+(0) = \operatorname{Proj} B$

are proved in a similar fashion to the affine case (cf. Lemma 2.1.1.2). The induced topology on $\operatorname{Proj} B$ is called the *Zariski topology*. Note that one also has

$$\mathrm{V}_+(I) = \mathrm{V}_+(\sqrt{I})$$

for any homogeneous ideal $I \subset B$.

LEMMA 2.5.1.7. *Let $I, J \subset B$ be homogeneous ideals. The following statements hold true.*

(i) $\mathrm{V}_+(I) \subset \mathrm{V}_+(J)$ *if and only if* $J \cap B_+ \subset \sqrt{I}$.

(ii) *One has* $\operatorname{Proj} B = \emptyset$ *if and only if* $B_+ \subset \sqrt{(0)}$.

(iii) $\mathrm{V}_+(I) = \emptyset$ *if and only if* $B_+ \subset \sqrt{I}$. *If* $\sqrt{I} = B_+$, *then* $\mathrm{V}_+(J) = \mathrm{V}_+(J \cap I)$.

Proof. We proceed step by step.

(i) Assume $J \cap B_+ \subset \sqrt{I}$, and fix a prime $\mathfrak{p} \in \mathrm{V}_+(I)$. Then

$$\mathfrak{p} \supset \sqrt{I} \supset J \cap B_+ \supset JB_+,$$

and since $\mathfrak{p} \not\supset B_+$ we must have $\mathfrak{p} \in \mathrm{V}_+(J)$.

Assume now that $\mathrm{V}_+(I) \subset \mathrm{V}_+(J)$. Recall that $\sqrt{I} = \bigcap_{\mathfrak{p} \in \mathrm{V}(I)} \mathfrak{p}$. Observe that any prime $\mathfrak{p} \in \mathrm{V}(I)$ satisfies (since I is homogeneous) $I = I^{\mathrm{h}} \subset \mathfrak{p}^{\mathrm{h}}$. Now there are two

possibilities. First: if $\mathfrak{p}^{\mathrm{h}} \not\supset B_+$, then by Lemma 2.5.1.6 $\mathfrak{p}^{\mathrm{h}}$ belongs to $\mathrm{V}_+(I)$, thus by assumption $\mathfrak{p} \supset \mathfrak{p}^{\mathrm{h}} \supset J \supset J \cap B_+$, which implies $J \cap B_+ \subset \sqrt{I}$. Second option: if $\mathfrak{p}^{\mathrm{h}} \supset B_+$, we still have $\mathfrak{p} \supset \mathfrak{p}^{\mathrm{h}} \supset B_+ \supset J \cap B_+$. Thus, also in this case, we have $J \cap B_+ \subset \sqrt{I}$.

(ii) We have $\operatorname{Proj} B = \emptyset$ if and only if every homogeneous prime ideal $\mathfrak{p} \subset B$ contains B_+, i.e. $\mathrm{V}_+(0) \subset \mathrm{V}_+(B_+)$, i.e. $B_+ \subset \sqrt{(0)}$ by (i).

(iii) Follows from (i) applied to $\mathrm{V}_+(I) \subset \emptyset = \mathrm{V}_+(B_+)$. Finally, $\mathrm{V}_+(J \cap I) = \mathrm{V}_+(J) \cup \mathrm{V}_+(I) = \mathrm{V}_+(J) \cup \mathrm{V}_+(B_+) = \mathrm{V}_+(J)$. □

Remark 2.5.1.8. Condition (iii) explains why B_+ is called *irrelevant*: the operation $\mathrm{V}_+(-)$ sends it to the empty set, and it does the same for every radical ideal containing it. One should keep in mind the case $B = \mathbf{k}[x_0, x_1, \ldots, x_n]$, where $B_+ = (x_0, x_1, \ldots, x_n)$ should 'correspond to the origin'. But there is no origin in $\mathbb{P}^n(\mathbf{k})$!

Notation 2.5.1.9. Let $f \in B$ be a homogeneous element. We call $\mathrm{D}_+(f) = \operatorname{Proj} B \setminus \mathrm{V}_+(f B)$ a *principal open set* in $\operatorname{Proj} B$. We write $\mathrm{V}_+(f)$ instead of $\mathrm{V}_+(f B)$.

Note that, as in the affine case, we have the identity

$$\mathrm{D}_+(fg) = \mathrm{D}_+(f) \cap \mathrm{D}_+(g)$$

for any two homogeneous elements $f, g \in B_+$.

Remark 2.5.1.10. Note that $\operatorname{Proj} B$ need not be quasicompact. For instance,

$$\operatorname{Proj} \mathbb{Z}[x_1, x_2, \ldots] = \bigcup_{i \geqslant 1} \mathrm{D}_+(x_i)$$

is an open cover admitting no finite subcover. This may sound counterintuitive: projective things 'should' be compact, affine things should not. But the Zariski topology is funny. When we have the correct notion of compactness, your intuition will get realigned.

There is a canonical inclusion

$$\varepsilon \colon \operatorname{Proj} B \hookrightarrow \operatorname{Spec} B$$

and for any $f \in B$ one has $\mathrm{V}(f) \cap \operatorname{Proj} B = \bigcap_{0 \leqslant i \leqslant e} \mathrm{V}_+(f_i)$ if $f = f_0 + f_1 + \cdots + f_e$ is the homogeneous decomposition of $f \in B$. Then $\mathrm{D}(f) \cap \operatorname{Proj} B = \bigcup_{0 \leqslant i \leqslant e} \mathrm{D}_+(f_i)$. This shows that the Zariski topology on $\operatorname{Proj} B$ is induced by the Zariski topology on $\operatorname{Spec} B$ (i.e. it agrees with the subspace topology), and moreover

$$\operatorname{Proj} B \setminus \mathrm{V}_+(I) = \bigcup_{\substack{f \in I \\ f \text{ homogeneous}}} \mathrm{D}_+(f)$$

for any homogeneous ideal $I \subset B$. In particular, the principal opens

$$\left\{ \mathrm{D}_+(f) \subset \operatorname{Proj} B \;\middle|\; f \text{ is homogeneous} \right\}$$

form a base for the Zariski topology. In fact, one can focus only on those $\mathrm{D}_+(f)$ where $f \in B_+$. The reason is the following: suppose $B_+ = (f_i \mid i \in I)$ with f_i homogeneous. Then,

$$\operatorname{Proj} B = \operatorname{Proj} B \setminus \emptyset = \operatorname{Proj} B \setminus \mathrm{V}_+(B_+) = \operatorname{Proj} B \setminus \bigcap_{i\in I} \mathrm{V}_+(f_i) = \bigcup_{i\in I} \mathrm{D}_+(f_i),$$

so that for any homogeneous $g \in B$ we have

$$\mathrm{D}_+(g) = \mathrm{D}_+(g) \cap \operatorname{Proj} B = \mathrm{D}_+(g) \cap \bigcup_{i\in I} \mathrm{D}_+(f_i) = \bigcup_{i\in I} \mathrm{D}_+(g) \cap \mathrm{D}_+(f_i) = \bigcup_{i\in I} \mathrm{D}_+(g\, f_i),$$

where of course $g\, f_i \in B_+$ for every $i \in I$. We will thus use

$$\mathcal{B} = \left\{ \mathrm{D}_+(f) \subset \operatorname{Proj} B \;\middle|\; f \in B_+ \text{ is homogeneous} \right\} \tag{2.5.2}$$

as a base of open sets for $\operatorname{Proj} B$.

2.5.2 Structure sheaf on $\operatorname{Proj} B$

Let B be a graded A-algebra as in the previous section. We want to define a sheaf of A-algebras $\mathscr{O}_X$ on $X = \operatorname{Proj} B$, making $(X, \mathscr{O}_X)$ into an A-scheme. Our working definition will be

$$\mathrm{D}_+(f) \longmapsto B_{(f)}, \quad f \in B_+ \text{ homogeneous.}$$

Here $B_{(f)}$ is the homogeneous principal localisation of Construction 2.5.1.4, which is an A-algebra by Diagram (2.5.1). In order to make sense of this and verify it is a $\mathcal{B}$-sheaf (i.e. a sheaf on the basis $\mathcal{B}$), we need some (algebraic) preparation.

LEMMA 2.5.2.1. *Let $f \in B_+$ be homogeneous of degree d. Set $B^{(d)} = \bigoplus_{e\geqslant 0} B_{de} \subset B$. Then, there is a ring isomorphism*

$$\alpha_f \colon B^{(d)}/(f-1)B^{(d)} \xrightarrow{\sim} B_{(f)}.$$

In particular, if $\deg f = 1$*, we have*

$$\alpha_f \colon B/(f-1)B \xrightarrow{\sim} B_{(f)}.$$

Proof. There is a surjective ring homomorphism $B^{(d)} \twoheadrightarrow B_{(f)}$ defined on homogeneous elements (and then extended additively) by sending $a \in B_{de}$ to a/f^e. This sends $f \in B_d = B_1^{(d)}$ to 1, so descends to a map α_f. The inverse is constructed as follows. Pick $w = z/f^n \in B_{(f)}$, so that z is homogeneous of degree $d\,n$. Send w to

$$\text{the image of } z \in B_{dn} \subset B^{(d)} \text{ along } B^{(d)} \twoheadrightarrow B^{(d)}/(f-1)B^{(d)}.$$

If $z/f^n = z'/f^{n'}$ in B_f, then for some $q \geqslant 0$,

$$f^q\left(f^{n'}z - f^n z'\right) = 0.$$

Modulo $(f-1)$ this equality becomes $z - z' = 0$. Hence the inverse is well defined. The two maps are visibly inverse ring homomorphisms. □

Terminology 2.5.2.2. The ring $B^{(d)}$ is called the *d-th Veronese ring* attached to B. It is an A-subalgebra of B.

Example 2.5.2.3. Let $B = A[x_0, x_1, \ldots, x_n]$ and $f = x_i$, which has degree 1. Then, $B^{(1)} = B$ and Lemma 2.5.2.1 yields

$$A[x_0, \ldots, x_n]_{(x_i)} \cong A[x_0, \ldots, x_n]/(x_i - 1) \cong A[x_0, \ldots, \widehat{x_i}, \ldots, x_n].^{12}$$

Let $\mathcal{B}$ be the base of the Zariski topology on $\operatorname{Proj} B$ as in Equation (2.5.2). Our next goal is to construct a $\mathcal{B}$-sheaf of rings on $X = \operatorname{Proj} B$. By Lemma 1.8.0.6, this will uniquely extend to a sheaf, which will be denoted $\mathscr{O}_X$.

If $f \in B_+$ is homogeneous, we have $\mathrm{D}_+(f) = \mathrm{D}(f) \cap \operatorname{Proj} B$. We next prove a few crucial properties of the composition

$$\begin{aligned} \theta \colon \mathrm{D}_+(f) &\hookrightarrow \mathrm{D}(f) = \operatorname{Spec} B_f \longrightarrow \operatorname{Spec} B_{(f)} \\ \mathfrak{p} &\longmapsto \mathfrak{p}B_f \cap B_{(f)}. \end{aligned}$$

LEMMA 2.5.2.4 ((De)homogenisation). *Let $f \in B_+$ be a homogeneous element.*

(i) *$\theta \colon \mathrm{D}_+(f) \to \operatorname{Spec} B_{(f)}$ is a homeomorphism.*

(ii) *If $\mathrm{D}_+(g) \subset \mathrm{D}_+(f)$ and $\alpha = g^{\deg f}/f^{\deg g} \in B_{(f)}$, then $\theta(\mathrm{D}_+(g)) = \mathrm{D}(\alpha)$.*

(iii) *If g and α are as in (ii), then there is a canonical homomorphism $B_{(f)} \to B_{(g)}$ inducing a ring isomorphism*

$$(B_{(f)})_\alpha \xrightarrow{\sim} B_{(g)}.$$

Proof. We proceed step by step.

(i)–(ii) The map θ is continuous, as we have already observed that the Zariski topology on $\operatorname{Proj} B$ is induced by that of $\operatorname{Spec} B$. We first need to prove it is bijective. Then, proving (ii) will show that it is open, hence a homeomorphism.

θ is injective: Suppose $\mathfrak{p}B_f \cap B_{(f)} = \mathfrak{p}'B_f \cap B_{(f)}$ for $\mathfrak{p}, \mathfrak{p}'$ two elements of $\mathrm{D}_+(f)$. Fix a homogeneous element $b \in \mathfrak{p}$, so that $b^{\deg f}/f^{\deg b} \in \mathfrak{p}B_f \cap B_{(f)} \subset \mathfrak{p}'B_f$. Then

[12] Notational warning: do not confuse $A[x_0, \ldots, x_n]_{(x_i)}$ (homogeneous localisation) with the localisation of $A[x_0, \ldots, x_n]$ at the prime ideal $(x_i) = x_i A[x_0, \ldots, x_n]$. Same potential problem when $(f) = fB \subset B$ is a prime ideal.

$b^{\deg f} \in \mathfrak{p}'$, and since $\mathfrak{p}'$ is prime we deduce $b \in \mathfrak{p}'$. Hence $\mathfrak{p} \subset \mathfrak{p}'$. Exchanging the roles of $\mathfrak{p}$ and $\mathfrak{p}'$ we obtain equality.

θ is surjective: Fix $\mathfrak{q} \in \operatorname{Spec} B_{(f)}$. Define a homogeneous ideal $\mathfrak{p} \subset B$ by declaring that $x \in B_d$ lies in $\mathfrak{p}$ if and only if $x^{\deg f}/f^d \in \mathfrak{q} \subset B_{(f)}$. It is a homogeneous ideal, and moreover it is prime. Indeed, pick two homogeneous elements $x \in B_d$ and $y \in B_e$ such that $xy \in \mathfrak{p}_{d+e}$. This means that

argument to be added

$$\frac{(xy)^{\deg f}}{f^{d+e}} = \frac{x^{\deg f}}{f^d}\frac{y^{\deg f}}{f^e} \in \mathfrak{q}.$$

Then either $x^{\deg f}/f^d$ or $y^{\deg f}/f^e$ lies in $\mathfrak{q}$, because $\mathfrak{q}$ is prime. But by definition this means that either $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. Moreover $f \notin \mathfrak{p}$, for otherwise we would have $1 \in \mathfrak{q}$. Finally, $\mathfrak{p}B_f \cap B_{(f)} = \mathfrak{q}$, which shows surjectivity.

(ii) holds: Fix $\mathfrak{p} \in \mathrm{D}_+(f)$. It is clear that $g \in \mathfrak{p}$ if and only if $\alpha \in \mathfrak{p}B_f \cap B_{(f)}$.

(iii) (sketch): If $\mathrm{D}_+(g) \subset \mathrm{D}_+(f)$, by Lemma 2.5.1.7(i) we have $gB = gB \cap B_+ \subset \sqrt{fB}$, i.e. $g^r = fb$ for some $b \in B$ and some $r > 0$. We may assume b is homogeneous by replacing it with its component of degree $r \cdot \deg g - \deg f$. Then $B_{(f)} \to B_{(g)}$ is defined by sending $a/f^n \mapsto ab^n/g^{rn}$. We are then in the situation

$$\begin{array}{ccc} B_{(f)} & \xrightarrow{\ell} & (B_{(f)})_\alpha \\ \downarrow & \swarrow & \\ B_{(g)} & & \end{array}$$

where ℓ is the (classical) localisation at $\alpha \in B_{(f)}$. Since α is invertible in $B_{(g)}$, the dotted arrow can be completed to a solid one. We leave it as an exercise to show that this map is an isomorphism. □

We are ready for the main theorem of this section.

THEOREM 2.5.2.5. *Let B be a graded A-algebra. Then $X = \operatorname{Proj} B$ is canonically an A-scheme, with the property that the principal open subset $\mathrm{D}_+(f) \subset X$ is an affine scheme isomorphic to $\operatorname{Spec} B_{(f)}$, for every homogeneous element $f \in B_+$.*

Moreover, the local ring $\mathcal{O}_{X,x}$ at a point $x \in X$ corresponding to a homogeneous prime $\mathfrak{p} \subset B$ is canonically isomorphic to the homogeneous localisation $B_{(\mathfrak{p})}$.

Proof. For $f \in B_+$ a homogeneous element, define

$$\mathcal{O}_X(\mathrm{D}_+(f)) = B_{(f)}. \tag{2.5.3}$$

We first confirm that this prescription defines a $\mathcal{B}$-presheaf on X. This, in fact, follows at once by Lemma 2.5.2.4(iii), which shows that a canonical restriction map $\mathcal{O}_X(\mathrm{D}_+(f)) \to \mathcal{O}_X(\mathrm{D}_+(g))$ exists whenever $\mathrm{D}_+(g) \subset \mathrm{D}_+(f)$, and that $B_{(f)}$ and $B_{(g)}$ are canonically isomorphic as soon as $\mathrm{D}_+(g) = \mathrm{D}_+(f)$.

In fact, (2.5.3) defines a $\mathcal{B}$-*sheaf* on X. This can be confirmed via the equaliser sequence. What we need to show is that for any $f \in B_+$ and any open cover $\mathrm{D}_+(f) = \bigcup_{i\in I} \mathrm{D}_+(f_i)$, the sequence

$$\mathscr{O}_X(\mathrm{D}_+(f)) \longrightarrow \prod_{i\in I} \mathscr{O}_X(\mathrm{D}_+(f_i)) \rightrightarrows \prod_{(i,j)\in I\times I} \mathscr{O}_X(\mathrm{D}_+(f_i f_j))$$

is an equaliser diagram in the category of A-algebras. This can be rewritten as the sequence

$$B_{(f)} \longrightarrow \prod_{i\in I} B_{(f_i)} \rightrightarrows \prod_{(i,j)\in I\times I} B_{(f_i f_j)}. \tag{2.5.4}$$

Let us define

$$\alpha_i = f_i^{\deg f}/f^{\deg f_i}, \quad \alpha_{ij} = (f_i f_j)^{\deg f}/f^{\deg f_i f_j}$$

in $B_{(f)}$. Then, we know that

$$\theta(\mathrm{D}_+(f_i)) = \mathrm{D}(\alpha_i), \quad \theta(\mathrm{D}_+(f_i f_j)) = \mathrm{D}(\alpha_{ij})$$

by Lemma 2.5.2.4(ii). Since $\theta\colon \mathrm{D}_+(f) \to \operatorname{Spec} B_{(f)}$ is a homeomorphism, we have an open covering

$$\operatorname{Spec} B_{(f)} = \bigcup_{i\in I} \theta(\mathrm{D}_+(f_i)) = \bigcup_{i\in I} \mathrm{D}(\alpha_i).$$

In particular, we have an equaliser sequence

$$B_{(f)} \longrightarrow \prod_{i\in I} (B_{(f)})_{\alpha_i} \rightrightarrows \prod_{(i,j)\in I\times I} (B_{(f)})_{\alpha_{ij}}$$

in the category of A-algebras, because $\mathscr{O}_{\operatorname{Spec} B_{(f)}}$ is a sheaf of A-algebras. But thanks to Lemma 2.5.2.4(iii) this is precisely the sequence (2.5.4). Therefore $\mathscr{O}_X$ is a $\mathcal{B}$-sheaf.

Let $\mathscr{O}_X$ denote the induced sheaf of A-algebras. The stalks are local rings. Indeed, if $x \in X$ corresponds to a homogeneous prime ideal $\mathfrak{p} \subset B$, one has a canonical isomorphism

$$\mathscr{O}_{X,x} = \varinjlim_{\substack{f \text{ homogeneous} \\ f\notin\mathfrak{p}}} B_{(f)} \xrightarrow{\sim} B_{(\mathfrak{p})}.$$

The proof is identical to the one we gave for Spec (cf. Theorem 2.1.4.3(c)).

It follows that the pair $(X, \mathscr{O}_X)$ defines a locally ringed space. Next, we confirm that the homeomorphism $\theta\colon \mathrm{D}_+(f) \to \operatorname{Spec} B_{(f)}$ extends to an isomorphism of locally ringed spaces

$$(\theta, \theta^\#)\colon (\mathrm{D}_+(f), \mathscr{O}_X|_{\mathrm{D}_+(f)}) \xrightarrow{\sim} (\operatorname{Spec} B_{(f)}, \mathscr{O}_{\operatorname{Spec} B_{(f)}}).$$

This shows that $(X, \mathscr{O}_X)$ is a scheme with the sought after property.

To construct $\theta^\#\colon \mathscr{O}_{\operatorname{Spec} B_{(f)}} \to \theta_*(\mathscr{O}_X|_{\mathrm{D}_+(f)})$, take a principal open $\mathrm{D}(\alpha) \subset \operatorname{Spec} B_{(f)}$.

If $\alpha = 0$, there is nothing to construct. Otherwise write $\alpha = a/f^r$, where a is homogeneous and $\deg a = r \deg f$. Set $d = \deg f$ and $g = af$. Then

$$\frac{g^d}{f^{\deg g}} = \frac{(af)^d}{f^{(r+1)d}} = \frac{a^d}{f^{rd}} = \alpha^d.$$

Lemma 2.5.2.4(ii) therefore gives $\theta^{-1}(\mathrm{D}(\alpha)) = D_+(g)$, since $\mathrm{D}(\alpha) = \mathrm{D}(\alpha^d)$. By Lemma 2.5.2.4(iii),

$$(B_{(f)})_\alpha = (B_{(f)})_{\alpha^d} \xrightarrow{\sim} B_{(g)}, \qquad g = af.$$

We take this canonical isomorphism to be $\theta^\#(\mathrm{D}(\alpha))$. This makes sense (i.e. the target is correct one), since

$$\theta_*(\mathscr{O}_X|_{\mathrm{D}_+(f)})(\mathrm{D}(\alpha)) = \mathscr{O}_X|_{\mathrm{D}_+(f)}(\theta^{-1}\,\mathrm{D}(\alpha)) = \mathscr{O}_X|_{\mathrm{D}_+(f)}(\mathrm{D}_+(g)) = \mathscr{O}_X(\mathrm{D}_+(g)) = B_{(g)}.$$

Finally, the A-scheme structure of X is given by the fact that each $B_{(f)}$ is naturally an A-algebra, combined with Corollary 2.4.3.4. □

Example 2.5.2.6. Let $B = A[x_0, x_1, \ldots, x_n]$, with the usual grading ($\deg x_i = 1$ for all i). Then

$$\operatorname{Proj} A[x_0, x_1, \ldots, x_n] = \mathbb{P}^n_A$$

where projective n-space over A was defined via gluing in Section 2.4.4. The structure morphism $\mathbb{P}^n_A \to \operatorname{Spec} A$ allows one to think of $\mathbb{P}^n_A$ as a 'family of projective spaces' parametrised by the points of $\operatorname{Spec} A$.

Example 2.5.2.7. We have $\mathbb{P}^0_A = \operatorname{Proj} A[x] \xrightarrow{\sim} \operatorname{Spec} A$. Indeed,

$$\operatorname{Proj} A[x] = \mathrm{D}_+(x) = \operatorname{Spec} A[x]_{(x)} = \operatorname{Spec} A[x]/(x-1) \xrightarrow{\sim} \operatorname{Spec} A,$$

using Example 2.5.2.3 for the identification $A[x]_{(x)} = A[x]/(x-1)$.

2.5.3 Proj is not a functor

One may think that, in analogy with the case of the affine spectrum, sending $B \mapsto \operatorname{Proj} B$ could be a functor from graded A-algebras to schemes. This is not the case. In this section we discuss why this fails and to what extent it can be remedied.



Caution 2.5.3.1. Proj is *not a functor*! It is not true that a morphism of graded A-algebras $\phi : B \to C$ induces a morphism of A-schemes $\operatorname{Proj} C \to \operatorname{Proj} B$ sending $\mathfrak{p} \mapsto \phi^{-1}\mathfrak{p}$. The problem is that

$$\mathfrak{p} \not\supset C_+ \text{ does not imply } \phi^{-1}\mathfrak{p} \not\supset B_+$$

for a homogeneous prime ideal $\mathfrak{p} \subset C$. See, however, Proposition 2.5.3.3 for the closest to a functor one can get.

Example 2.5.3.2. If $\phi : B = \mathbf{k}[x_0, x_1] \hookrightarrow \mathbf{k}[x_0, x_1, x_2] = C$ is the natural inclusion, then $C_+ = (x_0, x_1, x_2) \not\subset \mathfrak{p} = (x_0, x_1) \in \operatorname{Proj} C$, but $\phi^{-1}\mathfrak{p} = (x_0, x_1) = B_+$. This is the only 'problematic' point.

PROPOSITION 2.5.3.3. *Let $\phi : B \to C$ be a graded morphism of graded A-algebras. Then there is a canonical morphism of schemes*

$$f : \operatorname{Proj} C \setminus \mathrm{V}_+(B_+C) \longrightarrow \operatorname{Proj} B, \quad \mathfrak{p} \mapsto \phi^{-1}\mathfrak{p}$$

such that for any homogeneous $h \in B_+$ we have $f^{-1}(\mathrm{D}_+(h)) = \mathrm{D}_+(\phi(h))$, and the induced morphism $\mathrm{D}_+(\phi(h)) \to \mathrm{D}_+(h)$ of affine schemes corresponds to the canonical restriction $B_{(h)} \to C_{(\phi(h))}$.

Proof. If $\mathfrak{p} \subset C$ is a homogeneous prime ideal, then $\phi^{-1}\mathfrak{p} \subset B$ is a homogeneous prime ideal. We have

$$B_+ \not\subset \phi^{-1}\mathfrak{p} \iff \phi(B_+) \not\subset \mathfrak{p} \iff B_+C \not\subset \mathfrak{p},$$

therefore the association $\mathfrak{p} \mapsto \phi^{-1}\mathfrak{p}$ is well-defined precisely on the subset $\operatorname{Proj} C \setminus \mathrm{V}_+(B_+C) \subset \operatorname{Proj} C$, which, being open, has a canonical scheme structure inherited from $\operatorname{Proj} C$. Note that f is continuous, since the Zariski topology is induced by that of the affine spectrum, and f is the restriction of the same map $\mathfrak{p} \mapsto \phi^{-1}\mathfrak{p}$ going from $\operatorname{Spec} C$ to $\operatorname{Spec} B$.

We thus only need to construct the morphism at the level of structure sheaves. Since morphisms to a fixed target form a sheaf (cf. Lemma 2.4.3.1), it is enough to define the morphism on a base of open subsets of $\operatorname{Proj} C$. Consider principal open subsets

$$\mathrm{D}_+(h) \subset \operatorname{Proj} B, \quad \mathrm{D}_+(\phi(h)) \subset \operatorname{Proj} C, \quad h \in B_+.$$

As h runs in B_+, the opens $\mathrm{D}_+(h) \subset \operatorname{Proj} B$ cover the target $\operatorname{Proj} B$, and $\mathrm{D}_+(\phi(h)) \subset \operatorname{Proj} C$ cover $\operatorname{Proj} C \setminus \mathrm{V}_+(B_+C)$. In the commutative diagram

$$\begin{array}{ccc} B & \xrightarrow{\phi} & C \\ {\scriptstyle \ell_h}\downarrow & & \downarrow{\scriptstyle \ell_{\phi(h)}} \\ B_h & \xrightarrow{\phi_h} & C_{\phi(h)} \\ \uparrow & & \uparrow \\ B_{(h)} & \dashrightarrow & C_{(\phi(h))} \end{array}$$

the map ϕ_h is a graded morphism, therefore it preserves the degree 0 pieces, which induces $B_{(h)} \to C_{(\phi(h))}$. Taking Spec of this map recovers precisely the morphism of affine schemes $f_h : f^{-1}\mathrm{D}_+(h) = \mathrm{D}_+(\phi(h)) \to \mathrm{D}_+(h)$. These morphisms agree on the intersections (reason: the map $B_{hk} \to C_{\phi(hk)}$ agrees with both the localisation of $B_h \to C_{\phi(h)}$ and the localisation of $B_k \to C_{\phi(k)}$), and therefore glue to a global morphism f. □

Example 2.5.3.4 (Projection from a point). In the situation of Example 2.5.3.2, the morphism we obtain applying Proposition 2.5.3.3 is

$$\mathbb{P}^2_{\mathbf{k}} \setminus \mathrm{V}_+(x_0, x_1) = \mathbb{P}^2_{\mathbf{k}} \setminus \{(0:0:1)\} \longrightarrow \mathbb{P}^1_{\mathbf{k}}.$$

More generally, we have a morphism

$$\mathbb{P}^{n+1}_{\mathbf{k}} \setminus \{(0:\cdots:0:1)\} \longrightarrow \mathbb{P}^n_{\mathbf{k}}$$

defined by the inclusion of graded $\mathbf{k}$-algebras $\mathbf{k}[x_0,\ldots,x_n] \hookrightarrow \mathbf{k}[x_0,\ldots,x_n,x_{n+1}]$. This is called *projection from a point*. It sends the closed point $(a_0:\cdots:a_n:a_{n+1})$ to the closed point $(a_0:\cdots:a_n)$. In this example we chose to project from $(0:\cdots:0:1)$, but one can of course project from any other coordinate point in $\mathbb{P}^{n+1}_{\mathbf{k}}$.

Example 2.5.3.5. Consider the graded morphism $B = \mathbf{k}[x_0,x_1] \to \mathbf{k}[y_0,y_1] = C$ sending $x_i \mapsto y_i^n$ for $i = 0,1$. In this case, $\mathrm{V}_+(B_+C) = \mathrm{V}_+(y_0^n, y_1^n) = \mathrm{V}_+(y_0,y_1) = \emptyset$, so we get a well-defined morphism $\mathbb{P}^1_{\mathbf{k}} \to \mathbb{P}^1_{\mathbf{k}}$, which on closed points sends $(a_0:a_1) \mapsto (a_0^n : a_1^n)$.

Projective varieties

If $\phi : B \to C$ is a *surjective* graded morphism of graded A-algebras, we have $C_+ = \phi(B_+) = B_+C$, hence by Proposition 2.5.3.3 there is global A-morphism

$$f : \operatorname{Proj} C \to \operatorname{Proj} B,$$

locally given by the closed immersions

$$\operatorname{Spec} C_{(\phi(f))} \hookrightarrow \operatorname{Spec} B_{(f)}$$

induced by the natural surjections $B_{(f)} \twoheadrightarrow C_{(\phi(f))}$. Therefore f is a closed immersion of A-schemes. A special case of this will be recorded as the next corollary.

COROLLARY 2.5.3.6. *Let $I \subset B = A[x_0,x_1,\ldots,x_n]$ be a homogeneous ideal, and let $\phi : B \to B/I$ be the canonical surjection. Then $\phi(B_+) = (B/I)_+$. Therefore, the* Proj *construction yields a closed immersion*

$$\operatorname{Proj} B/I \hookrightarrow \mathbb{P}^n_A = \operatorname{Proj} B$$

over $\operatorname{Spec} A$*, with image homeomorphic to* $\mathrm{V}_+(I) \subset \operatorname{Proj} B$.

Terminology 2.5.3.7. Algebras of the form $A[x_0,x_1,\ldots,x_n]/I$ as in Corollary 2.5.3.6 are called *homogeneous A-algebras*. A scheme that is isomorphic to a closed subscheme of $\mathbb{P}^n_A$, for some $n \geqslant 0$, is called a *projective scheme over A*.

The converse of Corollary 2.5.3.6 is the content of the following exercise.



Exercise 2.5.3.8 (Different ideals, same closed subscheme). Let $S = \mathbb{F}[x_0,x_1]$, and set

$$I = (x_0^2, x_0x_1), \qquad J = (x_0).$$

Show that $I \neq J$ but that $\operatorname{Proj} S/I$ and $\operatorname{Proj} S/J$ define the same closed subscheme of $\mathbb{P}^1_{\mathbb{F}}$. (**Hint**: compare their restrictions to $\mathrm{D}_+(x_0)$ and $\mathrm{D}_+(x_1)$).

The converse statement — that every closed subscheme of projective space is defined by a homogeneous ideal, is the following (nontrivial) exercise.



This needs a hint at this point

Exercise 2.5.3.9. Let $X \hookrightarrow \mathbb{P}^n_A$ be a closed immersion. Show that there exists a homogeneous ideal $I \subset B = A[x_0, x_1, \ldots, x_n]$ such that X is isomorphic to $\operatorname{Proj} B/I$.

Important Definition 2.5.3.10 (Projective variety)**.** A *projective variety* over a field $\mathbb{F}$ is a projective scheme over $\mathbb{F}$, i.e. a closed subscheme of $\mathbb{P}^n_{\mathbb{F}}$ for some n. An algebraic variety is called *quasiprojective* (resp. *quasiaffine*) if it admits an open immersion into a projective (resp. affine) variety.

A morphism $X \to Y = \operatorname{Spec} A$ is said to be *projective* if it factors as a closed immersion $X \hookrightarrow \mathbb{P}^n_A$ followed by the canonical projection $\mathbb{P}^n_A \to Y$.

Remark 2.5.3.11. Sanity check: projective varieties are algebraic varieties in the sense of Important Definition 2.4.1.1, by the observation (cf. Remark 2.4.4.3) that $\mathbb{P}^n_{\mathbb{F}}$ is quasicompact. In a little more detail, use the finite standard affine open cover

$$\mathbb{P}^n_{\mathbb{F}} = \bigcup_{i=0}^{n} \mathrm{D}_+(x_i) = \bigcup_{i=0}^{n} \operatorname{Spec} \mathbb{F}[x_0, \ldots, x_n]_{(x_i)}$$

to induce, for a closed subscheme $X \hookrightarrow \mathbb{P}^n_{\mathbb{F}}$, the open cover

$$X = \bigcup_{i=0}^{n} X \cap \operatorname{Spec} \mathbb{F}[x_0, \ldots, x_n]_{(x_i)}.$$

Now, each $X_i = X \cap \operatorname{Spec} \mathbb{F}[x_0, \ldots, x_n]_{(x_i)}$ is an affine scheme (because it is closed in the affine scheme $\operatorname{Spec} \mathbb{F}[x_0, \ldots, x_n]_{(x_i)}$), and therefore its coordinate ring is a quotient of the finitely generated $\mathbb{F}$-algebra

$$\mathbb{F}[x_0, \ldots, x_n]_{(x_i)} \cong \mathbb{F}[x_0, \ldots, x_n]/(x_i - 1)\mathbb{F}[x_0, \ldots, x_n],$$

cf. Lemma 2.5.2.1, therefore it is itself a finitely generated $\mathbb{F}$-algebra.

Terminology 2.5.3.12. Let $B = \mathbf{k}[x_0, \ldots, x_n]$. Fix a homogeneous polynomial $f \in B$ of degree $d > 0$. The closed subscheme $X = \operatorname{Proj} B/(f) \hookrightarrow \mathbb{P}^n_{\mathbf{k}}$ is called a hypersurface of degree d. If $d = 1$ (resp. $d = 2, 3, 4, 5, \ldots$), we say X is a *hyperplane* (resp. a *quadric*, a *cubic*, a *quartic*, a *quintic*... hypersurface).



Caution 2.5.3.13. An isomorphism $\operatorname{Proj} B \cong \operatorname{Proj} C$ need not come from an isomorphism of graded A-algebras. For example, if $B = \mathbb{F}[x_0, x_1]$ with its standard grading, Proposition 2.5.4.9 gives $\operatorname{Proj} B \cong \operatorname{Proj} B^{(d)}$ for every $d \geqslant 1$. For $d > 1$, however, $B \ncong B^{(d)}$ as graded $\mathbb{F}$-algebras: indeed,

$$\dim_{\mathbb{F}} B_1 = 2, \qquad \dim_{\mathbb{F}} B_1^{(d)} = \dim_{\mathbb{F}} B_d = d + 1.$$



Exercise 2.5.3.14. Show that every morphism $\mathbb{P}^n_F \to \mathbb{A}^m_{\mathbb{F}}$ is constant. (**Hint**: use $\Gamma(\mathbb{P}^n_{\mathbb{F}}, \mathscr{O}) = \mathbb{F}$ and Theorem 2.4.3.3.)

not ideal here

Exercise 2.5.3.15. Show that there is no nonconstant morphism $\mathbb{P}^n_{\mathbb{F}} \to \mathbb{P}^m_{\mathbb{F}}$ if $m < n$.

2.5.4 Examples of projective schemes

Let $\mathbb{F}$ be a field. First of all, let us clarify the relationship between the scheme $\mathbb{P}^n_{\mathbb{F}}$ and classical projective space $\mathbb{P}^n(\mathbb{F}) = (\mathbb{F}^{n+1} \setminus 0)/\mathbb{F}^\times$. There is a set-theoretic map

$$\mathbb{P}^n(\mathbb{F}) \longrightarrow \mathbb{P}^n_{\mathbb{F}} = \operatorname{Proj} \mathbb{F}[x_0, x_1, \ldots, x_n] \tag{2.5.5}$$

sending a point $(a_0 : a_1 : \cdots : a_n)$ to the homogeneous prime ideal

$$(a_i x_j - a_j x_i \,|\, 0 \leqslant i, j \leqslant n) \subset \mathbb{F}[x_0, x_1, \ldots, x_n]. \tag{2.5.6}$$

Note that such ideal can be viewed as generated by the 2-minors of the matrix

$$\begin{pmatrix} x_0 & x_1 & \cdots & x_n \\ a_0 & a_1 & \cdots & a_n \end{pmatrix}.$$

The map (2.5.5) is a bijection onto the set of $\mathbb{F}$-rational points of $\mathbb{P}^n_{\mathbb{F}}$, namely those points x for which the structural map $\mathbb{F} \to \kappa(x)$ is an isomorphism. These points are all closed, but if $\mathbb{F}$ is not algebraically closed there may be *more* closed points; their residue field is going to be a finite extension of $\mathbb{F}$. The points of the form (2.5.6) will be referred to as the *classical points* of $\mathbb{P}^n_{\mathbb{F}}$.



Exercise 2.5.4.1 (Rational points and closed points of projective space). Let $\mathbb{F}$ be an arbitrary field.

1. Prove that $\mathbb{P}^n(\mathbb{F})$ is naturally identified with the set of $\mathbb{F}$-rational points of the scheme $\mathbb{P}^n_{\mathbb{F}}$.
2. Prove that a closed point $x \in \mathbb{P}^n_{\mathbb{F}}$ has residue field $\kappa(x)$ finite over $\mathbb{F}$.
3. Give a closed point of $\mathbb{P}^1_{\mathbb{R}}$ which is not $\mathbb{R}$-rational.

Terminology 2.5.4.2. If we set $\mathbb{P}^n_{\mathbb{F}} = \operatorname{Proj} \mathbb{F}[x_0, x_1, \ldots, x_n]$, we call $(x_0, x_1, \ldots, x_n)$ the homogeneous coordinates of $\mathbb{P}^n_{\mathbb{F}}$. The $\mathbb{F}$-rational point of $\mathbb{P}^n_{\mathbb{F}}$ corresponding to (2.5.6) is denoted $(a_0 : a_1 : \cdots : a_n)$. For instance, the 'classical point' $(0 : 0 : \cdots : 1) \in \mathbb{P}^n(\mathbb{F})$ corresponds to the ideal $(x_0, x_1, \ldots, x_{n-1}) \in \mathbb{P}^n_{\mathbb{F}}$.

Remark 2.5.4.3. Assume $\mathbf{k}$ is an algebraically closed field. In the case of $\mathbb{P}^1_{\mathbf{k}}$, just as for $\mathbb{A}^1_{\mathbf{k}}$, there is only one nonclassical point, namely the point ξ corresponding to the trivial ideal $(0) \subset \mathbf{k}[x_0, x_1]$. It is the generic point of $\mathbb{P}^1_{\mathbf{k}}$, and one has $\kappa(\xi) = \mathbf{k}(t)$, with $t = x_0/x_1$.

Remark 2.5.4.4. If $\mathfrak{p} \in X = \operatorname{Proj} B$, the stalk of the structure sheaf $\mathscr{O}_X$ at $\mathfrak{p}$ is the homogeneous localisation $B_{(\mathfrak{p})}$. Say $U = \operatorname{Spec} B_{(f)} \subset X$ is a principal open neighbourhood containing $\mathfrak{p}$. Set $\mathfrak{q} = \mathfrak{p} B_f \cap B_{(f)} \subset B_{(f)}$ and $R = B_{(f)}$. Then

$$\mathscr{O}_{X,\mathfrak{p}} = B_{(\mathfrak{p})} \cong R_{\mathfrak{q}} = \mathscr{O}_{U,\mathfrak{p}}.$$

The local ring at a point does not change in an open neighbourhood, but the rings we localise do change. Let us try and understand what ideal $\mathfrak{p} \subset B$ becomes when viewed in the ring $R = B_{(f)}$. We explain this via an example. Consider for instance the 'coordinate point'

$$z_i = (0 : \cdots : 0 : 1 : 0 : \cdots : 0) \in \mathbb{P}^n_{\mathbb{F}},$$

with 1 sitting in the $(i+1)$-st slot, corresponding to the homogeneous prime ideal

$$\mathfrak{p}_i = (x_0, \ldots, \widehat{x_i}, \ldots, x_n) \subset \mathbb{F}[x_0, x_1, \ldots, x_n].$$

Then, we have $z_i \in \mathrm{D}_+(x_i) = \operatorname{Spec} \mathbb{F}[x_0, x_1, \ldots, x_n]_{(x_i)}$, and

$$\mathbb{F}[x_0, x_1, \ldots, x_n]_{(x_i)} \cong \mathbb{F}[x_0, \ldots, \widehat{x_i}, \ldots, x_n]$$

by Lemma 2.5.2.1. Under this identification, z_i corresponds to the origin in

$$\operatorname{Spec} \mathbb{F}[x_0, \ldots, \widehat{x_i}, \ldots, x_n],$$

which in turn corresponds to the ideal $\mathfrak{q}_i = (x_0, \ldots, \widehat{x_i}, \ldots, x_n)$. Therefore

$$\mathcal{O}_{\mathbb{P}^n_{\mathbb{F}}, z_i} = \mathbb{F}[x_0, x_1, \ldots, x_n]_{(\mathfrak{p}_i)} = \mathbb{F}[x_0, \ldots, \widehat{x_i}, \ldots, x_n]_{\mathfrak{q}_i},$$

which consists of fractions f/g of polynomials in n variables, where $g(0, \ldots, 0) \neq 0 \in \mathbb{F}$.

Terminology 2.5.4.5. Let V be an $\mathbb{F}$-vector space of dimension $n+1$. After choosing a basis of V, the symmetric algebra $\operatorname{Sym} V^\vee$ is identified with a polynomial ring $\mathbb{F}[x_0, \ldots, x_n]$. Intrinsically, one defines

$$\mathbb{P}(V) = \operatorname{Proj} \operatorname{Sym} V^\vee$$

This is the projective space attached to V, whose $\mathbb{F}$-rational points correspond to lines in V. It is isomorphic to $\mathbb{P}^n_{\mathbb{F}}$.

Example 2.5.4.6 (Plane curves). Consider the same polynomial $xy - z^2 \in \mathbb{C}[x, y, z]$ of Example 2.3.3.15. Note that it is homogeneous. Now, its vanishing locus $\mathrm{V}_+(xy - z^2)$ is the topological image of a closed immersion into $\mathbb{P}^2_{\mathbb{C}} = \operatorname{Proj} \mathbb{C}[x, y, z]$, namely the morphism

$$\operatorname{Proj} \mathbb{C}[x, y, z]/(xy - z^2) \hookrightarrow \mathbb{P}^2_{\mathbb{C}}$$

induced by the surjection $\mathbb{C}[x, y, z] \twoheadrightarrow \mathbb{C}[x, y, z]/(xy - z^2)$. In general, the vanishing scheme of a degree 2 nonzero homogeneous polynomial $f \in \mathbb{C}[x, y, z]$ is called a *plane conic*. The vanishing scheme of an arbitrary nonzero homogeneous polynomial of degree d is called a *plane curve of degree d*.



Exercise 2.5.4.7. Show that all nondegenerate plane conics $\operatorname{Proj} \mathbb{C}[x, y, z]/(f) \hookrightarrow \mathbb{P}^2_{\mathbb{C}}$ are isomorphic to $\mathbb{P}^1_{\mathbb{C}}$ (**Hint**: Show that you can reduce to the normal form $f = xy - z^2$, or $f = x^2 + y^2 + z^2$ if you prefer; then show that this particular plane conic is isomorphic

to $\mathbb{P}^1_{\mathbb{C}}$). In fact, you may want to show this over an arbitrary algebraically closed field $\mathbf{k}$ of characteristic different from 2. Note that the statement is false over a general field: nonsplit conics exist!

Example 2.5.4.8. Let B be a graded A-algebra. Fix $d > 0$. Consider the Veronese ring $B^{(d)} = \bigoplus_{e \geqslant 0} B_e^{(d)}$, defined by $B_e^{(d)} = B_{de}$. We have an inclusion

$$\phi \colon B^{(d)} \hookrightarrow B,$$

which is a graded homomorphism of graded A-algebras. We claim that ϕ induces an A-morphism

$$v_d \colon \operatorname{Proj} B \longrightarrow \operatorname{Proj} B^{(d)}.$$

We have

$$B_+^{(d)} = \bigoplus_{e>0} B_{de}.$$

To prove that v_d is well-defined, fix a point $\mathfrak{p} \in \operatorname{Proj} B$. Assume, by contradiction, that $\phi^{-1}\mathfrak{p} \supset B_+^{(d)}$. Then $\mathfrak{p} \supset \phi(\phi^{-1}\mathfrak{p}) \supset \phi(B_+^{(d)}) = B_+ \cap \phi(B^{(d)})$. If $a \in B_+$, then $a^d \in B_+ \cap \phi(B^{(d)}) \subset \mathfrak{p}$, which implies $a \in \mathfrak{p}$, and in turn $B_+ \subset \mathfrak{p}$, whence a contradiction. Thus v_d is globally well-defined by Proposition 2.5.3.3.

PROPOSITION 2.5.4.9. *Let B be a graded A-algebra, and fix $d > 0$. Then the morphism*

$$v_d \colon \operatorname{Proj} B \longrightarrow \operatorname{Proj} B^{(d)}$$

is an isomorphism of A-schemes.

Proof. We first confirm that v_d, sending $\mathfrak{p} \mapsto \mathfrak{p} \cap B^{(d)}$, is a homeomorphism, and then we show that it is an isomorphism on an open cover of the source. To prove it is a homeomorphism, we prove it is a continuous and open bijection. Continuity is automatic given that v_d is in fact a morphism, by Example 2.5.4.8.

To see that v_d is injective, assume $\mathfrak{p} \cap B^{(d)} = \mathfrak{p}' \cap B^{(d)}$ for $\mathfrak{p}, \mathfrak{p}' \in \operatorname{Proj} B$. If $a \in B$ is a homogeneous element, we have $a \in \mathfrak{p}$ if and only if $a^d \in \mathfrak{p}$, if and only if $a^d \in \mathfrak{p}'$, if and only if $a \in \mathfrak{p}'$, thus $\mathfrak{p} = \mathfrak{p}'$.

To see that v_d is surjective, fix $\mathfrak{q} \in \operatorname{Proj} B^{(d)}$ and define

$$\mathfrak{p} = \bigoplus_{e \geqslant 0} \left\{ x \in B_e \;\middle|\; x^d \in \mathfrak{q} \right\} \subset B.$$

The same binomial/radical argument used in the surjectivity part of Lemma 2.5.2.4 shows that $\mathfrak{p}$ is closed under addition. Explicitly, if $x, y \in B_e$ and $x^d, y^d \in \mathfrak{q}$, then for every i,

$$(x^i y^{d-i})^d = (x^d)^i (y^d)^{d-i} \in \mathfrak{q},$$

hence $x^i y^{d-i} \in \mathfrak{q}$, and the binomial expansion gives $(x+y)^d \in \mathfrak{q}$. Closure under multiplication follows from $(zx)^d = z^d x^d$, and primality follows from $(xy)^d = x^d y^d$. Thus $\mathfrak{p}$ is a homogeneous prime ideal and $\mathfrak{p} \cap B^{(d)} = \mathfrak{q}$.

Thus v_d is bijective on points.

To see that v_d is a homeomorphism, it is now enough to observe that

$$v_d(\mathrm{D}_+(f)) = \mathrm{D}_+(f^d),$$

for any homogeneous element $f \in B_+$. This follows from the observation that, for any $\mathfrak{p} \in \operatorname{Proj} B$, one has $f \in \mathfrak{p}$ if and only if $f^d \in \mathfrak{p} \cap B^{(d)}$.

The homeomorphisms

$$\mathrm{D}_+(f) \longrightarrow \mathrm{D}_+(f^d)$$

extend to morphisms of affine schemes

$$\operatorname{Spec} B_{(f)} \longrightarrow \operatorname{Spec} B^{(d)}_{(f^d)},$$

each corresponding to the canonical A-algebra homomorphism

$$\psi_f \colon B^{(d)}_{(f^d)} \longrightarrow B_{(f)}. \tag{2.5.7}$$

Such morphism is an isomorphism. Indeed, it is clearly injective, and to check it is surjective we observe that if $a/f^n \in B_{(f)}$, we may choose an integer $m \in \mathbb{Z}_{\geqslant 0}$ such that $n+m$ is divisible by d and write

$$\frac{a}{f^n} = \frac{af^m}{f^{n+m}} = \frac{af^m}{f^{kd}}$$

for some k. The numerator and denominator now lie in the appropriate Veronese degrees, showing that ψ_f is onto. The isomorphisms (2.5.7) are compatible with restrictions to smaller principal opens, thus v_d is globally an isomorphism. □

Example 2.5.4.10 (Veronese embedding). Fix a pair of positive integers n, d. Set $B = \mathbb{F}[x_0, \ldots, x_n]$. We construct a closed immersion

$$v_{d,n} \colon \mathbb{P}^n_{\mathbb{F}} \hookrightarrow \mathbb{P}^N_{\mathbb{F}}, \quad N = \tbinom{n+d}{d} - 1$$

as follows. Let $\{ m_0, \ldots, m_N \} \subset B_d$ be linearly independent monomials of degree d in the variables $x_0, \ldots, x_n$. In other words, fix the standard monomial basis of B_d. Fix indeterminates $w_0, \ldots, w_N$. Define an $\mathbb{F}$-algebra homomorphism

$$\mathbb{F}[w_0, \ldots, w_N] \longrightarrow B, \quad w_i \mapsto m_i.$$

This morphism has image the Veronese subalgebra $B^{(d)} \hookrightarrow B$, therefore the factorisation

$$\mathbb{F}[w_0, \ldots, w_N] \twoheadrightarrow B^{(d)} \hookrightarrow B$$

induces

$$v_{d,n} \colon \mathbb{P}^n_{\mathbb{F}} \xrightarrow{\sim} \operatorname{Proj} B^{(d)} \hookrightarrow \mathbb{P}^N_{\mathbb{F}}.$$

This is the d-th *Veronese embedding* of $\mathbb{P}^n_{\mathbb{F}}$. It is also called the *d-uple embedding* of $\mathbb{P}^n_{\mathbb{F}}$. The next two examples are important special cases.

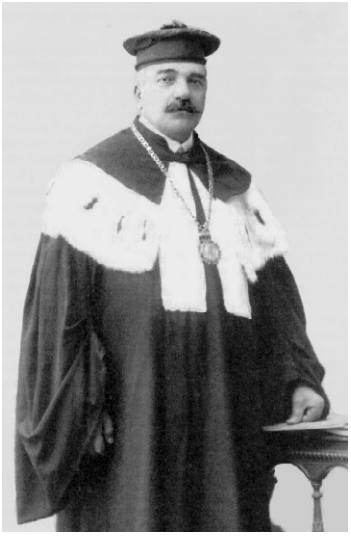


Figure 2.14: Giuseppe Veronese (1854–1917).

Example 2.5.4.11 (Rational normal curve)**.** Let $d > 0$ be an integer. The d-th Veronese embedding of $\mathbb{P}^1_{\mathbb{F}}$ is the map

$$\begin{aligned} \mathbb{P}^1_{\mathbb{F}} &\xhookrightarrow{\quad v_{d,1} \quad} \mathbb{P}^d_{\mathbb{F}} \\ (u:v) &\longmapsto (u^d : u^{d-1}v : \cdots : uv^{d-1} : v^d) \end{aligned}$$

defined by the map

$$\begin{aligned} \mathbb{F}[x_0, x_1, \ldots, x_d] &\longrightarrow \mathbb{F}[u, v] \\ x_i &\longmapsto u^{d-i}v^i. \end{aligned}$$

The image of this closed immersion is called the *rational normal curve* in $\mathbb{P}^d_{\mathbb{F}}$. If $d = 3$, the image of $\mathbb{P}^1_{\mathbb{F}} \hookrightarrow \mathbb{P}^3_{\mathbb{F}}$ is called a *twisted cubic in* $\mathbb{P}^3_{\mathbb{F}}$. Observe that the closed immersion defined above is cut out by the ideal J_d generated by the 2-minors of the matrix

$$\begin{pmatrix} x_0 & x_1 & \cdots & x_{d-1} \\ x_1 & x_2 & \cdots & x_d \end{pmatrix}.$$

For instance, the twisted cubic is cut out by the ideal

$$J_3 = (x_0x_3 - x_1x_2, x_1^2 - x_0x_2, x_2^2 - x_1x_3) \subset \mathbb{F}[x_0, x_1, x_2, x_3].$$

Example 2.5.4.12. Set $d = n = 2$. Then we have a factorisation

$$v_{2,2} \colon \mathbb{P}^2_{\mathbb{F}} \xrightarrow{\sim} S \hookrightarrow \mathbb{P}^5_{\mathbb{F}}$$

where S is called the *Veronese surface*. Consider the symmetric matrix

$$M = \begin{pmatrix} w_0 & w_1 & w_3 \\ w_1 & w_2 & w_4 \\ w_3 & w_4 & w_5 \end{pmatrix}.$$

One may show that, as schemes,

$$S = \operatorname{Proj} \mathbb{F}[w_0, \ldots, w_5]/J,$$

where J is the ideal generated by the 2-minors of M.

Example 2.5.4.13. Let B be a graded A-algebra with irrelevant ideal $B_+ \subset B$. For each homogeneous $f \in B_+$, the inclusion $B_{(f)} \hookrightarrow B_f$ induces a morphism

$$\mathrm{D}(f) = \operatorname{Spec} B_f \longrightarrow \mathrm{D}_+(f) = \operatorname{Spec} B_{(f)}.$$

These morphisms agree on overlaps and therefore glue to a morphism

$$\operatorname{Spec} B \setminus \mathrm{V}(B_+) \longrightarrow \operatorname{Proj} B.$$

Its underlying map sends $\mathfrak{p}$ to $\mathfrak{p}^{\mathrm{h}}$. For instance, we get a 'projection'

$$\mathbb{A}^{n+1}_A \setminus \mathrm{V}(x_0, x_1, \ldots, x_n) \longrightarrow \mathbb{P}^n_A.$$

If $\mathbb{F}$ is a field, this morphism is precisely, on $\mathbb{F}$-rational points, the *quotient* by the scaling action of $\mathbb{F}^\times$ on $\mathbb{F}^{n+1} \setminus \{0\}$.

2.5.5 (De)homogenisation and projective closure

Let $\mathbb{F}$ be a field, $n \geqslant 1$ an integer, and set

$$A = \mathbb{F}[y_1, \ldots, y_n], \qquad S = \mathbb{F}[x_0, x_1, \ldots, x_n], \qquad \deg x_i = 1.$$

The standard affine chart

$$U_0 = \mathrm{D}_+(x_0) \subset \mathbb{P}^n_{\mathbb{F}} = \operatorname{Proj} S$$

is canonically isomorphic to $\mathbb{A}^n_{\mathbb{F}} = \operatorname{Spec} A$ via

$$S_{(x_0)} = \left\{ \frac{G}{x_0^d} \;\middle|\; d \geqslant 0,\ g \in S_d \right\} \xrightarrow{\ \sim\ } A,$$

$$\frac{G}{x_0^d} \longmapsto G(1, y_1, \ldots, y_n).$$

Under this isomorphism, y_i corresponds to the affine coordinate x_i/x_0. Everything below has an analogous version on $\mathrm{D}_+(x_i)$, obtained by setting $x_i = 1$ and using the ratios x_j/x_i, for $j \neq i$.

Dehomogenising and homogenising polynomials

Definition 2.5.5.1 (Dehomogenisation)**.** Dehomogenisation with respect to x_0 is the surjective ring homomorphism

$$S \xrightarrow{\ \delta_0\ } A, \qquad G(x_0, x_1, \ldots, x_n) \longmapsto G(1, y_1, \ldots, y_n).$$

For $G \in S$, we also write $G^{\mathsf{deh}} = \delta_0(G)$. Although δ_0 is a ring homomorphism, it does not preserve the grading.

If $G \in S_d$ is homogeneous, then G^{deh} represents the regular function G/x_0^d on U_0.

Definition 2.5.5.2 (Homogenisation)**.** Let

$$f = \sum_{|\alpha| \leqslant d} c_\alpha y^\alpha \in A$$

be a nonzero polynomial of total degree d. Its *homogenisation* with respect to x_0 is

$$f^{\mathsf{hom}} = x_0^d f(x_1/x_0, \ldots, x_n/x_0) = \sum_{|\alpha| \leqslant d} c_\alpha x_0^{d-|\alpha|} x^\alpha \in S_d.$$

We set $0^{\mathsf{hom}} = 0$.

For nonzero $f, g \in A$, one has

$$(f^{\mathsf{hom}})^{\mathsf{deh}} = f, \qquad (fg)^{\mathsf{hom}} = f^{\mathsf{hom}} g^{\mathsf{hom}}, \qquad x_0 \nmid f^{\mathsf{hom}}.$$

The reverse operation loses precisely the largest power of x_0 dividing a homogeneous polynomial. Indeed, if $G \in S_d \setminus \{0\}$ and

$$G = x_0^r H, \qquad x_0 \nmid H,$$

then $\deg G^{\mathsf{deh}} = d - r$ and

$$\left(G^{\mathsf{deh}}\right)^{\mathsf{hom}} = H = G/x_0^r.$$

It is sometimes useful to homogenise to a prescribed degree. If $D \geqslant \deg f$, define

$$h_D(f) = x_0^D f(x_1/x_0, \ldots, x_n/x_0) = x_0^{D - \deg f} f^{\mathsf{hom}}.$$

For a fixed D, the operation h_D is $\mathbb{F}$-linear on the vector space of polynomials of degree at most D.



Warning 2.5.5.3 (Homogenisation is not a ring homomorphism)**.** Standard homogenisation uses the actual degree of each polynomial and is not additive. For example,

$$(y_1)^{\mathsf{hom}} = x_1, \qquad 1^{\mathsf{hom}} = 1, \qquad (y_1 + 1)^{\mathsf{hom}} = x_1 + x_0 \neq x_1 + 1.$$

Additivity is recovered only after choosing one common target degree D.

Homogeneous ideals

Let $I \subset A$ be an ideal. Its homogenisation is the homogeneous ideal

$$I^{\mathsf{hom}} = (f^{\mathsf{hom}} \mid f \in I) \subset S.$$

The phrase *for every* $f \in I$ is essential. In general, homogenising a chosen collection of generators does not yet produce I^{hom}.

For an ideal $J \subset S$, write

$$(J : x_0^N) = \left\{ G \in S \mid x_0^N G \in J \right\}, \qquad (J : x_0^\infty) = \bigcup_{N \geqslant 0} (J : x_0^N).$$

The latter, $(J : x_0^\infty)$, is called the x_0-*saturation* of J. It is an ideal: it equals $JS_{x_0} \cap S$.

PROPOSITION 2.5.5.4 (Homogenisation and saturation). *Let $I = (f_1, \dots, f_r) \subset A$, and put $J = (f_1^{\mathsf{hom}}, \dots, f_r^{\mathsf{hom}}) \subset S$. Then, the following statements hold true.*

(i) *There is an identity*
$$I^{\mathsf{hom}} = \bigoplus_{d \geqslant 0} \{G \in S_d \mid G^{\mathsf{deh}} \in I\}.$$
In particular, $(I^{\mathsf{hom}} : x_0) = I^{\mathsf{hom}} = (I^{\mathsf{hom}} : x_0^\infty)$.

(ii) *There is an identity* $I^{\mathsf{hom}} = (J : x_0^\infty)$.

(iii) *There is a canonical isomorphism* $(S/I^{\mathsf{hom}})_{(x_0)} \cong A/I$. *Thus, there is an isomorphism*
$$\operatorname{Proj} S/I^{\mathsf{hom}} \cap U_0 \cong \operatorname{Spec} A/I.$$

Proof. Suppose that $G \in S_d$ and $G^{\mathsf{deh}} \in I$. If $G \neq 0$, write $G = x_0^r H$, where $x_0 \nmid H$. Then
$$H = \left(G^{\mathsf{deh}}\right)^{\mathsf{hom}} \in I^{\mathsf{hom}},$$
and therefore $G \in I^{\mathsf{hom}}$. The converse follows by applying dehomogenisation to I^{hom}. This proves the description in (i). It also proves x_0-saturation, degree by degree.

Certainly $J \subset I^{\mathsf{hom}}$, so $(J : x_0^\infty) \subset I^{\mathsf{hom}}$ by (i). Conversely, take $f \in I$, write
$$f = \sum_{i=1}^{r} a_i f_i,$$
and omit the zero summands. Choose D at least as large as $\deg f$ and every $\deg a_i + \deg f_i$. Homogenising every term to this common degree gives
$$x_0^{D - \deg f} f^{\mathsf{hom}} = \sum_{i=1}^{r} x_0^{D - \deg a_i - \deg f_i} a_i^{\mathsf{hom}} f_i^{\mathsf{hom}} \in J.$$
Thus $f^{\mathsf{hom}} \in (J : x_0^\infty)$, proving (ii). Finally, (iii) follows by dehomogenising on $D_+(x_0)$. □

The scheme-theoretic projective closure

Let $i \colon Y \hookrightarrow P$ be a quasicompact immersion of schemes. We will see in **??** that the *scheme-theoretic closure* of Y in P is the smallest closed subscheme of P through which i factors. It is defined by the ideal sheaf
$$\mathscr{I} = \ker(\mathscr{O}_P \xrightarrow{i^\#} i_* \mathscr{O}_Y) \subset \mathscr{O}_P.$$

Going back to our setup, let
$$Y = \operatorname{Spec} A/I \xhookrightarrow{\iota} U_0 \cong \mathbb{A}^n_{\mathbb{F}}$$
be the closed subscheme defined by an ideal $I \subset A = \mathbb{F}[y_1, \dots, y_n]$, and let
$$U_0 \xhookrightarrow{j} \mathbb{P}^n_{\mathbb{F}}$$

denote the principal open immersion corresponding to $x_0 \neq 0$. We denote by $\overline{Y}$ the scheme-theoretic image of

$$i = j \circ \iota \colon Y \hookrightarrow \mathbb{P}^n_{\mathbb{F}},$$

namely the scheme-theoretic closure of Y in the projective space $\mathbb{P}^n_{\mathbb{F}}$. This is sometimes also called the *projective closure* of Y.

PROPOSITION 2.5.5.5 (Projective closure). *The scheme-theoretic closure of Y in $\mathbb{P}^n_{\mathbb{F}}$ is*

$$\overline{Y} = \operatorname{Proj} S/I^{\mathsf{hom}} \hookrightarrow \mathbb{P}^n_{\mathbb{F}}.$$

Moreover, there is a scheme-theoretic identity

$$\overline{Y} \cap U_0 = Y.$$

Proof. Set $Z = \operatorname{Proj} S/I^{\mathsf{hom}} \subset \mathbb{P}^n_{\mathbb{F}}$. By dehomogenising on $U_0 = \mathrm{D}_+(x_0)$, we obtain

$$Z \cap U_0 = \operatorname{Spec}(S/I^{\mathsf{hom}})_{(x_0)} \cong \operatorname{Spec} A/I = Y$$

as schemes.

It remains to prove minimality. If $i = j \circ \iota \colon Y \to \mathbb{P}^n_{\mathbb{F}}$, the ideal sheaf of the scheme-theoretic closure of Y is

$$\mathscr{I}_{\overline{Y}} = \ker(\mathscr{O}_{\mathbb{P}^n_{\mathbb{F}}} \xrightarrow{i^\#} i_*\mathscr{O}_Y).$$

On a standard chart $\mathrm{D}_+(x_i)$, it is represented by the homogeneous localisation

$$(I^{\mathsf{hom}})_{(x_i)} \subset S_{(x_i)}.$$

Let us prove minimality directly, as follows. Let $W = \operatorname{Proj} S/K \hookrightarrow \mathbb{P}^n_{\mathbb{F}}$ be a closed subscheme through which $Y \hookrightarrow \mathbb{P}^n_{\mathbb{F}}$ factors, where $K \subset S$ is a saturated homogeneous ideal. If $G \in K_d$, then the restriction of G to U_0 corresponds to

$$G^{\mathsf{deh}} = G(1, y_1, \ldots, y_n) \in A.$$

Since $Y \subset W \cap U_0$ scheme-theoretically, we have $G^{\mathsf{deh}} \in I$. By the characterisation

$$I^{\mathsf{hom}} \cap S_d = \{H \in S_d \mid H^{\mathsf{deh}} \in I\},$$

it follows that $G \in I^{\mathsf{hom}}$. Hence $K \subset I^{\mathsf{hom}}$, and therefore

$$Z = \operatorname{Proj} S/I^{\mathsf{hom}} \subset \operatorname{Proj} S/K = W.$$

Thus Z is the smallest closed subscheme of $\mathbb{P}^n_{\mathbb{F}}$ containing Y, so it is the scheme-theoretic closure of Y. □



Exercise 2.5.5.6. Let $I \subset A$ be an ideal, and set $Y = \operatorname{Spec} A/I$. Prove that

$$\sqrt{I^{\mathsf{hom}}} = (\sqrt{I})^{\mathsf{hom}},$$

so that, if $\overline{Y}$ is the scheme-theoretic closure of Y in $\mathbb{P}^n_{\mathbb{F}}$, one has

$$(\overline{Y})_{\mathsf{red}} = (Y_{\mathsf{red}})^{-}.$$

In particular, the scheme-theoretic projective closure of a reduced subscheme is reduced. (**Hint**: use $I^{\mathrm{hom}} = IS_{x_0} \cap S$ and the facts that radicals commute with localisation and with contraction.).

Example 2.5.5.7. Consider the cuspidal plane cubic (cf. Example 2.3.3.24)

$$C = \operatorname{Spec} \mathbb{C}[y_1, y_2]/(y_2^2 - y_1^3) \subset \mathbb{A}^2_{\mathbb{C}}.$$

Identify $\mathbb{A}^2_{\mathbb{C}}$ with $\mathbb{P}^2_{\mathbb{C}} \setminus \mathrm{V}_+(x_2) = \mathrm{D}_+(x_2) \subset \mathbb{P}^2_{\mathbb{C}}$ and let $\overline{C} \subset \mathbb{P}^2_{\mathbb{C}}$ be the closure of C in the projective plane. Then $\overline{C}$ is cut out by the homogeneisation of $f(y_1, y_2) = y_2^2 - y_1^3$ with respect to x_2, namely the polynomial

$$x_2^3 f(x_0/x_2, x_1/x_2) = x_2^3(x_1^2/x_2^2 - x_0^3/x_2^3) = x_1^2 x_2 - x_0^3.$$

Example 2.5.5.8 (The hyperbola)**.** Consider the closed subscheme

$$Y = \operatorname{Spec} \mathbb{F}[y_1, y_2]/(y_1 y_2 - 1) \hookrightarrow \mathbb{A}^2_{\mathbb{F}}.$$

Since $(y_1 y_2 - 1)^{\mathsf{hom}} = x_1 x_2 - x_0^2$, its projective closure is the conic

$$\overline{Y} = \operatorname{Proj} \mathbb{F}[x_0, x_1, x_2]/(x_1 x_2 - x_0^2) \hookrightarrow \mathbb{P}^2_{\mathbb{F}}.$$

Its scheme-theoretic intersection with the line at infinity $H_\infty = \operatorname{Proj} \mathbb{F}[x_0, x_1, x_2]/(x_0)$ is

$$\overline{Y} \cap H_\infty = \operatorname{Proj} \mathbb{F}[x_1, x_2]/(x_1 x_2),$$

namely the two reduced points $(0:1:0)$ and $(0:0:1)$.

Example 2.5.5.9 (A smooth curve can meet infinity nonreducedly)**.** Consider the closed subscheme

$$Y = \operatorname{Spec} \mathbb{F}[y_1, y_2]/(y_2 - y_1^2) \hookrightarrow \mathbb{A}^2_{\mathbb{F}}.$$

Its closure is the smooth conic

$$\overline{Y} = \operatorname{Proj} \mathbb{F}[x_0, x_1, x_2]/(x_0 x_2 - x_1^2) \hookrightarrow \mathbb{P}^2_{\mathbb{F}}.$$

Nevertheless,

$$\overline{Y} \cap H_\infty = \operatorname{Proj} \mathbb{F}[x_1, x_2]/(x_1^2)$$

is a double point supported at the rational point $(0:0:1)$. Thus a reduced, even smooth, projective closure may have a nonreduced scheme-theoretic intersection with the hyperplane at infinity. Geometrically, the conic is tangent to H_∞.

Example 2.5.5.10 (A nonreduced affine point). Consider, this time for $n = 1$ and $I = (y^2) \subset \mathbb{F}[y]$, the fat point

$$Y = \operatorname{Spec} \mathbb{F}[y]/(y^2) \subset \mathbb{A}^1_{\mathbb{F}}.$$

Here $I^{\mathsf{hom}} = (x_1^2) \subset F[x_0, x_1]$, and hence

$$\overline{Y} = \operatorname{Proj} \mathbb{F}[x_0, x_1]/(x_1^2) \subset \mathbb{P}^1_{\mathbb{F}}.$$

Its underlying space is the single point $(1:0)$, but its local ring there is $\mathbb{F}[y]/(y^2)$. Thus projective closure preserves the nonreduced scheme structure. Replacing I^{hom} by its radical would give the wrong closure.

Example 2.5.5.11 (Why the homogenised generators must be saturated). Consider the closed subscheme (isomorphic to $\mathbb{A}^1_{\mathbb{F}}$)

$$Y = \operatorname{Spec} \mathbb{F}[y_1, y_2, y_3]/(y_2 - y_1^2, y_3 - y_1^3) \hookrightarrow \mathbb{A}^3_{\mathbb{F}}.$$

Homogenising only the two displayed generators gives

$$J = (x_0 x_2 - x_1^2,\ x_0^2 x_3 - x_1^3) \subset \mathbb{F}[x_0, x_1, x_2, x_3].$$

This is not yet the ideal of the projective closure: if one uses J without x_0-saturation, then on $x_0 = 0$ its equations reduce to $x_1^2 = 0$. Thus $\operatorname{Proj} S/J$ contains an unwanted nonreduced line at infinity, although that line is invisible on U_0. We know that the projective closure of a reduced closed subscheme must be reduced, so J cannot be the ideal of the projective closure. To find the correct ideal, we compute

$$I^{\mathsf{hom}} = (J : x_0^\infty) = (x_0 x_3 - x_1 x_2, x_1^2 - x_0 x_2, x_2^2 - x_1 x_3).$$

The resulting projective curve is the *twisted cubic* (cf. Example 2.5.4.11, with $d = 3$).

Example 2.5.5.12 (Projecting the twisted cubic). Consider the twisted cubic $Y \subset \mathbb{P}^3_{\mathbf{k}} = \operatorname{Proj} \mathbf{k}[x_0, x_1, x_2, x_3]$. That is, Y is the curve in $\mathbb{P}^3_{\mathbf{k}}$ given parametrically by

$$(x_0, x_1, x_2, x_3) = (u^3, u^2 v, u v^2, v^3),$$

where $(u : v)$ are homogeneous coordinates on $\mathbb{P}^1_{\mathbf{k}}$. It can be defined by the homogeneous ideal

$$(x_0 x_3 - x_1 x_2, x_1^2 - x_0 x_2, x_2^2 - x_1 x_3) \subset \mathbf{k}[x_0, x_1, x_2, x_3].$$

The point $P = (0:0:1:0) \in \mathbb{P}^3_{\mathbf{k}}$ lies outside Y (look at the third equation). Consider the hyperplane cut out by x_2, namely the closed subscheme

$$H = \operatorname{Proj} \mathbf{k}[x_0, x_1, x_3] = \operatorname{Proj} \mathbf{k}[x_0, x_1, x_2, x_3]/(x_2) \hookrightarrow \mathbb{P}^3_{\mathbf{k}}.$$

Then, projection from P (cf. Example 2.5.3.4) is the morphism

$$\mathbb{P}^3_{\mathbf{k}} \setminus \{P\} \xrightarrow{\mathrm{pr}_P} H$$
$$(a:b:c:d) \longmapsto (a:b:d).$$

We claim that the scheme-theoretic image of the composite morphism

$$\nu\colon \mathbb{P}^1_{\mathbf{k}} \xrightarrow{\sim} Y \hookrightarrow \mathbb{P}^3_{\mathbf{k}} \setminus \{P\} \xrightarrow{\mathrm{pr}_P} H$$

is the (projective) *cuspidal plane cubic* $\overline{C}$ (cf. Example 2.5.5.7) defined as

$$\overline{C} = \operatorname{Proj} \mathbf{k}[x_0, x_1, x_3]/(x_1^3 - x_0^2 x_3) \hookrightarrow H,$$

namely the projective closure of the affine cuspidal cubic C along the open immersion $\mathrm{D}_+(x_3) \hookrightarrow \operatorname{Proj} \mathbf{k}[x_0, x_1, x_3] = H$.

Indeed, ν sends $(u:v) \mapsto (u^3 : u^2 v : v^3)$ and so its set-theoretic image is contained in $\overline{C}$ since $(u^2 v)^3 = (u^3)^2 v^3$. We now check that ν surjects onto $\overline{C}$. On the affine chart $\overline{C} \cap \mathrm{D}_+(x_3) = C = \overline{C} \setminus \{(1:0:0)\} \subset \overline{C}$, set

$$\mathsf{X} = \frac{x_1}{x_3}, \qquad \mathsf{Y} = \frac{x_0}{x_3}.$$

The equation of C becomes $\mathsf{X}^3 = \mathsf{Y}^2$. On the corresponding chart $v \neq 0$ of $\mathbb{P}^1_{\mathbf{k}}$, with coordinate $t = u/v$, the morphism ν is nothing but the morphism $\operatorname{Spec} \mathbf{k}[t] \to C$ defined by

$$\mathsf{X} \mapsto t^2, \qquad \mathsf{Y} \mapsto t^3.$$

Conversely, suppose that $\mathsf{X}^3 = \mathsf{Y}^2$. If $\mathsf{X} = 0$, then $\mathsf{Y} = 0$, and the point is the image of $t = 0$. If $\mathsf{X} \neq 0$, set $t = \mathsf{Y}/\mathsf{X}$. Then

$$t^2 = \frac{\mathsf{Y}^2}{\mathsf{X}^2} = \mathsf{X}, \qquad t^3 = t\, t^2 = \mathsf{Y}.$$

Hence every point of $C = \overline{C} \cap \mathrm{D}_+(x_3)$ lies in the image of ν.

Finally, the equation $x_1^3 = x_0^2 x_3$ shows that the only point of $\overline{C}$ with $x_3 = 0$ is $(1:0:0)$, and $\nu(1:0) = (1:0:0)$. Therefore ν surjects onto C at the level of $\mathbf{k}$-points.

To verify surjectivity on all scheme-theoretic points, let $q \in \overline{C}$ and consider the canonical point

$$\widetilde{q}\colon \operatorname{Spec} \kappa(q) \longrightarrow \overline{C}$$

with image q. The preceding calculation, applied over the field $\kappa(q)$, produces a point

$$\widetilde{p}\colon \operatorname{Spec} \kappa(q) \longrightarrow \mathbb{P}^1_{\mathbf{k}}$$

such that $\nu \circ \widetilde{p} = \widetilde{q}$. If p is the image of $\widetilde{p}$, then $\nu(p) = q$. Hence ν is surjective on the underlying topological spaces, i.e. it is surjective as a morphism of schemes.

In particular, $\nu\colon \mathbb{P}^1_{\mathbf{k}} \to C$ is dominant. Since $\overline{C}$ is integral, the morphism $\mathscr{O}_{\overline{C}} \to \nu_*\mathscr{O}_{\mathbb{P}^1_{\mathbf{k}}}$ is injective: at the generic point it is the injective homomorphism of function fields $\mathbf{k}(t^2, t^3) \hookrightarrow \mathbf{k}(t)$, and a regular function on an integral scheme which vanishes at the generic point is zero. Thus $\nu\colon \mathbb{P}^1_{\mathbf{k}} \to C$ is schematically dominant. It follows that the scheme-theoretic image of $\nu\colon \mathbb{P}^1_{\mathbf{k}} \to H$ is exactly C.

The morphism

$$Y \longrightarrow C = \operatorname{Proj} \mathbf{k}[x_0, x_1, x_3]/(x_1^3 - x_0^2 x_3)$$

is not an isomorphism: it is the normalisation[13] morphism!

[13] See **??**.

A | Basic category theory

A.1 Categories and functors

Definition A.1.0.1 (Category)**.** A *category* $\mathscr{C}$ is the datum of

(i) a class $\mathrm{Ob}(\mathscr{C})$ of 'objects',

(ii) a class $\mathrm{Hom}(\mathscr{C})$ of 'morphisms' (or 'arrows', or 'maps') between the objects,

(iii) class functions $d\colon \mathrm{Hom}(\mathscr{C}) \to \mathrm{Ob}(\mathscr{C})$ and $t\colon \mathrm{Hom}(\mathscr{C}) \to \mathrm{Ob}(\mathscr{C})$ specifying domain and target of every morphism,

(iv) for each pair of objects x and y, a subclass

$$\mathrm{Hom}_{\mathscr{C}}(x,y) = \left\{ f \in \mathrm{Hom}(\mathscr{C}) \,\middle|\, d(f) = x,\ t(f) = y \right\} \subset \mathrm{Hom}(\mathscr{C})$$

of morphisms with domain x and target y, partitioning the whole class $\mathrm{Hom}(\mathscr{C})$,

(v) a binary operation

$$\begin{aligned} \mathrm{Hom}_{\mathscr{C}}(x,y) \times \mathrm{Hom}_{\mathscr{C}}(y,z) &\xrightarrow{\ \circ\ } \mathrm{Hom}_{\mathscr{C}}(x,z) \\ (f,g) &\longmapsto g \circ f \end{aligned}$$

called 'composition' of morphisms, for every triple of objects x, y and z.

Such data must fulfill the following axioms:

(CAT1) For every $x \in \mathrm{Ob}(\mathscr{C})$, there is an identity morphism $\mathrm{id}_x \in \mathrm{Hom}_{\mathscr{C}}(x,x)$ enjoying the properties

$$f \circ \mathrm{id}_x = f, \quad \mathrm{id}_y \circ g = g$$

for every morphism f with domain x, and for every morphism g with target y.

(CAT2) The associativity relation

$$(h \circ g) \circ f = h \circ (g \circ f)$$

holds for every triple (f,g,h) of composable morphisms.

Definition A.1.0.2 (Small and locally small)**.** A category $\mathscr{C}$ is *small* if both $\mathrm{Ob}(\mathscr{C})$ and $\mathrm{Hom}(\mathscr{C})$ are sets, and not proper classes. We say that $\mathscr{C}$ is *locally small* if $\mathrm{Hom}_{\mathscr{C}}(x,y)$ is a set, and not a proper class, for every pair of objects x and y. For a locally small category $\mathscr{C}$, the sets $\mathrm{Hom}_{\mathscr{C}}(x,y)$ are called *Hom-sets*.

Example A.1.0.3. The following are familiar examples of categories:

- Sets, the category of sets whose morphisms are the functions between sets,
- Grp, the category of groups with morphisms the group homomorphisms,
- Ab, the category of abelian groups with morphisms the group homomorphisms,
- Rings, the category of rings with morphisms the ring homomorphisms,
- Fields, the category of fields with morphisms the field homomorphisms,
- $\mathsf{Vec}_{\mathbb{F}}$, the category of vector spaces over a field $\mathbb{F}$ with morphisms the $\mathbb{F}$-linear maps,
- Alg_R, the category of algebras over a ring R, with morphisms the R-algebra homomorphisms,
- Top, the category of topological spaces, with morphisms the continuous maps,
- Mod_R, the category of modules over a ring R, with morphisms the R-linear maps,
- Mfd, the category of smooth manifolds, with morphisms the C^∞ maps.

Remark A.1.0.4. The category Sets is locally small, but not small. This is related to Russell's Paradox, which roughly says that there is no set of all sets; large categories are handled using proper classes or universes. In fact, for the same reasons as for Sets, all of the categories in Example A.1.0.3 are locally small, but not small.

From now on, the reader is encouraged to think that all categories considered are locally small. Often, this will actually be necessary.

The following notion will be important when discussing abelian categories in Appendix A.6.

Definition A.1.0.5 (Preadditive category)**.** Let $\mathscr{C}$ be a locally small category. We say that $\mathscr{C}$ is *preadditive* if all Hom-sets $\mathrm{Hom}_{\mathscr{C}}(x,y)$ are abelian groups, and composition of morphisms is $\mathbb{Z}$-bilinear.

Example A.1.0.6. Let R be a ring. The category $\mathscr{C}$ with one object, $\mathrm{Ob}(\mathscr{C})=\{\bullet\}$, and morphisms indexed by R, i.e. $\mathrm{Hom}(\mathscr{C})=\mathrm{Hom}_{\mathscr{C}}(\bullet,\bullet)=R$. Composition is multiplication in R and the addition in R realises $\mathrm{Hom}_{\mathscr{C}}(\bullet,\bullet)$ as the underlying abelian group of R.

The categories Ab, $\mathsf{Vec}_{\mathbb{F}}$ and more generally Mod_R are also preadditive.

The following are standard, important notions of morphisms in an arbitrary category.

Definition A.1.0.7 (Mono, epi, iso)**.** Let $\mathscr{C}$ be a category, and fix a morphism $f: x \to y$. We have the following notions.

(i) We say that f is a *monomorphism* if it is left-cancellable, i.e. for every pair of morphisms $g_1, g_2: z \to x$ one has that $f \circ g_1 = f \circ g_2$ implies $g_1 = g_2$.

(ii) We say that f is an *epimorphism* if it is right-cancellable, i.e. for every pair of morphisms $g_1, g_2: y \to z$ one has that $g_1 \circ f = g_2 \circ f$ implies $g_1 = g_2$.

(iii) We say that f is an *isomorphism* if it is an invertible morphism, i.e. if there exists a morphism $g: y \to x$ satisfying $f \circ g = \mathrm{id}_y$ and $g \circ f = \mathrm{id}_x$.

Two objects $x, y \in \mathrm{Ob}(\mathscr{C})$ are said to be *isomorphic* when there is an isomorphism $x \to y$ (often denoted '$x \xrightarrow{\sim} y$').

Remark A.1.0.8. Every isomorphism is in particular both an epimorphism and a monomorphism, but the converse need not hold in general. Consider for instance the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ in the category of rings: it is a monomorphism and an epimorphism, but not an isomorphism.

Definition A.1.0.9 (Functor)**.** Let $\mathscr{C}$ and $\mathscr{C}'$ be two categories. A functor from $\mathscr{C}$ to $\mathscr{C}'$, denoted $\mathsf{F}: \mathscr{C} \to \mathscr{C}'$, is the assignment of

- an object $\mathsf{F}(x) \in \mathrm{Ob}(\mathscr{C}')$ for every $x \in \mathrm{Ob}(\mathscr{C})$, and
- a morphism $\mathsf{F}(f) \in \mathrm{Hom}_{\mathscr{C}'}(\mathsf{F}(x), \mathsf{F}(y))$ for every morphism $f \in \mathrm{Hom}_{\mathscr{C}}(x, y)$,

subject to the following axioms:

(1) $\mathsf{F}(\mathrm{id}_x) = \mathrm{id}_{\mathsf{F}(x)}$ for every $x \in \mathrm{Ob}(\mathscr{C})$,

(2) $\mathsf{F}(g \circ f) = \mathsf{F}(g) \circ \mathsf{F}(f)$ for every pair (f, g) of composable arrows.

Remark A.1.0.10. By the axioms, functors preserve isomorphisms.

A functor as in Definition A.1.0.9 is said to be *covariant*. On the other hand, a *contravariant* functor $\mathsf{F}: \mathscr{C} \to \mathscr{C}'$ assigns a morphism $\mathsf{F}(f) \in \mathrm{Hom}_{\mathscr{C}'}(\mathsf{F}(y), \mathsf{F}(x))$ for every morphism $f \in \mathrm{Hom}_{\mathscr{C}}(x, y)$, and condition (2) becomes $\mathsf{F}(g \circ f) = \mathsf{F}(f) \circ \mathsf{F}(g)$. For instance, taking an $\mathbb{F}$-vector space V to its dual $V^* = \mathrm{Hom}_{\mathbb{F}}(V, \mathbb{F})$ is a contravariant functor from $\mathsf{Vec}_{\mathbb{F}}$ to itself.

Example A.1.0.11. Every category $\mathscr{C}$ admits an *identity functor* $\mathrm{Id}_{\mathscr{C}}: \mathscr{C} \to \mathscr{C}$, sending every object and every morphism to itself. This will be essential in Definition A.3.0.4.

Define $\mathscr{C}^{\mathrm{op}}$ the *opposite category* of $\mathscr{C}$ to be the category with objects $\mathrm{Ob}(\mathscr{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathscr{C})$ and with

$$\mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(x,y) = \mathrm{Hom}_{\mathscr{C}}(y,x) \tag{A.1.1}$$

for every $x, y \in \mathrm{Ob}(\mathscr{C})$. Then a contravariant functor $\mathscr{C} \to \mathscr{C}'$ is the same as a covariant functor $\mathscr{C}^{\mathrm{op}} \to \mathscr{C}'$.

Notation A.1.0.12. From now on, we shall simply write $x \in \mathscr{C}$ to denote that x is an object of a category $\mathscr{C}$.

A.2 Final objects and universal properties

Definition A.2.0.1 (Initial, final, zero)**.** Let $\mathscr{C}$ be a category. An *initial object* is an object $\alpha \in \mathscr{C}$ such that for every object $x \in \mathscr{C}$ there is exactly one morphism $\alpha \to x$. A *final object* (also known as a *terminal object*) is an object $\omega \in \mathscr{C}$ such that for every object $x \in \mathscr{C}$ there is exactly one morphism $x \to \omega$. A *zero object* (also called a *null object*) is an object $0 \in \mathscr{C}$ that is simultaneously initial and final.

Remark A.2.0.2 (Dualisation of initial and final objects)**.** A final, resp. initial object in a category $\mathscr{C}$ is an initial, resp. final object in $\mathscr{C}^{\mathrm{op}}$.

Example A.2.0.3. The empty set $\emptyset$, resp. the singleton $\{*\}$, is the unique initial, resp. final, object in Sets. In categories such as Grp, Ab, Mod_R, the zero object is the trivial object (trivial group, trivial R-module). In Rings, the initial object is $\mathbb{Z}$, whereas the final object is the trivial ring (the one in which $1 = 0$). In Fields, there are no initial or final objects.

When an initial, final or zero object exists, it is unique up to a unique isomorphism. This is quite a useful fact, so we spend a few words on it. How do we use it? The point is that many important structures are defined by a universal property; one way to prove that an object defined by a universal property, when it exists, is unique up to unique isomorphism (compatible with the universal property), is to realise it as a final (or initial object) in some auxiliary category $\mathscr{U}$. To be able to apply this in the future, we need to convince ourselves that if a final object exists, it is unique up to a unique isomorphism. Say $\omega, \omega' \in \mathscr{C}$ are two final objects. There exists a unique morphism $f : \omega \to \omega'$ and a unique morphism $g : \omega' \to \omega$. For the same reason, $g \circ f : \omega \to \omega' \to \omega$ and $f \circ g : \omega' \to \omega \to \omega'$ are the identity, i.e. $\omega \cong \omega'$. But again, the isomorphism is necessarily unique!

Let us make an explicit example of the principle: if an object is defined by a universal property where the word "unique" is involved in a suitable way, then it will be unique up to a unique isomorphism. By 'involved in a suitable way' we mean: allowing us to see the universal object as a final object in an appropriate category $\mathscr{U}$. Let us consider *fibre*

products, since they are so important in algebraic geometry. These are special cases of limits (cf. Definition A.5.0.1 and Example A.5.0.4) but let us ignore this for now.

Definition A.2.0.4 (Fibre product)**.** The *fibre product* of a diagram

$$\begin{array}{ccc} & & x \\ & & \downarrow{\scriptstyle f} \\ y & \xrightarrow{g} & z \end{array}$$

in a category $\mathscr{C}$ is a triple $(x \times_z y, \mathrm{pr}_1, \mathrm{pr}_2)$, where

$$x \times_z y \xrightarrow{\mathrm{pr}_1} x, \qquad x \times_z y \xrightarrow{\mathrm{pr}_2} y$$

are morphisms in $\mathscr{C}$ such that $f \circ \mathrm{pr}_1 = g \circ \mathrm{pr}_2$, with the following universal property: for any other triple (a, p_1, p_2) where $p_1 : a \to x$ and $p_2 : a \to y$ are morphisms such that $f \circ p_1 = g \circ p_2$, there exists a **unique** morphism $\phi : a \to x \times_z y$ such that $p_i = \mathrm{pr}_i \circ \phi$ for $i = 1, 2$.

The relevant category here is the category $\mathscr{U}_{f,g}$ whose objects are

$$\mathrm{Ob}(\mathscr{U}_{f,g}) = \left\{ (a, p_1, p_2) \,\middle|\, f \circ p_1 = g \circ p_2 \right\},$$

and whose morphisms $(a, p_1, p_2) \to (a', p_1', p_2')$ are given by morphisms $\psi : a \to a'$ in $\mathscr{C}$ such that $p_i' \circ \psi = p_i$ for $i = 1, 2$. Thus we see that a fibre product of f and g exists if and only if $\mathscr{U}_{f,g}$ has a final object. By the reasoning above, it will be *automatically* unique up to unique isomorphism.

Definition A.2.0.5 (Pushout)**.** The *pushout* of a diagram

$$\begin{array}{ccc} z & \longrightarrow & x \\ \downarrow & & \\ y & & \end{array}$$

in a category $\mathscr{C}$ is the fibre product of the diagram

$$\begin{array}{ccc} & & x \\ & & \downarrow \\ y & \longrightarrow & z \end{array}$$

viewed in $\mathscr{C}^{\mathrm{op}}$ via (A.1.1).



Exercise A.2.0.6. Formulate the universal property of pushout in a category $\mathscr{C}$ and identify the auxiliary category $\mathscr{U}$ analogously to the case of fibre product. Prove that the pushout in category Alg_A of algebras over a ring A coincides with the tensor product over A. That is, given A-algebras

$$\begin{array}{ccc} A & \longrightarrow & R \\ \downarrow & & \\ S & & \end{array}$$

prove that their pushout is given by

$$\begin{array}{ccc} A & \longrightarrow & R \\ \downarrow & & \downarrow \\ S & \longrightarrow & R \otimes_A S \end{array}$$

where $R \to R \otimes_A S$ sends $r \mapsto r \otimes 1_S$ and similarly for $S \to R \otimes_A S$. If you have not seen tensor products, consult Appendix B.2 first.

A.3 Equivalences

Definition A.3.0.1 (Natural transformation)**.** Let $\mathscr{C}$, $\mathscr{C}'$ be two categories. A *natural transformation* $\eta\colon \mathsf{F} \Rightarrow \mathsf{G}$ between two functors $\mathsf{F}, \mathsf{G}\colon \mathscr{C} \to \mathscr{C}'$ is the datum, for every $x \in \mathscr{C}$, of a morphism $\eta_x\colon \mathsf{F}(x) \to \mathsf{G}(x)$ in $\mathscr{C}'$, such that for every $f \in \mathrm{Hom}_{\mathscr{C}}(x_1, x_2)$ the diagram

$$\begin{array}{ccc} \mathsf{F}(x_1) & \xrightarrow{\eta_{x_1}} & \mathsf{G}(x_1) \\ {\scriptstyle \mathsf{F}(f)}\downarrow & & \downarrow{\scriptstyle \mathsf{G}(f)} \\ \mathsf{F}(x_2) & \xrightarrow{\eta_{x_2}} & \mathsf{G}(x_2) \end{array}$$

is commutative in $\mathscr{C}'$.

Definition A.3.0.2 (Natural isomorphism)**.** Let $\mathscr{C}$, $\mathscr{C}'$ be two categories. Let $\mathsf{Fun}(\mathscr{C}, \mathscr{C}')$ be the category whose objects are functors $\mathscr{C} \to \mathscr{C}'$ and whose morphisms are the natural transformations. An isomorphism in the category $\mathsf{Fun}(\mathscr{C}, \mathscr{C}')$ is called a *natural isomorphism*.



Exercise A.3.0.3. Let $\mathbb{F}$ be a field, and $\mathsf{Vec}_{\mathbb{F}}$ the category of $\mathbb{F}$-vector spaces. Consider the identity functor $\mathrm{Id}_{\mathsf{Vec}_{\mathbb{F}}}$ and the double dual functor $(-)^{**}\colon \mathsf{Vec}_{\mathbb{F}} \to \mathsf{Vec}_{\mathbb{F}}$. Show that

(i) there is a natural transformation $\mathrm{Id}_{\mathsf{Vec}_{\mathbb{F}}} \Rightarrow (-)^{**}$, and in particular a natural morphism $V \to V^{**}$ for every $V \in \mathsf{Vec}_{\mathbb{F}}$, and

(ii) if $\mathsf{Vec}_{\mathbb{F}}^{\mathsf{f}}$ denotes the category of finite-dimensional $\mathbb{F}$-vector spaces, then the construction in (i) yields a natural isomorphism.

Definition A.3.0.4 (Equivalence of categories)**.** Let $\mathscr{C}$ and $\mathscr{C}'$ be categories. An *equivalence* between them is a pair of functors

$$\mathscr{C} \xrightarrow{\mathsf{F}} \mathscr{C}', \qquad \mathscr{C}' \xrightarrow{\mathsf{G}} \mathscr{C}$$

along with a pair of natural isomorphisms

$$\mathsf{F} \circ \mathsf{G} \xrightarrow{\sim} \mathrm{Id}_{\mathscr{C}'}, \qquad \mathsf{G} \circ \mathsf{F} \xrightarrow{\sim} \mathrm{Id}_{\mathscr{C}}.$$

Terminology A.3.0.5. One often says that a functor $\mathsf{F}\colon \mathscr{C} \to \mathscr{C}'$ is an equivalence when there exists a functor $\mathsf{G}\colon \mathscr{C}' \to \mathscr{C}$ along with a pair of natural isomorphisms as in Definition A.3.0.4.

Assumption A.3.0.6. From now on, all categories are tacitly assumed locally small.

Definition A.3.0.7 (Fully faithful, essentially surjective)**.** A (covariant) functor $\mathsf{F}\colon \mathscr{C} \to \mathscr{C}'$ between locally small categories is called:

(i) *full* (resp. *faithful*, resp. *fully faithful*) if for any two objects $x, y \in \mathscr{C}$ the map of sets

$$\mathrm{Hom}_{\mathscr{C}}(x,y) \longrightarrow \mathrm{Hom}_{\mathscr{C}'}(\mathsf{F}(x),\mathsf{F}(y))$$

is surjective (resp. injective, resp. bijective).

(ii) *essentially surjective* if every object of $\mathscr{C}'$ is isomorphic to an object of the form $\mathsf{F}(x)$ for some $x \in \mathscr{C}$.

Definition A.3.0.8 (Full subcategory)**.** Let $\mathscr{C}$ be a category. A *subcategory* $\mathscr{D} \subset \mathscr{C}$ is specified by the following data: a subclass of objects $\mathrm{Ob}(\mathscr{D}) \subset \mathrm{Ob}(\mathscr{C})$ and a subclass of morphisms $\mathrm{Hom}(\mathscr{D}) \subset \mathrm{Hom}(\mathscr{C})$ such that

- $\mathrm{id}_x \in \mathrm{Hom}(\mathscr{D})$ for every $x \in \mathrm{Ob}(\mathscr{D})$,
- for every $f \in \mathrm{Hom}(\mathscr{D})$ one has $d(f) \in \mathrm{Ob}(\mathscr{D})$ and $t(f) \in \mathrm{Ob}(\mathscr{D})$, and
- the $\mathscr{C}$-composition of every pair of (composable) morphisms in $\mathrm{Hom}(\mathscr{D})$ lies again in $\mathrm{Hom}(\mathscr{D})$.

We say that $\mathscr{D}$ is a *full subcategory* of $\mathscr{C}$ if $\mathrm{Hom}_{\mathscr{C}}(x,y) = \mathrm{Hom}_{\mathscr{D}}(x,y)$ for every pair of objects $x, y \in \mathrm{Ob}(\mathscr{D})$.

Note that the conditions ensure that $\mathscr{D}$ is itself a category (in particular $\mathrm{Hom}_{\mathscr{D}}(-,-)$ is well-defined in the definition of full subcategory), and moreover it comes with a faithful functor $j\colon \mathscr{D} \to \mathscr{C}$, injective on objects — therefore we will denote such functor by $\mathscr{D} \hookrightarrow \mathscr{C}$. Moreover, $\mathscr{D}$ is a full subcategory of $\mathscr{C}$ if and only if j is full.

For instance, $\mathsf{Ab} \hookrightarrow \mathsf{Grp}$ is a full subcategory.

The following property, left as an exercise, is quite useful in practice.



Exercise A.3.0.9. Prove that a fully faithful functor $\mathsf{F}\colon \mathscr{C} \to \mathscr{C}'$ induces an equivalence of $\mathscr{C}$ with the essential image of F, namely the full subcategory of $\mathscr{C}'$ consisting of objects isomorphic to objects of the form $\mathsf{F}(x)$ for some $x \in \mathscr{C}$. In particular, a functor $\mathsf{F}\colon \mathscr{C} \to \mathscr{C}'$ induces an equivalence of categories if and only if it is fully faithful and essentially surjective. (During the proof, you will have to construct a so-called 'quasi-inverse' — the functor back from the essential image of F to $\mathscr{C}$, and this requires choosing one preimage object for each isomorphism class in the target; luckily we assumed the axiom of choice from the very start!).

Definition A.3.0.10 (Concrete category)**.** A *concrete category* is a category $\mathscr{C}$ that is equipped with a faithful functor $\mathsf{F}\colon \mathscr{C} \to \mathsf{Sets}$ to the category of sets.

The existence of a faithful functor to $\mathscr{C} \to \mathsf{Sets}$ is a property of a category; specifying a particular such functor, thus realising concreteness, is additional structure.

A.4 Adjunctions

Another notion that is rather important in category theory is that of an adjoint pair of functors.

Definition A.4.0.1 (Adjoint pair)**.** Let $\mathscr{C}$ and $\mathscr{D}$ be (locally small) categories. Let $\mathsf{F}\colon \mathscr{C} \to \mathscr{D}$ and $\mathsf{G}\colon \mathscr{D} \to \mathscr{C}$ be functors. We say that (F,G) is an *adjoint pair* of functors if for every pair of objects $(c,d) \in \mathrm{Ob}(\mathscr{C}) \times \mathrm{Ob}(\mathscr{D})$ one has a bijection of sets

$$\mathrm{Hom}_{\mathscr{D}}(\mathsf{F}(c),d) \cong \mathrm{Hom}_{\mathscr{C}}(c,\mathsf{G}(d)),$$

natural in both c and d. We say, more precisely, that F is a *left adjoint* to G and that G is a *right adjoint* to F.

Sometimes, one uses the pictorial description

$$\mathscr{C} \underset{\mathsf{G}}{\overset{\mathsf{F}}{\rightleftarrows}} \mathscr{D}$$

to say that (F,G) is an adjoint pair.

Example A.4.0.2. Here are some examples of adjunctions.

(a) Let $\mathsf{F}\colon \mathsf{Sets} \to \mathsf{Grp}$ be the functor sending a set S to the free group generated by the elements of S. Let $\Phi\colon \mathsf{Grp} \to \mathsf{Sets}$ be the forgetful functor. Then (F,Φ) is an adjoint pair.

$$\mathsf{Sets} \underset{\Phi}{\overset{\mathsf{F}}{\rightleftarrows}} \mathsf{Grp}$$

(b) Let $\mathsf{j}\colon \mathsf{Ab} \hookrightarrow \mathsf{Grp}$ be the inclusion. It is right adjoint to the abelianisation functor $\mathsf{ab}\colon \mathsf{Grp} \to \mathsf{Ab}$ sending a group G to $G^{\mathrm{ab}} = G/[G,G]$. That is, (ab,j) is an adjoint pair.

$$\mathsf{Grp} \underset{\mathsf{j}}{\overset{\mathsf{ab}}{\rightleftarrows}} \mathsf{Ab}$$

(c) Let R be a ring. Consider the functor $\mathsf{Sym}_R\colon \mathsf{Mod}_R \to \mathsf{Alg}_R$ sending $M \mapsto \mathsf{Sym}_R(M)$. Consider the forgetful functor $\Phi_R\colon \mathsf{Alg}_R \to \mathsf{Mod}_R$ sending an R-algebra to its underlying R-module. Then (Sym_R,Φ_R) is an adjoint pair.

$$\mathsf{Mod}_R \underset{\Phi_R}{\overset{\mathsf{Sym}_R}{\rightleftarrows}} \mathsf{Alg}_R$$

(d) Let $\alpha\colon R \to S$ be a ring homomorphism. Then every S-module is naturally an R-module, thus we have a forgetful functor $\Phi_\alpha\colon \mathsf{Mod}_S \to \mathsf{Mod}_R$. On the other hand, we have a functor (called extension of scalars) $-\otimes_R S\colon \mathsf{Mod}_R \to \mathsf{Mod}_S$ sending an R-module M to the S-module $M \otimes_R S$. Then $(-\otimes_R S, \Phi_\alpha)$ is an adjoint pair.

$$\mathsf{Mod}_R \underset{\Phi_\alpha}{\overset{-\otimes_R S}{\rightleftarrows}} \mathsf{Mod}_S$$

(e) Let ID be the category of integral domains (with morphisms the injective ring homomorphisms), Fields the category of fields. We have a functor $\mathrm{Frac}\colon \mathsf{ID} \to \mathsf{Fields}$ sending a domain to its fraction field, and an inclusion functor $\mathsf{j}\colon \mathsf{Fields} \hookrightarrow \mathsf{ID}$. Then $(\mathrm{Frac}, \mathsf{j})$ is an adjoint pair.

$$\mathsf{ID} \underset{\mathsf{j}}{\overset{\mathrm{Frac}}{\rightleftarrows}} \mathsf{Fields}$$

(f) Let R be a ring, $M \in \mathsf{Mod}_R$ an R-module. Consider the endofunctors on the category Mod_R given by $-\otimes_R M$ and $\mathsf{h}^M = \mathrm{Hom}_{\mathsf{Mod}_R}(M, -)$. Then $(-\otimes_R M, \mathsf{h}^M)$ is an adjoint pair. The (natural) bijections

$$\mathrm{Hom}_{\mathsf{Mod}_R}(N \otimes_R M, P) \cong \mathrm{Hom}_{\mathsf{Mod}_R}(N, \mathrm{Hom}_{\mathsf{Mod}_R}(M, P))$$

induced by the adjunction

$$\mathsf{Mod}_R \underset{\mathsf{h}^M}{\overset{-\otimes_R M}{\rightleftarrows}} \mathsf{Mod}_R$$

are in fact isomorphisms of abelian groups (recall that Mod_R is preadditive, cf. Example A.1.0.6).

It is important to remember the following properties:

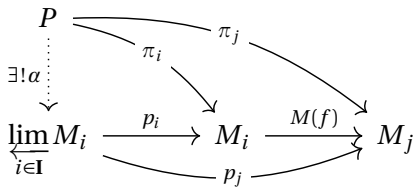
- every equivalence can be promoted to an adjoint equivalence,
- if a functor has two left (or right) adjoints, then they are naturally isomorphic.

A.5 Limits and colimits

Let $\mathscr{C}$ be a category, $\mathbf{I}$ a small category. Define an $\mathbf{I}$-diagram to be just a functor $M\colon \mathbf{I} \to \mathscr{C}$. Denote by $M_i = M(i)$ the object of $\mathscr{C}$ image of the object $i \in \mathbf{I}$ via M. If $f\colon i \to j$ is an arrow in $\mathbf{I}$, the induced arrow in $\mathscr{C}$ is denoted $M(f)\colon M_i \to M_j$.

Definition A.5.0.1 (Limit)**.** A *limit* of an $\mathbf{I}$-diagram $M\colon \mathbf{I} \to \mathscr{C}$ is an object $\varprojlim_{i\in\mathbf{I}} M_i$ of $\mathscr{C}$ along with an arrow $p_i\colon \varprojlim_{i\in\mathbf{I}} M_i \to M_i$ for every $i \in \mathbf{I}$, such that for every arrow $f\colon i \to j$ in $\mathbf{I}$ one has $p_j = M(f) \circ p_i$, and satisfying the following universal property: given an

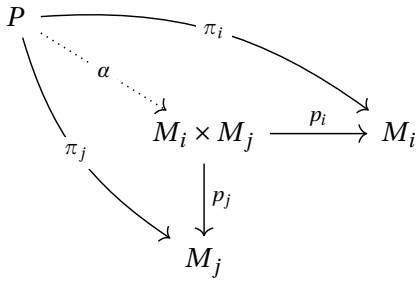
object P along with morphisms $\pi_i : P \to M_i$ such that $\pi_j = M(f) \circ \pi_i$ for every $f : i \to j$ in $\mathbf{I}$, there exists a unique arrow $\alpha : P \to \varprojlim_{i \in \mathbf{I}} M_i$ such that $\pi_i = p_i \circ \alpha$ for all $i \in \mathbf{I}$.





Exercise A.5.0.2. The limit over the empty diagram satisfies the universal property of a final object of $\mathscr{C}$.

Example A.5.0.3 (Products are limits). Let $\mathbf{I}$ be the category with two objects i, j and no morphisms between them. Then an $\mathbf{I}$-diagram $M : \mathbf{I} \to \mathscr{C}$ is just the choice of two objects M_i, M_j of $\mathscr{C}$. The limit of M satisfies the universal property of the product $M_i \times M_j$.



Example A.5.0.4 (Fibre products are limits). Let $\mathbf{I}$ be the category with three objects i, j, k and only two morphisms $i \to k$ and $j \to k$. Then an $\mathbf{I}$-diagram $M : \mathbf{I} \to \mathscr{C}$ is just the choice of three objects M_i, M_j and M_k of $\mathscr{C}$, along with a diagram

$$\begin{array}{ccc} & & M_i \\ & & \downarrow f \\ M_j & \xrightarrow{g} & M_k \end{array}$$

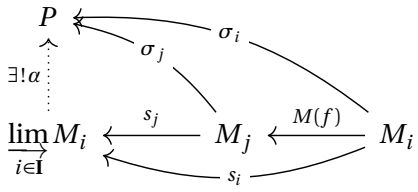
and it is clear that the limit of M satisfies the universal property of the product $(M_i \times_{M_k} M_j, \mathrm{pr}_1, \mathrm{pr}_2)$, see Definition A.2.0.4.

Example A.5.0.5 (Equalisers are limits). Let $\mathbf{I}$ be the category with two objects i, j and two arrows $i \rightrightarrows j$. Then an $\mathbf{I}$-diagram $M : \mathbf{I} \to \mathscr{C}$ is just the choice of two parallel arrows $\phi, \psi : M_i \rightrightarrows M_j$ in $\mathscr{C}$. The limit of M satisfies the universal property of the equaliser of (ϕ, ψ).

Example A.5.0.6 (Kernels are limits). This is because kernels are equalisers (in the previous example take $\psi = 0$ — for this to make sense of course you need to assume $\mathscr{C}$ to be preadditive).

Definition A.5.0.7 (Colimit). A *colimit* of an $\mathbf{I}$-diagram $M : \mathbf{I} \to \mathscr{C}$ is an object $\varinjlim_{i \in \mathbf{I}} M_i$ of $\mathscr{C}$ along with an arrow $s_i : M_i \to \varinjlim_{i \in \mathbf{I}} M_i$ for every $i \in \mathbf{I}$, such that for every arrow

$f: i \to j$ in $\mathbf{I}$ one has $s_i = s_j \circ M(f)$, and satisfying the following universal property: given an object P along with morphisms $\sigma_i: M_i \to P$ such that $\sigma_i = \sigma_j \circ M(f)$ for every $f: i \to j$ in $\mathbf{I}$, there exists a unique arrow $\alpha: \varinjlim_{i\in\mathbf{I}} M_i \to P$ such that $\sigma_i = \alpha \circ s_i$ for all $i \in \mathbf{I}$.



Exercise A.5.0.8. The colimit over the empty diagram satisfies the universal property of an initial object of $\mathscr{C}$ (cf. Exercise A.5.0.2).

Exercise A.5.0.9. Convince yourself that coproducts, coequalisers, cokernels and pushouts are examples of colimits, along the same lines of Examples A.5.0.3, A.5.0.5, A.5.0.6 and A.5.0.4. (**Hint**: work with $\mathscr{C}^{\mathrm{op}}$ using Remark A.2.0.2).

Definition A.5.0.10 (Filtered category)**.** A nonempty category $\mathbf{I}$ is *filtered* if for every two objects $i, j \in \mathbf{I}$ the following are true:

- there exists an object $k \in \mathbf{I}$ along with two morphisms $i \to k$ and $j \to k$, and
- for any two morphisms $f, g \in \mathrm{Hom}_{\mathbf{I}}(i, j)$ there exists an object $k \in \mathbf{I}$ along with a morphism $h: j \to k$ such that $h \circ f = h \circ g$ in $\mathrm{Hom}_{\mathbf{I}}(i, k)$.

The colimit of an $\mathbf{I}$-diagram $M: \mathbf{I} \to \mathscr{C}$ where $\mathbf{I}$ is a filtered category is a *filtered colimit*.

Example A.5.0.11. In the definition of stalk of a presheaf $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ at a point $x \in X$, we have been taking

$$\mathbf{I} = \{\, U \in \tau_X \mid x \in U \,\}^{\mathrm{op}}, \qquad M(U) = \mathcal{F}(U).$$

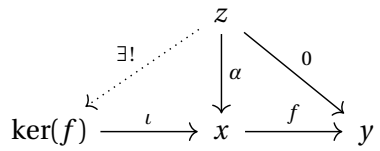
Exercise A.5.0.12. Prove that limits and colimits are unique up to unique isomorphisms, when they exist. (**Hint**: realise them as final objects in an appropriate category).

A.6 Abelian categories

Let $f: x \to y$ be an arrow in a preadditive category $\mathscr{C}$. The *kernel* of f is the final object $(\ker(f), \iota)$, if it exists, in the category[1] of pairs (z, α) where z is an object of $\mathscr{C}$ and $\alpha: z \to x$ is a morphism such that $f \circ \alpha = 0 \in \mathrm{Hom}_{\mathscr{C}}(z, y)$. The universal property of the kernel,

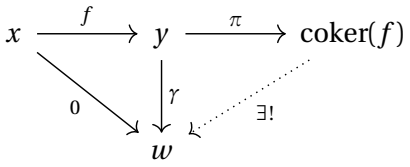
[1] As one might guess, a morphism $(z, \alpha) \to (z', \alpha')$ in this category is just a morphism $\phi: z \to z'$ such that $\alpha = \alpha' \circ \phi$.

translating the final object condition, can be represented by the diagram



where the two induced triangles commute.

The *cokernel* of f is the dual notion: it is the initial object $(\mathrm{coker}(f),\pi)$ in the category of pairs (w,γ) where w is an object of $\mathscr{C}$ and $\gamma\colon y\to w$ is a morphism such that $\gamma\circ f = 0\in\mathrm{Hom}_{\mathscr{C}}(x,w)$. The universal property of the cokernel, translating the initial object condition, can be represented by the diagram



where the two induced triangles commute.

Remark A.6.0.1. Let $f\colon x\to y$ be an arrow in a preadditive category $\mathscr{C}$. If $(\ker(f),\iota)$ is a kernel, then ι is a monomorphism, and if $(\mathrm{coker}(f),\pi)$ is a cokernel, then π is an epimorphism.

Terminology A.6.0.2. If $f\colon x\to y$ is a monomorphism in a preadditive category $\mathscr{C}$, resp. an epimorphism, we say that x is a *subobject* of y, resp. y is a *quotient object* of x.

Assume $\mathscr{C}$ has all kernels and cokernels, for simplicity. Define *image* and *coimage* of a morphism $f\colon x\to y$ to be

$$\mathrm{im}(f)=\ker(y\to\mathrm{coker}(f)),$$
$$\mathrm{coim}(f)=\mathrm{coker}(\ker(f)\to x).$$



Exercise A.6.0.3. Let $f\colon x\to y$ be an arrow in a preadditive category $\mathscr{C}$ admitting all kernels and cokernels. Show that f has a unique factorisation

$$x\longrightarrow\mathrm{coim}(f)\xrightarrow{\rho_f}\mathrm{im}(f)\longrightarrow y.$$

The following notion is extremely useful and of crucial importance in commutative algebra and algebraic geometry, as well as in the theory of sheaves.

Definition A.6.0.4 (Abelian category)**.** Let $\mathscr{C}$ be a preadditive category. We say that $\mathscr{C}$ is

- *additive*, if it admits all finite coproducts and a zero object $0\in\mathscr{C}$,
- *abelian*, if it is additive, has all kernels and cokernels and for every arrow $f\colon x\to y$ the natural map

  $$\mathrm{coim}(f)\xrightarrow{\rho_f}\mathrm{im}(f)$$

  is an isomorphism.

In an abelian category one can define a sequence

$$x \xrightarrow{f} y \xrightarrow{g} z$$

to be a complex if $g \circ f = 0$, and a complex to be *exact* (or exact *in the middle*) if the canonical map $\mathrm{im}(f) \to \ker(g)$ is an isomorphism (equivalently, if $\mathrm{im}(f)$ and $\ker(g)$ define the same subobject of y). This allows us to define $\mathbb{Z}$-indexed complexes

$$\cdots \longrightarrow x_{i-1} \xrightarrow{f_{i-1}} x_i \xrightarrow{f_i} x_{i+1} \xrightarrow{f_{i+1}} x_{i+2} \longrightarrow \cdots$$

of arbitrary length and their exactness (at each $i \in \mathbb{Z}$). A special case is given by *short exact sequences*, namely exact sequences of the form

$$0 \longrightarrow x \xrightarrow{f} y \xrightarrow{g} z \longrightarrow 0.$$

Note that, in such a sequence, $(z, g) = \mathrm{coker}(f)$ and $(x, f) = \ker(g)$.

Note also that every morphism $f : x \to y$ in an abelian category defines an exact sequence

$$0 \longrightarrow \ker(f) \longrightarrow x \xrightarrow{f} y \longrightarrow \mathrm{coker}(f) \longrightarrow 0,$$



Exercise A.6.0.5. Let $\mathscr{C}$ be an abelian category. Declare a morphism $f : x \to y$ in $\mathscr{C}$ to be injective, resp. surjective, if $\ker(f) = 0$, resp. $\mathrm{coker}(f) = 0$. Show that

(i) a morphism $f : x \to y$ is injective if and only if it is a monomorphism,

(ii) a morphism $f : x \to y$ is surjective if and only if it is an epimorphism,

(iii) a morphism $f : x \to y$ is an isomorphism if and only if it is both injective and surjective.



Exercise A.6.0.6. Let $\mathscr{C}$ be an abelian category. Prove that finite products and coproducts are the same, and deduce that $\mathscr{C}$ has all finite limits and finite colimits.

The following result is of crucial importance in the theory of abelian categories.

LEMMA A.6.0.7 (Snake lemma [10, Tag 010H]). *Given a commutative diagram*

$$\begin{array}{ccccccc} x & \xrightarrow{f} & y & \longrightarrow & z & \longrightarrow & 0 \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & u & \longrightarrow & v & \xrightarrow{h} & w \end{array}$$

with exact rows, there is an induced exact sequence

$$\ker(\alpha) \xrightarrow{f'} \ker(\beta) \longrightarrow \ker(\gamma) \longrightarrow \mathrm{coker}(\alpha) \longrightarrow \mathrm{coker}(\beta) \xrightarrow{h'} \mathrm{coker}(\gamma).$$

Furthermore, f' is injective if f is, and h' is surjective if h is. This property is known as the snake lemma.

Using the snake lemma (Lemma A.6.0.7), one may prove the following important result.

LEMMA A.6.0.8 (Four Lemma [10, Tag 05QA]). *Let $\mathscr{C}$ be an abelian category. Given a commutative diagram*

$$\begin{array}{ccccccc} w & \longrightarrow & x & \longrightarrow & y & \longrightarrow & z \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \delta} \\ w' & \longrightarrow & x' & \longrightarrow & y' & \longrightarrow & z' \end{array}$$

with exact rows, one has:

(i) if α, γ are surjective and δ is injective, then β is surjective.

(ii) if β, δ are injective and α is surjective, then γ is injective.

The Four Lemma, in turn, implies the following.

LEMMA A.6.0.9 (Five Lemma). *Given a commutative diagram*

$$\begin{array}{ccccccccc} v & \longrightarrow & w & \longrightarrow & x & \longrightarrow & y & \longrightarrow & z \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \delta} & & \downarrow{\scriptstyle \epsilon} \\ v' & \longrightarrow & w' & \longrightarrow & x' & \longrightarrow & y' & \longrightarrow & z' \end{array}$$

with exact rows, one has that if β, δ are isomorphisms, ϵ is injective, and α is surjective then γ is an isomorphism.

Definition A.6.0.10 (Additive functor)**.** Let $\mathsf{F}\colon \mathscr{C} \to \mathscr{C}'$ be a covariant functor between abelian categories. We say that F is *additive* if the natural map

$$\mathrm{Hom}_{\mathscr{C}}(x, y) \longrightarrow \mathrm{Hom}_{\mathscr{C}'}(\mathsf{F}(x), \mathsf{F}(y))$$

is a group homomorphism for every $x, y \in \mathscr{C}$.

Definition A.6.0.11 (Exactness)**.** Let $\mathsf{F}\colon \mathscr{C} \to \mathscr{C}'$ be an additive functor between abelian categories. We say that F is *left exact*, resp. *right exact*, if a short exact sequence

$$0 \longrightarrow x \longrightarrow y \longrightarrow z \longrightarrow 0$$

gets sent to an exact sequence

$$0 \longrightarrow \mathsf{F}(x) \longrightarrow \mathsf{F}(y) \longrightarrow \mathsf{F}(z),$$

resp. to an exact sequence

$$\mathsf{F}(x) \longrightarrow \mathsf{F}(y) \longrightarrow \mathsf{F}(z) \longrightarrow 0.$$

We say that F is *exact* if it is both left and right exact.



Exercise A.6.0.12 (Contravariant exactness). Formulate the notions of exactness introduced in Definition A.6.0.11 for a contravariant functor. Confirm that a covariant additive functor is left exact, resp. right exact, if and only if it turns kernels into kernels, resp. cokernels into cokernels. What about an exact contravariant functor?



Exercise A.6.0.13 (Exactness of adjoints). Let

$$\mathscr{C} \underset{\mathsf{G}}{\overset{\mathsf{F}}{\rightleftarrows}} \mathscr{D}$$

be an adjoint pair of additive functors between abelian categories, i.e. F is left adjoint to G. Prove that F (the left adjoint) preserves all colimits that exist and G (the right adjoint) preserves all limits that exist. Deduce that every right adjoint (resp. left adjoint) additive functor between two abelian categories is left exact (resp. right exact).

A.7 Yoneda Lemma

In this section we study representable functors and recall the statement of the Yoneda Lemma. This all connects again to our discussion on universal properties and final objects, cf. Appendix A.2. More details and examples can be found, for instance, in [13].

All categories are assumed to be *locally small* throughout. We ignore size issues, or equivalently work inside fixed Grothendieck universes.

Let $\mathscr{C}$ be a category. Consider the category of contravariant functors $\mathscr{C} \to \mathsf{Sets}$, i.e. the *functor category* (cf. Definition A.3.0.2)

$$\mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets}).$$

We denote its Hom-sets, consisting of natural transformations, by

$$\mathrm{Nat}(\mathsf{F},\mathsf{G}) = \mathrm{Hom}_{\mathsf{Fun}(\mathscr{C}^{\mathrm{op}},\mathsf{Sets})}(\mathsf{F},\mathsf{G}).$$

For every object x of $\mathscr{C}$ there is a functor $\mathsf{h}_x \colon \mathscr{C}^{\mathrm{op}} \to \mathsf{Sets}$ defined by

$$u \longmapsto \mathsf{h}_x(u) = \mathrm{Hom}_{\mathscr{C}}(u,x), \quad u \in \mathscr{C}.$$

A morphism $\phi \in \mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(u,v) = \mathrm{Hom}_{\mathscr{C}}(v,u)$ gets sent to the map of sets

$$\mathsf{h}_x(\phi) \colon \mathsf{h}_x(u) \longrightarrow \mathsf{h}_x(v), \quad \alpha \mapsto \alpha \circ \phi.$$

Consider the functor

$$\mathsf{h}_{\mathscr{C}} \colon \mathscr{C} \longrightarrow \mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets}), \quad x \mapsto \mathsf{h}_x. \tag{A.7.1}$$

This is, indeed, a functor: for every arrow $f \colon x \to y$ in $\mathscr{C}$ and object u of $\mathscr{C}$ we can define a map of sets

$$\mathsf{h}_f u \colon \mathsf{h}_x(u) \longrightarrow \mathsf{h}_y(u), \quad \alpha \mapsto f \circ \alpha,$$

with the property that for every morphism $\phi\colon v \to u$ in $\mathscr{C}$ there is a commutative diagram

$$\begin{array}{ccc} \mathsf{h}_x(u) & \xrightarrow{\mathsf{h}_f u} & \mathsf{h}_y(u) \\ {\scriptstyle \mathsf{h}_x(\phi)}\downarrow & & \downarrow{\scriptstyle \mathsf{h}_y(\phi)} \\ \mathsf{h}_x(v) & \xrightarrow{\mathsf{h}_f v} & \mathsf{h}_y(v) \end{array} \qquad \begin{array}{ccc} u \xrightarrow{\alpha} x & \longmapsto & u \xrightarrow{f\circ\alpha} y \\ \downarrow & & \downarrow \\ v \xrightarrow{\alpha\circ\phi} x & \longmapsto & v \xrightarrow{f\circ\alpha\circ\phi} y \end{array}$$

defining a natural transformation

$$\mathsf{h}_f\colon \mathsf{h}_x \Longrightarrow \mathsf{h}_y.$$

LEMMA A.7.0.1 (Strong Yoneda). *Let* $\mathsf{F} \in \mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets})$ *be a functor,* $x \in \mathscr{C}$ *an object. Then, there is a bijection*

$$\mathrm{Nat}(\mathsf{h}_x, \mathsf{F}) \xrightarrow{\sim} \mathsf{F}(x),$$

natural both in the object $x \in \mathscr{C}$ *and in the functor* F*.*

Proof. One can construct a map of sets

$$\mathrm{Nat}(\mathsf{h}_x, \mathsf{F}) \xrightarrow{g_x} \mathsf{F}(x)$$

by sending a natural transformation $\eta\colon \mathsf{h}_x \Rightarrow \mathsf{F}$ to the element

$$g_x(\eta) = \eta_x(\mathrm{id}_x) \in \mathsf{F}(x),$$

the image of $\mathrm{id}_x \in \mathsf{h}_x(x)$ via the map $\eta_x\colon \mathsf{h}_x(x) \to \mathsf{F}(x)$. The inverse of g_x is the map that assigns to an element $\xi \in \mathsf{F}(x)$ the natural transformation $\eta(x,\xi)\colon \mathsf{h}_x \Rightarrow \mathsf{F}$ defined as follows: for a given object $u \in \mathscr{C}$, we define

$$\mathsf{h}_x(u) \xrightarrow{\eta(x,\xi)_u} \mathsf{F}(u)$$

by sending a morphism $\alpha\colon u \to x$ to the image of ξ under $\mathsf{F}(\alpha)\colon \mathsf{F}(x) \to \mathsf{F}(u)$. This is indeed a natural transformation: we need to confirm that, given $\phi\colon v \to u$ in $\mathscr{C}$, the diagram

$$\begin{array}{ccc} \mathsf{h}_x(u) & \xrightarrow{\eta(x,\xi)_u} & \mathsf{F}(u) \\ {\scriptstyle \mathsf{h}_x(\phi)}\downarrow & & \downarrow{\scriptstyle \mathsf{F}(\phi)} \\ \mathsf{h}_x(v) & \xrightarrow{\eta(x,\xi)_v} & \mathsf{F}(v) \end{array}$$

commutes. If $\alpha \in \mathsf{h}_x(u)$, we have

$$\big(\mathsf{F}(\phi) \circ \eta(x,\xi)_u\big)(\alpha) = \mathsf{F}(\phi)(\mathsf{F}(\alpha)(\xi)) = \mathsf{F}(\alpha\circ\phi)(\xi),$$
$$\big(\eta(x,\xi)_v \circ \mathsf{h}_x(\phi)\big)(\alpha) = \eta(x,\xi)_v(\alpha\circ\phi) = \mathsf{F}(\alpha\circ\phi)(\xi),$$

as we wanted.

Finally to check that g_x is bijective with inverse $\xi \mapsto \eta(x,\xi)$, note that for every $\xi \in \mathsf{F}(x)$ and for every natural transformation $\eta\colon \mathsf{h}_x \Rightarrow \mathsf{F}$ one has

$$\begin{aligned} &g_x(\eta(x,\xi)) = \eta(x,\xi)_x(\mathrm{id}_x) = \mathsf{F}(\mathrm{id}_x)(\xi) = \mathrm{id}_{\mathsf{F}(x)}(\xi) = \xi, \\ &\eta(x, g_x(\eta)) = \eta. \end{aligned}$$

For the second identity, fix $u \in \mathscr{C}$ and $\alpha \in \mathsf{h}_x(u)$, and compute

$$\begin{aligned} \eta(x,g_x(\eta))_u(\alpha) &= \eta(x,\eta_x(\mathrm{id}_x))_u(\alpha) \\ &= \mathsf{F}(\alpha)(\eta_x(\mathrm{id}_x)) \\ &= \eta_u(\alpha). \end{aligned}$$

The last identity comes from naturality of $\eta\colon \mathsf{h}_x \Rightarrow \mathsf{F}$. Thus we have proved $\eta(x,g_x(\eta))_u = \eta_u$ for all $u \in \mathscr{C}$, which proves $\eta(x,g_x(\eta)) = \eta$ and confirms the bijectivity of g_x. We leave the naturality of this bijection as an exercise. □

The weak form of Yoneda's Lemma is the following statement about the functor $\mathsf{h}_{\mathscr{C}}$ defined above.

LEMMA A.7.0.2 (Weak Yoneda). *The functor* $\mathsf{h}_{\mathscr{C}}$ *defined in* (A.7.1) *is fully faithful.*



Exercise A.7.0.3. Show that Strong Yoneda, that we proved (Lemma A.7.0.1), implies Weak Yoneda (Lemma A.7.0.2).

Definition A.7.0.4 (Representable functor)**.** A functor $\mathsf{F} \in \mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets})$ is *representable* if it lies in the essential image of $\mathsf{h}_{\mathscr{C}}$, i.e. if it is isomorphic to a functor h_x for some $x \in \mathscr{C}$. In this case, we say that the object $x \in \mathscr{C}$ represents F.

Note that, combining Lemma A.7.0.2 and Exercise A.3.0.9, we have that $\mathscr{C}$ is equivalent to its essential image $\mathsf{h}_{\mathscr{C}}(\mathscr{C}) \subset \mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets})$.

Remark A.7.0.5. By Lemma A.7.0.2, if $x \in \mathscr{C}$ represents F, then x is unique up to isomorphism. Indeed, suppose we have isomorphisms

$$a\colon \mathsf{h}_x \xrightarrow{\sim} \mathsf{F}, \qquad b\colon \mathsf{h}_y \xrightarrow{\sim} \mathsf{F}$$

in the category $\mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets})$. By full faithfulness of Yoneda there is a unique isomorphism $f\colon x \xrightarrow{\sim} y$ inducing $\mathsf{h}_f = b^{-1} \circ a\colon \mathsf{h}_x \xrightarrow{\sim} \mathsf{h}_y$. Thus chosen representations, or universal pairs (x,a), are unique up to a unique compatible isomorphism. In particular, the underlying representing object (alone, without the datum of $a\colon \mathsf{h}_x \xrightarrow{\sim} \mathsf{F}$) is unique up to isomorphism.

Definition A.7.0.6 (Universal object)**.** Let $\mathsf{F}\colon \mathscr{C}^{\mathrm{op}} \to \mathsf{Sets}$ be a functor. A *universal object* for F is a pair (x,ξ) where $\xi \in \mathsf{F}(x)$, such that for every pair (u,σ) with $\sigma \in \mathsf{F}(u)$, there

exists a unique morphism $\alpha\colon u \to x$ with the property that $\mathsf{F}(\alpha)\colon \mathsf{F}(x) \to \mathsf{F}(u)$ sends ξ to σ.



Exercise A.7.0.7. Show that a pair (x, ξ) is a universal object for a functor $\mathsf{F}\colon \mathscr{C}^{\mathrm{op}} \to \mathsf{Sets}$ if and only if the natural transformation $\eta(x, \xi)$ defined in the proof of Lemma A.7.0.1 is a natural isomorphism. In particular, F is representable if and only if it has a universal object.

B | Commutative algebra

B.1 Frequently used theorems

B.1.1 Nakayama's Lemma(s)

LEMMA B.1.1.1 (Nakayama). *Let A be a ring, $I \subset A$ an ideal, M a finitely generated A-module. If $M = IM$, then there exists an element $r \in A$, with $1 - r \in I$, such that $rM = 0$. Equivalently, there exists $i \in I$ such that $im = m$ for every $m \in M$.*

The following weaker statements are also referred to as 'Nakayama's Lemma'.

LEMMA B.1.1.2. *Let A be a ring, $J_A \subset A$ its Jacobson radical (the intersection of all maximal ideals), M a finitely generated A-module. If $J_A M = M$, then $M = 0$.*

In particular, if M is a finitely generated A-module and $N \subset M$ is a submodule such that $M = N + J_A M$, then $M = N$.

LEMMA B.1.1.3. *Let $(A, \mathfrak{m}, \kappa)$ be a local ring, M a finitely generated A-module. Let $m_1, \dots, m_n \in M$ be elements such that their classes $\overline{m}_1, \dots, \overline{m}_n \in M/\mathfrak{m}M$ generate the κ-vector space $M/\mathfrak{m}M$. Then $m_1, \dots, m_n$ generate M as an A-module.*

B.1.2 Noether's Lemma

The original proof of the following fundamental result in commutative algebra can be found in Emmy Noether's work [8], published in 1926, exactly a century before this edition of the notes.

THEOREM B.1.2.1 (Noether's normalisation). *Let A be a nonzero finitely generated $\mathbb{F}$-algebra. Then, there exists an integer $d \geqslant 0$ and a finite injective ring homomorphism*

$$\mathbb{F}[x_1, \dots, x_d] \hookrightarrow A.$$

In particular, $\dim A = d$.

Proof. Write $A = \mathbb{F}[a_1, \dots, a_n]$. If $a_1, \dots, a_n$ are algebraically independent over $\mathbb{F}$, there is nothing to prove. So we assume there is a polynomial relation and we argue by induction

on n. There exists a nonzero polynomial

$$f = \sum_{\alpha \in \Lambda} c_\alpha T^\alpha \in \mathbb{F}[T_1, \ldots, T_n], \qquad T^\alpha = T_1^{\alpha_1} \cdots T_n^{\alpha_n}, \qquad c_\alpha \in \mathbb{F},$$

such that $f(a_1, \ldots, a_n) = 0$. Choose positive integers $N_1, \ldots, N_{n-1}$ such that the (finite) set

$$\left\{ \alpha_n + \sum_{i=1}^{n-1} N_i \alpha_i \;\middle|\; \alpha \in \Lambda \right\} \subset \mathbb{Z}$$

contains pairwise distinct integers. For instance, one may take the N_i to be sufficiently rapidly growing powers of a sufficiently large integer.

Set

$$b_i = a_i - a_n^{N_i}, \qquad i = 1, \ldots, n-1,$$

and put

$$B = \mathbb{F}[b_1, \ldots, b_{n-1}].$$

Consider the polynomial

$$f\big(b_1 + T^{N_1}, \ldots, b_{n-1} + T^{N_{n-1}}, T\big) \in B[T].$$

By the choice of the N_i, this polynomial has a unique term of highest degree in T, and its leading coefficient is a nonzero element of $\mathbb{F}$. After multiplying by a scalar in $\mathbb{F}^\times$, we may therefore assume that it is monic.

Since

$$f\big(b_1 + a_n^{N_1}, \ldots, b_{n-1} + a_n^{N_{n-1}}, a_n\big) = f(a_1, \ldots, a_n) = 0,$$

it follows that a_n is integral over B. Moreover,

$$a_i = b_i + a_n^{N_i} \qquad (i = 1, \ldots, n-1),$$

so

$$A = B[a_n].$$

Thus A is finite as a B-module; equivalently, the inclusion

$$B \hookrightarrow A$$

is finite.

Now B is generated as a $\mathbb{F}$-algebra by $n-1$ elements. By the induction hypothesis, there are algebraically independent elements $x_1, \ldots, x_d \in B$ such that

$$\mathbb{F}[x_1, \ldots, x_d] \hookrightarrow B$$

is finite. Since finiteness is transitive, the composite

$$\mathbb{F}[x_1, \ldots, x_d] \hookrightarrow B \hookrightarrow A$$

is finite, as required. □

Example B.1.2.2. For $A = F[x, y]/(xy - 1)$, and setting $t = x + y$, we have that A is finite over $\mathbb{F}[t]$: the element y satisfies $y^2 - ty + 1 = 0$ and $x = t - y$. This confirms $\dim A = 1$, which we also know from Krull's theorem.

B.1.3 Zariski's Lemma

LEMMA B.1.3.1 (Zariski). *Let $\mathbb{F}$ be a field and $\mathbb{E} \supset \mathbb{F}$ a field extension such that $\mathbb{E}$ is a finitely generated $\mathbb{F}$-algebra. Then $\mathbb{E}$ is finite dimensional over $\mathbb{F}$. In particular, $\mathbb{E} \supset \mathbb{F}$ is algebraic.*

Proof. This is a consequence of Noether's Normalisation Lemma. Since $\mathbb{E}$ is finitely generated as an $\mathbb{F}$-algebra, we have a finite injective homomorphism $B = \mathbb{F}[t_1, \dots, t_d] \hookrightarrow \mathbb{E}$. We claim that $d = 0$. If d were positive, we could take t_1 and note that $\alpha = 1/t_1 \in \mathbb{E}$ since $\mathbb{E}$ is a field. Then α is integral over B (since finite implies integral), which means there is a monic polynomial $q(y) \in B[y]$ such that $q(\alpha) = 0$. If we write $q(y) = y^m + b_{m-1}y^{m-1} + \cdots + b_1 y + b_0$, then we have

$$0 = \alpha^m + b_{m-1}\alpha^{m-1} + \cdots + b_1\alpha + b_0$$

in $\mathbb{E}$. But after multiplication by t_1^m, this becomes

$$0 = 1 + b_{m-1}t_1 + \cdots + b_1 t_1^{m-1} + b_0 t_1^m,$$

which says that t_1 is invertible in B, contradicting $d > 0$. □

Remark B.1.3.2. Note that the proof of Lemma B.1.3.1 actually solves **??**.

COROLLARY B.1.3.3. *If $\mathbb{F}$ is a field, A is a finitely generated $\mathbb{F}$-algebra and $\mathfrak{m} \subset A$ is a maximal ideal, the quotient $A/\mathfrak{m}$ is a finite extension of $\mathbb{F}$.*

Proof. Apply Zariski's Lemma to $\mathbb{E} = A/\mathfrak{m}$. □

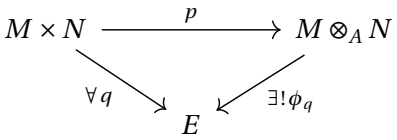
B.2 Tensor products

Definition B.2.0.1 (Tensor product of modules)**.** Let A be a ring, M and N two A-modules. The *tensor product* of M and N over A is defined to be a pair $(M \otimes_A N, p)$ where

- $M \otimes_A N$ is an A-module,
- $p : M \times N \to M \otimes_A N$ is a bilinear map,

such that the following universal property is satisfied: for every pair (E, q) where E is an A-module and $q : M \times N \to E$ is a bilinear map, there is exactly one A-linear homomorphism $\phi_q : M \otimes_A N \to E$ such that $q = \phi_q \circ p$.

The universal property of Definition B.2.0.1 can be depicted in the diagram



and, more importantly, can be rephrased by saying that there is a bijection

$$\mathsf{Bil}_A(M\times N,E)\xrightarrow{\sim}\mathrm{Hom}_{\mathsf{Mod}_A}(M\otimes_A N,E),\quad q\mapsto\phi_q,$$

functorial in E.

Regarding existence of an object $(M\otimes_A N,p)$ with the required universal property, one first considers the standard basis $\{\,e_{m,n}\mid m\in M,\ n\in N\,\}$ of the direct sum $A^{\oplus M\times N}$. One then constructs the quotient module

$$M\otimes_A N=A^{\oplus M\times N}/T$$

where $T\subset A^{\oplus M\times N}$ is the submodule generated by elements of the form

$$\begin{aligned}
&e_{m_1+m_2,n}-e_{m_1,n}-e_{m_2,n},\\
&e_{m,n_1+n_2}-e_{m,n_1}-e_{m,n_2},\\
&e_{am,n}-e_{m,an},\\
&a\,e_{m,n}-e_{am,n}.
\end{aligned}$$

The map $p\colon M\times N\to M\otimes_A N$ is defined by sending $(m,n)\mapsto[e_{m,n}]$, where the square bracket means equivalence class. One sets

$$m\otimes n=[e_{m,n}].$$

This is standard notation. Note that not all elements of $M\otimes_A N$ are of the form $m\otimes n$ for elements $m\in M$ and $n\in N$. However, every element $u\in M\otimes_A N$ can be written (non-uniquely) as a finite sum

$$u=\sum_{k=1}^{r}m_k\otimes n_k,\quad r>0.$$



Exercise B.2.0.2. Show that the pair $(M\otimes_A N,p)$ satisfies the universal property of Definition B.2.0.1.

Thanks to the universal property, $(M\otimes_A N,p)$ is unique up to a unique isomorphism. It is easy to check, using the universal property, that one has isomorphisms

$$\begin{aligned}
M\otimes_A A&\cong M,\\
M\otimes_A N&\cong N\otimes_A M,\\
(M\otimes_A N)\otimes_A P&\cong M\otimes_A(N\otimes_A P).
\end{aligned}$$

for A-modules M, N and P.



Exercise B.2.0.3. Let $m,n\in\mathbb{Z}$ be integers with $d=\gcd(m,n)$. Show that $\mathbb{Z}/m\mathbb{Z}\otimes_{\mathbb{Z}}\mathbb{Z}/n\mathbb{Z}\cong\mathbb{Z}/d\mathbb{Z}$. In particular, $\mathbb{Z}/m\mathbb{Z}\otimes_{\mathbb{Z}}\mathbb{Z}/n\mathbb{Z}=0$ when m and n are coprime.



Exercise B.2.0.4. Prove that tensor products of fields need not be fields or domains, by showing that

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C}.$$



Exercise B.2.0.5 ($\otimes$ commutes with $\oplus$)**.** If $(M_i)_{i\in I}$ is a family of A-modules, one has a canonical isomorphism

$$\bigoplus_{i\in I}(M_i \otimes_A N) \xrightarrow{\sim} \left(\bigoplus_{i\in I} M_i\right) \otimes_A N$$

for any A-module N.



Exercise B.2.0.6 (Tensoring is right exact)**.** Let A be a ring, M an A-module. Prove that the functor

$$\begin{array}{ccc} \mathsf{Mod}_A & \xrightarrow{M\otimes_A -} & \mathsf{Mod}_A \\ N & \longmapsto & M \otimes_A N \end{array}$$

is right exact, i.e. that an exact sequence $N_0 \to N_1 \to N_2 \to 0$ gets sent to an exact sequence $M \otimes_A N_0 \to M \otimes_A N_1 \to M \otimes_A N_2 \to 0$.



Exercise B.2.0.7 (Restriction to submodules)**.** Let M and N be A-modules, $i : N' \hookrightarrow N$ a submodule. Denote by $i_M : M \otimes_A N' \to M \otimes_A N$ the canonical map. Show that there is a canonical isomorphism of A-modules

$$(M \otimes_A N)/\mathrm{im}(i_M) \xrightarrow{\sim} M \otimes_A (N/N').$$

In particular, if $N = A$ and $N' = I \subset A$ is an ideal, then there is a canonical isomorphism

$$M/IM \xrightarrow{\sim} M \otimes_A (A/I).$$

(**Hint**: use the universal property, or tensor the right exact sequence $N' \to N \to N/N' \to 0$).

B.3 Localisation

B.3.1 General construction for modules

Let A be a ring, M an A-module. Fix a *multiplicative subset* $S \subset A$, i.e. a subset containing the identity $1 \in A$ and such that $s_1 s_2 \in S$ whenever $s_1, s_2 \in S$. We do *not* insist that $0 \notin S$.

Example B.3.1.1. The following are key examples of multiplicative subsets:

(i) $S = \{ f^n \mid n \geqslant 0 \}$ for some $f \in A$.

(ii) $S = A \setminus \mathfrak{p}$, where $\mathfrak{p} \subset A$ is a prime ideal.

(iii) $S = A \setminus 0$, if A is an integral domain.

(iv) $S = A \setminus \mathcal{Z}$, where $\mathcal{Z}$ is the set of all zero-divisors in A.

Consider the equivalence relation on $M \times S$ defined by

$$\boxed{(m,s) \sim (m',s') \Longleftrightarrow \text{there exists } u \in S \text{ such that } u(s'm - sm') = 0 \in M.}$$

We denote by m/s, or by $\frac{m}{s}$, the equivalence class of (m,s). The set of such equivalence classes

$$S^{-1}M = (M \times S)/\sim \tag{B.3.1}$$

is an abelian group via

$$\frac{m}{s} + \frac{m'}{s'} = \frac{sm' + s'm}{ss'},$$

and if $M = A$ then $S^{-1}A$ becomes a ring via

$$\frac{a}{s} \cdot \frac{a'}{s'} = \frac{aa'}{ss'}.$$

The $\mathbb{Z}$-module $S^{-1}M$ is an $S^{-1}A$-module via

$$\frac{a}{s} \cdot \frac{m}{s'} = \frac{am}{ss'}. \tag{B.3.2}$$

Here 'am' refers to the A-module structure on M.

Definition B.3.1.2 (Localisation of a module)**.** Let A be a ring, M an A-module. The localisation of M with respect to a multiplicative subset $S \subset A$ is the $S^{-1}A$-module $S^{-1}M$, where the linear structure is given by Equation (B.3.2).

Notation B.3.1.3. If $S = \{f^n \mid n \geqslant 0\}$ as in Example B.3.1.1(i) above, then we write M_f for the localisation. If $S = A \setminus \mathfrak{p}$ as in Example B.3.1.1(ii) above, then we write $M_{\mathfrak{p}}$ for the localisation. Do not confuse M_f and $M_{(f)}$ when $(f) = fA \subset A$ is a prime ideal!

Remark B.3.1.4. If $0 \in S$, then $S^{-1}M = 0$.

Remark B.3.1.5. The localisations of the form M_f and $M_{\mathfrak{p}}$ are crucial in algebraic geometry. In M_f, the equivalence relation defining the localisation reads

$$M_f \ni \frac{m}{f^d} = \frac{n}{f^e} \iff \text{there exists } k \geqslant 0 \text{ such that } f^k(f^e m - f^d n) = 0 \in M.$$

In particular, $m/f^d = 0 \in M_f$ if and only if there exists $k \geqslant 0$ such that $f^k m = 0 \in M$. In particular,

$$M_f = 0 \iff \text{for every } m \in M \text{ there exists } k \geqslant 0 \text{ such that } f^k m = 0 \in M.$$

If M is finitely generated, this is equivalent to the existence of a uniform $r \geqslant 0$ such that $f^r M = 0$. If $M = A$, one actually has that $A_f = 0$ if and only if f is nilpotent.[1]

[1] It is not true, for an arbitrary A-module, that $M_f = 0$ implies that f is nilpotent. For example, with $A = \mathbb{Z}$, $M = \mathbb{Z}/2\mathbb{Z}$, and $f = 2$, one has $M_f = 0$ although 2 is not nilpotent.

Remark B.3.1.6. We make the following observations.

- Note that $S^{-1}M$ comes with an A-linear map

$$\begin{aligned} M &\xrightarrow{\ell} S^{-1}M \\ m &\longmapsto \frac{m}{1}. \end{aligned}$$

  To see that ℓ is A-linear, simply note that the A-linear structure on $S^{-1}M$ is given by $a \cdot \frac{m}{s} = \frac{a}{1} \cdot \frac{m}{s} = \frac{am}{s}$, where the second identity uses (B.3.2), so

$$\ell(a \cdot m) = \frac{am}{1} = \frac{a}{1} \cdot \frac{m}{1} = a \cdot \ell(m).$$

- Localisation is functorial: if $\phi : N \to M$ is an A-linear map, there is an induced $S^{-1}A$-linear map

$$\begin{aligned} S^{-1}N &\xrightarrow{S^{-1}\phi} S^{-1}M, \\ \frac{n}{s} &\longmapsto \frac{\phi(n)}{s}. \end{aligned}$$

  To see that $S^{-1}\phi$ is $S^{-1}A$-linear, note that if $a/t \in S^{-1}A$ then

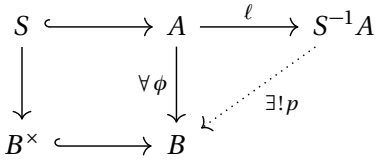
$$S^{-1}\phi\left(\frac{a}{t} \cdot \frac{n}{s}\right) = S^{-1}\phi\left(\frac{an}{ts}\right) = \frac{\phi(an)}{ts} = \frac{a \cdot \phi(n)}{ts} = \frac{a}{t} \cdot \frac{\phi(n)}{s} = \frac{a}{t} \cdot S^{-1}\phi\left(\frac{n}{s}\right).$$

B.3.2 Universal property of localisation

Set $M = A$. There is a canonical ring homomorphism

$$\ell : A \to S^{-1}A, \quad a \mapsto \frac{a}{1}$$

sending S inside the group of invertible elements of $S^{-1}A$ (the inverse of $s/1$ being $1/s$), and making the pair $(S^{-1}A, \ell)$ universal with this property: whenever one has a ring homomorphism $\phi : A \to B$ such that $\phi(S) \subset B^\times$, there is exactly one ring homomorphism $p : S^{-1}A \to B$ such that $\phi = p \circ \ell$.



Explicitly, the map p is defined by $p(a/s) = \phi(a)\phi(s)^{-1}$.

In fact, there is a universal property also for localised modules.

LEMMA B.3.2.1 ([10, Tag 07K0]). *Let A be a ring, $S \subset A$ a multiplicative subset, and fix an A-module M and an $S^{-1}A$-module N. Then, the canonical map*

$$\mathrm{Hom}_{S^{-1}A}(S^{-1}M, N) \longrightarrow \mathrm{Hom}_A(M, N)$$

sending $p : S^{-1}M \to N$ to the map $p \circ \ell : M \to S^{-1}M \to N$, where $\ell : M \to S^{-1}M$ is the localisation map, is an isomorphism.

In other words, for any A-linear map $\phi: M \to N$, there exists exactly one $S^{-1}A$-linear map $p: S^{-1}M \to N$ such that

$$\begin{array}{ccc} M & \xrightarrow{\ell} & S^{-1}M \\ {\scriptstyle \forall\, \phi}\big\downarrow & \swarrow_{\exists!\, p} & \\ N & & \end{array}$$

commutes. Explicitly, the map p is given by $p(m/s) = (1/s)\cdot\phi(m)$.

B.3.3 Ring Localisations

The following lemma is of key importance to us.

LEMMA B.3.3.1. *Let A be a ring, and $\ell: A \to S^{-1}A$ a localisation. Sending $\mathfrak{r} \mapsto \ell^{-1}(\mathfrak{r})$ establishes a bijection*

$$\begin{array}{ccc} \{\text{prime ideals } \mathfrak{r} \subset S^{-1}A\} & \xrightarrow{\simeq} & \{\text{prime ideals } \mathfrak{q} \subset A \text{ such that } \mathfrak{q} \cap S = \emptyset\} \\ \| & & \big\downarrow \\ \operatorname{Spec} S^{-1}A & & \operatorname{Spec} A \end{array}$$

having as inverse the extension operation, sending

$$\mathfrak{q} \longmapsto \mathfrak{q}S^{-1}A = \left\{ \frac{a}{f} \,\middle|\, a \in \mathfrak{q},\, f \in S \right\} \subset S^{-1}A.$$

COROLLARY B.3.3.2. *For any prime ideal $\mathfrak{p} \subset A$ the ring*

$$A_{\mathfrak{p}} = \left\{ \frac{a}{f} \,\middle|\, a \in A,\, f \notin \mathfrak{p} \right\}$$

is local, with maximal ideal

$$\mathfrak{p}A_{\mathfrak{p}} = \left\{ \frac{a}{f} \,\middle|\, a \in \mathfrak{p},\, f \notin \mathfrak{p} \right\} \subset A_{\mathfrak{p}}.$$

Proof. Indeed, the correspondence of Lemma B.3.3.1 becomes, in the case $S = A \setminus \mathfrak{p}$,

$$\{\text{prime ideals } \mathfrak{r} \subset A_{\mathfrak{p}}\} \xrightarrow{\simeq} \{\text{prime ideals } \mathfrak{q} \subset A \text{ such that } \mathfrak{q} \subset \mathfrak{p}\}$$

and since its inverse (extension along $A \to A_{\mathfrak{p}}$) is inclusion-preserving it follows that every prime ideal $\mathfrak{r} \subset A_{\mathfrak{p}}$ must be contained in $\mathfrak{p}A_{\mathfrak{p}}$. This means that $\mathfrak{p}A_{\mathfrak{p}}$ is the unique maximal ideal. □



Exercise B.3.3.3. If $(A, \mathfrak{m})$ is a local ring, then the canonical map $A \to A_{\mathfrak{m}}$ is an isomorphism.



Warning B.3.3.4. In the case when B is a graded ring and $\mathfrak{p}$ is a homogeneous prime ideal, we use the notation $B_{\mathfrak{p}}$ for the localisation of B at the multiplicative subset consisting of *homogeneous* elements that are not in $\mathfrak{p}$.

PROPOSITION B.3.3.5 ([6, Prop. 5.8]). *If $\mathfrak{m} \subset A$ is a maximal ideal and $k > 0$ is an integer, there is a natural ring isomorphism*

$$A/\mathfrak{m}^k \xrightarrow{\sim} A_{\mathfrak{m}}/(\mathfrak{m}A_{\mathfrak{m}})^k.$$

It induces isomorphisms

$$\mathfrak{m}^h/\mathfrak{m}^k \xrightarrow{\sim} (\mathfrak{m}A_{\mathfrak{m}})^h/(\mathfrak{m}A_{\mathfrak{m}})^k$$

for every $h \leqslant k$.

LEMMA B.3.3.6. *Let A be a ring, $S \subset A$ a multiplicative subset. Then $\ell : A \to S^{-1}A$ is injective if and only if S contains no zero divisors.*

Proof. Suppose $a/1 = 0/1$ in $S^{-1}A$. Then there is $u \in S$ such that $au = 0$. But u is not a zero divisor, thus $a = 0$ and ℓ is injective. Conversely, if $s \in S$ and $sa = 0$ with $a \neq 0$, then $a/1 = 0 \in S^{-1}A$, contradicting injectivity. □

Example B.3.3.7. Let A be an integral domain, which means that $(0) \subset A$ is prime. Then the localisation

$$A_{(0)} = \left\{ \frac{a}{b} \;\middle|\; a \in A,\ b \in A \setminus 0 \right\}$$

is a field, called the *fraction field* of A, that we denote by $\operatorname{Frac} A$. The canonical map $\ell : A \to \operatorname{Frac} A$ is injective by Lemma B.3.3.6.

Example B.3.3.8. Let A be a ring. Consider $S = A \setminus \mathcal{Z}$ as in Example B.3.1.1(iv). The localisation $S^{-1}A$ is called the *total ring of fractions* of A. By Lemma B.3.3.6, $S = A \setminus \mathcal{Z}$ is the largest multiplicative set such that $\ell : A \to S^{-1}A$ is injective.

Example B.3.3.9. Let $A = \mathbb{Z}$. Fix a prime number $p \in \mathbb{Z}$. Then the localisation map

$$\mathbb{Z} \hookrightarrow \mathbb{Z}_{(p)} = \left\{ \tfrac{n}{m} \;\middle|\; n \in \mathbb{Z},\ p \nmid m \right\}$$

is injective, and so is the localisation map

$$\mathbb{Z} \hookrightarrow \mathbb{Z}[1/p] = \left\{ \tfrac{n}{p^k} \;\middle|\; n \in \mathbb{Z},\ k \geqslant 0 \right\}.$$

LEMMA B.3.3.10. *If A is reduced and $S \subset A$ is a multiplicative subset, then $S^{-1}A$ is also reduced.*

Proof. Assume there exists $a \in A$, $s \in S$ and $r \in \mathbb{Z}_{>0}$ such that $0/1 = (a/s)^r = a^r/s^r \in S^{-1}A$. Then there exists $u \in S$ such that $ua^r = 0 \in A$, thus $(ua)^r = 0$, and hence $ua = 0$ by assumption. But this means $0/1 = a/1 \in S^{-1}A$, and thus $0/1 = (a/1)(1/s) = a/s \in S^{-1}A$. □

B.3.4 Exactness of localisation

LEMMA B.3.4.1. *Let A be a ring, $S \subset A$ a multiplicative subset, M an A-module. Then, there is a canonical isomorphism of $S^{-1}A$-modules*

$$\phi : S^{-1}M \xrightarrow{\sim} M \otimes_A S^{-1}A.$$

Proof. First of all, the $S^{-1}A$-module structure on $M \otimes_A S^{-1}A$ is defined (on generators) by

$$\frac{a}{t} \cdot \left(m \otimes \frac{b}{s} \right) = m \otimes \frac{ab}{ts}.$$

The map ϕ is defined by

$$\phi\left(\frac{m}{s}\right) = m \otimes \frac{1}{s}.$$

It is $S^{-1}A$-linear, since

$$\begin{aligned}
\phi\left(\frac{a}{t} \cdot \frac{m}{s}\right) &= \phi\left(\frac{am}{ts}\right) \\
&= am \otimes \frac{1}{ts} \\
&= m \otimes \frac{a}{ts} \\
&= \frac{a}{t} \cdot \left(m \otimes \frac{1}{s} \right) \\
&= \frac{a}{t} \cdot \phi\left(\frac{m}{s}\right).
\end{aligned}$$

Its inverse is given by $m \otimes (a/s) \mapsto (am)/s$. □

PROPOSITION B.3.4.2. *Let A be a ring, $S \subset A$ a multiplicative subset. Then, sending $M \mapsto S^{-1}M$ defines an exact functor from A-modules to $S^{-1}A$-modules.*

Proof. Fix a short exact sequence

$$0 \longrightarrow M \xrightarrow{\iota} N \xrightarrow{\pi} P \longrightarrow 0$$

of A-modules. We already know that

$$S^{-1}M \xrightarrow{S^{-1}\iota} S^{-1}N \xrightarrow{S^{-1}\pi} S^{-1}P \longrightarrow 0$$

is exact, since this sequence is isomorphic to

$$M \otimes_A S^{-1}A \longrightarrow N \otimes_A S^{-1}A \longrightarrow P \otimes_A S^{-1}A \longrightarrow 0$$

by Lemma B.3.4.1, and tensor product (by any A-module, e.g. $S^{-1}A$) is a right exact functor, cf. Exercise B.2.0.6. So we only need to show that

$$S^{-1}M \xrightarrow{S^{-1}\iota} S^{-1}N$$

is injective. Assume there is an element $m/s \in S^{-1}M$ such that $0 = 0/1 = S^{-1}\iota(m/s) = \iota(m)/s \in S^{-1}N$. Then there exists $u \in S$ such that $0 = u\iota(m) = \iota(um)$ in N. This implies $um = 0 \in M$, hence $m/s = um/us = 0/us = 0$. □

B.4 Length of modules

We recall some notions and facts from [10, Tag 00IU]. The theory of lengths is very important in the study of 0-dimensional coherent sheaves on varieties, and it will appear multiple times in our discussion on associated primes.

Let A be a ring, M an A-module. A chain of submodules of length n is a filtration

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M.$$

Definition B.4.0.1 (Length of a module)**.** Let A be a ring, M an A-module. Define the *length* of M to be

$$\mathsf{length}_A(M) = \sup_n \left\{ n \;\middle|\; \text{there exists a chain of submodules of length } n \right\}.$$

Remark B.4.0.2. If M has finite length, it is not obvious from the definition that all maximal chains have the same length. This is, however, true (cf. Proposition B.4.0.7).

The length is related to the notion of *support* of a module, which will also be central in later sections of this appendix.

Definition B.4.0.3 (Support of a module)**.** Let A be a ring, M an A-module. Define the *support* of M to be

$$\mathsf{Supp}_A(M) = \left\{ \mathfrak{p} \in \operatorname{Spec} A \;\middle|\; M_{\mathfrak{p}} \neq 0 \right\} \subset \operatorname{Spec} A.$$

Here are a series of easy, useful facts about length and support. Most proofs can be found in [10, Tag 00IU].

- If $\mathsf{length}_A(M) < \infty$, then M is finitely generated.
- If $0 \to M' \to M \to M'' \to 0$ is a short exact sequence of A-modules, then

$$\begin{aligned} \mathsf{length}_A(M) &= \mathsf{length}_A(M') + \mathsf{length}_A(M''), \\ \mathsf{Supp}_A(M) &= \mathsf{Supp}_A(M') \cup \mathsf{Supp}_A(M''). \end{aligned} \tag{B.4.1}$$

  The second identity is proved, for instance, in [10, Tag 00L3].

- If $(A, \mathfrak{m})$ is local and M has finite length, then $\mathfrak{m}^n M = 0$ for some $n \geqslant 0$. Conversely, if $\mathfrak{m}$ and M are finitely generated and $\mathfrak{m}^n M = 0$ for some n, then M has finite length.
- For a noetherian local ring $(A, \mathfrak{m})$ and a finitely generated module $M \neq 0$, this gives the useful equivalence

$$\mathsf{length}_A(M) < \infty \iff \mathfrak{m}^n M = 0 \text{ for some } n \iff \mathsf{Supp}_A(M) = \{\mathfrak{m}\}.$$

- If $A \to B$ is a ring homomorphism and M is a B-module, then $\mathsf{length}_A(M) \geqslant \mathsf{length}_B(M)$, with equality if $A \to B$ is surjective.
- In particular, if $B = S^{-1}A$ for some multiplicative subset S, since every B-submodule $N' \subset S^{-1}M$ is of the form $S^{-1}N$ for some A-submodule $N \subset M$, we deduce that $\mathsf{length}_A(M) \geqslant \mathsf{length}_{S^{-1}A}(S^{-1}M)$.
- Let $(A, \mathfrak{m})$ be a local ring, M an A-module with $\mathfrak{m}M = 0$. Then

$$\mathsf{length}_A(M) = \dim_{A/\mathfrak{m}} M,$$

and $\mathsf{length}_A(M) < \infty$ if and only if M is finitely generated.

Remark B.4.0.4. The implication

$$(A, \mathfrak{m}) \text{ local, } M \text{ finitely generated} \quad \Longrightarrow \quad \mathfrak{m}^n M = 0 \text{ for some } n.$$

is false. For example, take $M = A = \mathbb{F}[x]_{(x)}$. Then M is finitely generated over the local ring A, but $x^n A \neq 0$ for every n. One really needs finite length for the conclusion to hold.



Exercise B.4.0.5. Recall from **??** the characterisations of 0-dimensional noetherian local rings A. We propose now the following further characterisation: show that 0-dimensionality, for a noetherian local ring A, is equivalent to $\mathsf{length}_A(A) < \infty$.

Recall that a module $M \neq 0$ is *simple* if it has no proper nonzero submodules.

LEMMA B.4.0.6 (Length and simple modules [10, Tag 00J2]). *Let A be a ring, $M \neq 0$ an A-module. The following are equivalent:*

(i) *M is simple,*

(ii) *$\mathsf{length}_A(M) = 1$, and*

(iii) *$M \cong A/\mathfrak{m}$ for some maximal ideal $\mathfrak{m} \subset A$.*

The following proposition, in particular the last statement about the "local length", is tightly linked to Proposition B.5.1.13.

PROPOSITION B.4.0.7 ([10, Tag 00J3]). *Let A be a ring, M an A-module of finite length. Fix a maximal chain of submodules*

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M.$$

Then $n = \mathsf{length}_A(M)$, each subquotient M_i/M_{i-1} is simple, hence isomorphic to $A/\mathfrak{m}_i$ for some maximal ideal $\mathfrak{m}_i \subset A$, and for every maximal ideal $\mathfrak{m} \subset A$ one has that

$$\#\{\, i \mid \mathfrak{m}_i = \mathfrak{m} \,\} = \mathsf{length}_{A_\mathfrak{m}}(M_\mathfrak{m}).$$



Exercise B.4.0.8. Let M be a finitely generated module over a local ring $(A, \mathfrak{m}, \kappa)$. Assume $\mathfrak{m}^n M = 0$ for some $n \geqslant 0$. Prove that

$$\mathsf{length}_A(M) = \sum_{i=0}^{n-1} \dim_\kappa \left(\frac{\mathfrak{m}^i M}{\mathfrak{m}^{i+1} M} \right).$$

B.5 Associated points, and embedded components

For the theory of associated points we mainly follow Vakil's exposition [12, Ch. 6] and Eisenbud's [2, Ch. 3], whereas for Cohen–Macaulay schemes we follow [10].

What are associated points? Just like the generic points, they are very specific points of a (locally noetherian) scheme X. They are more relevant than all other points, in that they reveal part of the behaviour of the structure sheaf.

The generic points of the irreducible components of X are always associated points. There may also be nonminimal associated points, called embedded points. The closure $\overline{\{x\}} \subset X$ of an embedded point $x \in X$ is called an *embedded component*, and is always contained, set-theoretically, in an irreducible component of X. A reduced locally noetherian scheme never has embedded points. The presence of embedded points records additional scheme-theoretic structure concentrated on proper closed subsets. However, a nonreduced scheme need not have embedded points: its nilpotent structure may just be supported along an irreducible component.

In this section, the "best" results are obtained for *finitely generated* modules over a *noetherian* ring. We will treat the general theory without making these assumptions right away, but the reader can safely make them from the start.

B.5.1 Associated points: local case

Let A be a commutative ring, and let M be an A-module. If $m \in M$, we set

$$\begin{aligned}
\mathsf{Ann}_A(m) &= \{\, a \in A \mid am = 0 \,\} = \ker(A \xrightarrow{\cdot m} M) \subset A,\\
\mathsf{Ann}_A(M) &= \{\, a \in A \mid am = 0 \text{ for all } m \in M \,\} \subset A,\\
\mathsf{Supp}_A(m) &= \left\{\, \mathfrak{p} \in \operatorname{Spec} A \;\middle|\; m_{\mathfrak{p}} \neq 0 \,\right\} \subset \operatorname{Spec} A,\\
\mathsf{Supp}_A(M) &= \left\{\, \mathfrak{p} \in \operatorname{Spec} A \;\middle|\; M_{\mathfrak{p}} \neq 0 \,\right\} \subset \operatorname{Spec} A.
\end{aligned}$$

In the third identity, $m_{\mathfrak{p}}$ denotes the image of $m \in M$ in the localisation $M_{\mathfrak{p}}$. The support $\mathsf{Supp}_A(M)$ had already been introduced in Definition B.4.0.3.

We have the identity

$$\mathsf{Supp}_A(m) = \mathrm{V}(\mathsf{Ann}_A(m)),$$

and if $M = mA$ is principal, then

$$\mathsf{Supp}_A(M) = \mathsf{Supp}_A(m) = \mathrm{V}(\mathsf{Ann}_A(m)).$$

In general, it is clear that

$$\mathsf{Supp}_A(M) \subset \mathrm{V}(\mathsf{Ann}_A(M)) = \{\, \mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \supset \mathsf{Ann}_A(M) \,\}. \tag{B.5.1}$$

Indeed, fix $\mathfrak{p}$ such that $M_{\mathfrak{p}} \neq 0$. If $\mathfrak{p} \notin \mathrm{V}(\mathsf{Ann}_A(M))$, then there exists $a \in \mathsf{Ann}_A(M)$ such that $a \notin \mathfrak{p}$. But $a \notin \mathfrak{p}$ implies $\overline{a} = a/1 \in A_{\mathfrak{p}}^{\times} \subset A_{\mathfrak{p}}$ is a unit, whereas $a \in \mathsf{Ann}_A(M)$ implies

that $\overline{a}M_\mathfrak{p} = 0$. But a unit cannot annihilate $M_\mathfrak{p}$ unless $M_\mathfrak{p} = 0$, contradiction. To obtain the reverse inclusion $\mathsf{V}(\mathsf{Ann}_A(M)) \subset \mathsf{Supp}_A(M)$, on the other hand, one needs to assume M to be finitely generated, as the next exercise proposes to confirm.



Exercise B.5.1.1 ($\mathsf{Supp}_A(M) = \mathsf{V}(\mathsf{Ann}_A(M))$ if M is finite)**.** Assume M is a finitely generated A-module, and there exists $a \in A$ such that $a \in \mathfrak{p}$ for all $\mathfrak{p} \in \mathsf{Supp}_A(M)$. Then there exists $r > 0$ such that $a^r \in \mathsf{Ann}_A(M)$. In other words,

$$\bigcap_{\mathfrak{p} \in \mathsf{Supp}_A(M)} \subset \sqrt{\mathsf{Ann}_A(M)}.$$

Coupled with (B.5.1), we obtain, when M is finitely generated, the identity

$$\mathsf{Supp}_A(M) = \mathsf{V}(\mathsf{Ann}_A(M)).$$

Definition B.5.1.2 (Associated primes)**.** Let A be a commutative ring, and let M be an A-module. A prime ideal $\mathfrak{p} \in \operatorname{Spec} A$ is said to be *associated to* M if $\mathfrak{p} = \mathsf{Ann}_A(m)$ for some $m \in M$. The set of all primes associated to M is denoted

$$\mathsf{Ass}_A(M) = \{\, \mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \text{ is associated to } M \,\} \subset \operatorname{Spec} A.$$

We have the following characterisation of associated primes.

LEMMA B.5.1.3. *Let A be a ring, and let M be an A-module. Let $\mathfrak{p} \subset A$ be a prime ideal. Then $\mathfrak{p} \in \mathsf{Ass}_A(M)$ if and only if $A/\mathfrak{p}$ is an A-submodule of M.*

Proof. If $\mathfrak{p} = \mathsf{Ann}_A(m)$ for some $m \in M$, consider the map $\phi_m \colon A \to M$ defined by $\phi_m(a) = a \cdot m$. Since its kernel is by definition $\mathsf{Ann}_A(m)$, the quotient $A/\mathfrak{p}$ is an A-submodule of M. Conversely, given an A-linear inclusion $i \colon A/\mathfrak{p} \hookrightarrow M$, consider the composition $\phi \colon A \twoheadrightarrow A/\mathfrak{p} \hookrightarrow M$. Then $\phi_{i(1)}(a) = a \cdot i(1) = i(a + \mathfrak{p}) = \phi(a)$ for all $a \in A$, i.e. $\phi = \phi_{i(1)}$. □

Example B.5.1.4. If A is an integral domain, the only associated prime of $M = A$ is the trivial ideal $(0) \subset A$. If A is a UFD and $f \in A$ is a nonzero nonunit, the only associated primes in $M = A/fA$ are the prime factors of f. See [12, Ex. 6.6.E and 6.6.F].

Remark B.5.1.5. Let A be a ring, and let M be an A-module. If $\mathfrak{p} \in \mathsf{Ass}_A(M)$ then $\mathfrak{p} \supset \mathsf{Ann}_A(M)$. Thus, $\mathsf{Ass}_A(M) \subset \mathsf{V}(\mathsf{Ann}_A(M))$. In fact, something finer is true: we always have

$$\boxed{\mathsf{Ass}_A(M) \subset \mathsf{Supp}_A(M) \subset \mathsf{V}(\mathsf{Ann}_A(M)).}$$

Indeed, if $\mathfrak{p} = \mathsf{Ann}_A(m) \in \mathsf{Ass}_A(M)$, then $am \neq 0$ for every $a \in A \setminus \mathfrak{p}$, and this tells us that $0 \neq m/1 \in M_\mathfrak{p}$, in particular $\mathfrak{p} \in \mathsf{Supp}_A(M)$. The inclusion $\mathsf{Supp}_A(M) \subset \mathsf{V}(\mathsf{Ann}_A(M))$, on the other hand, was confirmed in (B.5.1).



Exercise B.5.1.6. An element $a \in A$ is said to be a zero-divisor on an A-module M if $am = 0$ for some nonzero $m \in M$. Denote by $\mathsf{Z}_A(M)$ the set of zero-divisors on M. Assume A is noetherian. Prove that

$$\mathsf{Z}_A(M) = \bigcup_{\mathfrak{p} \in \mathsf{Ass}_A(M)} \mathfrak{p}.$$

Note that this generalises Lemma 2.1.0.6.



Exercise B.5.1.7. Let A be a noetherian ring, M an A-module. Show that

(i) $\mathsf{Ass}_A(M) = \emptyset$ if and only if $M = 0$, and

(ii) the natural map $M \to \prod_{\mathfrak{p} \in \mathsf{Ass}_A(M)} M_{\mathfrak{p}}$ is injective,

For (i), you can look at [12, Prop. 6.6.10] or [2, Prop. 3.4] if you need a hint.

Notation B.5.1.8. We denote by $\mathsf{Min}(S) \subset S$ the subset of minimal elements, with respect to inclusion, in a set S.

Definition B.5.1.9 (Isolated primes)**.** Let A be a ring, M an A-module. The minimal elements (with respect to inclusion) in $\mathsf{Supp}_A(M)$ are called *isolated primes* of M. In symbols, this is the subset $\mathsf{Min}(\mathsf{Supp}_A(M)) \subset \mathsf{Supp}_A(M)$.

Note that it is not yet clear at this point that all isolated primes of M belong to $\mathsf{Ass}_A(M)$. All we have at this point is

$$\begin{array}{ccccccc} & & \mathsf{Ass}_A(M) & & & & \\ & \overset{\cdots\searrow}{\nearrow} & \downarrow & & & & \\ \mathsf{Min}(\mathsf{Supp}_A(M)) & \hookrightarrow & \mathsf{Supp}_A(M) & \hookrightarrow & \mathsf{V}(\mathsf{Ann}_A(M)) & \hookrightarrow & \operatorname{Spec} A \end{array}$$

but the inclusion $\mathsf{Min}(\mathsf{Supp}_A(M)) \subset \mathsf{Ass}_A(M)$ will be true for $M \neq 0$ finitely generated over a noetherian ring A by Theorem B.5.1.11, and the above picture will become

$$\begin{array}{ccccccc} & & \mathsf{Ass}_A(M) & \overset{\text{finite} \neq \emptyset}{\longleftarrow} & & & \\ & \overset{\text{Theorem B.5.1.11}}{\nearrow} & \downarrow & & & & \\ \mathsf{Min}(\mathsf{Supp}_A(M)) & \hookrightarrow & \mathsf{Supp}_A(M) & = & \mathsf{V}(\mathsf{Ann}_A(M)) & \hookrightarrow & \operatorname{Spec} A, \end{array}$$

at least very informally speaking.

The next result records the key steps in the general theory.

THEOREM B.5.1.10. *The following statements hold true.*

(1) *If* $0 \to M' \to M \to M'' \to 0$ *is a short exact sequence of* A*-modules, there are inclusions*

$$\mathsf{Ass}_A(M') \subset \mathsf{Ass}_A(M) \subset \mathsf{Ass}_A(M') \cup \mathsf{Ass}_A(M''). \tag{B.5.2}$$

(2) *If M is finitely generated, then $\overline{\mathrm{Ass}_A(M)} \subset \mathrm{Supp}_A(M)$, and if A is furthermore assumed to be noetherian, one has*

$$\overline{\mathrm{Ass}_A(M)} = \mathrm{Supp}_A(M).$$

(3) *If M is a finitely generated module over a noetherian ring A, and $S \subset A$ is a multiplicative subset, then*

$$\mathrm{Ass}_{S^{-1}A}(S^{-1}M) = \left\{ S^{-1}\mathfrak{p} \;\middle|\; \mathfrak{p} \in \mathrm{Ass}_A(M),\, \mathfrak{p} \cap S = \emptyset \right\} \subset \mathrm{Spec}\, S^{-1}A.$$

In particular, for $\mathfrak{p} \in \mathrm{Spec}\, A$, there is an identity

$$\mathrm{Ass}_{A_\mathfrak{p}}(M_\mathfrak{p}) = \left\{ \mathfrak{q}A_\mathfrak{p} \;\middle|\; \mathfrak{q} \in \mathrm{Ass}_A(M),\, \mathfrak{q} \subset \mathfrak{p} \right\}.$$

Proof. We provide references or give quick proofs for all the statements.

For part (1), the inclusions (B.5.2) are proved in [12, Prop. 6.6.15] (see also [2, Lemma 3.6] and [10, Tag 02M3]).

The inclusion $\overline{\mathrm{Ass}_A(M)} \subset \mathrm{Supp}_A(M)$ in part (2) follows from the inclusion $\mathrm{Ass}_A(M) \subset \mathrm{Supp}_A(M)$ of Remark B.5.1.5 combined with the closedness of $\mathrm{Supp}_A(M) = \mathrm{V}(\mathrm{Ann}_A(M))$, cf. Exercise B.5.1.1, whereas the identity $\overline{\mathrm{Ass}_A(M)} = \mathrm{Supp}_A(M)$ is proved in [12, Prop. 6.6.24] under the stated noetherian assumption. Part (3) is proved in [12, Prop. 6.6.26] or [2, Thm. 3.1]. □

From now on we assume A is noetherian and M is finitely generated. We have the following result.

THEOREM B.5.1.11. *Let A be a noetherian ring, $M \neq 0$ a finitely generated A-module. Then $\mathrm{Ass}_A(M)$ is a finite nonempty set containing all isolated primes.*

Proof. The fact that $\mathrm{Ass}_A(M) \neq \emptyset$ is just Exercise B.5.1.7(i).

Finiteness follows from the following standard fact in commutative algebra: every nonzero finitely generated module M over a noetherian ring A admits a finite 'prime filtration'

$$\boxed{0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_s = M, \qquad M_i/M_{i-1} \cong A/\mathfrak{p}_i.}$$

To obtain such a prime filtration, one proceeds as follows (as done e.g. in [2, Prop. 3.7]): Since $M \neq 0$, by Exercise B.5.1.7(i) we have at least one element $\mathfrak{p}_1 \in \mathrm{Ass}_A(M)$, so by Lemma B.5.1.3 we know $M_1 = A/\mathfrak{p}_1 \subset M$ is a submodule. Apply this reasoning again to M/M_1 to produce M_2, and use the fact that submodules of M satisfy the ascending chain condition to conclude that the process stops, i.e. $M_n = M$ for some integer n. See also [10, Tag 00L0].

Caveat: a prime filtration as above is not unique, and neither the full set of primes $\{\mathfrak{p}_i\}$ nor their multiplicities is invariant in general. What is always true is that, for any such filtration, giving rise to prime ideals $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_s\}$, one has

$$\mathrm{Ass}_A(M) \subset \{\mathfrak{p}_1, \ldots, \mathfrak{p}_s\} \subset \mathrm{Supp}_A(M).$$

The first inclusion is easily proved combining Equation (B.5.2) with an inductive argument, and implies that $\mathsf{Ass}_A(M)$ is a finite set.

It remains to confirm that $\mathsf{Ass}_A(M)$ contains all isolated primes. Let $\mathfrak{p} \in \mathsf{Min}(\mathsf{Supp}_A(M))$ be an isolated prime. Localising at $\mathfrak{p}$, we get

$$\mathsf{Supp}_{A_\mathfrak{p}}(M_\mathfrak{p}) = \{\mathfrak{p}A_\mathfrak{p}\}.$$

Since $M_\mathfrak{p} \neq 0$ is finitely generated over the noetherian ring $A_\mathfrak{p}$, we have

$$\emptyset \neq \mathsf{Ass}_{A_\mathfrak{p}}(M_\mathfrak{p}) \subset \mathsf{Supp}_{A_\mathfrak{p}}(M_\mathfrak{p}).$$

So necessarily

$$\mathfrak{p}A_\mathfrak{p} \in \mathsf{Ass}_{A_\mathfrak{p}}(M_\mathfrak{p}) = \left\{ \mathfrak{q}A_\mathfrak{p} \;\middle|\; \mathfrak{q} \in \mathsf{Ass}_A(M),\, \mathfrak{q} \subset \mathfrak{p} \right\},$$

where we used Theorem B.5.1.10(3). Thus $\mathfrak{p}A_\mathfrak{p} = \mathfrak{q}A_\mathfrak{p}$ for some $\mathfrak{q} \subset \mathfrak{p}$. Since $\mathsf{Ass}_A(M) \subset \mathsf{Supp}_A(M)$ and $\mathfrak{p}$ is minimal in $\mathsf{Supp}_A(M)$, we must have $\mathfrak{q} = \mathfrak{p}$, hence $\mathfrak{p} \in \mathsf{Ass}_A(M)$. □

Example B.5.1.12. Fix a nonzero finitely generated module M over a noetherian ring A, and consider a prime filtration

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_s = M, \qquad M_i/M_{i-1} \cong A/\mathfrak{p}_i. \tag{B.5.3}$$

Note that 'extra primes' may occur. For $A = \mathbb{F}[x] = M$, the filtration $0 \subsetneq A$ has the single factor $A/(0)$, giving rise to the single prime $\mathfrak{p}_1 = (0)$, whereas $0 \subsetneq xA \subsetneq A$ gives rise to primes $\mathfrak{p}_2 = xA$ (second inclusion) and $\mathfrak{p}_1 = (0)$ (since $xA/(0) = xA \cong A = A/(0)$). However, $\mathfrak{p}_1 = (0)$ is the only isolated prime.

There is nevertheless an invariant multiplicity at the *minimal* primes of the support. If $\mathfrak{p}$ is minimal in $\mathsf{Supp}_A(M)$, i.e. it is isolated, then the number of indices i with $\mathfrak{p}_i = \mathfrak{p}$ is independent of the prime filtration and equals

$$\mathsf{length}_{A_\mathfrak{p}} M_\mathfrak{p}.$$

The length of a module was introduced in Definition B.4.0.1.

It is worth stating this fact in a precise way.

PROPOSITION B.5.1.13. *Let $M \neq 0$ be a finitely generated module over a noetherian ring A. Then, the following sets of prime ideals in A coincide:*

(i) *the isolated primes of M, i.e.* $\mathsf{Min}(\mathsf{Supp}_A(M))$,

(ii) *the minimal primes in* $\mathsf{Ass}_A(M)$, *i.e.* $\mathsf{Min}(\mathsf{Ass}_A(M))$, *and*

(iii) *the minimal primes in $S = \{\mathfrak{p}_1, \ldots, \mathfrak{p}_s\}$, where S arises from any prime filtration as in* (B.5.3).

Furthermore, if $\mathfrak{p}$ belongs to any of these sets, then the number of times it occurs in a prime filtration as in (B.5.3) *is independent of the filtration, and it equals* $\mathsf{length}_{A_\mathfrak{p}} M_\mathfrak{p}$.

Proof. The sets in (i) and (iii) agree because $\mathrm{Supp}_A(M) = \bigcup_{1\leqslant i\leqslant s} \mathrm{V}(\mathfrak{p}_i)$. See [10, Tag 02CE] for the other equivalence.

To obtain the last statement, localise the filtration at $\mathfrak{p}$. A factor $(A/\mathfrak{p}_i)_\mathfrak{p}$ is nonzero exactly when $\mathfrak{p}_i \subset \mathfrak{p}$. Since every $\mathfrak{p}_i$ lies in $\mathrm{Supp}_A(M)$ and $\mathfrak{p}$ is minimal there, this forces $\mathfrak{p}_i = \mathfrak{p}$. Each surviving factor is the residue field $\kappa(\mathfrak{p})$, of length one. □

Definition B.5.1.14 (Embedded primes)**.** Let A be a noetherian ring, M a finitely generated A-module. The non-isolated primes in $\mathrm{Ass}_A(M)$ are called the *embedded primes* of M.

In particular, in the setup of Definition B.5.1.14, one has

$$\mathrm{Ass}_A(M) = \{\text{ embedded primes of } M\ \} \amalg \{\text{ isolated primes of } M\ \}.$$

The most boring situation is when A is an integral domain, in which case the generic point $\xi \in \operatorname{Spec} A$ is the only associated (and clearly isolated, being minimal) prime, cf. Example B.5.1.4. A similar situation occurs with reduced affine schemes, as the following exercise illustrates.



Exercise B.5.1.15 (Reduced implies no embedded points, I)**.** Let A be a reduced noetherian ring. Show that $M = A$ has no embedded associated primes. In other words, $\mathrm{Ass}_A(A) = \mathrm{Min}(\operatorname{Spec} A)$, i.e. associated primes precisely correspond to the irreducible components of $\operatorname{Spec} A$.

Remark B.5.1.16. We make the following observations. Let M be a finitely generated module over a noetherian ring A.

(i) We know that
$$\mathrm{Min}(\mathrm{Supp}_A(M)) \subset \mathrm{Ass}_A(M) \subset \mathrm{Supp}_A(M),$$
therefore $\mathrm{Min}(\mathrm{Supp}_A(M)) = \mathrm{Min}(\mathrm{Ass}_A(M))$. In other words, the isolated associated primes are precisely the minimal associated primes.

(ii) The isolated associated primes, i.e. the elements of $\mathrm{Min}(\mathrm{Supp}_A(M)) = \mathrm{Min}(\mathrm{Ass}_A(M))$, are precisely the generic points of the irreducible components of $\mathrm{Supp}_A(M)$, namely the minimal primes lying above the ideal $\mathrm{Ann}_A(M) \subset A$.

(iii) Let $\mathfrak{q} \in \mathrm{Ass}_A(M) \setminus \mathrm{Min}(\mathrm{Supp}_A(M))$ be an *embedded* prime. Then there is an isolated prime $\mathfrak{p} \in \mathrm{Min}(\mathrm{Supp}_A(M))$ such that $\mathfrak{p} \subsetneq \mathfrak{q}$. This means that $\mathrm{V}(\mathfrak{q})$ is entirely contained in some irreducible component $\mathrm{V}(\mathfrak{p}) \subset \mathrm{Supp}_A(M)$.

The important point in Remark B.5.1.16(iii) is that embedded primes are *not visible from the underlying topological space* $\mathrm{Supp}_A(M)$ alone. They record extra module-theoretic information, such as torsion or scheme-theoretic structure concentrated along a proper closed subset. We illustrate this with the following example.

Example B.5.1.17. Set $A=\mathbb{F}[x,y]$ and $M=A/(x)\oplus A/(x,y)$, so that $\mathsf{Supp}_A(M)=\mathrm{V}(x)\subset \mathbb{A}^2_{\mathbb{F}}$. However, $\mathsf{Ass}_A(M)=\{(x),(x,y)\}$. The prime (x) is isolated: it is the generic point of the unique irreducible component $\mathrm{V}(x)$ of $\mathsf{Supp}_A(M)=\mathrm{V}(\mathsf{Ann}_A(M))$. The prime (x,y) is embedded: geometrically it is the origin sitting inside the line $\mathrm{V}(x)$.

Of course, the A-module $M'=A/(x)$ has the same support as M, but the two modules are not the same! The difference is recorded by the presence, for M, of an embedded prime. Informally speaking, the support of a module tells us *where* the module lives, whereas its associated primes tell us *how* it lives there.

B.5.2 Primary decomposition and embedded components

Definition B.5.2.1 (Primary ideal)**.** Let A be a noetherian ring. A proper ideal $\mathfrak{q}\subsetneq A$ is called *primary* if whenever a product xy lies in $\mathfrak{q}$, either x or a power of y lies in $\mathfrak{q}$. Equivalently, $\mathfrak{q}\subsetneq A$ is primary if every zero-divisor in $A/\mathfrak{q}$ is nilpotent.

Let A be a noetherian ring. Every proper ideal $I\subset A$ has a (non-unique) *primary decomposition*, i.e. an expression as intersection

$$I=\mathfrak{q}_1\cap\cdots\cap\mathfrak{q}_r$$

of primary ideals. See [2, Prop. 3.10] for a proof. One verifies that the radical of a primary ideal is prime, and one says that $\mathfrak{q}$ is $\mathfrak{p}$-primary if $\sqrt{\mathfrak{q}}=\mathfrak{p}$. One can always ensure that the decomposition is irredundant, i.e. removing any $\mathfrak{q}_i$ changes the intersection, and $\sqrt{\mathfrak{q}_i}\neq\sqrt{\mathfrak{q}_j}$ for all $i\neq j$. Note that (even irredundant) primary decompositions are not unique! However, the *radicals* of the primary ideals appearing in an irredundant primary decomposition are intrinsic, as the next proposition will confirm.

PROPOSITION B.5.2.2 ([2, Prop. 3.10.b]). *Let $I\subset A$ be an ideal with irredundant primary decomposition $I=\mathfrak{q}_1\cap\cdots\cap\mathfrak{q}_r$. Then, the set*

$$\{\,\sqrt{\mathfrak{q}_i}\mid 1\leqslant i\leqslant r\,\}\subset\operatorname{Spec}A$$

is determined by I. More precisely, it coincides with $\mathsf{Ass}_A(A/I)$ and there are inclusions $\mathsf{Ass}_A(A/I)\subset\mathsf{Supp}_A(A/I)=\mathrm{V}(\mathsf{Ann}_A(A/I))=\mathrm{V}(I)$.



Exercise B.5.2.3. Set $A=\mathbb{F}[x,y]$ and $I=(xy,y^2)$. Show that $\mathsf{Ass}_A(A/I)=\{(y),(x,y)\}$.

Definition B.5.2.4 (Embedded component)**.** Let I be an ideal in a noetherian ring A. An *embedded component* of $\operatorname{Spec}A/I$ is a closed subset $Z\subset\operatorname{Spec}A/I$ of the form $Z=\mathrm{V}(\mathfrak{p})=\overline{\{\mathfrak{p}\}}$ for some embedded prime $\mathfrak{p}\in\mathsf{Ass}_A(A/I)\setminus\mathsf{Min}(\mathsf{Ass}_A(A/I))$.

Remark B.5.2.5. Let $I\subset A$ be an ideal with a chosen irredundant primary decomposition $I=\mathfrak{q}_1\cap\cdots\cap\mathfrak{q}_r$. Let

$$\mathsf{Min}(\mathsf{Ass}_A(A/I))\subset\mathsf{Ass}_A(A/I)=\{\,\sqrt{\mathfrak{q}_i}\mid 1\leqslant i\leqslant r\,\}\subset\operatorname{Spec}A$$

be the set of minimal primes, and choose an embedded prime $\sqrt{\mathfrak{q}_i} \in \mathsf{Ass}_A(A/I) \setminus \mathsf{Min}(\mathsf{Ass}_A(A/I))$. Then, as already mentioned, there exists $\mathfrak{p} \in \mathsf{Min}(\mathsf{Ass}_A(A/I))$ such that $\mathfrak{p} \subsetneq \sqrt{\mathfrak{q}_i}$. Set-theoretically, we have $\mathrm{V}(\mathfrak{q}_i) = \mathrm{V}(\sqrt{\mathfrak{q}_i}) \subsetneq \mathrm{V}(\mathfrak{p})$, but *scheme-theoretically* we do not have

$$\operatorname{Spec} A/\mathfrak{q}_i \subset \operatorname{Spec} A/\mathfrak{p}!$$

Example B.5.2.6. Consider $A = \mathbb{F}[x,y]$ and $I = (xy, y^2)$, as in Exercise B.5.2.3. A primary decomposition of I is

$$I = (x,y)^2 \cap (y).$$

However, the closed subscheme $\operatorname{Spec} A/(x,y)^2 \subset \operatorname{Spec} A = \mathbb{A}^2_{\mathbb{F}}$ is not scheme-theoretically contained in $\operatorname{Spec} A/(y)$.

One might be tempted to call a closed subscheme of the form

$$\operatorname{Spec} A/\mathfrak{q}_i \hookrightarrow \operatorname{Spec} A/I,$$

with $I \subset \mathfrak{q}_i \subset A$ as in Remark B.5.2.5, a *scheme-theoretic embedded component*.[2]

But these objects are not canonical, as they depend on the chosen (irredundant) primary decomposition; what is the canonical is the collection of embedded components in the (purely topological) sense of Definition B.5.2.4. For instance, going back to Example B.5.2.6, one has also the primary decompositions

$$(xy, y^2) = (x, y^2) \cap (y) = (x+y, y^2) \cap (y)$$

where of course both $\mathfrak{q} = (x, y^2)$ and $\mathfrak{q}' = (x+y, y^2)$ have radical (x,y). They correspond to the same embedded component, and they (incidentally) give rise to abstractly isomorphic schemes, as $\mathbb{F}[x,y]/(x,y^2) \cong \mathbb{F}[y]/y^2 \cong \mathbb{F}[x,y]/(x+y,y^2)$, but they are not isomorphic (even abstractly) to $\operatorname{Spec} \mathbb{F}[x,y]/(x,y)^2$, as the latter has length 3. To summarise, embedded components contain only topological information, whereas any 'scheme-theoretic embedded component' gives a closed *subscheme* $Y \subset \operatorname{Spec} A/I$ which remembers *how* the nonreduced structure 'causes' the presence of the (topological) embedded components.

B.5.3 Associated points: global case

Let X be a locally noetherian scheme and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module, so that $\mathsf{Supp}(\mathcal{F})$ is a closed subset of X. The set of associated points of $\mathcal{F}$ is the set

$$\mathsf{Ass}_X(\mathcal{F}) = \left\{ x \in X \;\middle|\; \mathfrak{m}_x \in \mathsf{Ass}_{\mathcal{O}_{X,x}}(\mathcal{F}_x) \right\},$$

where $\mathfrak{m}_x$ is the maximal ideal of the local ring $\mathcal{O}_{X,x}$. This set is locally finite,[3] i.e. its intersection with every affine open in X is finite. In fact, $\mathsf{Ass}_X(\mathcal{F})$ is finite if X is noetherian,

[2] This terminology is not standard.

[3] Note that it really is *not* finite in general: for instance, $X = \coprod_{n \geqslant 1} \operatorname{Spec} \mathbb{F}$ is locally noetherian and $\mathcal{O}_X$ is coherent, but $\mathsf{Ass}_X(\mathcal{O}_X) = X$ is infinite.

or more generally if $\mathsf{Supp}(\mathcal{F})$ is quasicompact. This definition of $\mathsf{Ass}_X(\mathcal{F})$ is affine-local: if $U = \operatorname{Spec} A \subset X$ is open and $\mathcal{F}|_U \cong \widetilde{M}$, with M finitely generated, then

$$\mathsf{Ass}_X(\mathcal{F}) \cap U = \mathsf{Ass}_A(M) \subset U.$$

An associated point $x \in \mathsf{Ass}_X(\mathcal{F})$ is isolated if it is minimal in the support in the affine-local sense. The isolated associated points are precisely the generic points $\{\eta_i\}_{i\in I}$ of the irreducible components $Z_i \subset \mathsf{Supp}(\mathcal{F})$. An associated point $x \in \mathsf{Ass}_X(\mathcal{F})$ is embedded if it is not isolated, i.e. it is not a generic point of an irreducible component of the support. Equivalently, in an affine neighbourhood $U = \operatorname{Spec} A$ with $\mathcal{F}|_U \cong \widetilde{M}$, the corresponding prime $\mathfrak{q}$ is an embedded associated prime of M. Geometrically, if x is embedded, then its closure $\overline{\{x\}}$ is properly contained in an irreducible component of $\mathsf{Supp}(\mathcal{F})$. The embedded points of $\mathscr{O}_X$ are simply called the embedded points of X. An *embedded component* of X is an irreducible closed subset of the form $\overline{\{x\}}$, where $x \in \mathsf{Ass}_X(\mathscr{O}_X)$ is an embedded point.

Let $Z \subset Y$ be a locally noetherian closed subscheme of a scheme Y. Then Z has an embedded associated point if and only if there exists a dense open $U \subset Y$ such that $Z \cap U$ is topologically dense in Z but not schematically dense.

Concretely, one has to keep in mind the following situation. Let X be locally noetherian and let $p \in X$ be a closed point which is not the generic point of an irreducible component. Then p is an embedded associated point of X if and only if $X \setminus \{p\}$ is topologically dense in X but is not schematically dense, equivalently if its scheme-theoretic closure in X is a proper closed subscheme.

For instance, if $X = \operatorname{Spec} \mathbb{F}[x,y]/(xy, y^2)$, the scheme-theoretic closure of $X \setminus \{0\} \subset X$ is $\operatorname{Spec} \mathbb{F}[x,y]/y \cong \mathbb{A}^1_{\mathbb{F}} \cong X_{\mathsf{red}}$, which is not equal to X as a scheme.

B.5.4 Links with finite length modules

In this section we will see (further) links between length, associated primes, and support.

THEOREM B.5.4.1. *Let A be a noetherian ring and let M be a finitely generated A-module. The following are equivalent:*

(i) $\mathsf{length}_A(M) < \infty$,

(ii) $\mathsf{Supp}_A(M)$ *is a finite set of maximal ideals, and*

(iii) $A/\mathsf{Ann}_A(M)$ *is an artinian ring.*

Proof. Set $B = A/\mathsf{Ann}_A(M)$. Notice first that, since M is finitely generated, Exercise B.5.1.1 gives $\mathsf{Supp}_A(M) = \mathrm{V}(\mathsf{Ann}_A(M))$. Therefore there is a canonical homeomorphism

$$\operatorname{Spec} B \xrightarrow{\simeq} \mathsf{Supp}_A(M).$$

We now proceed proving the equivalences.

(i) ⇒ (ii). Assume $\mathsf{length}_A(M) < \infty$. By Proposition B.4.0.7, M admits a composition series

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

whose factors are simple. By Lemma B.4.0.6, for every i there is a maximal ideal $\mathfrak{m}_i \subset A$ such that

$$M_i/M_{i-1} \cong A/\mathfrak{m}_i.$$

Applying repeatedly the support identity of (B.4.1), we obtain

$$\mathsf{Supp}_A(M) = \bigcup_{i=1}^{n} \mathsf{Supp}_A(M_i/M_{i-1}) = \bigcup_{i=1}^{n} \mathrm{V}(\mathfrak{m}_i) = \{\mathfrak{m}_1, \ldots, \mathfrak{m}_n\}.$$

Thus $\mathsf{Supp}_A(M)$ is a finite set consisting of maximal ideals.

(ii) ⇒ (iii). By the homeomorphism $\operatorname{Spec} B \simeq \mathsf{Supp}_A(M)$, every prime ideal of B is maximal. Hence $\dim B = 0$. Moreover, B is noetherian because it is a quotient of the noetherian ring A. Therefore the standard characterisation of 0-dimensional noetherian rings, cf. **??**, gives that B is artinian.

(iii) ⇒ (i). Since B is a quotient of the noetherian ring A, it is noetherian. Thus B is both noetherian and Artinian. Consequently B, viewed as a module over itself, has finite length.

The A-module structure on M factors through B, and M is finitely generated as a B-module. Hence there is a surjection $B^{\oplus r} \twoheadrightarrow M$ for some $r \geqslant 0$. Since B has finite length, so does $B^{\oplus r}$, and therefore, by additivity of length,

$$\mathsf{length}_B(M) < \infty.$$

Finally, since $A \to B$ is surjective, the A-submodules of M are precisely its B-submodules. Hence $\mathsf{length}_A(M) = \mathsf{length}_B(M) < \infty$, as desired. □

COROLLARY B.5.4.2. *Let A be noetherian and let M be a finitely generated A-module of finite length. Then*

$$\mathsf{Ass}_A(M) = \mathsf{Supp}_A(M),$$

and the canonical map

$$M \longrightarrow \bigoplus_{\mathfrak{m} \in \mathsf{Supp}_A(M)} M_{\mathfrak{m}}$$

is an isomorphism. In particular,

$$\mathsf{length}_A(M) = \sum_{\mathfrak{m} \in \mathsf{Supp}_A(M)} \mathsf{length}_{A_{\mathfrak{m}}}(M_{\mathfrak{m}}).$$

Proof. By Theorem B.5.4.1, $\mathsf{Supp}_A(M)$ consists of finitely many maximal ideals. Since every minimal element of the support is associated, and every element of the support is now minimal, we obtain $\mathsf{Supp}_A(M) \subset \mathsf{Ass}_A(M)$. The reverse inclusion always holds.

Next we prove that $M \to \bigoplus_{\mathfrak{m}\in \mathsf{Supp}_A(M)} M_{\mathfrak{m}}$ is bijective arguing by induction on $n = \mathsf{length}_A(M)$.

If $n = 0$, then $M = 0$, so there is nothing to prove. Assume $n = 1$. Then M is simple, hence by Lemma B.4.0.6 we have $M \cong A/\mathfrak{m}$ for some maximal ideal $\mathfrak{m} \subset A$. Therefore

$$\mathsf{Supp}_A(M) = \{\mathfrak{m}\}.$$

Moreover, the canonical map $M \to M_{\mathfrak{m}}$ is an isomorphism since every element of $A \setminus \mathfrak{m}$ acts invertibly on $A/\mathfrak{m}$.

Suppose now that $n > 1$. By Proposition B.4.0.7, we can choose a submodule $N \subsetneq M$ such that $Q = M/N$ is simple. By Lemma B.4.0.6, there is a maximal ideal $\mathfrak{m}_0 \subset A$ such that $Q \cong A/\mathfrak{m}_0$. Thus we have a short exact sequence

$$0 \longrightarrow N \longrightarrow M \longrightarrow Q \longrightarrow 0.$$

Set $S = \mathsf{Supp}_A(M)$. By the behaviour of support in short exact sequences,

$$S = \mathsf{Supp}_A(N) \cup \mathsf{Supp}_A(Q).$$

Moreover, $\mathsf{Supp}_A(Q) = \{\mathfrak{m}_0\}$, and Q has length 1 by Lemma B.4.0.6(ii). It follows from (B.4.1) that $\mathsf{length}_A(N) = n - 1$, so the induction hypothesis can be applied to both Q and N.

For every $\mathfrak{m} \in S$, exactness of localisation gives an exact sequence

$$0 \longrightarrow N_{\mathfrak{m}} \longrightarrow M_{\mathfrak{m}} \longrightarrow Q_{\mathfrak{m}} \longrightarrow 0.$$

Since S is finite, taking the direct sum over $\mathfrak{m} \in S$ gives an exact sequence

$$0 \longrightarrow \bigoplus_{\mathfrak{m}\in S} N_{\mathfrak{m}} \longrightarrow \bigoplus_{\mathfrak{m}\in S} M_{\mathfrak{m}} \longrightarrow \bigoplus_{\mathfrak{m}\in S} Q_{\mathfrak{m}} \longrightarrow 0.$$

We therefore obtain a commutative diagram with exact rows

$$\begin{array}{ccccccccc}
0 & \longrightarrow & N & \longrightarrow & M & \longrightarrow & Q & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \theta_N} & & \downarrow{\scriptstyle \theta_M} & & \downarrow{\scriptstyle \theta_Q} & & \\
0 & \longrightarrow & \displaystyle\bigoplus_{\mathfrak{m}\in S} N_{\mathfrak{m}} & \longrightarrow & \displaystyle\bigoplus_{\mathfrak{m}\in S} M_{\mathfrak{m}} & \longrightarrow & \displaystyle\bigoplus_{\mathfrak{m}\in S} Q_{\mathfrak{m}} & \longrightarrow & 0.
\end{array}$$

Both outer maps are isomorphisms by induction. Induction can be applied to both N and Q because, regarding N, we have $N_{\mathfrak{m}} = 0$ whenever $\mathfrak{m} \notin \mathsf{Supp}_A(N)$, so enlarging the index set from $\mathsf{Supp}_A(N)$ to S merely adds zero summands. Regarding Q, we already observed it has only $\mathfrak{m}_0$ in its support, so the same reasoning applies. Hence the statement follows from the Five Lemma (cf. Lemma A.6.0.9). □

Definition B.5.4.3 (Length of a 0-dimensional scheme)**.** Let Z be a 0-dimensional noetherian scheme. We define the *length* of Z by

$$\mathsf{length}(Z) = \sum_{z\in Z} \mathsf{length}_{\mathcal{O}_{Z,z}}(\mathcal{O}_{Z,z}).$$

Note that the definition makes sense: the right hand side is a finite sum of positive integers: indeed, a 0-dimensional noetherian scheme has finitely many points, and each local ring $\mathscr{O}_{Z,z}$ is a 0-dimensional noetherian local ring, hence an artinian local ring.

If $Z = \operatorname{Spec} B$, then B is artinian and

$$\mathsf{length}(Z) = \mathsf{length}_B(B).$$

If Z is finite over a field $\mathbb{F}$, we have

$$\dim_{\mathbb{F}} \mathrm{H}^0(Z, \mathscr{O}_Z) = \sum_{z \in Z} [\kappa(z) : \mathbb{F}] \, \mathsf{length}_{\mathscr{O}_{Z,z}}(\mathscr{O}_{Z,z}).$$

Therefore, if Z is finite over an algebraically closed field $\mathbf{k}$, we obtain

$$\mathsf{length}(Z) = \dim_{\mathbf{k}} \mathrm{H}^0(Z, \mathscr{O}_Z).$$

B.5.5 The cycle associated to a coherent sheaf

Let X be a noetherian scheme. We denote by

$$\mathrm{Z}_*(X)$$

the free abelian group generated by the irreducible closed subsets of X (equivalently, by the integral closed subschemes of X, where an irreducible closed subset is endowed with its reduced induced scheme structure). Its elements are called *cycles* on X, and can be written as

$$\sum_{1 \leqslant i \leqslant s} m_i [Y_i], \quad m_i \in \mathbb{Z},$$

for some integer $0 \leqslant s < \infty$.

Let $\mathcal{F}$ be a coherent $\mathscr{O}_X$-module. Since X is noetherian, the closed subset $\mathsf{Supp}(\mathcal{F}) \subset X$ has finitely many irreducible components, say $Z_1, \ldots, Z_r \subset X$. Let ξ_i be the generic point of Z_i. The points ξ_i are precisely the isolated associated points of $\mathcal{F}$.

LEMMA B.5.5.1. *Let X be a noetherian scheme, let $\mathcal{F}$ be a coherent $\mathscr{O}_X$-module, and let $\xi \in \mathsf{Supp}(\mathcal{F})$ be the generic point of an irreducible component of* $\mathsf{Supp}(\mathcal{F})$*. Then*

$$\mathsf{length}_{\mathscr{O}_{X,\xi}}(\mathcal{F}_\xi) < \infty.$$

Proof. Since ξ is a minimal point of $\mathsf{Supp}(\mathcal{F})$, after localising at ξ we have

$$\mathsf{Supp}_{\mathscr{O}_{X,\xi}}(\mathcal{F}_\xi) = \{\mathfrak{m}_\xi\},$$

where $\mathfrak{m}_\xi \subset \mathscr{O}_{X,\xi}$ is the maximal ideal. Indeed, the prime ideals in the support of $\mathcal{F}_\xi$ correspond to points of $\mathsf{Supp}(\mathcal{F})$ specialising to ξ, and by minimality of ξ there are no such points other than ξ itself.

Now $\mathcal{F}_\xi$ is a finitely generated module over the noetherian local ring $\mathcal{O}_{X,\xi}$. Hence, by the finite-length criterion of Theorem B.5.4.1, we have

$$\mathsf{length}_{\mathcal{O}_{X,\xi}}(\mathcal{F}_\xi) < \infty,$$

as desired. □

Definition B.5.5.2 (Cycle associated to a coherent sheaf)**.** Let X be a noetherian scheme and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. For every irreducible component $Z \subset \mathsf{Supp}(\mathcal{F})$, with generic point ξ, define the *multiplicity of Z in $\mathcal{F}$* to be

$$m_Z(\mathcal{F}) = \mathsf{length}_{\mathcal{O}_{X,\xi}}(\mathcal{F}_\xi).$$

The *cycle associated to $\mathcal{F}$* is[4]

$$[\mathcal{F}] = \sum_{\substack{\xi \in \mathsf{Ass}_X(\mathcal{F}) \\ \xi \text{ isolated}}} m_Z(\mathcal{F})[Z],$$

where in the sum Z is related to ξ via $Z = \overline{\{\xi\}}$.

Example B.5.5.3. Set $A = \mathbb{F}[x,y]$ and $M = A/(x^2y^3)$. Then $(x^2y^3) = (x^2) \cap (y^3)$ is an irredundant primary decomposition, and thus all associated primes of M are isolated,

$$\mathsf{Min}(\mathsf{Supp}_A(M)) = \mathsf{Ass}_A(M) = \{(x),(y)\}.$$

At (x) the element y is invertible, so $M_{(x)} \cong A_{(x)}/x^2A_{(x)}$ and

$$\mathsf{length}_{A_{(x)}}(M_{(x)}) = 2.$$

Similarly,

$$\mathsf{length}_{A_{(y)}}(M_{(y)}) = 3.$$

Consider $\mathcal{F} = \widetilde{M}$. The two irreducible components have generic multiplicities 2 and 3, and we obtain the cycle

$$[\mathcal{F}] = 2[\mathrm{V}(x)] + 3[\mathrm{V}(y)] \in \mathrm{Z}_*(\mathbb{A}^2_{\mathbb{F}}).$$



Warning B.5.5.4. Only the isolated associated points of $\mathcal{F}$ contribute to the cycle $[\mathcal{F}]$. Embedded associated points are *not recorded* by $[\mathcal{F}]$.

Thus the cycle associated to a coherent sheaf records its generic multiplicity along the irreducible components of its support, but does not remember scheme-theoretic structure concentrated in smaller closed subsets.

[4] See also [10, Tag 02QV] for a more general construction.

Example B.5.5.5. Returning to Example B.5.1.17, set $A = \mathbb{F}[x,y]$ and $M = A/(x) \oplus A/(x,y)$. Consider $\mathcal{F} = \widetilde{M}$. Then

$$\mathsf{Ass}_A(M) = \{(x),(x,y)\},$$

but (x,y) is embedded. The support has the unique irreducible component $\mathrm{V}(x)$, and $M_{(x)} \cong \mathrm{Frac}(A/(x))$. Thus

$$\mathsf{length}_{A_{(x)}} M_{(x)} = 1$$

and we obtain the cycle

$$[\mathcal{F}] = [\mathrm{V}(x)] \in \mathrm{Z}_*(\mathbb{A}^2_{\mathbb{F}}).$$

In particular, the embedded summand $A/(x,y)$ is invisible to the cycle.



Warning B.5.5.6. Our definition of the cycle associated to a coherent sheaf does not satisfy the natural additivity property on short exact sequences. This is already clear from Example B.5.5.5. If one wants additivity on short exact sequences, a slighter finer theory has to be developed, see [10, Tag 02QV].

B.6 Depth and Cohen–Macaulay schemes

We recall some notions, mainly from [10, Tag 00LE].

Definition B.6.0.1 (Regular sequence [10, Tag 00LF])**.** Let M be a module over a ring A. Then a sequence $f_1,\dots,f_r$ of elements in A is called an *M-regular sequence* (of length r) if, for every i, multiplication by f_i is injective on $M/(f_1,\dots,f_{i-1})M$ and $M/(f_1,\dots,f_r)M \neq 0$. If $M = A$, we simply call it a *regular sequence*.

Definition B.6.0.2 (Depth of a module [10, Tag 00LI])**.** Let M be a finitely generated module over a ring A, and let $I \subset A$ be an ideal. If $IM \neq M$, define the *I-depth* of M as

$$\mathrm{depth}_I(M) = \sup_r \left\{ r \;\middle|\; \text{there exists an } M\text{-regular sequence } f_1,\dots,f_r \in I \right\}.$$

If $IM = M$, define $\mathrm{depth}_I(M) = +\infty$. In particular, $\mathrm{depth}_A(M) = +\infty$ for all M.

If $(A,\mathfrak{m})$ is a local ring, we call $\mathrm{depth}_{\mathfrak{m}}(M)$ simply the *depth* of M.

If A is noetherian and $IM \neq M$, then $\mathrm{depth}_I(M) < +\infty$, see [10, Tag 0AUJ].

The following list of facts reflects the fairly good behaviour of depth for finite modules over *local noetherian* rings. Also note that item (ii) connects to the theory of associated points. Since depth is used to define Cohen–Macaulay schemes, it will not be surprising when we will see that Cohen–Macaulay schemes can be described via embedded points, see Lemma B.6.0.9.

Fact B.6.0.3. Assume $(A,\mathfrak{m})$ is noetherian and local, and $M \neq 0$ is finitely generated. Then, the following statements hold true.

(i) By [10, Tag 00LW], one has

$$\operatorname{depth}_{\mathfrak{m}}(M) = \min_i \left\{ i \in \mathbb{N} \,\middle|\, \operatorname{Ext}^i(A/\mathfrak{m}, M) \neq 0 \right\}.$$

(ii) By [10, Tag 0BK4], one has

$$\operatorname{depth}_{\mathfrak{m}}(M) \leqslant \dim A/\mathfrak{p}, \qquad \text{for all } \mathfrak{p} \in \operatorname{Ass}_A(M).$$

(iii) For an arbitrary prime $\mathfrak{p} \in \operatorname{Spec} A$, by [10, Tag 0FCC] one has

$$\operatorname{depth}_{\mathfrak{m}}(M) \leqslant \dim A/\mathfrak{p} + \operatorname{depth}_{\mathfrak{p}A_{\mathfrak{p}}}(M_{\mathfrak{p}}).$$

(iv) By [10, Tag 00LK], one has

$$\dim \operatorname{Supp}_A(M) \geqslant \operatorname{depth}_{\mathfrak{m}}(M).$$



Exercise B.6.0.4. Let $(A, \mathfrak{m}, \kappa)$ be a local noetherian ring and $M \neq 0$ a finitely generated A-module. Prove that the following conditions are equivalent:

(i) $\mathfrak{m} \in \operatorname{Ass}_A(M)$

(ii) $\operatorname{Hom}_A(\kappa, M) \neq 0$, and

(iii) $\operatorname{depth}_{\mathfrak{m}}(M) = 0$.

The A-module $\operatorname{Hom}_A(\kappa, M)$ is called the *socle* of M.

Definition B.6.0.5 (Cohen–Macaulay module)**.** A finitely generated module $M \neq 0$ over a noetherian local ring $(A, \mathfrak{m})$ is said to be *Cohen–Macaulay* if $\dim \operatorname{Supp}_A(M) = \operatorname{depth}_{\mathfrak{m}}(M)$. A noetherian local ring $(A, \mathfrak{m})$ is said to be *Cohen–Macaulay* if it is Cohen–Macaulay as a module over itself.

Definition B.6.0.6 (Cohen–Macaulay scheme)**.** Let X be a locally noetherian scheme. We say that X is *Cohen–Macaulay* if for every $x \in X$ the local ring $\mathcal{O}_{X,x}$ is Cohen–Macaulay, i.e. if

$$\operatorname{depth}_{\mathfrak{m}_x}(\mathcal{O}_{X,x}) = \dim \mathcal{O}_{X,x}.$$

Example B.6.0.7. Consider the rings

$$A_1 = \mathbb{F}[x, y]/(xy), \qquad A_2 = \mathbb{F}[x, y]/(x^2), \qquad A_3 = \mathbb{F}[x, y]/(x^2, xy).$$

Set $X_i = \operatorname{Spec} A_i$. Then X_1 is reduced Cohen–Macaulay with two isolated associated primes, X_2 is nonreduced Cohen–Macaulay with one isolated associated prime, and finally X_3 has the embedded prime (x, y) and is not Cohen–Macaulay.

Lastly, we recall Serre's conditions and basic results from [10, Tag 033P]. Let $k \in \mathbb{Z}_{\geqslant 0}$ be a nonnegative integer. We say that a locally noetherian scheme X satisfies

(i) *Serre's condition* S_k if for every point $x \in X$, we have

$$\operatorname{depth}_{\mathfrak{m}_x}(\mathscr{O}_{X,x}) \geqslant \min(k, \dim \mathscr{O}_{X,x}).$$

(ii) condition R_k if for every point $x \in X$ we have that $\dim \mathscr{O}_{X,x} \leqslant k$ implies that $\mathscr{O}_{X,x}$ is regular.

LEMMA B.6.0.8 ([10, Tag 033P]). *Let X be a locally noetherian scheme. Then,*

(a) *X is regular if and only if it satisfies R_k for every $k \geqslant 0$.*

(b) *X is Cohen–Macaulay if and only if it satisfies S_k for every $k \geqslant 0$.*

(c) *X is reduced if and only if it satisfies S_1 and R_0.*

(d) *X is normal if and only if it satisfies S_2 and R_1.*

The notion of embedded point connects to the notion of Cohen–Macaulay scheme through the following result.

LEMMA B.6.0.9 ([10, Tag 05AJ]). *Let X be a locally noetherian scheme. Then,*

(i) *X has no embedded points if and only if it satisfies Serre's condition S_1.*

(ii) *If $\dim X \leqslant 1$, then X is Cohen–Macaulay if and only if it has no embedded points.*

(iii) *Let $U \subset X$ be an open subscheme. Then, the following are equivalent:*

- *For every open subscheme $V \subset X$, the scheme-theoretic closure of $U \cap V$ in V is equal to V,*[5]
- *U is dense in X and U contains all embedded points of X, and*
- *$\operatorname{Ass}_X(\mathscr{O}_X) \subset U$.*

We propose part of the previous lemma as an exercise.



Exercise B.6.0.10. Let X be a locally noetherian scheme and $j\colon U \hookrightarrow X$ an open immersion. Prove that $\mathscr{O}_X \to j_*\mathscr{O}_U$ is injective if and only if $\operatorname{Ass}_X(\mathscr{O}_X) \subset U$.

We close this section with a couple of useful remarks.

Remark B.6.0.11 (Reduced implies no embedded points, II)**.** Let X be a reduced locally noetherian scheme. Then X has no embedded points. This follows combining Lemma B.6.0.9(i) and Lemma B.6.0.8(c) or, more simply, from the affine-local description of $\operatorname{Ass}_X(\mathscr{O}_X)$ combined with Exercise B.5.1.15.

[5] This condition is saying that U is *schematically dense* in X, see [10, Tag 01RB].

Remark B.6.0.12 (Cohen–Macaulay does not imply reduced)**.** By Lemma B.6.0.9(ii), we know that a reduced 1-dimensional locally noetherian scheme is Cohen–Macaulay. However, there can be nonreduced Cohen–Macaulay curves: for example, curves with a nonreduced component, such as the affine plane curve $\operatorname{Spec}\mathbb{F}[x,y]/x^2 \subset \mathbb{A}^2$, to be visualised as a "double y-axis". In this example, one indeed has $\mathsf{Ass}_A(A) = \{(\overline{x})\}$, where $A = \mathbb{F}[x,y]/x^2$.

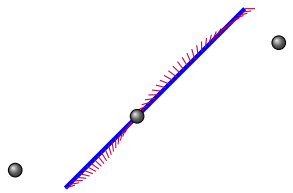


Figure B.1: A thickened (non-Cohen–Macaulay) curve with an embedded point and two isolated (possibly fat) points.

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